```
# ==== run.py ====
# -*- coding: utf-8 -*-
"""系统启动入口。
运行：
    python run.py                启动并自动打开浏览器
    python run.py --no-browser   只启动服务，不打开浏览器
    python run.py --dev          开发模式（开启代码热重载与调试器）
    python run.py --port 5001    指定端口
默认以 127.0.0.1:5000 启动，就绪后自动打开浏览器。
"""
import os
import sys
import threading
import time
import webbrowser
if hasattr(sys.stdout, "reconfigure"):
    sys.stdout.reconfigure(encoding="utf-8")   # 控制台中文正常显示
from app import create_app
from app.models import db
PORT = 5000
for i, a in enumerate(sys.argv):
    if a == "--port" and i + 1 < len(sys.argv):
        PORT = int(sys.argv[i + 1])
URL = "http://127.0.0.1:%d" % PORT
NO_BROWSER = "--no-browser" in sys.argv
DEV = "--dev" in sys.argv
app = create_app()
def open_browser_later(delay=1.2):
    """等服务工作起来再打开浏览器，避免打开时页面还没就绪。
    放在后台线程里，不用 sleep 阻塞服务启动；线程设为守护线程，
    用户 Ctrl+C 停止服务时不会残留。
    """
    def run():
        time.sleep(delay)
        try:
            webbrowser.open(URL)
        except Exception:
            pass            # 打开失败不影响服务本身
    t = threading.Thread(target=run, daemon=True)
    t.start()
if __name__ == "__main__":
    with app.app_context():
        db.create_all()     # 首次运行时自动建表
        from app.utils.schema import ensure_schema
        added = ensure_schema(db)     # 旧库补齐新增字段
        if added:
            print("已补充字段：%s" % "、".join(added))
    print("=" * 58)
    print("   历史遗留废弃矿山自然恢复适宜性评价系统  Ｖ１．０")
    print("=" * 58)
```

```
    print("   访问地址：%s" % URL)
    print("   演示账号：admin      密码：admin123")
    print("   停止运行：按 Ctrl+C，或直接关闭本窗口")
    print("=" * 58)
    if not NO_BROWSER and not DEV:
        print("   正在打开浏览器……")
    print()
    # 开发模式才启用热重载（重载会二次启动进程，故此时不自动开浏览器）
    if not NO_BROWSER and not DEV:
        open_browser_later()
    app.run(host="127.0.0.1", port=PORT, debug=DEV,
    use_reloader=DEV)
# ==== app/__init__.py ====
# -*- coding: utf-8 -*-
"""Flask 应用工厂。"""
from datetime import date
from flask import Flask
from flask_login import LoginManager
from app.config import Config
from app.models import db, User
from app.utils.constants import ROLE_LABELS
login_manager = LoginManager()
login_manager.login_view = "auth.login"
login_manager.login_message = "请先登录"
@login_manager.user_loader
def load_user(user_id):
    return db.session.get(User, int(user_id))
def create_app(config_class=Config):
    app = Flask(__name__)
    app.config.from_object(config_class)
    db.init_app(app)
    login_manager.init_app(app)
    # 注册视图（各模块路由）
    from app.views import auth, project, unit, standard,
    constraint, \
        ahp, indicator, evaluate, report, search
    app.register_blueprint(auth.bp)
    app.register_blueprint(project.bp)
    app.register_blueprint(unit.bp)
    app.register_blueprint(standard.bp)
    app.register_blueprint(constraint.bp)
    app.register_blueprint(ahp.bp)
    app.register_blueprint(indicator.bp)
    app.register_blueprint(evaluate.bp)
    app.register_blueprint(report.bp)
    app.register_blueprint(search.bp)
    # 全局登录保护：除登录页与静态资源外，均需登录
    from flask import request, redirect, url_for,
    send_from_directory
    from flask_login import current_user, login_user
```

```
PUBLIC_ENDPOINTS = {"auth.login", "static", "favicon"}
#  浏览器默认会请求根路径下的 /favicon.ico。
#  页面里虽已用 <link rel="icon"> 指向 static 目录，但出错页等场景仍会
#  回退到这个路径，补一条路由免得日志里一直刷 404。
@app.route("/favicon.ico")
def favicon():
    return send_from_directory(app.static_folder,
"favicon.ico",
                               mimetype="image/vnd.micro
soft.icon")
@app.before_request
def require_login():
    #  截图模式（仅用于生成说明书配图）：自动以管理员身份登录
    if app.config.get("SCREENSHOT_MODE"):
        if (request.endpoint not in ("auth.login",
"static")
                and not current_user.is_authenticated):
            admin = User.query.filter_by(role="admin").f
irst()
            if admin:
                login_user(admin)
        return None
    if request.endpoint in PUBLIC_ENDPOINTS or
request.endpoint is None:
        return None
    if not current_user.is_authenticated:
        return redirect(url_for("auth.login"))
    return None
#  侧边栏导航：按论文评价流程的顺序编排，分为「评价流程」与「系统管理」两组。
#  不使用模块编号，避免出现与实际流程不一致的代号。
#  每项带图标名（内联 SVG，见 templates/_macros.html 的 ICONS）
#  与可选的待办角标（该步骤还没做完的图斑数）。
FLOW_LINKS = [
    ("项目与矿山档案", "/project/", "folder"),
    ("政策与标准知识库", "/standard/", "book"),
    ("限制性指标判定", "/constraint/", "shield"),
    ("微地质单元", "/unit/", "layers"),
    ("指标体系与权重", "/ahp/", "sliders"),
    ("指标数据采集", "/indicator/", "database"),
    ("适宜性定量评价", "/evaluate/", "target"),
    ("报告与结果输出", "/report/", "file"),
]
ADMIN_LINKS = [
    ("个人设置", "/profile", "user"),
    ("用户与操作日志", "/users", "users"),
]
@app.context_processor
def inject_nav():
    path = request.path or "/"
    def hit(prefix):
```

```
            """当前路径是否属于某个入口（/unit/ 也匹配 /unit/3）。"""
            return path.startswith(prefix.rstrip("/"))
      badges = {}
      if current_user.is_authenticated:
            from app.services import dashboard as dash
            badges = dash.nav_badges()
      def build(links):
            return [{"label": label, "href": href, "icon": icon,
                     "active": hit(href), "badge": badges.get(href, 0)}
                    for label, href, icon in links]
      groups = [
            {"title": "评价流程", "links": build(FLOW_LINKS)},
            {"title": "系统管理", "links": build(ADMIN_LINKS)},
      ]
      #  面包屑用的当前页名：按路径命中的入口取，未命中时给个通用名
      if path == "/":
            page_title = "工作台"
      elif path.startswith("/profile"):
            page_title = "个人设置"
      elif path.startswith(("/users", "/logs")):
            page_title = "用户与操作日志"
      elif path.startswith("/search"):
            page_title = "搜索结果"
      else:
            page_title = "工作台"
            for label, href, _icon in FLOW_LINKS:
                  if hit(href):
                        page_title = label
                        break
      #  层级引导用：项目是归档单位，图斑才是评价对象。
      #  下游各页面按图斑组织数据，因此需要提示「哪些项目下还没有图斑」。
      from app.models import Project, MinePatch
      projects = Project.query.order_by(Project.id).all()
      return {
            "nav_groups": groups,
            "home_active": path == "/",
            "page_title": page_title,
            "today": date.today().strftime("%Y-%m-%d"),
            "search_q": request.args.get("q", ""),
            "role_labels": ROLE_LABELS,
            "project_count": len(projects),
            "patch_count": MinePatch.query.count(),
            "projects_without_patch": [p for p in projects
                                       if p.patches.count() == 0],
      }
#  操作审计：统一在响应阶段记录成功的写操作（操作日志）
from app.services import audit
```

```
    @app.after_request                                                  201
    def audit_writes(response):                                         202
        if request.method in ("POST", "PUT", "PATCH",                   203
    "DELETE"):                                                          204
            audit.record(request.endpoint,                              205
    response.status_code)                                               206
        return response                                                 207
    # 错误页：给出可读提示，而不是默认的英文错误页                      208
    from flask import render_template                                   209
    @app.errorhandler(403)                                              210
    def forbidden(e):                                                   211
        return render_template("error.html", code=403,                  212
                               message="没有权限执行该操作"), 403       213
    @app.errorhandler(404)                                              214
    def not_found(e):                                                   215
        return render_template("error.html", code=404,                  216
                               message="请求的页面不存在"), 404         217
    return app                                                          218
# ==== app/config.py ====                                             219
# -*- coding: utf-8 -*-                                               220
"""应用配置。"""                                                        221
import os                                                               222
from datetime import timedelta                                          223
BASE_DIR = os.path.abspath(os.path.dirname(__file__))                   224
class Config:                                                           225
    SECRET_KEY = os.environ.get("SECRET_KEY",                           226
    "mine-restoration-dev-key")                                         227
    # 单机部署，SQLite 数据库文件放在应用目录下                         228
    SQLALCHEMY_DATABASE_URI = "sqlite:///" +                            229
    os.path.join(BASE_DIR, "app.db")                                    230
    SQLALCHEMY_TRACK_MODIFICATIONS = False                              231
    # 禁止跨站请求伪造（CSRF），登录与表单提交依赖此项                  232
    WTF_CSRF_ENABLED = True                                             233
    # 「记住我」：登录时勾选则下发持久 cookie，关闭浏览器后仍可自动登录 234
    REMEMBER_COOKIE_DURATION = timedelta(days=30)                       235
    REMEMBER_COOKIE_HTTPONLY = True                                     236
    REMEMBER_COOKIE_SAMESITE = "Lax"                                    237
    # 影像与数据文件上传（遥感影像挂接、批量导入）                     238
    UPLOAD_FOLDER = os.path.join(BASE_DIR, "static",                    239
    "uploads")                                                          240
    MAX_CONTENT_LENGTH = 32 * 1024 * 1024          # 单次上传上限       241
    32 MB                                                               242
    ALLOWED_IMAGE_EXT = {"png", "jpg", "jpeg", "gif",                   243
    "webp", "bmp", "tif", "tiff"}                                       244
    ALLOWED_DATA_EXT = {"csv", "xlsx", "xls"}                           245
    # 头像只用浏览器能直接显示的格式（不含 bmp / tif），并单独限制体积 246
    ALLOWED_AVATAR_EXT = {"png", "jpg", "jpeg", "gif",                  247
    "webp"}                                                             248
    MAX_AVATAR_LENGTH = 4 * 1024 * 1024             # 4 MB              249
# ==== app/models.py ====                                             250
```

```
# -*- coding: utf-8 -*-
"""数据模型：系统全部数据表。
本文件由甲在阶段 0 一次性写完，作为两人独立开发的地基。
约定：两人只读此文件；确需改表时，由乙提需求、甲统一修改。
"""
import json
from datetime import datetime
from flask_login import UserMixin
from flask_sqlalchemy import SQLAlchemy
db = SQLAlchemy()
class User(UserMixin, db.Model):
    """用户（用户与权限管理）。"""
    __tablename__ = "users"
    id = db.Column(db.Integer, primary_key=True)
    username = db.Column(db.String(64), unique=True,
    nullable=False, index=True)
    password_hash = db.Column(db.String(256),
    nullable=False)
    # 角色：管理员 admin / 评价员 evaluator / 查看者 viewer
    role = db.Column(db.String(20), nullable=False,
    default="viewer")
    active = db.Column(db.Boolean, nullable=False,
    default=True)
    created_at = db.Column(db.DateTime,
    default=datetime.utcnow)
    # ---- 个人资料（用户可在「个人设置」中自行维护）----
    display_name = db.Column(db.String(64), nullable=True)
    # 显示名称（姓名）
    title = db.Column(db.String(64), nullable=True)
    # 职务 / 岗位
    email = db.Column(db.String(128), nullable=True)
    phone = db.Column(db.String(32), nullable=True)
    avatar = db.Column(db.String(256), nullable=True)
    # 头像，相对 static 的路径
    @property
    def shown_name(self):
        """界面上显示的名字：优先用显示名称，没填就用账号名。"""
        return self.display_name or self.username
    @property
    def initial(self):
        """没有头像时用首字母占位。"""
        return (self.shown_name or "?")[0].upper()
    def to_dict(self):
        return {"id": self.id, "username": self.username,
                "display_name": self.display_name,
                "role": self.role, "active": self.active}
class OperationLog(db.Model):
    """操作日志。"""
    __tablename__ = "operation_logs"
    id = db.Column(db.Integer, primary_key=True)
```

```
    user_id = db.Column(db.Integer, db.ForeignKey("users.id"
    ), nullable=True)
    action = db.Column(db.String(128), nullable=False)
    # 操作类型
    target_type = db.Column(db.String(64), nullable=True)
    # 对象类型
    target_id = db.Column(db.Integer, nullable=True)
    # 对象主键
    detail = db.Column(db.Text, nullable=True)
    # 操作详情
    created_at = db.Column(db.DateTime,
    default=datetime.utcnow, index=True)
class Project(db.Model):
    """评价项目（项目与矿山档案）。"""
    __tablename__ = "projects"
    id = db.Column(db.Integer, primary_key=True)
    name = db.Column(db.String(128), nullable=False)
    code = db.Column(db.String(64), nullable=True)
    # 项目编号
    description = db.Column(db.Text, nullable=True)
    status = db.Column(db.String(20), default="进行中")
    # 进行中 / 已完成 / 已归档
    created_by = db.Column(db.Integer,
    db.ForeignKey("users.id"), nullable=True)
    created_at = db.Column(db.DateTime,
    default=datetime.utcnow)
    patches = db.relationship("MinePatch",
    backref="project", lazy="dynamic",
                              cascade="all, delete-orphan")
class MinePatch(db.Model):
    """矿山图斑。"""
    __tablename__ = "mine_patches"
    id = db.Column(db.Integer, primary_key=True)
    project_id = db.Column(db.Integer,
    db.ForeignKey("projects.id"), nullable=False)
    name = db.Column(db.String(128), nullable=False)
    admin_region = db.Column(db.String(128), nullable=True)
    # 所在行政区
    center_x = db.Column(db.Float, nullable=True)
    # 中心经度 / 横坐标
    center_y = db.Column(db.Float, nullable=True)
    # 中心纬度 / 纵坐标
    area_ha = db.Column(db.Float, nullable=True)
    # 图斑面积（公顷）
    mineral_type = db.Column(db.String(64), nullable=True)
    # 开采矿种
    mining_method = db.Column(db.String(64), nullable=True)
    # 开采方式
    stop_years = db.Column(db.Float, nullable=True)
    # 停产年限
```

```
    responsible_entity_missing = db.Column(db.Boolean,
    default=True)   # 责任主体是否缺失
    # 遥感底图 / 无人机正射影像挂接
    image_path = db.Column(db.String(256), nullable=True)
    # 相对 static 的路径
    image_note = db.Column(db.String(256), nullable=True)
    # 影像说明（如来源、时相）
    image_uploaded_at = db.Column(db.DateTime,
    nullable=True)
    created_at = db.Column(db.DateTime,
    default=datetime.utcnow)
    units = db.relationship("MicroUnit", backref="patch",
    lazy="dynamic",
                            cascade="all, delete-orphan")
class ReferenceEcosystem(db.Model):
    """参照生态系统：标准化基准值，与图斑绑定。"""
    __tablename__ = "reference_ecosystems"
    id = db.Column(db.Integer, primary_key=True)
    patch_id = db.Column(db.Integer, db.ForeignKey("mine_pat
    ches.id"), nullable=False)
    coverage_benchmark = db.Column(db.Float, default=75.0)
    # 植被覆盖度基准 %
    shannon_benchmark = db.Column(db.Float, default=0.7)
    # Shannon-Wiener 基准
    dominant_species = db.Column(db.String(256),
    nullable=True)   # 建群种
    soil_benchmark = db.Column(db.String(256),
    nullable=True)     # 土壤基准
class Standard(db.Model):
    """地方标准 / 政策文件（政策与标准知识库）。"""
    __tablename__ = "standards"
    id = db.Column(db.Integer, primary_key=True)
    province = db.Column(db.String(32), nullable=False)
    # 省份
    name = db.Column(db.String(256), nullable=False)
    # 名称
    doc_number = db.Column(db.String(128), nullable=True)
    # 标准编号 / 文号
    publish_date = db.Column(db.String(32), nullable=True)
    implement_date = db.Column(db.String(32), nullable=True)
    # 类别：地方标准 / 政策文件
    category = db.Column(db.String(32), nullable=False,
    default="地方标准")
    summary = db.Column(db.Text, nullable=True)
    # 特点概述
    rules = db.relationship("StandardRule",
    backref="standard",
                            cascade="all, delete-orphan")
class StandardRule(db.Model):
    """结构化规则：将标准条文拆解为可执行规则。"""
```

```
    __tablename__ = "standard_rules"                                    401
    id = db.Column(db.Integer, primary_key=True)                        402
    standard_id = db.Column(db.Integer,                                 403
    db.ForeignKey("standards.id"), nullable=False)                      404
    item = db.Column(db.String(128), nullable=False)                    405
    # 判定项                                                            406
    operator = db.Column(db.String(8), nullable=False)                  407
    # 比较符：> >= < <= ==                                              408
    threshold = db.Column(db.String(64), nullable=True)                 409
    # 阈值                                                              410
    condition_type = db.Column(db.String(32),                           411
    default="必要条件")  # 必要条件 / 充分条件                          412
    region = db.Column(db.String(64), nullable=True)                    413
    # 适用分区（宁夏分区差异化）                                       414
    note = db.Column(db.Text, nullable=True)                            415
class ConstraintCheck(db.Model):                                        416
    """限制性判定记录。"""                                              417
    __tablename__ = "constraint_checks"                                 418
    id = db.Column(db.Integer, primary_key=True)                        419
    patch_id = db.Column(db.Integer, db.ForeignKey("mine_pat            420
    ches.id"), nullable=False)                                          421
    policy_pass = db.Column(db.Boolean, nullable=True)                  422
    # 政策约束                                                          423
    location_pass = db.Column(db.Boolean, nullable=True)                424
    # 区位约束                                                          425
    pollution_pass = db.Column(db.Boolean, nullable=True)               426
    # 土壤污染                                                          427
    geo_pass = db.Column(db.Boolean, nullable=True)                     428
    # 地质灾害                                                          429
    farmland_pass = db.Column(db.Boolean, nullable=True)                430
    # 耕地属性                                                          431
    passed = db.Column(db.Boolean, nullable=True)                       432
    # 总体是否通过                                                      433
    reason = db.Column(db.Text, nullable=True)                          434
    # 不通过原因                                                        435
    checked_at = db.Column(db.DateTime,                                 436
    default=datetime.utcnow)                                            437
class MicroUnit(db.Model):                                              438
    """微地质单元：岩质边坡 / 碎石边坡 / 平台。"""                      439
    __tablename__ = "micro_units"                                       440
    id = db.Column(db.Integer, primary_key=True)                        441
    patch_id = db.Column(db.Integer, db.ForeignKey("mine_pat            442
    ches.id"), nullable=False)                                          443
    unit_type = db.Column(db.String(32), nullable=False)                444
    # 单元类型                                                          445
    area = db.Column(db.Float, nullable=True)                           446
    # 面积                                                              447
    ratio = db.Column(db.Float, nullable=True)                          448
    # 面积占比（自动计算）                                              449
    description = db.Column(db.Text, nullable=True)                     450
```

```
    #  现场描述
class Indicator(db.Model):
    """指标定义：指标树，含角色与启用单元类型。"""
    __tablename__ = "indicators"
    id = db.Column(db.Integer, primary_key=True)
    code = db.Column(db.String(32), unique=True,
    nullable=False)   #  指标编码
    name = db.Column(db.String(64), nullable=False)
    #  指标名称
    level = db.Column(db.Integer, nullable=False)
    #  1/2/3 级
    parent_code = db.Column(db.String(32), nullable=True)
    #  父指标编码
    #  角色：土壤污染指标同时作为限制性指标与评价指标
    is_constraint = db.Column(db.Boolean, default=False)
    is_evaluative = db.Column(db.Boolean, default=True)
    unit = db.Column(db.String(32), nullable=True)
    #  单位
    ideal_value = db.Column(db.String(64), nullable=True)
    #  理想值（可配置）
    ideal_type = db.Column(db.String(32), nullable=True)
    #  benefit/cost/mid/discrete
    #  启用该指标的微地质单元（逗号分隔）。岩质边坡不启用土壤指标、平台不启用地形指标
    enabled_units = db.Column(db.String(128), nullable=True)
    description = db.Column(db.Text, nullable=True)
    #  取值依据说明
    def units_list(self):
        return [s for s in (self.enabled_units or
    "").split(",") if s]
class WeightScheme(db.Model):
    """权重方案：按微地质单元类型分别保存。"""
    __tablename__ = "weight_schemes"
    id = db.Column(db.Integer, primary_key=True)
    unit_type = db.Column(db.String(32), nullable=False)
    #  所属单元类型
    name = db.Column(db.String(64), nullable=True)
    #  方案名称
    level = db.Column(db.Integer, nullable=False,
    default=1)         #  1=一级指标  2=二级指标
    is_default = db.Column(db.Boolean, default=False)
    #  是否默认（论文表 3）
    created_at = db.Column(db.DateTime,
    default=datetime.utcnow)
class WeightItem(db.Model):
    """权重明细：方案内的各项权重。"""
    __tablename__ = "weight_items"
    id = db.Column(db.Integer, primary_key=True)
    scheme_id = db.Column(db.Integer,
    db.ForeignKey("weight_schemes.id"), nullable=False)
    indicator_code = db.Column(db.String(32),
```

```
    nullable=False)        #  指标编码                                    501
    weight = db.Column(db.Float, nullable=False)                          502
    #  权重值                                                             503
class ExpertMatrix(db.Model):                                             504
    """专家判断矩阵：多专家打分的原始矩阵。"""                              505
    __tablename__ = "expert_matrices"                                     506
    id = db.Column(db.Integer, primary_key=True)                          507
    scheme_id = db.Column(db.Integer,                                     508
    db.ForeignKey("weight_schemes.id"), nullable=True)                    509
    #  直接标注适用对象，便于无需先建权重方案即可录入专家矩阵             510
    unit_type = db.Column(db.String(32), nullable=True)                   511
    #  微地质单元类型                                                     512
    l1_name = db.Column(db.String(64), nullable=True)                     513
    #  所属一级指标（二级矩阵用）                                         514
    expert_name = db.Column(db.String(64), nullable=True)                 515
    #  专家姓名                                                           516
    level = db.Column(db.Integer, nullable=False,                         517
    default=1)          #  矩阵对应指标层级                               518
    matrix_json = db.Column(db.Text, nullable=False)                      519
    #  判断矩阵（ＪＳＯＮ 二维数组）                                      520
    cr = db.Column(db.Float, nullable=True)                               521
    #  该专家矩阵的一致性比例                                             522
    created_at = db.Column(db.DateTime,                                   523
    default=datetime.utcnow)                                              524
class IndicatorData(db.Model):                                            525
    """指标实测数据：按微地质单元组织。"""                                  526
    __tablename__ = "indicator_data"                                      527
    id = db.Column(db.Integer, primary_key=True)                          528
    unit_id = db.Column(db.Integer, db.ForeignKey("micro_uni              529
    ts.id"), nullable=False, unique=True)                                 530
    soil_ratio = db.Column(db.Float, nullable=True)                       531
    #  土石比                                                             532
    thickness = db.Column(db.Float, nullable=True)                        533
    #  土层厚度  cm                                                       534
    ph = db.Column(db.Float, nullable=True)                               535
    organic_matter = db.Column(db.Float, nullable=True)                   536
    #  有机质  g/kg                                                       537
    #  重金属含量，ＪＳＯＮ 字符串：{"Hg": 0.05, ...}，未检测项缺失或为 null 538
    heavy_metals = db.Column(db.Text, nullable=True)                      539
    coverage = db.Column(db.Float, nullable=True)                         540
    #  植被覆盖度  %                                                      541
    shannon = db.Column(db.Float, nullable=True)                          542
    #  Shannon-Wiener 多样性指数                                          543
    slope = db.Column(db.Float, nullable=True)                            544
    #  坡度  °                                                            545
    aspect = db.Column(db.String(16), nullable=True)                      546
    #  坡向：阳坡/阴阳坡/阴坡                                             547
    def heavy_metals_dict(self):                                          548
        return json.loads(self.heavy_metals) if                           549
    self.heavy_metals else {}                                             550
```

```
class EvaluationResult(db.Model):
    """评价结果。"""
    __tablename__ = "evaluation_results"
    id = db.Column(db.Integer, primary_key=True)
    patch_id = db.Column(db.Integer, db.ForeignKey("mine_pat
    ches.id"), nullable=False)
    final_score = db.Column(db.Float, nullable=True)
    # 最终评价因子 F
    grade = db.Column(db.String(32), nullable=True)
    # 适宜性等级
    measures = db.Column(db.Text, nullable=True)
    # 推荐措施
    unit_scores = db.Column(db.Text, nullable=True)
    # 各单元得分 JSON
    created_at = db.Column(db.DateTime,
    default=datetime.utcnow)
class ScoreDetail(db.Model):
    """得分明细：每一层指标的权重、标准化值、加权贡献，供追溯与复核。"""
    __tablename__ = "score_details"
    id = db.Column(db.Integer, primary_key=True)
    result_id = db.Column(db.Integer,
    db.ForeignKey("evaluation_results.id"), nullable=False)
    unit_id = db.Column(db.Integer, nullable=True)
    l1_name = db.Column(db.String(64), nullable=True)
    # 一级指标
    l2_name = db.Column(db.String(64), nullable=True)
    # 二级指标
    raw_value = db.Column(db.String(64), nullable=True)
    # 实测值
    std_value = db.Column(db.Float, nullable=True)
    # 标准化值
    weight = db.Column(db.Float, nullable=True)
    # 权重
    contribution = db.Column(db.Float, nullable=True)
    # 加权贡献
# ==== app/forms.py ====
# -*- coding: utf-8 -*-
"""表单定义（共享，由甲统一维护）。
各模块的 Flask-WTF 表单集中于此。
"""
import re
from flask_wtf import FlaskForm
from wtforms import SelectField, SubmitField, StringField,
    PasswordField, BooleanField
from wtforms.validators import (DataRequired, Length,
    Optional, EqualTo,
                                ValidationError)
from app.utils.constants import UNIT_TYPES, ASPECT_TYPES
# 邮箱格式：够用的轻量校验。
# WTForms 自带的 Email() 需要额外安装 email_validator，
```

```
# 为一个字段引入依赖不划算，这里自己用正则判断。
EMAIL_RE = re.compile(r"^[^@\s]+@[^@\s]+\.[A-Za-z]{2,}$")
def email_format(form, field):
    v = (field.data or "").strip()
    if v and not EMAIL_RE.match(v):
        raise ValidationError("邮箱格式不正确")
class AHPSelectForm(FlaskForm):
    """选择微地质单元类型与指标层级，以确定判断矩阵规模。"""
    unit_type = SelectField("微地质单元类型",
    validators=[DataRequired()])
    level = SelectField("指标层级",
                        choices=[("1", "一级指标"), ("2",
    "二级指标")],
                        validators=[DataRequired()])
    l1 = SelectField("一级指标", validators=[],
    validate_choice=False)  # 动态填充，层级=2 时使用
    submit = SubmitField("生成判断矩阵")
    def __init__(self, *args, **kwargs):
        super().__init__(*args, **kwargs)
        self.unit_type.choices = [(t, t) for t in
    UNIT_TYPES]
class EmptyForm(FlaskForm):
    """仅用于承载 CSRF 令牌的空表单（矩阵录入等自定义表单使用）。"""
    pass
class LoginForm(FlaskForm):
    """登录表单。支持「记住我」以在关闭浏览器后保持登录状态。"""
    username = StringField("用户名", validators=[DataRequired()
    , Length(max=64)])
    password = PasswordField("密码", validators=[DataRequired(
    )])
    remember = BooleanField("记住我")
    submit = SubmitField("登录")
class ProfileForm(FlaskForm):
    """个人资料：显示名称、职务与联系方式。字段都可留空。"""
    display_name = StringField("显示名称",
    validators=[Optional(), Length(max=64)])
    title = StringField("职务", validators=[Optional(),
    Length(max=64)])
    email = StringField("邮箱", validators=[Optional(),
    email_format,
                                            Length(max=128)])
    phone = StringField("手机号", validators=[Optional(),
    Length(max=32)])
    submit_info = SubmitField("保存资料")
class PasswordForm(FlaskForm):
    """修改密码：需要先验证当前密码。"""
    old_password = PasswordField("当前密码",
    validators=[DataRequired()])
    new_password = PasswordField("新密码",
                                 validators=[DataRequired(),
```

```
        Length(min=6, max=64,                                                   651
                                                                                652
        message="新密码至少 6 位")])                                             653
        confirm = PasswordField("确认新密码",                                    654
                                validators=[DataRequired(),                     655
                                                                                656
        EqualTo("new_password",                                                 657
                                                                                658
        message="两次输入的新密码不一致")])                                      659
        submit_pwd = SubmitField("修改密码")                                    660
# ==== app/utils/constants.py ====                                              661
# -*- coding: utf-8 -*-                                                         662
"""评价指标常量表。                                                             663
全部数值取自论文，作为系统计算的唯一基准：                                      664
- 论文 4.2.2 节：指标标准化理想极值与取值依据                                   665
- 论文 4.2.3 节 / 表 4：适宜性等级划分                                          666
- 论文表 3：按微地质单元分别确定的一级、二级指标权重                            667
- 论文 4.2.1 节：平均随机一致性指标 RI                                          668
"""                                                                             669
# ------------------------------------------------------------------------      670
        ------------------                                                      671
# 一、微地质单元类型                                                            672
# ------------------------------------------------------------------------      673
        ------------------                                                      674
# 论文 4.1 节："按照地形地貌划分，将其划分为平台、碎石边坡和岩质边坡              675
# 三种地质微单元类型"                                                           676
UNIT_TYPES = ("岩质边坡", "碎石边坡", "平台")                                   677
# 每个单元类型启用的指标类别（论文表 3 隐含：岩质边坡无土壤指标，平台无地形指标）  678
# 供数据录入界面与评价引擎判断"该单元是否需要某类指标"                          679
UNIT_ENABLED_INDICATORS = {                                                     680
        "岩质边坡": ("植被指标", "地形指标"),                  # 不设土壤指标     681
        "碎石边坡": ("土壤指标", "植被指标", "地形指标"),                       682
        "平台": ("土壤指标", "植被指标"),                      # 不设地形指标     683
}                                                                               684
# ------------------------------------------------------------------------      685
        ------------------                                                      686
# 二、指标标准化理想极值（论文 4.2.2 节）                                       687
# ------------------------------------------------------------------------      688
        ------------------                                                      689
IDEAL_SOIL_RATIO = 0.8          # 土石比理想值                                  690
IDEAL_THICKNESS = 20.0          # 土层厚度理想值 （cm）                         691
IDEAL_PH_RANGE = (5.0, 9.0)     # pH 理想区间（适中型指标）                     692
IDEAL_ORGANIC = 30.0            # 有机质理想值                                  693
        （g/kg），全国第二次土壤普查养分等级划分标准二等最低值                  694
IDEAL_COVERAGE = 75.0           # 植被覆盖度参照生态系统 （%）                  695
IDEAL_SHANNON = 0.7             # Shannon-Wiener 多样性指数参照生态系统         696
# 重金属极值 （mg/kg），《绿化种植土壤》CJ/T 340—2016 土壤重金属含量 IV 级最高值  697
# 论文原文：镉、汞、铅、铬、砷、镍、铜、锌分别为 2、2、530、400、45、220、600、800  698
HEAVY_METALS = ("Cd", "Hg", "Pb", "Cr", "As", "Ni", "Cu",                       699
        "Zn")                                                                   700
```

```
HEAVY_METAL_LIMIT = {
    "Cd": 2.0, "Hg": 2.0, "Pb": 530.0, "Cr": 400.0,
    "As": 45.0, "Ni": 220.0, "Cu": 600.0, "Zn": 800.0,
}
# 污染指标下 8 种重金属"重要性相当，权重平均分配"（论文 4.2.1 节）
HEAVY_METAL_EQUAL_WEIGHT = 1.0 / 8.0
# 坡度分界，论文 4.2.2 节引《新一轮退耕还林还草总体方案》发改西部〔2014〕1772 号
SLOPE_BREAK = 25.0
# 坡向离散赋值，论文 4.2.2 节："阳坡、阴阳坡、阴坡分别赋值 0.2、0.6、1"
ASPECT_SCORES = {"阳坡": 0.2, "阴阳坡": 0.6, "阴坡": 1.0}
ASPECT_TYPES = ("阳坡", "阴阳坡", "阴坡")
# pH 适中型标准化的容差边界（区间外线性衰减，避免得分突变）
PH_TOL_LOW = 4.0
PH_TOL_HIGH = 10.0
# ---------------------------------------------------------------------
# 三、适宜性等级划分（论文表 4）
# ---------------------------------------------------------------------
GRADE_LEVELS = (
    {"grade": "不适宜", "min": None, "max": 0.3,
     "meaning": "存在限制性条件或区域生态本底较差，自然恢复影响因子严重制约自然恢复",
     "measure": "生态重建、人工辅助"},
    {"grade": "基本适宜", "min": 0.3, "max": 0.7,
     "meaning": "生态本底良好，核心自然恢复影响因子基本满足自然恢复需求",
     "measure": "自然恢复为主，可辅以封闭场地和少量补种、补植、飞播等"},
    {"grade": "适宜", "min": 0.7, "max": None,
     "meaning": "各自然恢复影响因子均能满足自然恢复需求，自然演替速度快",
     "measure": "自然恢复"},
)
def grade_of(score):
    """按论文表 4 判定适宜性等级。"""
    for g in GRADE_LEVELS:
        lo, hi = g["min"], g["max"]
        if lo is None and score < hi:
            return g
        if hi is None and score >= lo:
            return g
        if lo is not None and hi is not None and lo <= score < hi:
            return g
    # score 恰好等于 0.7 时落入"适宜"
    return GRADE_LEVELS[-1]
# ---------------------------------------------------------------------
# 四、平均随机一致性指标 RI（论文 4.2.1 节给出 n=2,3,4,5，其余为常用扩展值）
# ---------------------------------------------------------------------
RI_TABLE = {1: 0.0, 2: 0.0, 3: 0.58, 4: 0.90, 5: 1.12, 6: 1.24,
```

```
                7：  1．32，  8：  1．41，  9：  1．45，  10：  1．49｝
#  一致性比例阈值（论文：当  CR＜0．1  时，判断矩阵满足一致性要求）
CR_THRESHOLD  =  0．1
#  ----------------------------------------------------------------------
      ------------------
#  五、默认权重方案（论文表  3）
#  ----------------------------------------------------------------------
      ------------------
#  论文  4．2．2  节："划分岩质边坡、碎石土边坡、平台三类微地质单元开展独立评价，
#  分别计算评价指标权重"（论文摘要、图  2  流程与表  5  均作"碎石边坡"，
#  系统统一采用"碎石边坡"）。权重由  10  位专家打分后经  AHP  方法确定。
#  下表为论文表  3  所载的归一化权重，作为系统默认方案。
DEFAULT_WEIGHTS  =  {
      "岩质边坡"：  {
            "l1"：  {"植被指标"：  0．45，  "地形指标"：  0．55}，
            "l2"：  {
                  "植被指标"：  {"植被覆盖度"：  0．67，  "生物多样性"：  0．33}，
                  "地形指标"：  {"坡度"：  0．64，  "坡向"：  0．36}，
            }，
      }，
      "碎石边坡"：  {
            "l1"：  {"土壤指标"：  0．44，  "植被指标"：  0．31，  "地形指标"：  0．25}，
            "l2"：  {
                  "土壤指标"：  {"土石比"：  0．25，  "土层厚度"：  0．25，  "pH"：  0．17，
                                    "有机质"：  0．17，  "污染指标"：  0．17}，
                  "植被指标"：  {"植被覆盖度"：  0．64，  "生物多样性"：  0．36}，
                  "地形指标"：  {"坡度"：  0．67，  "坡向"：  0．33}，
            }，
      }，
      "平台"：  {
            "l1"：  {"土壤指标"：  0．54，  "植被指标"：  0．46}，
            "l2"：  {
                  "土壤指标"：  {"土石比"：  0．25，  "土层厚度"：  0．25，  "pH"：  0．17，
                                    "有机质"：  0．17，  "污染指标"：  0．17}，
                  "植被指标"：  {"植被覆盖度"：  0．61，  "生物多样性"：  0．39}，
            }，
      }，
}
#  ----------------------------------------------------------------------
      ------------------
#  六、限制性指标（论文表  2  +  4．1  节流程）
#  ----------------------------------------------------------------------
      ------------------
#  论文摘要与  4．1  节：四项限制性指标；流程段另明确"是否为耕地"为约束条件
CONSTRAINT_ITEMS  =  （"政策"，  "区位约束"，  "土壤污染"，  "地质灾害"，  "耕地属性"）
#  ----------------------------------------------------------------------
      ------------------
#  七、用户角色
#  ----------------------------------------------------------------------
      ------------------
```

```
ROLE_LABELS = {"admin": "管理员", "evaluator": "评价员",
    "viewer": "查看者"}
# ==== app/utils/schema.py ====
# -*- coding: utf-8 -*-
"""轻量结构迁移。
SQLite 下 `db.create_all()` 只建缺失的表，不会给已有表补列。
本模块在启动时检查目标列是否存在，缺失则 ALTER TABLE 补上，
使旧数据库无需重建即可用上新字段。
"""
from sqlalchemy import text
# （表名， 列名， 列定义）
REQUIRED_COLUMNS = [
    # 遥感影像挂接
    ("mine_patches", "image_path", "VARCHAR(256)"),
    ("mine_patches", "image_note", "VARCHAR(256)"),
    ("mine_patches", "image_uploaded_at", "DATETIME"),
    # 指标维护
    ("indicators", "enabled_units", "VARCHAR(128)"),
    ("indicators", "description", "TEXT"),
    # 多专家矩阵
    ("expert_matrices", "unit_type", "VARCHAR(32)"),
    ("expert_matrices", "l1_name", "VARCHAR(64)"),
    ("expert_matrices", "cr", "FLOAT"),
    # 用户个人资料（个人设置页）
    ("users", "display_name", "VARCHAR(64)"),
    ("users", "title", "VARCHAR(64)"),
    ("users", "email", "VARCHAR(128)"),
    ("users", "phone", "VARCHAR(32)"),
    ("users", "avatar", "VARCHAR(256)"),
]
def ensure_schema(db):
    """为已有表补齐缺失列。返回补列清单。"""
    added = []
    try:
        with db.engine.connect() as conn:
            for table, column, ddl in REQUIRED_COLUMNS:
                rows = conn.execute(text("PRAGMA
    table_info(%s)" % table)).fetchall()
                if not rows:
                    continue                                #
    表还不存在，create_all 会建
                existing = {r[1] for r in rows}
                if column in existing:
                    continue
                conn.execute(text(
                    "ALTER TABLE %s ADD COLUMN %s %s" %
    (table, column, ddl)))
                conn.commit()
                added.append("%s.%s" % (table, column))
    except Exception as e:                                  #
```

```
        迁移失败不应阻断启动
            print("[schema] 结构检查跳过：%s" % e)
        return added
# ==== app/utils/uploads.py ====
# -*- coding: utf-8 -*-
"""上传文件处理（遥感影像、批量导入）。"""
import os
import re
import uuid
from datetime import datetime
from werkzeug.utils import secure_filename
def _safe_ext(filename):
    return os.path.splitext(filename)[1].lower().lstrip(".")
def ext_allowed(filename, allowed):
    return _safe_ext(filename) in allowed
def save_upload(file_storage, subdir, allowed_ext,
    upload_root):
    """保存上传文件，返回相对 static 的路径（如 uploads/patches/xxx.png）。
    文件名重命名为「时间戳_随机串.扩展名」，避免中文名与重名问题。
    """
    if file_storage is None or not file_storage.filename:
        return None, "未选择文件"
    ext = _safe_ext(file_storage.filename)
    if ext not in allowed_ext:
        return None, "不支持的文件类型：.%s" % ext
    target_dir = os.path.join(upload_root, subdir)
    os.makedirs(target_dir, exist_ok=True)
    # 仅保留安全字符，并加时间戳与随机串防重名
    base = secure_filename(os.path.splitext(file_storage.fil
    ename)[0]) or "file"
    base = re.sub(r"[^A-Za-z0-9_\-]", "", base)[:40] or
    "file"
    name = "%s_%s.%s" % (datetime.now().strftime("%Y%m%d%H%M
    %S"),
                         uuid.uuid4().hex[:6], ext)
    abs_path = os.path.join(target_dir, name)
    file_storage.save(abs_path)
    rel = os.path.join("uploads", subdir,
    name).replace("\\", "/")
    return rel, None
def remove_upload(rel_path, upload_root):
    """删除已保存的上传文件（相对 static 的路径）。"""
    if not rel_path:
        return
    abs_path = os.path.join(upload_root,
    rel_path.replace("uploads/", "", 1))
    try:
        if os.path.isfile(abs_path):
            os.remove(abs_path)
    except OSError:
```

```
            pass
# ==== app/utils/charts.py ====
# -*- coding: utf-8 -*-
"""轻量图表。
用纯 SVG 输出，不依赖任何绘图库或前端脚本：
- 服务端渲染，页面与打印（ＰＤＦ）中都能稳定显示
- 每个图元都带数值标签，黑白打印时仍可读（不单靠颜色区分）
- 单色方案，符合界面「一个视图一个强调色」的约定
"""
from html import escape
ACCENT = "#1f3a5f"
ACCENT_SOFT = "#7f93ad"
INK = "#23262b"
MUTED = "#6b7280"
LINE = "#dfe3e8"
# 灰度梯度：黑白打印时相邻扇区 / 条柱仍可区分
GRAYS = ["#1f3a5f", "#456188", "#6d86a8", "#96a9c4",
    "#bcc9da", "#dbe3ec"]
def _fmt(v, digits=4):
    try:
        return ("%%.%df" % digits) % float(v)
    except (TypeError, ValueError):
        return str(v)
def hbars(items, width=620, bar_h=22, gap=12, label_w=150,
    value_w=76):
    """水平条形图。items: [(标签, 数值, 显示文本)]"""
    if not items:
        return '<p class="muted">暂无数据</p>'
    n = len(items)
    height = n * (bar_h + gap) + 10
    max_v = max((abs(float(v)) for _, v, _ in items),
    default=1) or 1
    body_w = width - label_w - value_w
    parts = ['<svg viewBox="0 0 %d %d" width="100%%"
    height="%d" '
             'role="img" aria-label="条形图" '
             'style="max-width:%dpx;font-family:inherit;">'
    % (width, height, height, width)]
    for i, (label, value, disp) in enumerate(items):
        y = i * (bar_h + gap) + 5
        frac = (abs(float(value)) / max_v) if max_v else 0
        w = max(2.0, body_w * frac)
        fill = GRAYS[i % len(GRAYS)]
        parts.append(
            '<text x="%d" y="%d" text-anchor="end"
    font-size="12" fill="%s">%s</text>'
            % (label_w - 10, y + bar_h * 0.72, INK,
    escape(str(label))))
        parts.append(
            '<rect x="%d" y="%d" width="%.1f" height="%d"
```

```
        rx="3" fill="%s"/>'                                                  951
                 % (label_w, y, w, bar_h, fill))                             952
            parts.append(                                                    953
                 '<text x="%d" y="%d" font-size="12" fill="%s" '             954
                 'style="font-variant-numeric:tabular-nums;">%s</           955
        text>'                                                               956
                 % (label_w + w + 8, y + bar_h * 0.72, MUTED,                957
        escape(str(disp))))                                                  958
        parts.append('</svg>')                                               959
        return "".join(parts)                                                960
def donut(items, size=210, thickness=34):                                    961
        """环形图（权重分布）。items: [(标签, 数值, 显示文本)]                 962
        图例中给出数值与占比，黑白打印时靠图例与灰度深浅区分。                 963
        """                                                                  964
        vals = [(l, float(v), d) for l, v, d in items if                     965
        float(v) > 0]                                                        966
        if not vals:                                                         967
            return '<p class="muted">暂无数据</p>'                           968
        total = sum(v for _, v, _ in vals) or 1                              969
        r = (size - thickness) / 2                                           970
        cx = cy = size / 2                                                   971
        import math                                                          972
        parts = ['<svg viewBox="0 0 %d %d" width="%d"                        973
        height="%d" role="img" '                                             974
                 'aria-label="权重分布环形图">' % (size, size, size,         975
        size)]                                                               976
        angle = -math.pi / 2                                                 977
        for i, (label, v, disp) in enumerate(vals):                          978
            sweep = 2 * math.pi * (v / total)                                979
            a0, a1 = angle, angle + sweep                                    980
            x0, y0 = cx + r * math.cos(a0), cy + r *                         981
        math.sin(a0)                                                         982
            x1, y1 = cx + r * math.cos(a1), cy + r *                         983
        math.sin(a1)                                                         984
            large = 1 if sweep > math.pi else 0                              985
            parts.append(                                                    986
                 '<path d="M %.2f %.2f A %.2f %.2f 0 %d 1 %.2f               987
        %.2f" '                                                              988
                 'fill="none" stroke="%s" stroke-width="%d"/>'               989
                % (x0, y0, r, r, large, x1, y1, GRAYS[i %                    990
        len(GRAYS)], thickness))                                             991
            angle = a1                                                       992
        parts.append('<text x="%d" y="%d" text-anchor="middle"               993
        font-size="12" '                                                     994
                     'fill="%s">合计</text>' % (cx, cy - 2,                  995
        MUTED))                                                              996
        parts.append('<text x="%d" y="%d" text-anchor="middle"               997
        font-size="15" '                                                     998
                     'fill="%s" style="font-variant-numeric:tabu             999
        lar-nums;">%s</text>'                                               1000
```

```
                % (cx, cy + 18, INK, _fmt(total, 4)))
    parts.append('</svg>')
    legend = ['<table style="width:auto;margin-top:10px;"><t
    head><tr>'
              '<th scope="col">项目</th><th
    scope="col">数值</th>'
              '<th scope="col">占比</th></tr></thead><tbody>']
    for i, (label, v, disp) in enumerate(vals):
        legend.append(
            '<tr><td style="text-align:left;">'
            '<span style="display:inline-block;width:10px;he
    ight:10px;'
            'border-radius:2px;background:%s;margin-right:6p
    x;"></span>%s</td>'
            '<td class="mono">%s</td><td
    class="mono">%.1f%%</td></tr>'
            % (GRAYS[i % len(GRAYS)], escape(str(label)),
    escape(str(disp)),
               100.0 * v / total))
    legend.append('</tbody></table>')
    return ('<div style="display:flex;flex-wrap:wrap;gap:20p
    x;align-items:center;">'
            + "".join(parts) + "".join(legend) + '</div>')
# ==== app/utils/browser.py ====
# -*- coding: utf-8 -*-
"""HTML 转 PDF（报告导出）。
复用本机已安装的 Chrome/Edge 无头模式打印，避免引入 reportlab、
weasyprint 等额外依赖。生成过程使用系统临时目录中的 ASCII 文件名，
避免中文路径在 file:// URL 下出问题。
"""
import os
import shutil
import subprocess
import tempfile
import uuid
CANDIDATES = [
    r"C:\Program Files\Google\Chrome\Application\chrome.exe"
    ,
    r"C:\Program Files (x86)\Google\Chrome\Application\chrom
    e.exe",
    r"C:\Program Files (x86)\Microsoft\Edge\Application\msed
    ge.exe",
    r"C:\Program Files\Microsoft\Edge\Application\msedge.exe
    ",
]
# 允许通过环境变量覆盖，便于在其他机器上指定浏览器
ENV_KEY = "DSH_BROWSER_PATH"
def browser_path():
    """定位可用的浏览器可执行文件，找不到返回 None。"""
    override = os.environ.get(ENV_KEY)
```

```
    if override and os.path.isfile(override):                          1051
        return override                                                1052
    for p in CANDIDATES:                                               1053
        if os.path.isfile(p):                                          1054
            return p                                                   1055
    for name in ("chrome", "msedge", "chromium"):                      1056
        found = shutil.which(name)                                     1057
        if found:                                                      1058
            return found                                               1059
    return None                                                        1060
def html_to_pdf(html, timeout=90):                                     1061
    """把 HTML 字符串渲染为 PDF，返回字节；失败返回 (None, 原因)。"""          1062
    exe = browser_path()                                               1063
    if exe is None:                                                    1064
        return None, ("未找到可用的浏览器（Chrome/Edge），无法生成 PDF。"      1065
                      "可设置环境变量 %s 指向浏览器可执行文件。" % ENV_KEY)      1066
    tmpdir = tempfile.mkdtemp(prefix="dshpdf_")                        1067
    tag = uuid.uuid4().hex[:8]                                         1068
    html_path = os.path.join(tmpdir, "r_%s.html" % tag)                1069
    pdf_path = os.path.join(tmpdir, "r_%s.pdf" % tag)                  1070
    try:                                                               1071
        with open(html_path, "w", encoding="utf-8") as f:              1072
            f.write(html)                                              1073
        cmd = [                                                        1074
            exe, "--headless=new", "--disable-gpu",                    1075
    "--no-sandbox",                                                    1076
            "--no-first-run", "--no-default-browser-check",            1077
            "--no-pdf-header-footer",                                  1078
            "--virtual-time-budget=4000",                              1079
            "--print-to-pdf=%s" % pdf_path,                            1080
            "file:///" + html_path.replace("\\", "/"),                 1081
        ]                                                              1082
        p = subprocess.run(cmd, capture_output=True,                   1083
    text=True,                                                         1084
                           timeout=timeout,                            1085
    encoding="utf-8", errors="replace")                                1086
        if not os.path.isfile(pdf_path) or                             1087
    os.path.getsize(pdf_path) < 800:                                   1088
            return None, "浏览器未能生成 PDF(%s)" % (p.stderr or       1089
    "").strip()[:200]                                                  1090
        with open(pdf_path, "rb") as f:                                1091
            return f.read(), None                                      1092
    except subprocess.TimeoutExpired:                                  1093
        return None, "生成 PDF 超时，请稍后重试"                           1094
    except Exception as e:                                             1095
        return None, "生成 PDF 失败：%s" % e                             1096
    finally:                                                           1097
        shutil.rmtree(tmpdir, ignore_errors=True)                      1098
# ==== app/services/ahp_service.py ====                               1099
# -*- coding: utf-8 -*-                                              1100
```

```
"""AHP 权重引擎。
严格按论文 4.2.1 节实现：
- 式(2)(3)(4)：特征根法求权重（行元素乘积 → n 次方根 → 归一化）
- 式(5)(6)(7)：一致性检验（λmax、CI、CR）
- 论文要求 CR < 0.1；RI 按阶数查表
"""
import numpy as np
from app.utils.constants import RI_TABLE, CR_THRESHOLD
def matrix_from_upper(upper):
    """由判断矩阵上三角构造完整互反矩阵。
    参数
    ----
    upper : list[list[float]] 或 np.ndarray，上三角元素（不含对角线）。
        例：3 阶时传入 [[a12, a13], [a23]]。
        对角线为 1，下三角取倒数（a_ji = 1/a_ij）。
    """
    upper = np.asarray(upper, dtype=float)
    k = upper.shape[0]
    n = int(round((1 + (1 + 8 * k) ** 0.5) / 2))
    A = np.ones((n, n), dtype=float)
    idx = 0
    for i in range(n):
        for j in range(i + 1, n):
            A[i, j] = upper[idx]
            A[j, i] = 1.0 / upper[idx]
            idx += 1
    return A
def weights_and_consistency(A):
    """由判断矩阵 A 计算权重与一致性指标。
    返回
    ----
    dict：{weights: np.ndarray, lambda_max, ci, cr, n,
    consistent: bool}
    """
    A = np.asarray(A, dtype=float)
    n = A.shape[0]
    # 式(2) 行元素乘积
    row_prod = np.prod(A, axis=1)
    # 式(3) 开 n 次方根
    root = row_prod ** (1.0 / n)
    # 式(4) 归一化
    weights = root / root.sum()
    # 式(5) 最大特征值 λmax = (1/n) Σ (A·W)_i / W_i
    aw = A.dot(weights)
    lambda_max = float(np.sum(aw / weights) / n)
    # 式(6) 一致性指标 CI
    ci = (lambda_max - n) / (n - 1) if n > 1 else 0.0
    # 式(7) 一致性比例 CR = CI / RI
    ri = RI_TABLE.get(n, 1.49)
    cr = ci / ri if ri else 0.0
```

```
    return {
        "weights": weights,
        "lambda_max": lambda_max,
        "ci": ci,
        "cr": cr,
        "n": n,
        "consistent": cr < CR_THRESHOLD,
    }
def find_most_inconsistent(A):
    """定位判断矩阵中最不一致的一对比较。
    依据：完全一致矩阵应满足 a_ij = w_i / w_j。
    计算每个上三角元素与其理想值的相对偏差，返回偏差最大者。
    返回
    ----
    dict：{i, j, value, expected, deviation}，偏差最大的 (i, j)
    及对应取值。
    """
    A = np.asarray(A, dtype=float)
    r = weights_and_consistency(A)
    w = r["weights"]
    n = A.shape[0]
    best = None
    for i in range(n):
        for j in range(i + 1, n):
            expected = w[i] / w[j]
            if expected == 0:
                continue
            deviation = abs(A[i, j] - expected) / expected
            if best is None or deviation >
    best["deviation"]:
                best = {"i": i, "j": j, "value": A[i, j],
                        "expected": expected, "deviation":
    deviation}
    return best
def aggregate_matrices(matrices):
    """多专家判断矩阵聚合：几何平均（保持互反性）。
    参数
    ----
    matrices : list[np.ndarray]，多个同阶判断矩阵。
    """
    stack = np.stack([np.asarray(m, dtype=float) for m in
    matrices])
    return np.prod(stack, axis=0) ** (1.0 / len(stack))
def judge_from_weights(weights):
    """由权重向量构造完全一致判断矩阵并检验。
    用途：验证给定权重向量能否作为某个完全一致判断矩阵的主特征向量
    （A_ij = w_i / w_j 时 λmax = n、CR = 0）。
    """
    w = np.asarray(weights, dtype=float)
    n = w.shape[0]
```

```
    A = np.outer(w, 1.0 / w)
    r = weights_and_consistency(A)
    r["matrix"] = A
    return r
# ==== app/services/standardization.py ====
# -*- coding: utf-8 -*-
"""指标标准化（的一部分）。
按论文 4.2.2 节式(8) 实现：
    x' = (x - x_min) / (x_max - x_min)
按指标性质分为四类：越大越优 / 越小越优 / 适中 / 离散赋值。
所有理想极值取自 app.utils.constants，可配置。
"""
from app.utils.constants import (
    IDEAL_SOIL_RATIO, IDEAL_THICKNESS, IDEAL_PH_RANGE,
    IDEAL_ORGANIC,
    IDEAL_COVERAGE, IDEAL_SHANNON, HEAVY_METAL_LIMIT,
    HEAVY_METAL_EQUAL_WEIGHT, SLOPE_BREAK, ASPECT_SCORES,
    PH_TOL_LOW, PH_TOL_HIGH,
)
def std_benefit(value, ideal):
    """越大越优指标标准化：以理想值为上限，按比例给分，饱和于 [0,1]。"""
    if value is None:
        return None
    return max(0.0, min(1.0, value / ideal))
def std_cost(value, extreme):
    """越小越优指标标准化：以极值为参考，线性折减至 [0,1]。"""
    if value is None:
        return None
    return max(0.0, min(1.0, 1.0 - value / extreme))
def std_mid_ph(value, lo, hi, tol_lo=PH_TOL_LOW,
    tol_hi=PH_TOL_HIGH):
    """适中型指标标准化（pH）：理想区间内取 1，区间外线性衰减。"""
    if value is None:
        return None
    if lo <= value <= hi:
        return 1.0
    if value > hi:
        return max(0.0, (tol_hi - value) / (tol_hi - hi))
    return max(0.0, (value - tol_lo) / (lo - tol_lo))
def std_slope(value, break_point=SLOPE_BREAK, upper=90.0):
    """坡度标准化。
    论文 4.2.2 节：以 25° 为陡坡重要分界。25° 及以下记满分 1，
    超过 25° 线性折减，90° 记 0。
    """
    if value is None:
        return None
    if value <= break_point:
        return 1.0
    return max(0.0, (upper - value) / (upper - break_point))
def aspect_score(aspect):
```

```
    """坡向离散赋值：阳坡 0.2 / 阴阳坡 0.6 / 阴坡 1.0（论文 4.2.2 节）。"""
    if aspect is None:
        return None
    return ASPECT_SCORES.get(aspect)
def pollution_score(heavy_metals):
    """污染指标：8 种重金属等权（各 0.125）合成。
    论文注：未检测 Cr、Zn 的，从矿山类型看未受污染，按无污染（=1）处理。
    heavy_metals : dict，键为金属名，值为含量 mg/kg；缺失键视为未检测。
    """
    total = 0.0
    for metal, extreme in HEAVY_METAL_LIMIT.items():
        value = heavy_metals.get(metal)
        total += std_cost(value, extreme) if value is not
    None else 1.0
    return total / len(HEAVY_METAL_LIMIT)
def standardize_unit(unit_type, data, weights=None):
    """对单个微地质单元的全部指标做标准化。
    参数
    ----
    unit_type : str，岩质边坡 / 碎石边坡 / 平台
    data : dict，实测数据，字段见接口契约：
        soil_ratio, thickness, ph, organic_matter,
    heavy_metals(dict),
        coverage, shannon, slope, aspect
    weights : dict，该单元类型的权重（app.utils.constants.DEFAULT_WEIG
    HTS 结构），
                未提供时用默认值。
    返回
    ----
    dict：{ 一级指标： { 二级指标： （实测值， 标准化值， 权重， 贡献） } }
    标准化值可能为 None，表示该指标不适用或方法待定。
    """
    from app.utils.constants import DEFAULT_WEIGHTS
    if weights is None:
        weights = DEFAULT_WEIGHTS[unit_type]
    l1_enabled = weights["l1"].keys()
    l2w = weights["l2"]
    result = {}
    # ---- 土壤指标 ----
    if "土壤指标" in l1_enabled:
        result["土壤指标"] = {
            "土石比": (data.get("soil_ratio"),
                       std_benefit(data.get("soil_ratio"),
    IDEAL_SOIL_RATIO),
                       l2w["土壤指标"]["土石比"]),
            "土层厚度": (data.get("thickness"),
                         std_benefit(data.get("thickness"),
    IDEAL_THICKNESS),
                         l2w["土壤指标"]["土层厚度"]),
            "pH": (data.get("ph"),
```

```
                        std_mid_ph(data.get("ph"),                          1301
    *IDEAL_PH_RANGE),                                                       1302
                        l2w["土壤指标"]["pH"]),                             1303
                "有机质": (data.get("organic_matter"),                      1304
                           std_benefit(data.get("organic_matter")           1305
    , IDEAL_ORGANIC),                                                       1306
                           l2w["土壤指标"]["有机质"]),                      1307
                "污染指标": (data.get("heavy_metals") or {},                1308
                             pollution_score(data.get("heavy_meta           1309
    ls") or {}),                                                            1310
                             l2w["土壤指标"]["污染指标"]),                  1311
            }                                                               1312
        # ---- 植被指标 ----                                               1313
        if "植被指标" in l1_enabled:                                        1314
            result["植被指标"] = {                                          1315
                "植被覆盖度": (data.get("coverage"),                        1316
                               std_benefit(data.get("coverage"),            1317
    IDEAL_COVERAGE),                                                        1318
                               l2w["植被指标"]["植被覆盖度"]),              1319
                "生物多样性": (data.get("shannon"),                         1320
                               std_benefit(data.get("shannon"),             1321
    IDEAL_SHANNON),                                                         1322
                               l2w["植被指标"]["生物多样性"]),              1323
            }                                                               1324
        # ---- 地形指标 ----                                               1325
        if "地形指标" in l1_enabled:                                        1326
            result["地形指标"] = {                                          1327
                "坡度": (data.get("slope"),                                 1328
                         std_slope(data.get("slope")),                      1329
                         l2w["地形指标"]["坡度"]),                          1330
                "坡向": (data.get("aspect"),                                1331
                         aspect_score(data.get("aspect")),                  1332
                         l2w["地形指标"]["坡向"]),                          1333
            }                                                               1334
        return result                                                       1335
# ==== app/services/weight_service.py ====                                  1336
# -*- coding: utf-8 -*-                                                     1337
"""权重方案的存取与生效权重解析。                                           1338
系统有两套权重来源：                                                       1339
1. 论文表 3 给出的默认权重（app.utils.constants.DEFAULT_WEIGHTS），        1340
   是**权威取值来源**，案例复算与报告结论以此为准；                        1341
2. 通过层次分析法在本系统中实际算出的权重，保存在                          1342
   weight_schemes / weight_items 两张表中，可标记为「生效」。              1343
本模块负责把这两者合成为计算引擎直接可用的嵌套权重结构：                   1344
以论文表 3 为底，用生效方案中已保存的条目覆盖。                            1345
没有任何生效方案时，即完全使用论文表 3 的权重。                            1346
这样既保证默认行为与论文一致，又使 AHP 计算结果真正参与评价，              1347
而不是算完即弃。                                                           1348
"""                                                                        1349
import copy                                                                1350
```

```
from app.models import db, WeightScheme, WeightItem
from app.utils.constants import DEFAULT_WEIGHTS
from app.services.indicator_service import TREE
#  方案名称：每个微地质单元类型固定一个可生效的方案
ACTIVE_NAME = "自定义权重方案（AHP 计算结果）"
def _code_maps():
    """由指标树导出 编码 → 名称 的映射，避免名称在两处各写一遍。"""
    l1, l2 = {}, {}
    for row in TREE:
        code, name, level, parent = row[0], row[1], row[2],
    row[3]
        if level == 1:
            l1[code] = name
        elif level == 2:
            l2[code] = (l1.get(parent, parent), name)
    return l1, l2
def effective_weights(unit_type):
    """返回该单元类型当前生效的权重。
    返回 (weights, source_label, scheme)
        weights       嵌套权重结构，可直接交给标准化与评价引擎
        source_label  权重来源说明，用于界面与报告标注
        scheme        生效的权重方案对象，无自定义方案时为 None
    """
    base = DEFAULT_WEIGHTS.get(unit_type)
    if base is None:
        return None, "未定义该微地质单元类型的权重", None
    scheme = WeightScheme.query.filter_by(unit_type=unit_typ
    e,

    name=ACTIVE_NAME).first()
    if scheme is None:
        return copy.deepcopy(base), "论文表 3 默认权重", None
    merged = copy.deepcopy(base)
    l1_map, l2_map = _code_maps()
    applied = 0
    for item in WeightItem.query.filter_by(scheme_id=scheme.
    id).all():
        code, w = item.indicator_code, item.weight
        if w is None or w <= 0:
            continue
        if code in l1_map and l1_map[code] in merged["l1"]:
            merged["l1"][l1_map[code]] = w
            applied += 1
        elif code in l2_map:
            l1_name, l2_name = l2_map[code]
            if l1_name in merged["l2"] and l2_name in
    merged["l2"][l1_name]:
                merged["l2"][l1_name][l2_name] = w
                applied += 1
    if applied == 0:
```

```
        return copy.deepcopy(base), "论文表 3 默认权重", None
    return merged, "自定义方案（%d 项覆盖论文表 3）" % applied, scheme
def save_weights(unit_type, code_weight_pairs, note=None):
    """把一组「指标编码 → 权重」写入生效方案（已存在则更新对应条目）。
    仅覆盖本次传入的条目，其余条目保留，便于分层次（一级 / 二级）
    分次录入而无需一次填完。
    """
    scheme = WeightScheme.query.filter_by(unit_type=unit_type,
                                          name=ACTIVE_NAME).first()
    if scheme is None:
        scheme = WeightScheme(unit_type=unit_type, name=ACTIVE_NAME,
                              level=1, is_default=True)
        db.session.add(scheme)
        db.session.flush()
    if note:
        scheme.level = scheme.level or 1
    if not scheme.is_default:
        scheme.is_default = True
    written = 0
    for code, w in code_weight_pairs:
        if w is None or w <= 0:
            continue
        item = WeightItem.query.filter_by(scheme_id=scheme.id,
                                          indicator_code=code).first()
        if item is None:
            item = WeightItem(scheme_id=scheme.id, indicator_code=code)
            db.session.add(item)
        item.weight = float(w)
        written += 1
    db.session.commit()
    return scheme, written
def reset_weights(unit_type):
    """删除该单元类型的自定义方案，恢复为论文表 3 的默认权重。"""
    schemes = WeightScheme.query.filter_by(unit_type=unit_type,
                                           name=ACTIVE_NAME).all()
    n = 0
    for s in schemes:
        WeightItem.query.filter_by(scheme_id=s.id).delete(
            synchronize_session=False)
        db.session.delete(s)
        n += 1
    db.session.commit()
```

```
    return n
def scheme_items(unit_type):
    """返回该单元类型已保存的自定义权重条目 {指标编码： 权重}。"""
    scheme = WeightScheme.query.filter_by(unit_type=unit_type,
                                          name=ACTIVE_NAME).first()
    if scheme is None:
        return {}, None
    items = {i.indicator_code: i.weight
             for i in WeightItem.query.filter_by(scheme_id=scheme.id).all()}
    return items, scheme
def summary():
    """各单元类型的权重来源概览，供权重首页展示。"""
    rows = []
    for unit_type in DEFAULT_WEIGHTS:
        weights, label, scheme = effective_weights(unit_type)
        rows.append({
            "unit_type": unit_type,
            "label": label,
            "custom": scheme is not None,
        })
    return rows
# ==== app/services/evaluation_service.py ====
# -*- coding: utf-8 -*-
"""适宜性评价引擎。
按论文式（9） 实现：
- 单元级：F' = Σ（二级指标权重 × 该二级指标下三级指标标准化值）
- 矿山级：F = Σ（Si × F'i），Si 为各微地质单元面积占比
- 等级判定按论文表 4。
"""
from app.utils.constants import DEFAULT_WEIGHTS, grade_of
from app.services import standardization as std
def score_unit(unit_type, data, weights=None):
    """计算单个微地质单元的评价因子 F'。
    返回
    ----
    dict：{ l1_scores: {一级指标： 得分}, unit_score: F', details:
    [...] }
    若某一级指标下存在"不适用"（标准化值为 None）的二级指标，
    则按其余项权重归一化后计算，保证单元得分始终有定义。
    """
    if weights is None:
        weights = DEFAULT_WEIGHTS[unit_type]
    std_result = std.standardize_unit(unit_type, data,
    weights)
    l1_weights = weights["l1"]
    l1_scores = {}
```

```
    details = []
    for l1_name, items in std_result.items():
        wsum = 0.0
        acc = 0.0
        for l2_name, (raw, std_value, w2) in items.items():
            if std_value is None:
                # 不适用项：不计入
                details.append({
                    "l1": l1_name, "l2": l2_name, "raw":
raw,
                    "std_value": None, "weight": w2,
"contribution": None,
                })
                continue
            contribution = std_value * w2
            acc += contribution
            wsum += w2
            details.append({
                "l1": l1_name, "l2": l2_name, "raw": raw,
                "std_value": std_value, "weight": w2,
                "contribution": contribution,
            })
        l1_scores[l1_name] = acc / wsum if wsum > 0 else 0.0
    # 一级指标权重按实际参与计算的指标归一化。
    # 论文表 3 的权重本身已归一（和为 1），此处为恒等变换；
    # 若某层指标缺失或自定义权重不完整，可避免单元得分被整体压低。
    lw_sum = sum(l1_weights[k] for k in l1_scores if k in
l1_weights)
    if lw_sum > 0:
        unit_score = sum(l1_weights[k] * l1_scores[k]
                         for k in l1_scores if k in
l1_weights) / lw_sum
    else:
        unit_score = 0.0
    return {"l1_scores": l1_scores, "unit_score":
unit_score, "details": details}
def score_mine(units, weights_map=None):
    """计算整个矿山的评价因子 F 与适宜性等级。
    参数
    ----
    units : list[dict]，每个元素：
        {unit_type, ratio, data}
        ratio 为该单元面积占比（0~1），data 为实测数据 dict。
    weights_map : dict，{单元类型： 权重结构}。未提供的单元类型
        使用论文表 3 的默认权重。权重可来自 AHP 计算并经
        weight_service 保存生效，从而真正参与评价。
    返回
    ----
    dict:{ final_score, grade, grade_info, unit_results }
    """
```

```
    weights_map = weights_map or {}
    total = 0.0
    unit_results = []
    for u in units:
        r = score_unit(u["unit_type"], u["data"],
                       weights_map.get(u["unit_type"]))
        weighted = u["ratio"] * r["unit_score"]
        total += weighted
        unit_results.append({
            "unit_type": u["unit_type"],
            "ratio": u["ratio"],
            "unit_score": r["unit_score"],
            "weighted": weighted,
            "l1_scores": r["l1_scores"],
            "details": r["details"],
        })
    grade_info = grade_of(total)
    return {
        "final_score": total,
        "grade": grade_info["grade"],
        "grade_info": grade_info,
        "unit_results": unit_results,
    }
# ==== app/services/report_service.py ====
# -*- coding: utf-8 -*-
"""报告生成：生成 Word 评价报告。
依赖 python-docx。报告结构：
标题 → 图斑信息 → 评价结论 → 各微地质单元得分 → 完整得分明细 → 生成时间。
"""
import io
from datetime import datetime
from docx import Document
from docx.shared import Pt, Cm
from docx.oxml.ns import qn
from docx.enum.text import WD_ALIGN_PARAGRAPH
def _set_run(run, text=None, size=12, bold=False):
    if text is not None:
        run.text = text
    run.font.name = "Times New Roman"
    run.font.size = Pt(size)
    run.font.bold = bold
    # 中文字体
    run._element.rPr.rFonts.set(qn("w:eastAsia"), "宋体")
    return run
def _para(doc, text, size=12, bold=False, align=None,
    space_after=6):
    p = doc.add_paragraph()
    if align is not None:
        p.alignment = align
    p.paragraph_format.space_after = Pt(space_after)
```

```
    _set_run(p.add_run(), text, size, bold)                          1601
    return p                                                         1602
def _heading(doc, text, level=1):                                    1603
    h = doc.add_heading(text, level=level)                           1604
    for run in h.runs:                                               1605
        run.font.name = "Times New Roman"                            1606
        run._element.rPr.rFonts.set(qn("w:eastAsia"), "黑体")        1607
    return h                                                         1608
def _table(doc, headers, rows, widths=None):                         1609
    t = doc.add_table(rows=1, cols=len(headers))                     1610
    t.style = "Table Grid"                                           1611
    hdr = t.rows[0].cells                                            1612
    for i, htext in enumerate(headers):                              1613
        _set_run(hdr[i].paragraphs[0].add_run(), htext,              1614
    10.5, bold=True)                                                 1615
    for row in rows:                                                 1616
        cells = t.add_row().cells                                    1617
        for i, val in enumerate(row):                                1618
            _set_run(cells[i].paragraphs[0].add_run(),               1619
    str(val), 10.5)                                                  1620
    return t                                                         1621
def build_report(patch, mine_result):                                1622
    """生成评价报告 Word 文档，返回 BytesIO。                        1623
    参数                                                             1624
    ----                                                             1625
    patch : MinePatch，图斑对象                                      1626
    mine_result : evaluation_service.score_mine 的返回 dict          1627
    """                                                              1628
    doc = Document()                                                 1629
    # 页边距                                                         1630
    for section in doc.sections:                                     1631
        section.top_margin = Cm(2.5)                                 1632
        section.bottom_margin = Cm(2.5)                              1633
        section.left_margin = Cm(2.5)                                1634
        section.right_margin = Cm(2.5)                               1635
    _heading(doc, "历史遗留废弃矿山自然恢复适宜性评价报告", 0)       1636
    _para(doc, "软件：历史遗留废弃矿山自然恢复适宜性评价系统 Ｖ1．0",  1637
            size=10.5, align=WD_ALIGN_PARAGRAPH.CENTER)              1638
    _para(doc, "生成时间：%s" % datetime.now().strftime("%Y-%m-%d    1639
    %H:%M:%S"),                                                      1640
            size=10.5, align=WD_ALIGN_PARAGRAPH.CENTER,              1641
    space_after=12)                                                  1642
    # 图斑信息                                                       1643
    _heading(doc, "一、图斑基本信息", 1)                             1644
    info = [                                                         1645
        ("图斑名称", patch.name),                                    1646
        ("所在行政区", patch.admin_region or "—"),                   1647
        ("图斑面积（公顷）", str(patch.area_ha) if patch.area_ha is  1648
    not None else "—"),                                              1649
        ("开采矿种", patch.mineral_type or "—"),                     1650
```

```
        ("开采方式", patch.mining_method or "—"),
    ]
    _table(doc, ["项目", "内容"], info)
    # 评价结论
    _heading(doc, "二、评价结论", 1)
    gi = mine_result["grade_info"]
    concl = [
        ("最终评价因子 F", "%.4f" % mine_result["final_score"]),
        ("适宜性等级", mine_result["grade"]),
        ("判定依据", gi["meaning"]),
        ("推荐措施", gi["measure"]),
    ]
    _table(doc, ["项目", "结果"], concl)
    # 各单元得分
    _heading(doc, "三、各微地质单元得分", 1)
    rows = []
    for u in mine_result["unit_results"]:
        rows.append((u["unit_type"], "%.2f%%" % (u["ratio"] * 100),
                     "%.4f" % u["unit_score"], "%.4f" % u["weighted"]))
    _table(doc, ["微地质单元", "面积占比", "单元评价因子 F'", "加权贡献"], rows)
    # 完整得分明细
    _heading(doc, "四、得分明细", 1)
    for u in mine_result["unit_results"]:
        _para(doc, "%s（占比 %.2f%%）" % (u["unit_type"], u["ratio"] * 100),
              size=11, bold=True, space_after=4)
        detail_rows = []
        for d in u["details"]:
            raw = "—" if d["raw"] is None else d["raw"]
            std = "—" if d["std_value"] is None else "%.4f" % d["std_value"]
            w = "%.4f" % d["weight"]
            c = "—" if d["contribution"] is None else "%.4f" % d["contribution"]
            detail_rows.append((d["l1"], d["l2"], raw, std, w, c))
        _table(doc, ["一级指标", "二级指标", "实测值", "标准化值", "权重", "贡献"],
               detail_rows)
        # 一级指标得分
        l1_row = []
        for k, v in u["l1_scores"].items():
            l1_row.append("%s=%.4f" % (k, v))
        _para(doc, "一级指标得分：" + "，".join(l1_row), size=10.5, space_after=10)
    buf = io.BytesIO()
    doc.save(buf)
```

```
    buf.seek(0)
    return buf
# ==== app/services/indicator_service.py ====
# -*- coding: utf-8 -*-
"""指标定义维护。
把论文表 2/表 3 的指标体系写入 indicators 表，使指标名称、单位、
理想值与取值依据可在界面上查看维护，而不再只存在于代码常量中。
代码常量（app.utils.constants）仍是计算时的**权威取值来源**；
本表用于展示、核对与记录取值依据。二者由 seed_indicators() 保持同步。
"""
from app.models import db, Indicator
from app.utils.constants import (
    UNIT_TYPES, UNIT_ENABLED_INDICATORS, IDEAL_SOIL_RATIO,
    IDEAL_THICKNESS,
    IDEAL_PH_RANGE, IDEAL_ORGANIC, IDEAL_COVERAGE,
    IDEAL_SHANNON,
    HEAVY_METAL_LIMIT, SLOPE_BREAK, ASPECT_SCORES,
)
ALL_UNITS = ",".join(UNIT_TYPES)
# (code, name, level, parent, unit, ideal_value,
    ideal_type, enabled_units, desc, is_constraint)
TREE = [
    # ---- 一级指标 ----
    ("soil", "土壤指标", 1, None, None, None, None, "碎石边坡,平台",
    "论文表 2 评价指标之一级指标", False),
    ("veg", "植被指标", 1, None, None, None, None, ALL_UNITS,
    "论文表 2 评价指标之一级指标", False),
    ("terrain", "地形指标", 1, None, None, None, None,
    "岩质边坡,碎石边坡", "论文表 2 评价指标之一级指标", False),
    # ---- 二级指标：土壤 ----
    ("soil_ratio", "土石比", 2, "soil", "—",
    str(IDEAL_SOIL_RATIO), "benefit", "碎石边坡,平台",
     "论文 4.2.2 节：理想值 0.8，根据矿山修复实际情况确定", False),
    ("thickness", "土层厚度", 2, "soil", "cm",
    str(IDEAL_THICKNESS), "benefit", "碎石边坡,平台",
     "论文 4.2.2 节：理想值 20 cm", False),
    ("ph", "pH", 2, "soil", "—", "%g~%g" % IDEAL_PH_RANGE,
    "mid", "碎石边坡,平台",
     "论文 4.2.2 节：pH 用适中指标的标准化方法，理想区间 %g~%g" %
    IDEAL_PH_RANGE, False),
    ("organic_matter", "有机质", 2, "soil", "g/kg",
    str(IDEAL_ORGANIC), "benefit", "碎石边坡,平台",
     "《全国第二次土壤普查养分等级划分标准》二等（较丰富）最低值 30 g/kg", False),
    ("pollution", "污染指标", 2, "soil", "mg/kg", "见三级指标",
    "cost", "碎石边坡,平台",
     "同时作为限制性指标与评价指标；8 种重金属等权合成（各 0.125）", True),
    # ---- 二级指标：植被 ----
    ("coverage", "植被覆盖度", 2, "veg", "%",
    str(IDEAL_COVERAGE), "benefit", ALL_UNITS,
     "论文 4.2.2 节：参照生态系统取值 75%", False),
```

```
        ("shannon", "生物多样性", 2, "veg", "—", str(IDEAL_SHANNON),
        "benefit", ALL_UNITS,
         "Shannon-Wiener 多样性指数，参照生态系统取值 0.7", False),
        # ---- 二级指标：地形 ----
        ("slope", "坡度", 2, "terrain", "°", "%g° 分界" %
        SLOPE_BREAK, "cost", "岩质边坡,碎石边坡",
         "论文 4.2.2 节引《新一轮退耕还林还草总体方案》发改西部〔2014〕1772 号：25°
        为陡坡重要分界", False),
        ("aspect", "坡向", 2, "terrain", "—",
        "阳坡0.2/阴阳坡0.6/阴坡1.0", "discrete", "岩质边坡,碎石边坡",
         "论文 4.2.2 节：阴坡及半阴坡水分条件更优，故赋值高于阳坡", False),
]
# 污染指标下的三级指标（8 种重金属）
for _m, _limit in HEAVY_METAL_LIMIT.items():
    TREE.append((
        "hm_" + _m, _m, 3, "pollution", "mg/kg",
    str(_limit), "cost",
        "碎石边坡,平台",
        "《绿化种植土壤》CJ/T 340—2016 土壤重金属含量 IV 级最高值 %g mg/kg" %
    _limit,
        False,
    ))
def seed_indicators():
    """写入/更新指标定义（幂等）。返回写入条数。"""
    code_map = {
        "benefit": "越大越优", "cost": "越小越优",
        "mid": "适中型", "discrete": "离散赋值",
    }
    n = 0
    for (code, name, level, parent, unit, ideal, itype,
    units, desc, is_con) in TREE:
        ind = Indicator.query.filter_by(code=code).first()
        if ind is None:
            ind = Indicator(code=code)
            db.session.add(ind)
        ind.name = name
        ind.level = level
        ind.parent_code = parent
        ind.unit = unit
        ind.ideal_value = ideal
        ind.ideal_type = itype
        ind.enabled_units = units
        ind.description = desc
        ind.is_constraint = is_con
        ind.is_evaluative = (level in (2, 3))
        n += 1
    db.session.commit()
    return n
def ideal_type_label(code):
    return {"benefit": "越大越优", "cost": "越小越优",
```

```
                "mid": "适中型", "discrete": "离散赋值"}.get(code,
    code or "—")
# ==== app/services/import_service.py ====
# -*- coding: utf-8 -*-
"""指标数据批量导入。
支持 CSV 与 Excel（.xlsx/.xls）。一行对应一个微地质单元，
按「单元类型」匹配图斑内已有的微地质单元后写入指标数据。
表头约定（缺失列按不适用处理）：
    单元类型， 土石比， 土层厚度， pH， 有机质，
    Hg, As, Ni, Cu, Cd, Pb, Cr, Zn,
    植被覆盖度， Shannon， 坡度， 坡向
"""
import csv
import io
import json
from app.utils.constants import HEAVY_METALS, ASPECT_TYPES,
    UNIT_TYPES
# 表头（顺序即导出模板顺序）
HEADERS = (["单元类型", "土石比", "土层厚度", "pH", "有机质"]
           + list(HEAVY_METALS)
           + ["植被覆盖度", "Shannon", "坡度", "坡向"])
FIELD_ALIASES = {
    "单元类型": "unit_type",
    "土石比": "soil_ratio",
    "土层厚度": "thickness",
    "pH": "ph", "PH": "ph", "ph": "ph",
    "有机质": "organic_matter",
    "植被覆盖度": "coverage",
    "Shannon": "shannon", "shannon": "shannon", "生物多样性":
    "shannon",
    "坡度": "slope",
    "坡向": "aspect",
}
for _m in HEAVY_METALS:
    FIELD_ALIASES[_m] = "hm_" + _m
def build_template_csv():
    """生成导入模板（含表头与示例行）。"""
    buf = io.StringIO()
    w = csv.writer(buf)
    w.writerow(HEADERS)
    # 示例：碎石边坡一行，说明每列含义
    row = {h: "" for h in HEADERS}
    row.update({"单元类型": "碎石边坡", "土石比": "0.34", "土层厚度": "10",
                "pH": "9.24", "有机质": "4.89", "Hg": "0.05",
    "As": "5.17",
                "Ni": "12.60", "Cu": "11.70", "Cd": "0.04",
    "Pb": "6.50",
                "植被覆盖度": "87.92", "Shannon": "0.60",
                "坡度": "28.75", "坡向": "阳坡"})
    w.writerow([row[h] for h in HEADERS])
```

```
    return buf.getvalue()
def _read_rows(filename, stream):
    """读取上传文件为「表头 -> 值」的字典列表。"""
    ext = filename.rsplit(".", 1)[-1].lower()
    if ext == "csv":
        text = stream.read().decode("utf-8-sig",
errors="replace")
        reader = csv.DictReader(io.StringIO(text))
        return [dict(r) for r in reader]
    if ext in ("xlsx", "xls"):
        from openpyxl import load_workbook
        wb = load_workbook(io.BytesIO(stream.read()),
data_only=True)
        ws = wb.active
        rows = list(ws.iter_rows(values_only=True))
        if not rows:
            return []
        header = [str(c).strip() if c is not None else ""
for c in rows[0]]
        out = []
        for r in rows[1:]:
            if all(c is None or str(c).strip() == "" for c
in r):
                continue
            out.append({header[i]: (r[i] if i < len(r) else
None)
                        for i in range(len(header)) if
header[i]})
        return out
    raise ValueError("不支持的文件类型：.%s" % ext)
def _num(value):
    if value is None:
        return None, None
    s = str(value).strip()
    if s == "":
        return None, None
    try:
        return float(s), None
    except ValueError:
        return None, "「%s」不是有效数值" % s
def parse_upload(filename, stream, unit_type_map):
    """解析上传文件。
    参数
    ----
    unit_type_map : dict, {单元类型: MicroUnit 对象}
    返回
    ----
    (parsed, errors)
      parsed : list[dict], 每项 {unit_type, fields{...},
    heavy_metals{}}
```

```
    errors : list[str]，逐行问题说明
"""
rows = _read_rows(filename, stream)
parsed, errors = [], []
if not rows:
    return [], ["文件为空或没有数据行"]
# 表头校验
unknown = [h for h in (rows[0].keys() if rows else [])
if h not in FIELD_ALIASES]
if unknown:
    errors.append("存在无法识别的列，将被忽略：%s" %
"、".join(map(str, unknown)))
for idx, raw in enumerate(rows, start=2):       # 第 1 行是表头
    rec = {}
    for k, v in raw.items():
        key = FIELD_ALIASES.get(str(k).strip())
        if key:
            rec[key] = v
    unit_type = str(rec.get("unit_type") or "").strip()
    if not unit_type:
        errors.append("第 %d 行：缺少「单元类型」" % idx)
        continue
    if unit_type not in unit_type_map:
        errors.append("第 %d 行：图斑内不存在单元类型「%s」（现有：%s）"
                      % (idx, unit_type,
"、".join(unit_type_map) or "无"))
        continue
    if unit_type not in UNIT_TYPES:
        errors.append("第 %d 行：单元类型「%s」不是系统支持的三种之一"
                      % (idx, unit_type))
        continue
    fields, heavy = {}, {}
    ok = True
    for key, val in rec.items():
        if key in ("unit_type",):
            continue
        if key.startswith("hm_"):
            n, err = _num(val)
            if err:
                errors.append("第 %d 行：%s %s" % (idx,
key[3:], err))
                ok = False
            elif n is not None:
                heavy[key[3:]] = n
            continue
        if key == "aspect":
            s = str(val).strip() if val is not None
else ""
            if s == "":
                fields["aspect"] = None
```

```
                    elif s in ASPECT_TYPES:
                        fields["aspect"] = s
                    else:
                        errors.append("第 %d 行：坡向「%s」无效（应为 %s）"
                                      % (idx, s,
    "、".join(ASPECT_TYPES)))
                        ok = False
                    continue
                n, err = _num(val)
                if err:
                    errors.append("第 %d 行：%s %s" % (idx, key,
    err))
                    ok = False
                else:
                    fields[key] = n
            if ok:
                parsed.append({"unit_type": unit_type,
    "fields": fields,
                               "heavy_metals": heavy})
    return parsed, errors
# ==== app/services/workflow.py ====
# -*- coding: utf-8 -*-
"""评价流程前置条件（定量评价与报告输出的流程门禁）。
论文 4.1 节确定的评价流程存在严格的先后关系：
    建立图斑 → 限制性条件判定（通过） → 划分微地质单元并计算面积占比
    → 逐单元录入指标数据 → 计算评价因子并判定等级 → 输出报告
论文原文：「先进行政策、位置约束地质灾害判断……对于满足限制性条件的
废弃矿山，按照地形地貌划分，将其划分为平台、碎石边坡和岩质边坡三种
地质微单元类型，并计算各种地形面积占比。」限制性条件判定（一票否决）
在前，微地质单元划分在后，顺序不可颠倒。
本模块把该顺序固化为可检查的状态。后续模块在入口处调用它，
前置步骤未完成时直接拦截并给出「缺什么、去哪补」的提示，
避免在条件不具备时产出看似合理的评价结论。
"""
from app.models import (db, MicroUnit, IndicatorData,
    ConstraintCheck,
                        EvaluationResult)
# 指标数据非空字段（用于判断某单元是否已录入数据）
DATA_FIELDS = ("soil_ratio", "thickness", "ph",
    "organic_matter",
                "coverage", "shannon", "slope", "aspect")
def _has_data(ind):
    if ind is None:
        return False
    if any(getattr(ind, f, None) is not None for f in
    DATA_FIELDS):
        return True
    return bool(ind.heavy_metals_dict())
def patch_progress(patch_id):
    """检查某图斑的评价流程完成情况。
```

```
    返回 dict:                                                          2001
        units          {done, count, ratio_sum, detail}                 2002
        constraint     {done, passed, detail}                           2003
        data           {done, ready, missing, detail}                   2004
        blocking       未满足的前置步骤列表（按流程顺序）                2005
        ready          是否可进入定量评价                               2006
    """                                                                 2007
    units = MicroUnit.query.filter_by(patch_id=patch_id).all()          2008
                                                                        2009
    ratio_sum = sum(u.ratio or 0.0 for u in units)                      2010
    # ---- 步骤一：微地质单元 ----                                       2011
    units_step = {                                                      2012
        "done": bool(units),                                            2013
        "count": len(units),                                            2014
        "ratio_sum": ratio_sum,                                         2015
        "detail": ("已划分 %d 个微地质单元，面积占比合计 %.2f%%"          2016
                   % (len(units), ratio_sum * 100))                     2017
                  if units else "尚未划分微地质单元",                   2018
    }                                                                   2019
    # ---- 步骤二：限制性条件判定 ----                                  2020
    cc = ConstraintCheck.query.filter_by(patch_id=patch_id)             2021
    \                                                                   2022
        .order_by(ConstraintCheck.id.desc()).first()                    2023
    if cc is None:                                                      2024
        constraint_step = {"done": False, "passed": False,              2025
                           "detail": "尚未执行限制性条件判定"}          2026
    elif cc.passed:                                                     2027
        constraint_step = {"done": True, "passed": True,                2028
                           "detail": "限制性条件判定已通过"}            2029
    else:                                                               2030
        constraint_step = {"done": True, "passed": False,               2031
                           "detail": "限制性条件判定未通过（%s）" %     2032
    (cc.reason or "")}                                                  2033
    # ---- 步骤三：指标数据 ----                                        2034
    missing = []                                                        2035
    for u in units:                                                     2036
        ind = IndicatorData.query.filter_by(unit_id=u.id).fi            2037
    rst()                                                               2038
        if not _has_data(ind):                                          2039
            missing.append(u.unit_type)                                 2040
    if not units:                                                       2041
        data_step = {"done": False, "ready": False,                     2042
    "missing": [],                                                      2043
                     "detail": "尚无可录入数据的微地质单元"}            2044
    elif missing:                                                       2045
        data_step = {"done": False, "ready": False,                     2046
    "missing": missing,                                                 2047
                     "detail": "以下微地质单元尚未录入指标数据：%s"     2048
                               % "、".join(missing)}                    2049
    else:                                                               2050
```

```
        data_step = {"done": True, "ready": True,
"missing": [],
                     "detail": "全部 %d 个微地质单元的指标数据已录入" %
len(units)}
ratio_ok = ratio_sum > 0.999 if units else False
if units and not ratio_ok:
        units_step["detail"] += "（占比合计不等于 100%，请检查面积录入）"
#  ---- 汇总未满足的前置条件（按论文 4.1 节的流程顺序：限制性判定优先） ----
#  reason 只写「缺什么」这一个事实，不写为什么重要的说明 ——
#  缺少什么由 label 指明，怎么补由 fix_label 给出，页面不必再解释一遍。
blocking = []
if not constraint_step["done"]:
        blocking.append({
            "step": "constraint", "label": "限制性条件判定未执行",
            "reason": "五项限制性条件尚未逐项判定。",
            "fix_url": "/constraint/%d" % patch_id,
"fix_label": "去执行判定",
        })
elif not constraint_step["passed"]:
        blocking.append({
            "step": "constraint", "label": "限制性条件判定未通过",
            "reason": cc.reason or "",
            "fix_url": "/constraint/%d" % patch_id,
"fix_label": "去复核判定结果",
        })
if not units_step["done"]:
        blocking.append({
            "step": "units", "label": "未划分微地质单元",
            "reason": "该图斑下没有任何微地质单元。",
            "fix_url": "/unit/%d" % patch_id, "fix_label":
"去划分",
        })
elif not ratio_ok:
        blocking.append({
            "step": "units", "label": "面积占比合计不为 100%",
            "reason": "当前合计 %.2f%%。" % (ratio_sum * 100),
            "fix_url": "/unit/%d" % patch_id, "fix_label":
"去核对面积",
        })
if units_step["done"] and not data_step["ready"]:
        blocking.append({
            "step": "data", "label": "指标数据未录入完整",
            "reason": data_step["detail"],
            "fix_url": "/indicator/", "fix_label": "去录入",
        })
return {
        "units": units_step,
        "constraint": constraint_step,
        "data": data_step,
        "blocking": blocking,
```

```
            "ready": not blocking,
        }
def latest_result(patch_id):
        """取该图斑最近一次的评价记录。"""
        return EvaluationResult.query.filter_by(patch_id=patch_id) \
            .order_by(EvaluationResult.id.desc()).first()
def report_progress(patch_id):
        """报告输出的前置条件：评价流程齐备，且已有评价记录。
        报告本质上是评价结论的呈现，因此必须先真实执行过定量评价，
        导出时读取已保存的评价结果，而不是在导出环节重新计算。
        否则评价完成后再修改指标数据，报告会与系统记录的评价结论不一致，
        同一份评价出现两个数字，结论无法追溯。
        返回 dict：{ready, blocking, result, flow}
            result 为最近一次评价记录（EvaluationResult），无记录时为空。
        """
        flow = patch_progress(patch_id)
        blocking = list(flow["blocking"])
        record = latest_result(patch_id)
        if not blocking and record is None:
            blocking.append({
                "step": "evaluate", "label": "尚未执行适宜性评价",
                "reason": "该图斑还没有评价记录。",
                "fix_url": "/evaluate/%d" % patch_id,
                "fix_label": "去执行评价",
            })
        return {
            "ready": not blocking,
            "blocking": blocking,
            "result": record,
            "flow": flow,
        }
# ==== app/services/dashboard.py ====
# -*- coding: utf-8 -*-
"""工作台数据：统计数字、评价流程步骤状态、侧边栏待办角标。
首页（工作台）需要把「现在有哪些数据」「流程走到哪一步」「哪一步还差活」
一次算清楚，供模板直接渲染。集中在此，避免在视图与模板里各算一遍。
"""
from flask import g
from app.models import (db, Project, MinePatch, MicroUnit,
        IndicatorData,
                                ConstraintCheck, EvaluationResult,
        Standard)
from app.services import workflow as wf
#  评价流程的八个步骤（顺序即论文 4.1 节的作业顺序）
#     key          步骤标识
#     no           步骤编号（01~08）
#     title        名称
#     icon         图标名（见 templates/_macros.html 的 ICONS）
#     desc         一句话说明
```

```
#     href          入口地址                                                    2151
STEPS = [                                                                       2152
      dict(key="project", no="01", title="项目与矿山档案",                      2153
      icon="folder",                                                            2154
            href="/project/",                                                   2155
            desc="建立评价项目、录入矿山图斑、挂接遥感影像、设置参照生态系统"),  2156
      dict(key="standard", no="02", title="政策与标准知识库",                   2157
      icon="book",                                                              2158
            href="/standard/",                                                  2159
            desc="收录地方标准与政策文件，结构化判定规则，支持省际口径比对"),    2160
      dict(key="constraint", no="03", title="限制性指标判定",                   2161
      icon="shield",                                                            2162
            href="/constraint/",                                                2163
            desc="政策、区位约束、土壤污染、地质灾害、耕地属性五项一票否决"),    2164
      dict(key="unit", no="04", title="微地质单元", icon="layers",              2165
            href="/unit/",                                                      2166
            desc="划分岩质边坡、碎石边坡、平台三类单元，自动计算面积占比"),      2167
      dict(key="ahp", no="05", title="指标体系与权重",                          2168
      icon="sliders",                                                           2169
            href="/ahp/",                                                       2170
            desc="指标定义维护，判断矩阵录入、权重计算与一致性检验、多专家汇总"),2171
      dict(key="indicator", no="06", title="指标数据采集",                      2172
      icon="database",                                                          2173
            href="/indicator/",                                                 2174
            desc="按微地质单元录入土壤、植被、地形指标，支持 CSV / Excel 批量导入"),2175
      dict(key="evaluate", no="07", title="适宜性定量评价",                     2176
      icon="target",                                                            2177
            href="/evaluate/",                                                  2178
            desc="指标标准化、分单元赋分、面积加权合成，判定适宜性等级"),        2179
      dict(key="report", no="08", title="报告与结果输出",                       2180
      icon="file",                                                              2181
            href="/report/",                                                    2182
            desc="结果图表与逐层得分明细，导出 PDF 与 Word 评价报告"),          2183
]                                                                               2184
def _has_data(ind):                                                             2185
      return wf._has_data(ind)                                                  2186
#  系统的九个功能模块（登录页介绍用，含不在评价流程内的用户与权限管理）       2187
MODULES = [                                                                     2188
      ("01", "用户与权限管理", "登录注销、三种角色权限、写操作自动留痕"),      2189
      ("02", "项目与矿山档案", "建立项目、录入图斑、挂接遥感影像、设置参照生态系统"),2190
      ("03", "政策与标准知识库", "标准判定条件结构化为规则，支持省际口径比对"),2191
      ("04", "限制性指标判定", "政策、区位、污染、地质灾害、耕地五项一票否决"),2192
      ("05", "微地质单元管理", "划分岩质边坡、碎石边坡、平台，自动算面积占比"),2193
      ("06", "指标体系与权重", "指标定义维护、AHP 判断矩阵与一致性检验、多专家汇总"),2194
      ("07", "指标数据采集", "按单元类型录入土壤、植被、地形指标，支持批量导入"),2195
      ("08", "适宜性定量评价", "四类标准化、分单元赋分、面积加权、等级判定"),2196
      ("09", "报告与结果输出", "结果图表与逐层得分明细，导出 PDF 与 Word 报告"),2197
]                                                                               2198
def snapshot():                                                                 2199
      """一次算出工作台需要的全部数据（同一请求内只算一次）。                  2200
```

```
    返回 dict:                                                          2201
        stats       {projects, patches, unit_types, standards}      2202
        steps       八个步骤，附 available / pending / note             2203
        badges      {key: 待办数量}，仅含有待办的项                    2204
        progress    {patches, units, data_units, ready,             2205
    evaluated}                                                          2206
    """                                                                 2207
    cached = getattr(g, "_dash_snapshot", None)                         2208
    if cached is None:                                                  2209
        cached = _compute()                                             2210
        g._dash_snapshot = cached                                       2211
    return cached                                                       2212
def _compute():                                                         2213
    projects = Project.query.count()                                    2214
    patches = MinePatch.query.all()                                     2215
    patch_ids = [p.id for p in patches]                                 2216
    units = MicroUnit.query.filter(                                     2217
        MicroUnit.patch_id.in_(patch_ids)).all() if                     2218
    patch_ids else []                                                   2219
    unit_by_patch = {}                                                  2220
    for u in units:                                                     2221
        unit_by_patch.setdefault(u.patch_id, []).append(u)              2222
    unit_ids = [u.id for u in units]                                    2223
    data_unit_ids = set()                                               2224
    if unit_ids:                                                        2225
        for ind in IndicatorData.query.filter(                          2226
                IndicatorData.unit_id.in_(unit_ids)).all():             2227
            if _has_data(ind):                                          2228
                data_unit_ids.add(ind.unit_id)                          2229
    passed_constraint = set()                                           2230
    if patch_ids:                                                       2231
        for c in ConstraintCheck.query.filter(                          2232
                ConstraintCheck.patch_id.in_(patch_ids)).all2233
    ():                                                                 2234
            if c.passed:                                                2235
                passed_constraint.add(c.patch_id)                       2236
    evaluated = set()                                                   2237
    if patch_ids:                                                       2238
        for r in EvaluationResult.query.filter(                         2239
                EvaluationResult.patch_id.in_(patch_ids)).al2240
    l():                                                                2241
            evaluated.add(r.patch_id)                                   2242
    # ---- 各步骤的待办数量 ----                                        2243
    pending = {                                                         2244
        "constraint": sum(1 for p in patches if p.id not in 2245
    passed_constraint),                                                 2246
        "unit": sum(1 for p in patches if not                           2247
    unit_by_patch.get(p.id)),                                           2248
        "indicator": 0,                                                 2249
        "evaluate": 0,                                                  2250
```

```
    }
    ready_patches = 0
    for p in patches:
        us = unit_by_patch.get(p.id, [])
        if us and any(u.id not in data_unit_ids for u in
us):
            pending["indicator"] += 1
        pr = wf.patch_progress(p.id)
        if pr["ready"]:
            ready_patches += 1
            if p.id not in evaluated:
                pending["evaluate"] += 1
    # ---- 步骤可用性：前置数据不具备时置灰 ----
    avail = {
        "project": True,
        "standard": True,
        "constraint": bool(patches),
        "unit": bool(patches),
        "ahp": True,
        "indicator": bool(units),
        "evaluate": ready_patches > 0,
        "report": bool(evaluated),
    }
    notes = {
        "constraint": "需先建立矿山图斑",
        "unit": "需先建立矿山图斑",
        "indicator": "需先划分微地质单元",
        "evaluate": "需前置步骤齐备的图斑",
        "report": "需先执行适宜性评价",
    }
    steps = []
    for s in STEPS:
        steps.append(dict(
            s,
            available=avail[s["key"]],
            pending=pending.get(s["key"], 0),
            note=None if avail[s["key"]] else
notes.get(s["key"], ""),
        ))
    unit_types = len({u.unit_type for u in units})
    stats = {
        "projects": projects,
        "patches": len(patches),
        "unit_types": unit_types,
        "standards": Standard.query.count(),
    }
    return {
        "stats": stats,
        "steps": steps,
        "badges": {k: v for k, v in pending.items() if v},
```

```
            "progress": {
                "patches": len(patches),
                "units": len(units),
                "data_units": len(data_unit_ids),
                "ready": ready_patches,
                "evaluated": len(evaluated),
            },
        }
def nav_badges():
    """侧边栏待办角标：{入口地址前缀： 数量}，只含有待办的项。"""
    snap = snapshot()
    return {
        "/constraint/": snap["badges"].get("constraint", 0),
        "/unit/": snap["badges"].get("unit", 0),
        "/indicator/": snap["badges"].get("indicator", 0),
        "/evaluate/": snap["badges"].get("evaluate", 0),
    }
# ==== app/services/audit.py ====
# -*- coding: utf-8 -*-
"""操作审计。
在应用工厂中以 after_request 钩子统一埋点：凡是成功的写操作（POST/PUT/DELETE）
且已登录，即写入 operation_logs 表，无需在每个视图里手工调用。
动作名称由「端点 → 中文动作」映射表给出，未收录的端点回退为方法 + 路径。
"""
from flask import request
from flask_login import current_user
from app.models import db, OperationLog
# 端点 → 可读动作名
ACTION_NAMES = {
    "auth.login": "登录系统",
    "auth.logout": "退出登录",
    "auth.add_user": "新增用户",
    "auth.edit_user": "编辑用户",
    "auth.delete_user": "删除用户",
    "auth.reset_password": "重置用户密码",
    "auth.profile": "修改个人资料",
    "auth.upload_avatar": "更新头像",
    "auth.delete_avatar": "移除头像",
    "project.add": "新建项目",
    "project.edit": "编辑项目",
    "project.delete": "删除项目",
    "project.patch_add": "新建矿山图斑",
    "project.patch_edit": "编辑矿山图斑",
    "project.patch_delete": "删除矿山图斑",
    "project.upload_image": "上传遥感影像",
    "unit.add": "添加微地质单元",
    "unit.edit": "编辑微地质单元",
    "unit.delete": "删除微地质单元",
    "standard.add": "添加标准/政策文件",
    "standard.edit": "编辑标准/政策文件",
```

```
    "standard.delete": "删除标准/政策文件",
    "standard.rules": "维护标准规则",
    "standard.rule_delete": "删除标准规则",
    "constraint.check": "执行限制性条件判定",
    "ahp.save_expert": "录入专家判断矩阵",
    "ahp.delete_expert": "删除专家判断矩阵",
    "ahp.aggregate": "汇总多专家矩阵",
    "indicator.edit": "保存指标数据",
    "indicator.import_data": "批量导入指标数据",
    "indicator.update_meta": "维护指标定义",
    "evaluate.run": "执行适宜性评价",
}
# 无需记录的操作（查看类或纯导航）
SKIP_ENDPOINTS = {"static", "auth.index"}
def action_name(endpoint):
    # 个人设置页的两个表单共用一个端点，按提交的动作区分
    if endpoint == "auth.profile":
        return "修改密码" if request.form.get("action") ==
    "password" \
            else "修改个人资料"
    if endpoint in ACTION_NAMES:
        return ACTION_NAMES[endpoint]
    return "写操作 %s %s" % (request.method, request.path)
def extract_target_id():
    """从视图参数中取出被操作对象的主键，取不到则返回 None。"""
    args = request.view_args or {}
    for key in ("pid", "patch_id", "unit_id", "sid", "rid",
    "result_id", "user_id"):
        if key in args:
            try:
                return int(args[key])
            except (TypeError, ValueError):
                return None
    return None
def record(endpoint, status_code):
    """记录一次写操作。失败（4xx/5xx）也记录，便于追溯。"""
    if endpoint in SKIP_ENDPOINTS:
        return
    try:
        detail = "路径 %s" % request.path
        if status_code >= 400:
            detail += "；结果 HTTP %d" % status_code
        db.session.add(OperationLog(
            user_id=current_user.id if
    current_user.is_authenticated else None,
            action=action_name(endpoint),
            target_type=(endpoint or "").split(".")[0] or
    None,
            target_id=extract_target_id(),
            detail=detail,
```

```
            ))                                                          2401
            db.session.commit()                                         2402
        except Exception:                                               2403
            #  审计失败不应影响业务响应                                 2404
            db.session.rollback()                                       2405
#  ====  app/views/auth.py  ====                                        2406
#  -*-  coding:  utf-8  -*-                                            2407
"""用户与权限管理  ——  登录、注销、用户维护、操作日志。"""              2408
from flask import (Blueprint, render_template, redirect,               2409
    url_for, request,                                                   2410
                       flash, abort, current_app)                       2411
from flask_login import login_user, logout_user,                       2412
    login_required, current_user                                        2413
from werkzeug.security import check_password_hash,                     2414
    generate_password_hash                                              2415
from app.forms import LoginForm, EmptyForm, ProfileForm,               2416
    PasswordForm                                                        2417
from app.models import db, User, OperationLog                          2418
from app.utils import uploads as up                                    2419
from app.utils.constants import ROLE_LABELS                            2420
bp = Blueprint("auth", __name__)                                       2421
def require_admin():                                                   2422
    """非管理员直接  403，避免误操作基础数据。"""                      2423
    if current_user.role != "admin":                                   2424
        abort(403)                                                     2425
@bp.route("/")                                                         2426
def index():                                                           2427
    if current_user.is_authenticated:                                  2428
        from app.services import dashboard as dash                     2429
        return render_template("auth/home.html",                       2430
    dash=dash.snapshot())                                              2431
    return redirect(url_for("auth.login"))                             2432
@bp.route("/login", methods=["GET", "POST"])                           2433
def login():                                                           2434
    if current_user.is_authenticated:                                  2435
        return redirect(url_for("auth.index"))                         2436
    form = LoginForm()                                                 2437
    if form.validate_on_submit():                                      2438
        user = User.query.filter_by(username=form.username.d           2439
    ata).first()                                                       2440
        if user and check_password_hash(user.password_hash,            2441
    form.password.data):                                               2442
            if not user.active:                                        2443
                flash("该账号已停用，请联系管理员", "error")           2444
                return render_template("auth/login.html",              2445
    form=form,                                                         2446
                                                **_intro_data())       2447
            login_user(user, remember=form.remember.data)              2448
            #  用  toast  类别：渲染为页面正中的浮层，淡入停留后淡出。 2449
            #  登录成功只是告知性反馈，不需要用户停下来阅读。          2450
```

```
                flash("登录成功", "toast")                                  2451
                return redirect(url_for("auth.index"))                      2452
            flash("用户名或密码错误", "error")                               2453
        return render_template("auth/login.html", form=form,                2454
        **_intro_data())                                                    2455
def _intro_data():                                                          2456
    """登录页下方的系统介绍所用的静态内容。"""                                2457
    from app.services import dashboard as dash                              2458
    return {"modules": dash.MODULES}                                        2459
@bp.route("/logout")                                                        2460
@login_required                                                             2461
def logout():                                                               2462
    logout_user()                                                           2463
    flash("已退出登录", "ok")                                                2464
    return redirect(url_for("auth.login"))                                  2465
# ---------------------------- 个人设置 ----------------------------2466
    ----                                                                    2467
# 与「用户管理」的区别：那里是管理员维护别人，这里是任何登录用户维护自己。          2468
@bp.route("/profile", methods=["GET", "POST"])                              2469
@login_required                                                             2470
def profile():                                                              2471
    # 两个表单放在同一页，用字段前缀区分，避免 submit / csrf_token 重名          2472
    info_form = ProfileForm(prefix="info", obj=current_user)2473
    pwd_form = PasswordForm(prefix="pwd")                                   2474
    if request.method == "POST":                                            2475
        action = request.form.get("action")                                 2476
        if action == "info" and info_form.validate_on_submit2477
    ():                                                                     2478
            current_user.display_name =                                     2479
    (info_form.display_name.data or "").strip() or None                     2480
            current_user.title = (info_form.title.data or                   2481
    "").strip() or None                                                     2482
            current_user.email = (info_form.email.data or                   2483
    "").strip() or None                                                     2484
            current_user.phone = (info_form.phone.data or                   2485
    "").strip() or None                                                     2486
            db.session.commit()                                             2487
            flash("个人资料已保存", "ok")                                    2488
            return redirect(url_for("auth.profile"))                        2489
        if action == "password" and pwd_form.validate_on_sub2490
    mit():                                                                  2491
            if not check_password_hash(current_user.password2492
    _hash,                                                                  2493
                                                                            2494
    pwd_form.old_password.data):                                            2495
                pwd_form.old_password.errors.append("当前密码不正确2496
    ")                                                                      2497
            else:                                                           2498
                current_user.password_hash =                                2499
    generate_password_hash(                                                 2500
```

```
                pwd_form.new_password.data)
            db.session.commit()
            flash("密码已修改", "ok")
            return redirect(url_for("auth.profile"))
    return render_template("auth/profile.html",
    info_form=info_form,
                           pwd_form=pwd_form,
    avatar_form=EmptyForm(),
                           role_labels=ROLE_LABELS)
def _avatar_length_ok(file_storage):
    """按字节检查头像大小。超限时 werkzeug 读到内存里的内容也能量出来。"""
    limit = current_app.config.get("MAX_AVATAR_LENGTH", 4 *
    1024 * 1024)
    stream = file_storage.stream
    pos = stream.tell()
    stream.seek(0, 2)
    size = stream.tell()
    stream.seek(pos)
    return size <= limit, size
@bp.route("/profile/avatar", methods=["POST"])
@login_required
def upload_avatar():
    form = EmptyForm()
    if not form.validate_on_submit():
        flash("表单校验失败，请重试", "error")
        return redirect(url_for("auth.profile"))
    file = request.files.get("avatar")
    if file is None or not file.filename:
        flash("请选择头像图片", "error")
        return redirect(url_for("auth.profile"))
    ok, size = _avatar_length_ok(file)
    if not ok:
        flash("头像不能超过 %.0f MB（当前 %.1f MB）"
              % (current_app.config["MAX_AVATAR_LENGTH"] /
    1024 / 1024,
                 size / 1024 / 1024), "error")
        return redirect(url_for("auth.profile"))
    rel, err = up.save_upload(file, "avatars",
                              current_app.config["ALLOWED_AV
    ATAR_EXT"],
                              current_app.config["UPLOAD_FOL
    DER"])
    if err:
        flash(err, "error")
        return redirect(url_for("auth.profile"))
    # 换新头像时删掉旧文件，避免残留
    up.remove_upload(current_user.avatar,
    current_app.config["UPLOAD_FOLDER"])
    current_user.avatar = rel
    db.session.commit()
```

```
    flash("头像已更新", "ok")
    return redirect(url_for("auth.profile"))
@bp.route("/profile/avatar/delete", methods=["POST"])
@login_required
def delete_avatar():
    form = EmptyForm()
    if form.validate_on_submit():
        up.remove_upload(current_user.avatar,
    current_app.config["UPLOAD_FOLDER"])
        current_user.avatar = None
        db.session.commit()
        flash("头像已移除", "ok")
    return redirect(url_for("auth.profile"))
# ---------------------------- 用户维护 ----------------------------
    ----
@bp.route("/users")
@login_required
def users():
    all_users = User.query.order_by(User.id).all()
    return render_template("auth/users.html",
    users=all_users,
                           role_labels=ROLE_LABELS,
    form=EmptyForm())
@bp.route("/users/add", methods=["GET", "POST"])
@login_required
def add_user():
    require_admin()
    if request.method == "POST":
        username = request.form.get("username", "").strip()
        password = request.form.get("password", "")
        role = request.form.get("role", "viewer")
        if not username or not password:
            flash("用户名和密码不能为空", "error")
            return redirect(url_for("auth.add_user"))
        if User.query.filter_by(username=username).first():
            flash("用户名已存在", "error")
            return redirect(url_for("auth.add_user"))
        db.session.add(User(username=username, role=role,
                            password_hash=generate_password_
    hash(password)))
        db.session.commit()
        flash("用户已添加", "ok")
        return redirect(url_for("auth.users"))
    return render_template("auth/add_user.html")
@bp.route("/users/<int:user_id>/edit", methods=["GET",
    "POST"])
@login_required
def edit_user(user_id):
    require_admin()
    user = db.get_or_404(User, user_id)
```

```
    if request.method == "POST":                                    2601
        role = request.form.get("role", user.role)                  2602
        active = request.form.get("active") == "1"                  2603
        # 保护：不能把最后一个管理员降级或停用，否则无人能管理系统          2604
        if user.role == "admin" and (role != "admin" or not         2605
    active):                                                        2606
            others = User.query.filter(User.role == "admin",        2607
                                       User.id != user.id,          2608
                                                                    2609
    User.active.is_(True)).count()                                  2610
            if others == 0:                                         2611
                flash("系统至少需保留一名启用状态的管理员", "error")      2612
                return redirect(url_for("auth.edit_user",           2613
    user_id=user.id))                                               2614
        user.role = role                                            2615
        user.active = active                                        2616
        new_password = request.form.get("password",                 2617
    "").strip()                                                     2618
        if new_password:                                            2619
            user.password_hash = generate_password_hash(new_        2620
    password)                                                       2621
        db.session.commit()                                         2622
        flash("用户信息已保存", "ok")                                 2623
        return redirect(url_for("auth.users"))                      2624
    return render_template("auth/edit_user.html", user=user,        2625
                           role_labels=ROLE_LABELS,                 2626
    form=EmptyForm())                                               2627
@bp.route("/users/<int:user_id>/delete", methods=["POST"])          2628
@login_required                                                     2629
def delete_user(user_id):                                           2630
    require_admin()                                                 2631
    user = db.get_or_404(User, user_id)                             2632
    if user.id == current_user.id:                                  2633
        flash("不能删除当前登录的账号", "error")                      2634
        return redirect(url_for("auth.users"))                      2635
    if user.role == "admin":                                        2636
        others = User.query.filter(User.role == "admin",            2637
                                   User.id != user.id,              2638
                                   User.active.is_(True)).co        2639
    unt()                                                           2640
        if others == 0:                                             2641
            flash("系统至少需保留一名启用状态的管理员", "error")          2642
            return redirect(url_for("auth.users"))                  2643
    db.session.delete(user)                                         2644
    db.session.commit()                                             2645
    flash("用户「%s」已删除" % user.username, "ok")                   2646
    return redirect(url_for("auth.users"))                          2647
# ------------------------------ 操作日志 ---------------------------2648
    ----                                                            2649
@bp.route("/logs")                                                  2650
```

```
@login_required                                                    2651
def logs():                                                        2652
    page = request.args.get("page", 1, type=int)                   2653
    per_page = 50                                                  2654
    q = OperationLog.query.order_by(OperationLog.id.desc())        2655
    total = q.count()                                              2656
    records = q.offset((page - 1) * per_page).limit(per_page       2657
    ).all()                                                        2658
    users_map = {u.id: u.username for u in User.query.all()}       2659
    pages = max(1, (total + per_page - 1) // per_page)             2660
    return render_template("auth/logs.html",                       2661
    records=records, page=page,                                    2662
                                pages=pages, total=total,          2663
    users_map=users_map)                                           2664
# ==== app/views/project.py ====                                   2665
# -*- coding: utf-8 -*-                                            2666
"""项目与矿山档案 —— 项目、图斑、参照生态系统、遥感影像挂接。"""                2667
from datetime import datetime                                      2668
from flask import (Blueprint, render_template, request,            2669
    redirect, url_for,                                             2670
                        flash, current_app)                        2671
from flask_login import login_required, current_user               2672
from app.forms import EmptyForm                                    2673
from app.models import db, Project, MinePatch,                     2674
    ReferenceEcosystem                                             2675
bp = Blueprint("project", __name__, url_prefix="/project")         2676
@bp.route("/")                                                     2677
@login_required                                                    2678
def index():                                                       2679
    projects = Project.query.order_by(Project.id.desc()).all       2680
    ()                                                             2681
    return render_template("project/index.html",                   2682
    projects=projects,                                             2683
                                form=EmptyForm())                  2684
@bp.route("/add", methods=["GET", "POST"])                         2685
@login_required                                                    2686
def add():                                                         2687
    form = EmptyForm()                                             2688
    if request.method == "POST" and form.validate_on_submit(       2689
    ):                                                             2690
        p = Project(                                               2691
            name=request.form.get("name", "").strip(),             2692
            code=request.form.get("code", "").strip() or           2693
    None,                                                          2694
            description=request.form.get("description",            2695
    "").strip() or None,                                           2696
            created_by=current_user.id,                            2697
        )                                                          2698
        if not p.name:                                             2699
            flash("项目名称不能为空", "error")                              2700
```

```
            else:                                                    2701
                db.session.add(p)                                    2702
                db.session.commit()                                  2703
                flash("项目已创建", "ok")                              2704
                return redirect(url_for("project.index"))            2705
        return render_template("project/edit.html", form=form,       2706
        project=None)                                                2707
@bp.route("/<int:pid>/edit", methods=["GET", "POST"])                2708
@login_required                                                      2709
def edit(pid):                                                       2710
        p = db.get_or_404(Project, pid)                              2711
        form = EmptyForm()                                           2712
        if request.method == "POST" and form.validate_on_submit(     2713
        ):                                                           2714
            p.name = request.form.get("name", "").strip()            2715
            p.code = request.form.get("code", "").strip() or         2716
        None                                                         2717
            p.description = request.form.get("description",          2718
        "").strip() or None                                          2719
            p.status = request.form.get("status", "进行中")           2720
            db.session.commit()                                      2721
            flash("项目已保存", "ok")                                  2722
            return redirect(url_for("project.index"))                2723
        return render_template("project/edit.html", form=form,       2724
        project=p)                                                   2725
@bp.route("/<int:pid>/delete", methods=["POST"])                     2726
@login_required                                                      2727
def delete(pid):                                                     2728
        p = db.get_or_404(Project, pid)                              2729
        db.session.delete(p)                                         2730
        db.session.commit()                                          2731
        flash("项目已删除", "ok")                                      2732
        return redirect(url_for("project.index"))                    2733
# ------------------------------ 矿山图斑 ------------------------------2734
        ----                                                         2735
@bp.route("/<int:pid>/patches")                                      2736
@login_required                                                      2737
def patches(pid):                                                    2738
        p = db.get_or_404(Project, pid)                              2739
        return render_template("project/patches.html",               2740
        project=p,                                                   2741
                                    form=EmptyForm())                2742
@bp.route("/patch/<int:patch_id>/delete", methods=["POST"])          2743
@login_required                                                      2744
def patch_delete(patch_id):                                          2745
        """删除矿山图斑，及其微地质单元与指标数据。"""                          2746
        from app.models import MicroUnit, IndicatorData,             2747
        EvaluationResult, \                                          2748
            ScoreDetail, ConstraintCheck, ReferenceEcosystem         2749
        patch = db.get_or_404(MinePatch, patch_id)                   2750
```

```
    pid = patch.project_id                                                    2751
    unit_ids = [u.id for u in MicroUnit.query.filter_by(patc                  2752
    h_id=patch.id).all()]                                                     2753
    if unit_ids:                                                              2754
        IndicatorData.query.filter(IndicatorData.unit_id.in_                  2755
    (unit_ids)).delete(                                                       2756
            synchronize_session=False)                                        2757
    for r in EvaluationResult.query.filter_by(patch_id=patch                  2758
    .id).all():                                                               2759
        ScoreDetail.query.filter_by(result_id=r.id).delete(                   2760
            synchronize_session=False)                                        2761
    EvaluationResult.query.filter_by(patch_id=patch.id).dele                  2762
    te(                                                                       2763
        synchronize_session=False)                                            2764
    ConstraintCheck.query.filter_by(patch_id=patch.id).delet                  2765
    e(                                                                        2766
        synchronize_session=False)                                            2767
    MicroUnit.query.filter_by(patch_id=patch.id).delete(sync                  2768
    hronize_session=False)                                                    2769
    ReferenceEcosystem.query.filter_by(patch_id=patch.id).de                  2770
    lete(                                                                     2771
        synchronize_session=False)                                            2772
    db.session.delete(patch)                                                  2773
    db.session.commit()                                                       2774
    flash("图斑「%s」及其数据已删除" % patch.name, "ok")                                2775
    return redirect(url_for("project.patches", pid=pid))                      2776
@bp.route("/<int:pid>/patch/add", methods=["GET", "POST"])                    2777
@login_required                                                               2778
def patch_add(pid):                                                           2779
    p = db.get_or_404(Project, pid)                                           2780
    form = EmptyForm()                                                        2781
    if request.method == "POST" and form.validate_on_submit(                  2782
    ):                                                                        2783
        def fv(name, cast=str):                                               2784
            v = request.form.get(name, "").strip()                            2785
            return cast(v) if v != "" else None                               2786
        patch = MinePatch(                                                    2787
            project_id=p.id,                                                  2788
            name=request.form.get("name", "").strip(),                        2789
            admin_region=fv("admin_region"),                                  2790
            area_ha=fv("area_ha", float),                                     2791
            mineral_type=fv("mineral_type"),                                  2792
            mining_method=fv("mining_method"),                                2793
            stop_years=fv("stop_years", float),                               2794
            responsible_entity_missing=bool(request.form.get                  2795
    ("responsible_entity_missing")),                                          2796
        )                                                                     2797
        if not patch.name:                                                    2798
            flash("图斑名称不能为空", "error")                                      2799
        else:                                                                 2800
```

```
                db.session.add(patch)                                    2801
                db.session.flush()                                       2802
                #  参照生态系统默认取论文值                                2803
                db.session.add(ReferenceEcosystem(                       2804
                     patch_id=patch.id, coverage_benchmark=75.0,         2805
    shannon_benchmark=0.7))                                              2806
                db.session.commit()                                      2807
                flash("图斑已创建", "ok")                                 2808
                return redirect(url_for("project.patches",               2809
    pid=p.id))                                                           2810
    return render_template("project/patch_edit.html",                    2811
    form=form, project=p, patch=None)                                    2812
@bp.route("/patch/<int:patch_id>/edit", methods=["GET",                  2813
    "POST"])                                                             2814
@login_required                                                          2815
def patch_edit(patch_id):                                                2816
    patch = db.get_or_404(MinePatch, patch_id)                           2817
    form = EmptyForm()                                                   2818
    if request.method == "POST" and form.validate_on_submit(             2819
    ):                                                                   2820
        def fv(name, cast=str):                                          2821
            v = request.form.get(name, "").strip()                       2822
            return cast(v) if v != "" else None                          2823
        patch.name = request.form.get("name", "").strip()                2824
        patch.admin_region = fv("admin_region")                          2825
        patch.area_ha = fv("area_ha", float)                             2826
        patch.mineral_type = fv("mineral_type")                          2827
        patch.mining_method = fv("mining_method")                        2828
        patch.stop_years = fv("stop_years", float)                       2829
        patch.responsible_entity_missing =                               2830
    bool(request.form.get("responsible_entity_missing"))                 2831
        #  参照生态系统                                                  2832
        ref = ReferenceEcosystem.query.filter_by(patch_id=pa             2833
    tch.id).first()                                                      2834
        if ref is None:                                                  2835
            ref = ReferenceEcosystem(patch_id=patch.id)                  2836
            db.session.add(ref)                                          2837
        ref.coverage_benchmark = fv("coverage_benchmark",                2838
    float) or 75.0                                                       2839
        ref.shannon_benchmark = fv("shannon_benchmark",                  2840
    float) or 0.7                                                        2841
        ref.dominant_species = fv("dominant_species")                    2842
        patch.image_note = fv("image_note")                              2843
        db.session.commit()                                              2844
        flash("图斑已保存", "ok")                                         2845
        return redirect(url_for("project.patches",                       2846
    pid=patch.project_id))                                               2847
    return render_template("project/patch_edit.html",                    2848
    form=form,                                                           2849
                              project=patch.project,                     2850
```

```
    patch=patch,                                                    2851
                                    reference=ReferenceEcosystem.quer2852
    y.filter_by(                                                    2853
                                        patch_id=patch.id).first())  2854
# ------------------------------ 遥感影像挂接                        2855
    ------------------------------                                  2856
@bp.route("/patch/<int:patch_id>/image", methods=["POST"])          2857
@login_required                                                     2858
def upload_image(patch_id):                                         2859
    """上传遥感底图 / 无人机正射影像并挂接到图斑。"""                  2860
    from app.utils import uploads as up                             2861
    patch = db.get_or_404(MinePatch, patch_id)                      2862
    form = EmptyForm()                                              2863
    if not form.validate_on_submit():                               2864
        flash("表单校验失败，请重试", "error")                       2865
        return redirect(url_for("project.patch_edit",               2866
    patch_id=patch.id))                                             2867
    rel, err = up.save_upload(request.files.get("image"),           2868
    "patches",                                                      2869
                                    current_app.config["ALLOWED_IM2870
    AGE_EXT"],                                                      2871
                                    current_app.config["UPLOAD_FOL2872
    DER"])                                                          2873
    if err:                                                         2874
        flash(err, "error")                                         2875
        return redirect(url_for("project.patch_edit",               2876
    patch_id=patch.id))                                             2877
    # 替换旧影像时清理原文件，避免残留                              2878
    if patch.image_path:                                            2879
        up.remove_upload(patch.image_path,                          2880
    current_app.config["UPLOAD_FOLDER"])                            2881
    patch.image_path = rel                                          2882
    patch.image_uploaded_at = datetime.utcnow()                     2883
    note = request.form.get("image_note", "").strip()               2884
    if note:                                                        2885
        patch.image_note = note                                     2886
    db.session.commit()                                             2887
    flash("遥感影像已挂接", "ok")                                    2888
    return redirect(url_for("project.patch_edit",                   2889
    patch_id=patch.id))                                             2890
@bp.route("/patch/<int:patch_id>/image/delete",                     2891
    methods=["POST"])                                               2892
@login_required                                                     2893
def delete_image(patch_id):                                         2894
    from app.utils import uploads as up                             2895
    patch = db.get_or_404(MinePatch, patch_id)                      2896
    form = EmptyForm()                                              2897
    if form.validate_on_submit():                                   2898
        up.remove_upload(patch.image_path,                          2899
    current_app.config["UPLOAD_FOLDER"])                            2900
```

```
            patch.image_path = None                                    2901
            patch.image_uploaded_at = None                             2902
            db.session.commit()                                        2903
            flash("遥感影像已移除", "ok")                                2904
        return redirect(url_for("project.patch_edit",                  2905
        patch_id=patch.id))                                            2906
# ==== app/views/unit.py ====                                          2907
# -*- coding: utf-8 -*-                                                2908
"""微地质单元管理 —— 岩质边坡 / 碎石边坡 / 平台的划分与面积占比。"""          2909
from flask import Blueprint, render_template, request,                 2910
    redirect, url_for, flash                                           2911
from flask_login import login_required                                 2912
from app.forms import EmptyForm                                        2913
from app.models import db, MinePatch, MicroUnit,                       2914
    IndicatorData                                                      2915
from app.utils.constants import UNIT_TYPES                             2916
bp = Blueprint("unit", __name__, url_prefix="/unit")                   2917
def recalc_ratio(patch_id):                                            2918
    """按面积重算该图斑各单元面积占比。"""                                2919
    units = MicroUnit.query.filter_by(patch_id=patch_id).all           2920
    ()                                                                 2921
    total = sum(u.area or 0.0 for u in units)                          2922
    if total > 0:                                                      2923
        for u in units:                                                2924
            u.ratio = round((u.area or 0.0) / total, 6)                2925
    return total                                                       2926
@bp.route("/")                                                         2927
@login_required                                                        2928
def list_patches():                                                    2929
    """微地质单元入口：列出各图斑及其单元划分情况。"""                    2930
    patches = MinePatch.query.order_by(MinePatch.id.desc()).           2931
    all()                                                              2932
    rows = []                                                          2933
    for p in patches:                                                  2934
        units = MicroUnit.query.filter_by(patch_id=p.id).all           2935
    ()                                                                 2936
        rows.append({                                                  2937
            "patch": p,                                                2938
            "count": len(units),                                       2939
            "ratio_sum": sum(u.ratio or 0.0 for u in units),           2940
        })                                                             2941
    return render_template("unit/list.html", rows=rows)                2942
@bp.route("/<int:patch_id>")                                           2943
@login_required                                                        2944
def index(patch_id):                                                   2945
    patch = db.get_or_404(MinePatch, patch_id)                         2946
    units = MicroUnit.query.filter_by(patch_id=patch_id).all           2947
    ()                                                                 2948
    total = sum(u.area or 0.0 for u in units)                          2949
    ratio_sum = sum(u.ratio or 0.0 for u in units)                     2950
```

```
    return render_template("unit/index.html", patch=patch,
    units=units,
                           total_area=total,
    ratio_sum=ratio_sum,
                           unit_types=UNIT_TYPES,
    form=EmptyForm())
@bp.route("/<int:patch_id>/add", methods=["POST"])
@login_required
def add(patch_id):
    patch = db.get_or_404(MinePatch, patch_id)
    form = EmptyForm()
    if form.validate_on_submit():
        unit_type = request.form.get("unit_type")
        if unit_type not in UNIT_TYPES:
            flash("微地质单元类型不正确", "error")
            return redirect(url_for("unit.index",
    patch_id=patch_id))
        # 论文 4.1 节将图斑分为平台、碎石边坡、岩质边坡三类，每类至多一个。
        # 允许重复会让面积占比与按单元类型组织的指标数据对不上号。
        if MicroUnit.query.filter_by(patch_id=patch.id,
                                     unit_type=unit_type).fi
    rst():
            flash("该图斑已存在「%s」，同一类型只能划分一个" % unit_type,
    "error")
            return redirect(url_for("unit.index",
    patch_id=patch_id))
        area = request.form.get("area", "").strip()
        u = MicroUnit(patch_id=patch.id,
    unit_type=unit_type,
                      area=float(area) if area else None,
                      description=request.form.get("descript
    ion", "").strip() or None)
        db.session.add(u)
        db.session.flush()
        recalc_ratio(patch.id)
        db.session.commit()
        flash("微地质单元已添加", "ok")
    return redirect(url_for("unit.index",
    patch_id=patch_id))
@bp.route("/edit/<int:unit_id>", methods=["GET", "POST"])
@login_required
def edit(unit_id):
    """单个微地质单元的编辑页。"""
    u = db.get_or_404(MicroUnit, unit_id)
    form = EmptyForm()
    if request.method == "POST" and form.validate_on_submit(
    ):
        new_type = request.form.get("unit_type",
    u.unit_type)
        if new_type not in UNIT_TYPES:
```

```
                flash("微地质单元类型不正确", "error")                    3001
                return redirect(url_for("unit.edit",                    3002
    unit_id=u.id))                                                      3003
        # 改类型时同样不允许与本图斑已有的其他单元重复                    3004
        dup = MicroUnit.query.filter_by(patch_id=u.patch_id,            3005
                                                                        3006
    unit_type=new_type).first()                                         3007
        if dup is not None and dup.id != u.id:                          3008
            flash("该图斑已存在「%s」，同一类型只能划分一个" % new_type,  3009
    "error")                                                            3010
            return redirect(url_for("unit.edit",                        3011
    unit_id=u.id))                                                      3012
        u.unit_type = new_type                                          3013
        area = request.form.get("area", "").strip()                     3014
        u.area = float(area) if area else None                          3015
        u.description = request.form.get("description",                 3016
    "").strip() or None                                                 3017
        recalc_ratio(u.patch_id)                                        3018
        db.session.commit()                                             3019
        flash("微地质单元已保存", "ok")                                  3020
        return redirect(url_for("unit.index",                           3021
    patch_id=u.patch_id))                                               3022
    return render_template("unit/edit.html", unit=u,                    3023
    patch=u.patch,                                                      3024
                                unit_types=UNIT_TYPES, form=form)       3025
@bp.route("/delete/<int:unit_id>", methods=["POST"])                    3026
@login_required                                                         3027
def delete(unit_id):                                                    3028
    u = db.get_or_404(MicroUnit, unit_id)                               3029
    pid = u.patch_id                                                    3030
    name = u.unit_type                                                  3031
    # 一并删除该单元的指标数据，避免残留                                3032
    IndicatorData.query.filter_by(unit_id=u.id).delete(synch            3033
    ronize_session=False)                                               3034
    db.session.delete(u)                                                3035
    db.session.flush()                                                  3036
    recalc_ratio(pid)                                                   3037
    db.session.commit()                                                 3038
    flash("微地质单元「%s」已删除" % name, "ok")                         3039
    return redirect(url_for("unit.index", patch_id=pid))                3040
# ==== app/views/standard.py ====                                       3041
# -*- coding: utf-8 -*-                                                 3042
"""政策与标准知识库 —— 标准 / 政策文件的结构化规则与省际口径比对。"""    3043
from flask import Blueprint, render_template, request,                  3044
    redirect, url_for, flash                                            3045
from flask_login import login_required                                  3046
from app.forms import EmptyForm                                         3047
from app.models import db, Standard, StandardRule                       3048
bp = Blueprint("standard", __name__, url_prefix="/standard")            3049
@bp.route("/")                                                          3050
```

```
@login_required                                                    3051
def index():                                                       3052
    standards = Standard.query.order_by(Standard.category,         3053
    Standard.province).all()                                       3054
    local = [s for s in standards if s.category == "地方标准"]     3055
    policy = [s for s in standards if s.category != "地方标准"]    3056
    return render_template("standard/index.html",                  3057
    local=local, policy=policy,                                    3058
                                form=EmptyForm())                  3059
@bp.route("/add", methods=["GET", "POST"])                         3060
@login_required                                                    3061
def add():                                                         3062
    form = EmptyForm()                                             3063
    if request.method == "POST" and form.validate_on_submit(       3064
    ):                                                             3065
        s = Standard(                                              3066
            province=request.form.get("province",                  3067
    "").strip(),                                                   3068
            name=request.form.get("name", "").strip(),             3069
            doc_number=request.form.get("doc_number",              3070
    "").strip() or None,                                           3071
            publish_date=request.form.get("publish_date",          3072
    "").strip() or None,                                           3073
            implement_date=request.form.get("implement_date"       3074
    , "").strip() or None,                                         3075
            category=request.form.get("category", "地方标准"),     3076
            summary=request.form.get("summary", "").strip()        3077
    or None,                                                       3078
        )                                                          3079
        if not s.province or not s.name:                           3080
            flash("省份与名称不能为空", "error")                   3081
        else:                                                      3082
            db.session.add(s)                                      3083
            db.session.commit()                                    3084
            flash("标准已添加", "ok")                              3085
            return redirect(url_for("standard.index"))             3086
    return render_template("standard/edit.html", form=form,        3087
    standard=None)                                                 3088
@bp.route("/<int:sid>/edit", methods=["GET", "POST"])              3089
@login_required                                                    3090
def edit(sid):                                                     3091
    s = db.get_or_404(Standard, sid)                               3092
    form = EmptyForm()                                             3093
    if request.method == "POST" and form.validate_on_submit(       3094
    ):                                                             3095
        s.province = request.form.get("province",                  3096
    "").strip()                                                    3097
        s.name = request.form.get("name", "").strip()              3098
        s.doc_number = request.form.get("doc_number",              3099
    "").strip() or None                                            3100
```

```
            s.publish_date = request.form.get("publish_date",
        "").strip() or None
            s.implement_date = request.form.get("implement_date"
        , "").strip() or None
            s.category = request.form.get("category", "地方标准")
            s.summary = request.form.get("summary", "").strip()
        or None
            db.session.commit()
            flash("已保存", "ok")
            return redirect(url_for("standard.rules", sid=s.id))
        return render_template("standard/edit.html", form=form,
        standard=s)
@bp.route("/<int:sid>/delete", methods=["POST"])
@login_required
def delete(sid):
        s = db.get_or_404(Standard, sid)
        StandardRule.query.filter_by(standard_id=sid).delete()
        db.session.delete(s)
        db.session.commit()
        flash("已删除", "ok")
        return redirect(url_for("standard.index"))
@bp.route("/<int:sid>/rules", methods=["GET", "POST"])
@login_required
def rules(sid):
        s = db.get_or_404(Standard, sid)
        form = EmptyForm()
        if request.method == "POST" and form.validate_on_submit(
        ):
            r = StandardRule(
                standard_id=s.id,
                item=request.form.get("item", "").strip(),
                operator=request.form.get("operator", ">="),
                threshold=request.form.get("threshold",
        "").strip() or None,
                condition_type=request.form.get("condition_type"
        , "必要条件"),
                region=request.form.get("region", "").strip()
        or None,
                note=request.form.get("note", "").strip() or
        None,
            )
            if not r.item:
                flash("判定项不能为空", "error")
            else:
                db.session.add(r)
                db.session.commit()
                flash("规则已添加", "ok")
            return redirect(url_for("standard.rules", sid=s.id))
        rs = StandardRule.query.filter_by(standard_id=s.id).all(
        )
```

```
    return render_template("standard/rules.html",
    form=form, standard=s, rules=rs)
@bp.route("/rule/<int:rid>/delete", methods=["POST"])
@login_required
def rule_delete(rid):
    r = db.get_or_404(StandardRule, rid)
    sid = r.standard_id
    db.session.delete(r)
    db.session.commit()
    flash("规则已删除", "ok")
    return redirect(url_for("standard.rules", sid=sid))
@bp.route("/compare")
@login_required
def compare():
    """省际口径比对：同一指标在不同省份的阈值横向对比。"""
    rules = StandardRule.query.join(Standard).all()
    groups = {}
    for r in rules:
        groups.setdefault(r.item, []).append(r)
    return render_template("standard/compare.html",
    groups=groups)
# ==== app/views/constraint.py ====
# -*- coding: utf-8 -*-
"""限制性指标判定 —— 一票否决判定。
按论文表 2 与 4.1 节流程，共 5 项：政策、区位约束、土壤污染、地质灾害、耕地属性。
任一项不通过即判定为"人工辅助或生态重建"，不再进入定量评价。
"""
from flask import Blueprint, render_template, request,
    redirect, url_for, flash
from flask_login import login_required
from app.forms import EmptyForm
from app.models import db, MinePatch, ConstraintCheck
from app.utils.constants import CONSTRAINT_ITEMS
bp = Blueprint("constraint", __name__,
    url_prefix="/constraint")
# 字段名 -> 中文名
FIELDS = [
    ("policy_pass", "政策约束"),
    ("location_pass", "区位约束"),
    ("pollution_pass", "土壤污染"),
    ("geo_pass", "地质灾害"),
    ("farmland_pass", "耕地属性"),
]
@bp.route("/")
@login_required
def index():
    patches = MinePatch.query.order_by(MinePatch.id.desc()).
    all()
    latest = {}
    for p in patches:
```

```
            latest[p.id] = ConstraintCheck.query.filter_by(patch
        _id=p.id) \
                .order_by(ConstraintCheck.id.desc()).first()
        return render_template("constraint/index.html",
        patches=patches, latest=latest)
@bp.route("/<int:patch_id>", methods=["GET", "POST"])
@login_required
def check(patch_id):
        patch = db.get_or_404(MinePatch, patch_id)
        form = EmptyForm()
        record = ConstraintCheck.query.filter_by(patch_id=patch_
        id) \
            .order_by(ConstraintCheck.id.desc()).first()
        if request.method == "POST" and form.validate_on_submit(
        ):
            if record is None:
                record = ConstraintCheck(patch_id=patch.id)
                db.session.add(record)
            for field, _ in FIELDS:
                setattr(record, field, request.form.get(field)
        == "1")
            record.passed = all(getattr(record, f) for f, _ in
        FIELDS)
            failed = [name for f, name in FIELDS if not
        getattr(record, f)]
            record.reason = ("全部通过" if record.passed
                             else "未通过：" + "、".join(failed))
            db.session.commit()
            flash("判定结果已保存", "ok")
            return redirect(url_for("constraint.check",
        patch_id=patch.id))
        # 预处理各条件的当前取值，供模板直接使用（Jinja 无 getattr）
        field_states = [(f, name, None if record is None else
        getattr(record, f))
                        for f, name in FIELDS]
        return render_template("constraint/check.html",
        patch=patch, form=form,
                               fields=FIELDS,
        field_states=field_states,
                               record=record)
# ==== app/views/ahp.py ====
# -*- coding: utf-8 -*-
"""指标体系与 AHP 权重 —— 路由与页面逻辑。"""
import json
from flask import Blueprint, render_template, request,
    redirect, url_for, flash
from app.forms import AHPSelectForm, EmptyForm
from app.models import db, Indicator, ExpertMatrix,
    WeightScheme
from app.utils.constants import DEFAULT_WEIGHTS, UNIT_TYPES
```

```
from app.services import ahp_service as ahp
from app.services import indicator_service as ind_svc
from app.services import weight_service as ws
bp = Blueprint("ahp", __name__, url_prefix="/ahp")
#  一次计算的权重结果对应的指标编码：名称 → 编码
_L1_CODE = {"土壤指标": "soil", "植被指标": "veg", "地形指标":
    "terrain"}
_L2_CODE = {
    ("土壤指标", "土石比"): "soil_ratio",
    ("土壤指标", "土层厚度"): "thickness",
    ("土壤指标", "pH"): "ph",
    ("土壤指标", "有机质"): "organic_matter",
    ("土壤指标", "污染指标"): "pollution",
    ("植被指标", "植被覆盖度"): "coverage",
    ("植被指标", "生物多样性"): "shannon",
    ("地形指标", "坡度"): "slope",
    ("地形指标", "坡向"): "aspect",
}
def to_code_pairs(unit_type, level, l1, names, weights):
    """把一次 AHP 计算的权重结果转成「指标编码 → 权重」，便于落库。"""
    pairs = []
    for name, w in zip(names, weights):
        if level == 1:
            code = _L1_CODE.get(name)
        else:
            code = _L2_CODE.get((l1, name))
        if code:
            pairs.append((code, w))
    return pairs
def normalize_l1(unit_type, level, l1):
    """把 l1 参数规整为该单元类型下确实存在的一级指标名。
    level=2 的页面（判断矩阵、多专家矩阵）都按 l1 取二级指标。
    直接带 ?level=2 访问而没给 l1 时，原先会 KeyError: None 直接 500；
    这里退回该单元类型的第一个一级指标，页面照常可用。
    """
    if level != 2:
        return None
    valid = list(DEFAULT_WEIGHTS.get(unit_type,
    {}).get("l2", {}))
    if not valid:
        return None
    return l1 if l1 in valid else valid[0]
def indicator_list(unit_type, level, l1=None):
    """返回 (指标名称列表, 默认权重列表)。
    level=1 时返回一级指标；level=2 时返回指定一级指标下的二级指标。
    l1 缺失或不属于该单元类型时，退回第一个一级指标，不抛异常。
    """
    weights = DEFAULT_WEIGHTS[unit_type]
    if level == 1:
        names = list(weights["l1"].keys())
```

```
            vec = [weights["l1"][n] for n in names]
            return names, vec
        l2_all = weights["l2"]
        if l1 not in l2_all:
            l1 = next(iter(l2_all))
        l2 = l2_all[l1]
        names = list(l2.keys())
        vec = [l2[n] for n in names]
        return names, vec
def l1_choices(unit_type):
    """某单元类型的一级指标名称列表（用于二级指标录入时选择父指标）。"""
    return list(DEFAULT_WEIGHTS[unit_type]["l1"].keys())
@bp.route("/", methods=["GET", "POST"])
def index():
    form = AHPSelectForm()
    # 先根据请求中的单元类型填充 l1 选项，再进行表单校验
    unit = request.form.get("unit_type") or
    form.unit_type.choices[0][0]
    form.l1.choices = [(n, n) for n in l1_choices(unit)]
    if form.validate_on_submit():
        unit_type = form.unit_type.data
        level = int(form.level.data)
        if level == 1:
            return redirect(url_for("ahp.matrix",
    unit_type=unit_type, level=1))
        if not form.l1.data:
            flash("选择「二级指标」时，请同时选择所属的一级指标", "error")
            return render_template("ahp/index.html",
    form=form,
                                   weight_rows=ws.summary())
        return redirect(url_for("ahp.matrix",
    unit_type=unit_type,
                                level=2, l1=form.l1.data))
    return render_template("ahp/index.html", form=form,
                           weight_rows=ws.summary())
@bp.route("/matrix", methods=["GET", "POST"])
def matrix():
    unit_type = request.values.get("unit_type")
    level = int(request.values.get("level", 1))
    l1 = normalize_l1(unit_type, level,
    request.values.get("l1"))
    if unit_type not in DEFAULT_WEIGHTS:
        flash("无效的微地质单元类型", "error")
        return redirect(url_for("ahp.index"))
    names, vec = indicator_list(unit_type, level, l1)
    n = len(names)
    form = EmptyForm()
    # 预填默认值：论文表 3 权重对应的完全一致矩阵 a_ij = w_i / w_j
    default = [[round(vec[i] / vec[j], 4) for j in
    range(n)] for i in range(n)]
```

```
    def compute(upper):
        A = ahp.matrix_from_upper(upper)
        r = ahp.weights_and_consistency(A)
        bad = ahp.find_most_inconsistent(A)
        return {
            "weights": [round(v, 4) for v in r["weights"]],
            "lambda_max": round(r["lambda_max"], 4),
            "ci": r["ci"],
            "cr": r["cr"],
            "consistent": r["consistent"],
            "matrix": [[round(A[i, j], 4) for j in
range(n)] for i in range(n)],
            "most_inconsistent": bad,
        }
    result = None
    if request.method == "POST" and form.validate_on_submit(
):
        action = request.form.get("action", "calc")
        try:
            upper = [float(request.form.get("m_%d_%d" % (i,
j), "1"))
                     for i in range(n) for j in range(i +
1, n)]
            result = compute(upper)
        except (ValueError, ZeroDivisionError):
            flash("矩阵输入有误：请填入正数（对角线自动为 1，下三角自动取倒数）",
"error")
            return redirect(url_for("ahp.matrix",
unit_type=unit_type,
                                    level=level, l1=l1))
        if action == "save":
            if not result["consistent"]:
                flash("一致性比例 CR=%.4f ≥ 0.1，判断矩阵不满足一致性要求，"
                      "请调整后再保存" % result["cr"], "error")
            else:
                pairs = to_code_pairs(unit_type, level, l1,
names,
                                      result["weights"])
                scheme, cnt = ws.save_weights(unit_type,
pairs)
                flash("已保存并为「%s」生效 %d 项权重（CR=%.4f）。"
                      "此后该单元类型的评价将以此权重计算。"
                      % (unit_type, cnt, result["cr"]),
"ok")
                return redirect(url_for("ahp.matrix",
unit_type=unit_type,
                                        level=level, l1=l1))
    elif request.values.get("calc") == "1":
        # 按默认（论文表 3）矩阵直接试算，便于一键查看权重与一致性
        result = compute([default[i][j] for i in range(n)
```

```
    for j in range(i + 1, n)])                                          3401
    _, source_label, custom_scheme = ws.effective_weights(un            3402
    it_type)                                                            3403
    return render_template(                                             3404
        "ahp/matrix.html",                                              3405
        form=form, unit_type=unit_type, level=level, l1=l1,             3406
        names=names, vec=vec, default=default,                          3407
    result=result,                                                      3408
        source_label=source_label, custom_scheme=custom_sche            3409
    me,                                                                 3410
    )                                                                   3411
@bp.route("/weights/reset", methods=["POST"])                           3412
def weights_reset():                                                    3413
    """删除自定义权重方案，恢复为该单元类型的论文表 3 默认权重。"""                    3414
    unit_type = request.form.get("unit_type")                           3415
    form = EmptyForm()                                                  3416
    if form.validate_on_submit():                                       3417
        if unit_type not in DEFAULT_WEIGHTS:                            3418
            flash("无效的微地质单元类型", "error")                            3419
        else:                                                           3420
            n = ws.reset_weights(unit_type)                             3421
            if n:                                                       3422
                flash("已恢复「%s」为论文表 3 默认权重" % unit_type,              3423
    "ok")                                                               3424
            else:                                                       3425
                flash("「%s」本就是论文表 3 默认权重，无需恢复" % unit_type,        3426
    "ok")                                                               3427
    return redirect(url_for("ahp.index"))                               3428
# ------------------------------ 指标维护 ---------------------------   3429
    ----                                                                3430
@bp.route("/indicators")                                                3431
def indicators():                                                       3432
    """指标定义维护：查看指标树、单位、理想值与取值依据。"""                          3433
    if Indicator.query.count() == 0:                                    3434
        ind_svc.seed_indicators()                                       3435
    all_ind = Indicator.query.order_by(Indicator.level,                 3436
    Indicator.id).all()                                                 3437
    by_parent = {}                                                      3438
    for i in all_ind:                                                   3439
        by_parent.setdefault(i.parent_code, []).append(i)               3440
    return render_template("ahp/indicators.html",                       3441
    by_parent=by_parent,                                                3442
                                all_ind=all_ind,                        3443
    unit_types=UNIT_TYPES,                                              3444
                                type_label=ind_svc.ideal_type_lab       3445
    el,                                                                 3446
                                form=EmptyForm())                       3447
@bp.route("/indicators/<code>/edit", methods=["GET",                    3448
    "POST"])                                                            3449
def indicator_edit(code):                                               3450
```

```
    ind = Indicator.query.filter_by(code=code).first_or_404(3451
    )                                                          3452
    form = EmptyForm()                                         3453
    if request.method == "POST" and form.validate_on_submit(3454
    ):                                                         3455
        name = request.form.get("name", "").strip()            3456
        if not name:                                           3457
            flash("指标名称不能为空", "error")                  3458
            return redirect(url_for("ahp.indicator_edit",      3459
    code=code))                                                3460
        ind.name = name                                        3461
        ind.unit = request.form.get("unit", "").strip() or     3462
    None                                                       3463
        ind.ideal_value = request.form.get("ideal_value",      3464
    "").strip() or None                                        3465
        ind.ideal_type = request.form.get("ideal_type") or     3466
    ind.ideal_type                                             3467
        units = request.form.getlist("enabled_units")          3468
        ind.enabled_units = ",".join(units) if units else      3469
    None                                                       3470
        ind.is_constraint = request.form.get("is_constraint"3471
    ) == "1"                                                   3472
        ind.description = request.form.get("description",      3473
    "").strip() or None                                        3474
        db.session.commit()                                    3475
        flash("指标「%s」已保存" % ind.name, "ok")              3476
        return redirect(url_for("ahp.indicators"))             3477
    return render_template("ahp/indicator_edit.html",          3478
    ind=ind,                                                   3479
                           unit_types=UNIT_TYPES,              3480
                           type_label=ind_svc.ideal_type_lab3481
    el, form=form)                                             3482
@bp.route("/indicators/reset", methods=["POST"])               3483
def indicators_reset():                                        3484
    """按论文取值重置指标定义。"""                              3485
    form = EmptyForm()                                         3486
    if form.validate_on_submit():                              3487
        n = ind_svc.seed_indicators()                          3488
        flash("已按论文取值重置 %d 项指标定义" % n, "ok")        3489
    return redirect(url_for("ahp.indicators"))                 3490
# ---------------------------- 多专家矩阵 ------------------------3491
    -----                                                      3492
def expert_scope():                                            3493
    """从请求中取（并规整）单元类型、层级与所属一级指标。"""    3494
    unit_type = request.values.get("unit_type") or             3495
    UNIT_TYPES[0]                                              3496
    if unit_type not in DEFAULT_WEIGHTS:                       3497
        unit_type = UNIT_TYPES[0]                              3498
    level = int(request.values.get("level", 1))                3499
    l1 = normalize_l1(unit_type, level,                        3500
```

```
    request.values.get("l1"))                                          3501
    return unit_type, level, l1                                        3502
def expert_query(unit_type, level, l1):                                3503
    q = ExpertMatrix.query.filter_by(unit_type=unit_type,              3504
    level=level)                                                       3505
    if level == 2:                                                     3506
        q = q.filter_by(l1_name=l1)                                    3507
    else:                                                              3508
        q = q.filter(ExpertMatrix.l1_name.is_(None))                   3509
    return q.order_by(ExpertMatrix.id)                                 3510
def default_scheme(unit_type, level, l1=None):                         3511
    """取（或建）该范围的权重方案，专家矩阵统一挂到它下面。"""                3512
    name = "%s · %s%s" % (unit_type,                                   3513
                          "一级指标" if level == 1 else "二级指标",          3514
                          "(%s)" % l1 if l1 else "")                   3515
    scheme = WeightScheme.query.filter_by(unit_type=unit_typ           3516
    e,                                                                 3517
                                          level=level,                 3518
    name=name).first()                                                 3519
    if scheme is None:                                                 3520
        scheme = WeightScheme(unit_type=unit_type,                     3521
    level=level, name=name,                                            3522
                              is_default=False)                        3523
        db.session.add(scheme)                                         3524
        db.session.flush()                                             3525
    return scheme                                                      3526
@bp.route("/experts", methods=["GET", "POST"])                         3527
def experts():                                                         3528
    """多专家判断矩阵录入与汇总。"""                                        3529
    unit_type, level, l1 = expert_scope()                              3530
    if unit_type not in DEFAULT_WEIGHTS:                               3531
        flash("无效的微地质单元类型", "error")                              3532
        return redirect(url_for("ahp.index"))                          3533
    names, vec = indicator_list(unit_type, level, l1)                  3534
    n = len(names)                                                     3535
    form = EmptyForm()                                                 3536
    if request.method == "POST" and form.validate_on_submit(           3537
    ):                                                                 3538
        action = request.form.get("action")                            3539
        if action == "add":                                            3540
            expert_name = request.form.get("expert_name",              3541
    "").strip() or "未具名专家"                                           3542
            try:                                                       3543
                upper = [float(request.form.get("m_%d_%d" %            3544
    (i, j), "1"))                                                      3545
                         for i in range(n) for j in range(i            3546
    + 1, n)]                                                           3547
            except ValueError:                                         3548
                flash("矩阵输入有误，请填入正数", "error")                      3549
                return redirect(url_for("ahp.experts",                 3550
```

```
    unit_type=unit_type,                                                3551
                                                level=level, l1=l1))    3552
            A = ahp.matrix_from_upper(upper)                            3553
            r = ahp.weights_and_consistency(A)                          3554
            scheme = default_scheme(unit_type, level, l1)               3555
            db.session.add(ExpertMatrix(                                3556
                scheme_id=scheme.id,                                    3557
                unit_type=unit_type, l1_name=(l1 if level               3558
    == 2 else None),                                                    3559
                expert_name=expert_name, level=level,                   3560
                matrix_json=json.dumps([list(map(float,                 3561
    row)) for row in A]),                                               3562
                cr=round(r["cr"], 6)))                                  3563
            db.session.commit()                                         3564
            flash("已录入专家「%s」的判断矩阵（CR=%.4f）" % (expert_name,  3565
    r["cr"]), "ok")                                                     3566
        elif action == "save_agg":                                      3567
            # 把汇总权重落库并生效，使多专家结果真正参与评价计算              3568
            weights = _aggregate(unit_type, level, l1)                  3569
            if weights is None:                                         3570
                flash("暂无可汇总的专家矩阵", "error")                    3571
            elif not weights["consistent"]:                             3572
                flash("汇总矩阵一致性比例 CR=%.4f ≥ 0.1，不满足要求，"      3573
                      "请先检查或调整专家矩阵" % weights["cr"],             3574
    "error")                                                            3575
            else:                                                       3576
                names_, _ = indicator_list(unit_type,                   3577
    level, l1)                                                          3578
                pairs = to_code_pairs(unit_type, level, l1,             3579
    names_,                                                             3580
                                      weights["weights"])               3581
                _, cnt = ws.save_weights(unit_type, pairs)              3582
                flash("已保存并为「%s」生效 %d 项汇总权重（CR=%.4f）"          3583
                      % (unit_type, cnt, weights["cr"]),                3584
    "ok")                                                               3585
        return redirect(url_for("ahp.experts",                          3586
    unit_type=unit_type,                                                3587
                                level=level, l1=l1))                    3588
    rows = expert_query(unit_type, level, l1).all()                     3589
    default = [[round(vec[i] / vec[j], 4) for j in                      3590
    range(n)] for i in range(n)]                                        3591
    return render_template("ahp/experts.html",                          3592
    unit_type=unit_type, level=level,                                   3593
                               l1=l1, names=names, rows=rows,           3594
    default=default,                                                    3595
                               n=n, form=form)                          3596
@bp.route("/experts/<int:mid>/delete", methods=["POST"])                3597
def expert_delete(mid):                                                 3598
    row = db.get_or_404(ExpertMatrix, mid)                              3599
    unit_type, level, l1 = row.unit_type, row.level,                    3600
```

```
    row.l1_name                                                    3601
    form = EmptyForm()                                             3602
    if form.validate_on_submit():                                  3603
        db.session.delete(row)                                     3604
        db.session.commit()                                        3605
        flash("已删除专家矩阵", "ok")                                3606
    return redirect(url_for("ahp.experts",                         3607
    unit_type=unit_type,                                           3608
                                    level=level, l1=l1))           3609
def _aggregate(unit_type, level, l1):                              3610
    """按几何平均汇总多专家矩阵并做一致性检验。无矩阵时返回 None。"""          3611
    rows = expert_query(unit_type, level, l1).all()                3612
    mats = []                                                      3613
    for r in rows:                                                 3614
        try:                                                       3615
            mats.append([list(map(float, line))                    3616
                         for line in json.loads(r.matrix_jso3617
    n)])                                                           3618
        except Exception:                                          3619
            continue                                               3620
    if not mats:                                                   3621
        return None                                                3622
    agg = ahp.aggregate_matrices(mats)                             3623
    res = ahp.weights_and_consistency(agg)                         3624
    return {                                                       3625
        "count": len(mats),                                        3626
        "weights": res["weights"],                                 3627
        "lambda_max": res["lambda_max"],                           3628
        "ci": res["ci"], "cr": res["cr"],                          3629
        "consistent": res["consistent"],                           3630
        "most_inconsistent": ahp.find_most_inconsistent(agg)3631
    ,                                                              3632
        "expert_crs": [r.cr for r in rows if r.cr is not       3633
    None],                                                         3634
    }                                                              3635
@bp.route("/experts/aggregate")                                    3636
def experts_aggregate():                                           3637
    """按几何平均汇总多专家矩阵，并做一致性检验。"""                          3638
    unit_type, level, l1 = expert_scope()                          3639
    if unit_type not in DEFAULT_WEIGHTS:                           3640
        flash("无效的微地质单元类型", "error")                          3641
        return redirect(url_for("ahp.index"))                      3642
    names, vec = indicator_list(unit_type, level, l1)              3643
    rows = expert_query(unit_type, level, l1).all()                3644
    result = _aggregate(unit_type, level, l1)                      3645
    if result is not None:                                         3646
        result["weights"] = [round(v, 4) for v in              3647
    result["weights"]]                                             3648
        result["lambda_max"] = round(result["lambda_max"],     3649
    4)                                                             3650
```

```
    return render_template("ahp/experts_aggregate.html",
    unit_type=unit_type,
                           level=level, l1=l1, names=names,
    rows=rows,
                           result=result, form=EmptyForm())
# ==== app/views/indicator.py ====
# -*- coding: utf-8 -*-
"""指标数据采集与管理 —— 路由与页面逻辑。
按微地质单元组织录入；系统按单元类型自动决定应录入哪些指标
（岩质边坡无土壤指标、平台无地形指标）。
"""
import io
import json
from flask import (Blueprint, render_template, request,
    redirect, url_for,
                   flash, send_file)
from app.forms import EmptyForm
from app.models import db, MinePatch, MicroUnit,
    IndicatorData
from app.utils.constants import HEAVY_METALS, ASPECT_TYPES,
    UNIT_ENABLED_INDICATORS
from app.services import import_service as imp
bp = Blueprint("indicator", __name__,
    url_prefix="/indicator")
# 指标字段定义：字段名 -> (中文名, 单位, 是否可选)
SOIL_FIELDS = [
    ("soil_ratio", "土石比", ""),
    ("thickness", "土层厚度", "cm"),
    ("ph", "pH", ""),
    ("organic_matter", "有机质", "g/kg"),
]
VEG_FIELDS = [
    ("coverage", "植被覆盖度", "%"),
    ("shannon", "Shannon-Wiener 多样性指数", ""),
]
TERRAIN_FIELDS = [
    ("slope", "坡度", "°"),
    ("aspect", "坡向", ""),
]
def unit_field_config(unit_type):
    """根据单元类型返回需录入的字段分组。"""
    enabled = UNIT_ENABLED_INDICATORS.get(unit_type, ())
    groups = []
    if "土壤指标" in enabled:
        groups.append(("土壤指标", SOIL_FIELDS, True))    # True
    = 含重金属
    if "植被指标" in enabled:
        groups.append(("植被指标", VEG_FIELDS, False))
    if "地形指标" in enabled:
        groups.append(("地形指标", TERRAIN_FIELDS, False))
```

```
    return groups
@bp.route("/")
def index():
    patches = MinePatch.query.order_by(MinePatch.id.desc()).all()
    return render_template("indicator/index.html", patches=patches)
@bp.route("/<int:unit_id>", methods=["GET", "POST"])
def edit(unit_id):
    unit = db.get_or_404(MicroUnit, unit_id)
    ind = IndicatorData.query.filter_by(unit_id=unit.id).first()
    form = EmptyForm()
    groups = unit_field_config(unit.unit_type)
    if request.method == "POST" and form.validate_on_submit():
        def f(name, cast=float):
            v = request.form.get(name, "").strip()
            return cast(v) if v != "" else None
        heavy = {}
        for m in HEAVY_METALS:
            v = f("hm_" + m)
            if v is not None:
                heavy[m] = v
        if ind is None:
            ind = IndicatorData(unit_id=unit.id)
            db.session.add(ind)
        ind.soil_ratio = f("soil_ratio")
        ind.thickness = f("thickness")
        ind.ph = f("ph")
        ind.organic_matter = f("organic_matter")
        ind.heavy_metals = json.dumps(heavy, ensure_ascii=False) if heavy else None
        ind.coverage = f("coverage")
        ind.shannon = f("shannon")
        ind.slope = f("slope")
        ind.aspect = request.form.get("aspect", "").strip() or None
        db.session.commit()
        flash("指标数据已保存", "ok")
        return redirect(url_for("indicator.index"))
    # 组装当前值供模板回显
    current = {}
    if ind:
        current = {
            "soil_ratio": ind.soil_ratio, "thickness": ind.thickness,
            "ph": ind.ph, "organic_matter": ind.organic_matter,
            "coverage": ind.coverage, "shannon":
```

```
    ind.shannon,                                                    3751
            "slope": ind.slope, "aspect": ind.aspect,               3752
        }                                                           3753
        current["heavy"] = ind.heavy_metals_dict()                  3754
    return render_template("indicator/edit.html",                   3755
                           unit=unit, form=form,                    3756
    groups=groups,                                                  3757
                           heavy_metals=HEAVY_METALS,               3758
                           aspect_types=ASPECT_TYPES,               3759
    current=current)                                                3760
# ---------------------------- 批量导入 ----------------------------3761
    ----                                                            3762
@bp.route("/import", methods=["GET", "POST"])                       3763
def import_data():                                                  3764
    """批量导入指标数据（CSV / Excel）。"""                          3765
    patches = MinePatch.query.order_by(MinePatch.id.desc()).3766
    all()                                                           3767
    form = EmptyForm()                                              3768
    report = None                                                   3769
    if request.method == "POST" and form.validate_on_submit(3770
    ):                                                              3771
        patch_id = request.form.get("patch_id", type=int)           3772
        patch = db.session.get(MinePatch, patch_id) if              3773
    patch_id else None                                              3774
        if patch is None:                                           3775
            flash("请选择矿山图斑", "error")                         3776
            return redirect(url_for("indicator.import_data")3777
    )                                                               3778
        file = request.files.get("datafile")                        3779
        if file is None or not file.filename:                       3780
            flash("请选择要导入的文件", "error")                     3781
            return redirect(url_for("indicator.import_data")3782
    )                                                               3783
        ext = file.filename.rsplit(".", 1)[-1].lower()              3784
        if ext not in ("csv", "xlsx", "xls"):                       3785
            flash("仅支持 CSV 或 Excel（.xlsx/.xls）文件", "error")  3786
            return redirect(url_for("indicator.import_data")3787
    )                                                               3788
        units = MicroUnit.query.filter_by(patch_id=patch.id)3789
    .all()                                                          3790
        unit_map = {}                                               3791
        for u in units:                                             3792
            unit_map.setdefault(u.unit_type, u)                     3793
        if not unit_map:                                            3794
            flash("该图斑尚未划分微地质单元，请先划分后再导入", "error")  3795
            return redirect(url_for("indicator.import_data")3796
    )                                                               3797
        try:                                                        3798
            parsed, errors = imp.parse_upload(file.filename,3799
    file.stream, unit_map)                                          3800
```

```
        except Exception as e:
            flash("文件解析失败：%s" % e, "error")
            return redirect(url_for("indicator.import_data")
    )
        saved = 0
        for rec in parsed:
            unit = unit_map[rec["unit_type"]]
            ind = IndicatorData.query.filter_by(unit_id=unit
    .id).first()
            if ind is None:
                ind = IndicatorData(unit_id=unit.id)
                db.session.add(ind)
            f = rec["fields"]
            for key in ("soil_ratio", "thickness", "ph",
    "organic_matter",
                        "coverage", "shannon", "slope",
    "aspect"):
                if key in f:
                    setattr(ind, key, f[key])
            if rec["heavy_metals"]:
                ind.heavy_metals = json.dumps(rec["heavy_met
    als"],

    ensure_ascii=False)
            saved += 1
        db.session.commit()
        report = {"patch": patch.name, "saved": saved,
    "errors": errors,
                  "total": len(parsed) + len(
                      [e for e in errors if e.startswith("第
    ")])}
        if saved:
            flash("已导入 %d 个微地质单元的指标数据" % saved, "ok")
        else:
            flash("没有可导入的数据行，请检查文件内容", "error")
    return render_template("indicator/import.html",
    patches=patches,
                           form=form, report=report,
    headers=imp.HEADERS)
@bp.route("/import/template")
def import_template():
    """下载导入模板（CSV，含一行示例）。"""
    csv_text = imp.build_template_csv()
    data = csv_text.encode("utf-8-sig")       # 带 BOM，便于
    Excel 正确识别中文
    return send_file(io.BytesIO(data), as_attachment=True,
                     download_name="指标数据导入模板.csv",
                     mimetype="text/csv")
# ==== app/views/evaluate.py ====
# -*- coding: utf-8 -*-
```

```
"""适宜性定量评价引擎 —— 路由与页面逻辑。
从数据库加载某图斑的微地质单元与指标数据，调用评价引擎计算，
展示各单元得分、最终评价因子 F、适宜性等级与推荐措施。
入口处按论文 4.1 节的流程顺序检查前置条件（见 app/services/workflow.py），
前置步骤未完成时不予计算，避免在条件不具备时产出评价结论。
"""
import json
from flask import Blueprint, render_template, request,
    redirect, url_for, flash
from app.forms import EmptyForm
from app.models import (db, MinePatch, MicroUnit,
    IndicatorData,
                        EvaluationResult, ScoreDetail)
from app.services import evaluation_service as ev
from app.services import workflow as wf
bp = Blueprint("evaluate", __name__, url_prefix="/evaluate")
def load_patch_input(patch_id):
    """从数据库组装评价引擎输入。"""
    units = MicroUnit.query.filter_by(patch_id=patch_id).all
    ()
    result = []
    for u in units:
        ind = IndicatorData.query.filter_by(unit_id=u.id).fi
    rst()
        data = {
            "soil_ratio": ind.soil_ratio if ind else None,
            "thickness": ind.thickness if ind else None,
            "ph": ind.ph if ind else None,
            "organic_matter": ind.organic_matter if ind
    else None,
            "heavy_metals": ind.heavy_metals_dict() if ind
    else {},
            "coverage": ind.coverage if ind else None,
            "shannon": ind.shannon if ind else None,
            "slope": ind.slope if ind else None,
            "aspect": ind.aspect if ind else None,
        }
        result.append({
            "unit_id": u.id,
            "unit_type": u.unit_type,
            "ratio": u.ratio if u.ratio is not None else
    0.0,
            "data": data,
        })
    return result
def _round(v, n=6):
    return None if v is None else round(v, n)
def load_weights(units):
    """解析各单元类型当前生效的权重与来源说明。
    权重可来自 AHP 计算并保存的生效方案；未配置时用论文表 3 的默认值。
```

```
    来源一并记录，便于报告标注与事后复核。
    """
    from app.services import weight_service as ws
    weights, sources = {}, {}
    for u in units:
        ut = u["unit_type"]
        if ut not in weights:
            w, label, _ = ws.effective_weights(ut)
            weights[ut] = w
            sources[ut] = label
    return weights, sources
def pack_units(unit_results, sources=None):
    """把评价结果整理为可持久化的 JSON。
    完整保留各单元得分、一级指标得分与逐条二级指标明细，
    使结果页与报告能够复现「已记录的那一次评价」，
    而不是每次打开时重新推算。
    """
    sources = sources or {}
    out = []
    for u in unit_results:
        out.append({
            "unit_type": u["unit_type"],
            "ratio": _round(u["ratio"]),
            "unit_score": _round(u["unit_score"]),
            "weighted": _round(u["weighted"]),
            "weight_source": sources.get(u["unit_type"]),
            "l1_scores": {k: _round(v) for k, v in
    u["l1_scores"].items()},
            "details": [{
                "l1": d["l1"], "l2": d["l2"],
                "raw": None if d["raw"] is None else
    str(d["raw"]),
                "std_value": _round(d["std_value"]),
                "weight": _round(d["weight"]),
                "contribution": _round(d["contribution"]),
            } for d in u["details"]],
        })
    return out
def save_evaluation(patch_id, units):
    """执行评价并把结果与逐层明细落库，返回评价记录。
    权重取各单元类型当前生效的方案（AHP 计算结果），未配置时用
    论文表 3 的默认权重；来源一并记录。结果 JSON 保留各单元得分、
    一级指标得分与逐条二级指标明细，结果页与报告据此复现
    记录下来的那次评价，不再重新推算。
    """
    weights, sources = load_weights(units)
    res = ev.score_mine(units, weights)
    saved = EvaluationResult(
        patch_id=patch_id,
        final_score=round(res["final_score"], 4),
```

```
            grade=res["grade"],                                          3951
            measures=res["grade_info"]["measure"],                       3952
            unit_scores=json.dumps(pack_units(res["unit_results"]        3953
        ], sources),                                                     3954
                                        ensure_ascii=False),             3955
        )                                                                3956
        db.session.add(saved)                                            3957
        db.session.flush()        # 先取得 result.id，再写明细             3958
        unit_ids = {u["unit_type"]: u["unit_id"] for u in units}          3959
        for u in res["unit_results"]:                                    3960
            for d in u["details"]:                                       3961
                db.session.add(ScoreDetail(                              3962
                    result_id=saved.id,                                  3963
                    unit_id=unit_ids.get(u["unit_type"]),                3964
                    l1_name=d["l1"], l2_name=d["l2"],                    3965
                    raw_value=None if d["raw"] is None else              3966
        str(d["raw"]),                                                   3967
                    std_value=d["std_value"],                            3968
        weight=d["weight"],                                              3969
                    contribution=d["contribution"]))                     3970
        db.session.commit()                                              3971
        return saved                                                     3972
def block_reasons(progress):                                             3973
        """把未满足的前置条件转成可读的拦截原因。"""                              3974
        return ["%s：%s" % (b["label"], b["reason"]) for b in            3975
        progress["blocking"]]                                            3976
@bp.route("/")                                                           3977
def index():                                                             3978
        patches = MinePatch.query.order_by(MinePatch.id.desc()).         3979
        all()                                                            3980
        latest = {}                                                      3981
        progress = {}                                                    3982
        for p in patches:                                                3983
            latest[p.id] = EvaluationResult.query.filter_by(patc         3984
        h_id=p.id) \                                                     3985
                .order_by(EvaluationResult.id.desc()).first()            3986
            progress[p.id] = wf.patch_progress(p.id)                     3987
        return render_template("evaluate/index.html",                    3988
        patches=patches,                                                 3989
                                    latest=latest, progress=progress)    3990
@bp.route("/<int:patch_id>", methods=["GET", "POST"])                    3991
def run(patch_id):                                                       3992
        patch = db.get_or_404(MinePatch, patch_id)                       3993
        form = EmptyForm()                                               3994
        progress = wf.patch_progress(patch_id)                           3995
        if request.method == "POST" and form.validate_on_submit(         3996
        ):                                                               3997
            if not progress["ready"]:                                    3998
                for reason in block_reasons(progress):                   3999
                    flash("无法执行评价 —— " + reason, "error")            4000
```

```
            return redirect(url_for("evaluate.run",
    patch_id=patch_id))
        units = load_patch_input(patch_id)
        saved = save_evaluation(patch.id, units)
        return redirect(url_for("evaluate.result",
    result_id=saved.id))
    # 展示将采用的权重来源，便于在执行前确认
    units_now = load_patch_input(patch_id)
    _, sources = load_weights(units_now) if units_now else
    ({}, {})
    return render_template("evaluate/run.html",
    patch=patch, form=form,
                           progress=progress,
    weight_sources=sources)
@bp.route("/result/<int:result_id>")
def result(result_id):
    r = db.get_or_404(EvaluationResult, result_id)
    patch = db.session.get(MinePatch, r.patch_id)
    unit_scores = json.loads(r.unit_scores) if
    r.unit_scores else []
    return render_template("evaluate/result.html", r=r,
    patch=patch,
                           unit_scores=unit_scores)
# ==== app/views/report.py ====
# -*- coding: utf-8 -*-
"""报告与结果输出 —— 得分明细、图表、Word / PDF 导出。"""
import io
import json
from datetime import datetime
from flask import Blueprint, render_template, send_file,
    flash, redirect, url_for
from app.models import db, MinePatch, ReferenceEcosystem
from app.services import report_service as rs
from app.services import workflow as wf
from app.utils import charts
from app.utils import browser
from app.utils.constants import grade_of
bp = Blueprint("report", __name__, url_prefix="/report")
def _prepare(patch_id):
    """加载图斑并取出已记录的评价结果，供明细页与导出复用。
    报告是评价结论的呈现，因此这里读取最近一次保存的评价记录，
    而不是重新计算：评价完成后再改动指标数据，也不会让报告与
    系统记录的评价结论对不上。没有任何评价记录时，返回流程检查
    结果，由调用方拦截并提示先去执行评价。
    """
    patch = db.get_or_404(MinePatch, patch_id)
    prog = wf.report_progress(patch_id)
    if not prog["ready"]:
        return patch, None, None, None, prog
    rec = prog["result"]
```

```
    result = {                                                          4051
        "final_score": rec.final_score,                                 4052
        "grade": rec.grade,                                             4053
        "grade_info": grade_of(rec.final_score),                        4054
        "unit_results": json.loads(rec.unit_scores) if                  4055
    rec.unit_scores else [],                                            4056
    }                                                                   4057
    reference = ReferenceEcosystem.query.filter_by(patch_id=            4058
    patch_id).first()                                                   4059
    return patch, reference, result, rec, prog                          4060
def _deny(progress, patch_id, endpoint="report.detail"):                4061
    """前置条件未满足时的统一处理：逐条提示并退回报告列表。"""                4062
    for b in progress["blocking"]:                                      4063
        flash("暂不能生成报告 —— %s：%s" % (b["label"],                  4064
    b["reason"]), "error")                                              4065
    if progress["blocking"]:                                            4066
        flash("请先完成上述步骤后再生成报告。", "error")                  4067
    return redirect(url_for("report.index"))                            4068
def _charts(result):                                                    4069
    """生成两张图表：单元得分对比 + 一级指标平均得分分布。"""                4070
    units = [(u["unit_type"], u["unit_score"],                          4071
              "%.4f" % u["unit_score"]) for u in                        4072
    result["unit_results"]]                                             4073
    chart_units = charts.hbars(units)                                   4074
    # 各一级指标在各单元上的平均得分，反映其对评价结果的实际影响              4075
    acc, cnt = {}, {}                                                   4076
    for u in result["unit_results"]:                                    4077
        for k, v in (u.get("l1_scores") or {}).items():                 4078
            acc[k] = acc.get(k, 0.0) + v                                4079
            cnt[k] = cnt.get(k, 0) + 1                                  4080
    items = [(k, acc[k] / cnt[k], "%.4f" % (acc[k] /                    4081
    cnt[k])) for k in acc]                                              4082
    items.sort(key=lambda x: -x[1])                                     4083
    chart_weight = charts.donut(items)                                  4084
    return chart_units, chart_weight                                    4085
@bp.route("/")                                                          4086
def index():                                                            4087
    patches = MinePatch.query.order_by(MinePatch.id.desc()).            4088
    all()                                                               4089
    progress = {p.id: wf.report_progress(p.id) for p in                 4090
    patches}                                                            4091
    return render_template("report/index.html",                         4092
    patches=patches,                                                    4093
                                progress=progress)                      4094
@bp.route("/<int:patch_id>")                                            4095
def detail(patch_id):                                                   4096
    patch, reference, result, record, prog =                            4097
    _prepare(patch_id)                                                  4098
    if result is None:                                                  4099
        return _deny(prog, patch_id)                                    4100
```

```
    chart_units, chart_weight = _charts(result)                    4101
    return render_template("report/detail.html",                   4102
    patch=patch, result=result,                                    4103
                                record=record,                     4104
    reference=reference,                                           4105
                                chart_units=chart_units,           4106
    chart_weight=chart_weight)                                     4107
@bp.route("/<int:patch_id>/word")                                  4108
def word(patch_id):                                                4109
    patch, reference, result, record, prog =                       4110
    _prepare(patch_id)                                             4111
    if result is None:                                             4112
        return _deny(prog, patch_id)                               4113
    buf = rs.build_report(patch, result)                           4114
    filename = "自然恢复适宜性评价报告_%s.docx" % patch.name          4115
    return send_file(buf, as_attachment=True,                      4116
    download_name=filename,                                        4117
                        mimetype="application/vnd.openxmlformat4118
    s-officedocument."                                             4119
                                    "wordprocessingml.document")   4120
@bp.route("/<int:patch_id>/pdf")                                   4121
def pdf(patch_id):                                                 4122
    """导出 PDF 报告：服务端渲染打印模板，再交浏览器无头打印。"""          4123
    patch, reference, result, record, prog =                       4124
    _prepare(patch_id)                                             4125
    if result is None:                                             4126
        return _deny(prog, patch_id)                               4127
    chart_units, chart_weight = _charts(result)                    4128
    html = render_template(                                        4129
        "report/print.html", patch=patch,                          4130
    reference=reference, result=result,                            4131
        chart_units=chart_units, chart_weight=chart_weight,        4132
        generated_at=datetime.now().strftime("%Y-%m-%d             4133
    %H:%M"))                                                       4134
    data, err = browser.html_to_pdf(html)                          4135
    if err:                                                        4136
        flash("PDF 导出失败：%s" % err, "error")                     4137
        return redirect(url_for("report.detail",                   4138
    patch_id=patch_id))                                            4139
    return send_file(io.BytesIO(data), as_attachment=True,         4140
                        download_name="自然恢复适宜性评价报告_%s.pdf" %  4141
    patch.name,                                                    4142
                        mimetype="application/pdf")                4143
# ==== app/views/search.py ====                                    4144
# -*- coding: utf-8 -*-                                            4145
"""全局搜索：在评价项目、矿山图斑、标准与政策文件中查找关键词。                  4146
顶栏的搜索框指向本模块。只做关键词包含匹配，不做分词，                          4147
但覆盖名称、编号、行政区、矿种、文号、摘要等常用检索字段。                      4148
"""                                                                4149
from sqlalchemy import or_                                         4150
```

```
from flask import Blueprint, render_template, request
from app.models import db, Project, MinePatch, Standard
bp = Blueprint("search", __name__)
LIMIT = 50              # 每类结果上限，避免一次渲染过多
@bp.route("/search")
def index():
    q = (request.args.get("q") or "").strip()
    projects, patches, standards = [], [], []
    if q:
        like = "%" + q + "%"
        projects = Project.query.filter(or_(
            Project.name.like(like),
            Project.code.like(like),
            Project.description.like(like),
        )).order_by(Project.id.desc()).limit(LIMIT).all()
        patches = MinePatch.query.filter(or_(
            MinePatch.name.like(like),
            MinePatch.admin_region.like(like),
            MinePatch.mineral_type.like(like),
        )).order_by(MinePatch.id.desc()).limit(LIMIT).all()
        standards = Standard.query.filter(or_(
            Standard.name.like(like),
            Standard.doc_number.like(like),
            Standard.province.like(like),
            Standard.summary.like(like),
        )).order_by(Standard.id).limit(LIMIT).all()
    total = len(projects) + len(patches) + len(standards)
    return render_template("search.html", q=q,
    projects=projects,
                           patches=patches,
    standards=standards, total=total)
# ==== app/templates/base.html ====
{% import "_macros.html" as ui %}<!DOCTYPE html>
<html lang="zh-CN">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width,
    initial-scale=1.0">
  <link rel="icon" href="{{ url_for('static',
    filename='favicon.ico') }}">
  <title>{% block title %}历史遗留废弃矿山自然恢复适宜性评价系统{% endblock
    %}</title>
  <style>
    /* ---------- 设计变量：绿色主色 + 中性灰阶 ---------- */
    :root {
      --accent: #1f6b45;                /* 主色：深绿，呼应自然恢复主题 */
      --accent-hover: #185738;
      --accent-soft: #e9f4ee;
      --side-1: #17503a;                /* 侧边栏渐变起止色 */
      --side-2: #113c2b;
```

```
    --danger: #b3261e;
    --danger-soft: #fdecea;
    --ok: #1e6b3a;
    --ok-soft: #eaf6ef;
    --todo: #d98324;                    /* 待办角标 */
    --ink: #23262b;
    --ink-soft: #6b7280;
    --line: #e3e7ea;
    --surface: #ffffff;
    --canvas: #f4f7f5;
    --radius: 10px;
    --shadow: 0 1px 2px rgba(0, 0, 0, .05);
    --header-h: 60px;
    --nav-w: 226px;
}
* { box-sizing: border-box; }
html, body { margin: 0; padding: 0; }
body {
    font-family: "Microsoft YaHei", "PingFang SC",
"SimSun", sans-serif;
    font-size: 14px;
    line-height: 1.6;
    color: var(--ink);
    background: var(--canvas);
}
/* ---------- 跳到主要内容（键盘可达性） ---------- */
.skip-link {
    position: absolute;
    left: 12px;
    top: -60px;
    z-index: 30;
    padding: 8px 14px;
    background: var(--accent);
    color: #fff;
    border-radius: var(--radius);
    text-decoration: none;
    transition: top .15s ease-out;
}
.skip-link:focus { top: 12px; }
/* ---------- 整体骨架：左侧深绿导航 + 右侧内容 ---------- */
.app-shell {
    display: flex;
    align-items: stretch;
    min-height: 100dvh;
}
/* ---------- 侧边栏 ---------- */
.app-side {
    flex: 0 0 var(--nav-w);
    width: var(--nav-w);
    display: flex;
```

```
    flex-direction: column;                                          4251
    background: linear-gradient(180deg, var(--side-1) 0%,           4252
var(--side-2) 100%);                                                 4253
    color: #fff;                                                     4254
}                                                                    4255
.app-brand {                                                         4256
    display: flex;                                                   4257
    align-items: center;                                             4258
    gap: 11px;                                                       4259
    padding: 17px 16px 15px;                                         4260
    border-bottom: 1px solid rgba(255, 255, 255, .10);               4261
}                                                                    4262
.app-brand__mark {                                                   4263
    flex: none;                                                      4264
    display: grid;                                                   4265
    place-items: center;                                             4266
    width: 36px;                                                     4267
    height: 36px;                                                    4268
    border-radius: 10px;                                             4269
    background: rgba(255, 255, 255, .15);                            4270
}                                                                    4271
.app-brand__name {                                                   4272
    font-size: 14.5px;                                               4273
    font-weight: 600;                                                4274
    line-height: 1.35;                                               4275
    letter-spacing: .2px;                                            4276
}                                                                    4277
.app-brand__sub {                                                    4278
    font-size: 11.5px;                                               4279
    line-height: 1.35;                                               4280
    color: rgba(255, 255, 255, .58);                                 4281
}                                                                    4282
.app-nav { flex: 1; padding: 6px 12px 14px; overflow-y:              4283
auto; }                                                              4284
.app-nav__group {                                                    4285
    margin: 12px 0 5px;                                              4286
    padding: 0 10px;                                                 4287
    font-size: 11.5px;                                               4288
    letter-spacing: .05em;                                           4289
    color: rgba(255, 255, 255, .48);                                 4290
}                                                                    4291
.app-nav ul { margin: 0; padding: 0; list-style: none; }             4292
.app-nav a {                                                         4293
    display: flex;                                                   4294
    align-items: center;                                             4295
    gap: 10px;                                                       4296
    padding: 9px 11px;                                               4297
    margin-bottom: 2px;                                              4298
    border-radius: 8px;                                              4299
    color: rgba(255, 255, 255, .80);                                 4300
```

```
    text-decoration: none;
    transition: background-color .16s ease, color .16s ease;
}
.app-nav a:hover { background: rgba(255, 255, 255, .10); color: #fff; }
.app-nav a:focus-visible {
    outline: 2px solid rgba(255, 255, 255, .60);
    outline-offset: 1px;
}
.app-nav a[aria-current="page"] {
    background: rgba(255, 255, 255, .17);
    color: #fff;
    font-weight: 600;
}
.app-nav .ico { flex: none; opacity: .92; }
.app-nav__badge {
    margin-left: auto;
    min-width: 20px;
    padding: 0 6px;
    font-size: 11px;
    font-weight: 600;
    line-height: 18px;
    text-align: center;
    border-radius: 9px;
    background: var(--todo);
    color: #fff;
    font-variant-numeric: tabular-nums;
}
.app-user {
    display: flex;
    align-items: center;
    gap: 8px;
    padding: 10px 14px;
    border-top: 1px solid rgba(255, 255, 255, .10);
}
.app-user__me {
    display: flex;
    flex: 1;
    min-width: 0;
    align-items: center;
    gap: 10px;
    padding: 4px 6px;
    border-radius: 8px;
    color: inherit;
    text-decoration: none;
    transition: background-color .16s ease;
}
.app-user__me:hover { background: rgba(255, 255, 255, .10); }
```

```
.app-user__me:focus-visible {
  outline: 2px solid rgba(255, 255, 255, .60);
  outline-offset: 1px;
}
.app-user__avatar {
  flex: none;
  display: grid;
  place-items: center;
  width: 32px;
  height: 32px;
  border-radius: 50%;
  background: rgba(255, 255, 255, .16);
  font-size: 13px;
  font-weight: 600;
  object-fit: cover;
}
.app-user__name {
  display: block;
  font-size: 13px;
  line-height: 1.3;
  white-space: nowrap;
  overflow: hidden;
  text-overflow: ellipsis;
}
.app-user__state {
  display: flex;
  align-items: center;
  gap: 5px;
  font-size: 11.5px;
  line-height: 1.3;
  color: rgba(255, 255, 255, .60);
}
.app-user__dot {
  width: 6px;
  height: 6px;
  border-radius: 50%;
  background: #6ed49a;
}
.app-user__out {
  flex: none;
  padding: 6px;
  border-radius: 6px;
  color: rgba(255, 255, 255, .68);
  text-decoration: none;
}
.app-user__out:hover { color: #fff; background:
rgba(255, 255, 255, .12); }
/* ---------- 右侧内容区 ---------- */
.app-content {
  flex: 1;
```

```
    min-width: 0;
    display: flex;
    flex-direction: column;
}
/* ---------- 顶栏：面包屑 + 搜索 ---------- */
.app-header {
    display: flex;
    align-items: center;
    gap: 14px;
    height: var(--header-h);
    padding: 0 22px;
    background: var(--surface);
    border-bottom: 1px solid var(--line);
}
.crumb {
    display: flex;
    align-items: center;
    gap: 7px;
    font-size: 13px;
    color: var(--ink-soft);
    white-space: nowrap;
}
.crumb a { color: var(--ink-soft); text-decoration:
none; }
.crumb a:hover { color: var(--accent); }
.crumb strong { color: var(--ink); font-weight: 600; }
.crumb .ico { color: #c3c9d0; }
.app-header__spacer { flex: 1; }
.search-box {
    display: flex;
    align-items: center;
    gap: 8px;
    width: 268px;
    padding: 7px 12px;
    background: var(--canvas);
    border: 1px solid var(--line);
    border-radius: 9px;
    color: var(--ink-soft);
    transition: border-color .16s ease, background-color
.16s ease;
}
.search-box:focus-within { border-color: var(--accent);
background: #fff; }
.search-box input {
    flex: 1;
    min-width: 0;
    border: none;
    outline: none;
    background: transparent;
    font-family: inherit;
```

```
    font-size: 13px;                                                    4451
    color: var(--ink);                                                  4452
}                                                                       4453
.search-box button {                                                    4454
    display: grid;                                                      4455
    place-items: center;                                                4456
    padding: 0;                                                         4457
    border: none;                                                       4458
    background: transparent;                                            4459
    color: inherit;                                                     4460
    cursor: pointer;                                                    4461
}                                                                       4462
.icon-btn {                                                             4463
    display: grid;                                                      4464
    place-items: center;                                                4465
    width: 34px;                                                        4466
    height: 34px;                                                       4467
    border-radius: 9px;                                                 4468
    border: 1px solid transparent;                                      4469
    color: var(--ink-soft);                                             4470
    text-decoration: none;                                              4471
    transition: background-color .16s ease, color .16s                  4472
ease,                                                                   4473
                border-color .16s ease;                                 4474
}                                                                       4475
.icon-btn:hover {                                                       4476
    color: var(--accent);                                               4477
    background: var(--accent-soft);                                     4478
    border-color: rgba(31, 107, 69, .18);                               4479
}                                                                       4480
.app-main {                                                             4481
    flex: 1;                                                            4482
    min-width: 0;                                                       4483
    padding: 22px 26px 26px;                                            4484
}                                                                       4485
.container { max-width: 1120px; }                                       4486
/* ---------- 页脚 ---------- */                                        4487
.app-foot {                                                             4488
    display: flex;                                                      4489
    align-items: center;                                                4490
    justify-content: space-between;                                     4491
    gap: 14px;                                                          4492
    padding: 0 26px 18px;                                               4493
    font-size: 12px;                                                    4494
    color: var(--ink-soft);                                             4495
}                                                                       4496
/* ---------- 未登录时的裸页面（登录页） ---------- */                  4497
.login-shell {                                                          4498
    display: flex;                                                      4499
    flex-direction: column;                                             4500
```

```
  min-height: 100dvh;
}
/* ---------- 卡片 ---------- */
.card {
  background: var(--surface);
  border: 1px solid var(--line);
  border-radius: var(--radius);
  box-shadow: var(--shadow);
  padding: 22px 24px;
  margin-bottom: 18px;
}
.card > h2 {
  margin: 0 0 14px;
  font-size: 16px;
  font-weight: 600;
  color: var(--accent);
  text-wrap: balance;
}
.card > h3 {
  margin: 20px 0 10px;
  font-size: 14px;
  font-weight: 600;
  color: var(--accent);
}
/* ---------- 表格 ---------- */
table {
  width: 100%;
  border-collapse: collapse;
  margin: 10px 0 16px;
  font-variant-numeric: tabular-nums;
}
caption {
  text-align: left;
  color: var(--ink-soft);
  font-size: 12px;
  padding-bottom: 6px;
}
th, td {
  border: 1px solid var(--line);
  padding: 8px 12px;
  text-align: center;
  vertical-align: middle;
}
th {
  background: var(--accent-soft);
  font-weight: 600;
  white-space: nowrap;
}
tbody tr:hover { background: #fafbfc; }
/* ---------- 表单控件 ---------- */
```

```
label { display: inline-block; }
.field { margin-bottom: 12px; }
.field__label {
  display: block;
  margin-bottom: 4px;
  color: var(--ink-soft);
  font-size: 13px;
}
input[type="text"], input[type="number"],
input[type="password"],
select, textarea {
  padding: 6px 9px;
  font-family: inherit;
  font-size: 14px;
  color: var(--ink);
  background: #fff;
  border: 1px solid #c6ccd4;
  border-radius: 4px;
  max-width: 100%;
}
input[type="number"] {
  width: 104px;
  text-align: center;
  font-variant-numeric: tabular-nums;
}
textarea { width: 100%; min-height: 64px; resize:
vertical; }
input:focus-visible, select:focus-visible,
textarea:focus-visible {
  outline: 2px solid var(--accent);
  outline-offset: 1px;
  border-color: var(--accent);
}
.checkbox {
  display: inline-flex;
  align-items: center;
  gap: 7px;
  cursor: pointer;
  user-select: none;
}
.checkbox input { width: 16px; height: 16px; margin: 0;
cursor: pointer; }
/* ---------- 按钮 ----------
   统一为圆角矩形：默认半透明底色，鼠标悬停时填充为实色。 */
.btn {
  display: inline-flex;
  align-items: center;
  justify-content: center;
  gap: 6px;
  padding: 8px 18px;
```

```
    font-family: inherit;                                                4601
    font-size: 14px;                                                     4602
    line-height: 1.4;                                                    4603
    color: #fff;                                                         4604
    background: rgba(31, 107, 69, .80);                                  4605
    border: 1px solid rgba(31, 107, 69, .30);                            4606
    border-radius: 8px;                                                  4607
    cursor: pointer;                                                     4608
    text-decoration: none;                                               4609
    transition: background-color .16s ease, border-color                 4610
.16s ease,                                                               4611
                color .16s ease, box-shadow .16s ease,                   4612
transform .16s ease;                                                     4613
}                                                                        4614
.btn:hover {                                                             4615
    background: var(--accent);                                           4616
    border-color: var(--accent);                                         4617
    box-shadow: 0 3px 12px rgba(31, 107, 69, .22);                       4618
}                                                                        4619
.btn:active { transform: translateY(1px); box-shadow:                    4620
none; }                                                                  4621
.btn:focus-visible { outline: 2px solid var(--accent);                   4622
outline-offset: 2px; }                                                   4623
.btn--ghost {                                                            4624
    color: var(--accent);                                                4625
    background: rgba(31, 107, 69, .06);                                  4626
    border-color: rgba(31, 107, 69, .22);                                4627
}                                                                        4628
.btn--ghost:hover {                                                      4629
    color: #fff;                                                         4630
    background: var(--accent);                                           4631
    border-color: var(--accent);                                         4632
}                                                                        4633
.btn--danger {                                                           4634
    color: var(--danger);                                                4635
    background: rgba(179, 38, 30, .07);                                  4636
    border-color: rgba(179, 38, 30, .26);                                4637
}                                                                        4638
.btn--danger:hover {                                                     4639
    color: #fff;                                                         4640
    background: var(--danger);                                           4641
    border-color: var(--danger);                                         4642
    box-shadow: 0 3px 12px rgba(179, 38, 30, .20);                       4643
}                                                                        4644
.btn--sm { padding: 5px 13px; font-size: 13px;                           4645
border-radius: 7px; }                                                    4646
.btn:disabled { opacity: .45; cursor: not-allowed; }                     4647
.btn:disabled:hover {                                                    4648
    background: rgba(31, 107, 69, .80);                                  4649
    border-color: rgba(31, 107, 69, .30);                                4650
```

```
    box-shadow: none;                                                    4651
}                                                                        4652
.btn--ghost:disabled:hover { background: rgba(31, 107,                   4653
69, .06); color: var(--accent); }                                        4654
/* ---------- 表格操作列的行内小按钮 ----------                          4655
   取代原来「文字 | 文字 | 文字」的写法：                                4656
   圆角矩形 + 半透明底色，悬停时变实色。 */                              4657
.row-actions {                                                           4658
  display: inline-flex;                                                  4659
  flex-wrap: nowrap;          /* 按钮不折行；宁可让旁边长名称换行 */     4660
  align-items: center;                                                   4661
  gap: 6px;                                                              4662
  white-space: nowrap;                                                   4663
}                                                                        4664
.row-actions form { display: inline-flex; margin: 0; }                   4665
.link-btn {                                                              4666
  display: inline-flex;                                                  4667
  align-items: center;                                                   4668
  padding: 4px 11px;                                                     4669
  font-family: inherit;                                                  4670
  font-size: 13px;                                                       4671
  line-height: 1.5;                                                      4672
  color: var(--accent);                                                  4673
  background: rgba(31, 107, 69, .07);                                    4674
  border: 1px solid rgba(31, 107, 69, .20);                              4675
  border-radius: 7px;                                                    4676
  cursor: pointer;                                                       4677
  text-decoration: none;                                                 4678
  white-space: nowrap;                                                   4679
  transition: background-color .16s ease, border-color                   4680
.16s ease,                                                               4681
              color .16s ease, box-shadow .16s ease,                     4682
transform .16s ease;                                                     4683
}                                                                        4684
.link-btn:hover {                                                        4685
  color: #fff;                                                           4686
  background: var(--accent);                                             4687
  border-color: var(--accent);                                           4688
  box-shadow: 0 2px 8px rgba(31, 107, 69, .20);                          4689
}                                                                        4690
.link-btn:active { transform: translateY(1px);                           4691
box-shadow: none; }                                                      4692
.link-btn:focus-visible { outline: 2px solid                             4693
var(--accent); outline-offset: 2px; }                                    4694
.link-btn--danger {                                                      4695
  color: var(--danger);                                                  4696
  background: rgba(179, 38, 30, .07);                                    4697
  border-color: rgba(179, 38, 30, .24);                                  4698
}                                                                        4699
.link-btn--danger:hover {                                                4700
```

```
    color: #fff;
    background: var(--danger);
    border-color: var(--danger);
    box-shadow: 0 2px 8px rgba(179, 38, 30, .22);
}
.actions { display: flex; flex-wrap: wrap; align-items: center; gap: 8px; }
.actions form { display: inline; }
/* ---------- 提示条 ---------- */
.flash {
    padding: 10px 14px;
    margin-bottom: 14px;
    border-radius: var(--radius);
    border-left: 4px solid;
}
.flash.error {
    color: var(--danger);
    background: var(--danger-soft);
    border-left-color: var(--danger);
}
.flash.ok {
    color: var(--ok);
    background: var(--ok-soft);
    border-left-color: var(--ok);
}
/* ---------- 居中浮层提示（登录成功、各类操作完成） ----------
   外层 .toast-stack 固定在页面正中心并纵向排列，
   内层每条各自淡入 → 停留 → 淡出，多条提示会依次排开而不重叠。
   半透明底 + 背景模糊做出毛玻璃质感，不做成实心块；
   pointer-events: none 让它不拦截鼠标，出现期间也能正常操作。 */
.toast-stack {
    position: fixed;
    top: 50%;
    left: 50%;
    z-index: 100;
    display: flex;
    flex-direction: column;
    align-items: center;
    gap: 10px;
    transform: translate(-50%, -50%);
    pointer-events: none;
}
.toast-center {
    max-width: min(560px, 78vw);
    padding: 13px 28px;
    border-radius: 999px;
    background: rgba(31, 107, 69, .62);          /* 半透明，非实体 */
    color: #fff;
    font-size: 16px;
```

```
    font-weight: 600;
    line-height: 1.65;
    letter-spacing: .03em;
    text-align: center;
    text-wrap: pretty;
    border: 1px solid rgba(255, 255, 255, .28);
    backdrop-filter: blur(9px) saturate(1.3);
    -webkit-backdrop-filter: blur(9px) saturate(1.3);
    box-shadow: 0 10px 34px rgba(19, 36, 58, .18);
    animation: toast-fade 2.6s ease-in-out forwards;
}
@keyframes toast-fade {
    0%          { opacity: 0; transform: scale(.90); }
    14%, 74%    { opacity: 1; transform: scale(1); }
    100%        { opacity: 0; transform: scale(1); }
}
/* 系统设置里要求「减少动态效果」时，去掉缩放，只做淡入淡出 */
@media (prefers-reduced-motion: reduce) {
    .toast-center { animation: toast-fade-plain 2.6s
linear forwards; }
    @keyframes toast-fade-plain {
        0%          { opacity: 0; }
        14%, 74%    { opacity: 1; }
        100%        { opacity: 0; }
    }
}
/* 不支持背景模糊时，加深底色保证文字仍然清楚 */
@supports not (backdrop-filter: blur(1px)) {
    .toast-center { background: rgba(31, 107, 69, .88); }
}
/* ---------- 文字工具类 ---------- */
.muted { color: var(--ink-soft); font-size: 12.5px; }
.mono { font-family: Consolas, "Courier New",
monospace; font-variant-numeric: tabular-nums; }
.num { font-variant-numeric: tabular-nums; }
.warn { color: var(--danger); font-weight: 600; }
.ok { color: var(--ok); font-weight: 600; }
.nowrap { white-space: nowrap; }
/* 表格操作列里常把删除按钮放在 <form> 中。form 默认是块级元素，
   会把「删除」挤到下一行，导致操作列参差不齐，这里让它留在行内。 */
.nowrap form { display: inline; margin: 0; }
.stack { display: flex; flex-direction: column; gap:
14px; }
/* 状态标签：不依赖颜色单独表意 */
.tag {
    display: inline-block;
    padding: 1px 9px;
    font-size: 12.5px;
    border-radius: 10px;
    border: 1px solid;
```

```
  white-space: nowrap;
}
.tag--ok { color: var(--ok); background:
var(--ok-soft); border-color: #b8ddc6; }
.tag--warn { color: var(--danger); background:
var(--danger-soft); border-color: #eec4c0; }
.tag--muted { color: var(--ink-soft); background:
#f1f3f6; border-color: #dde1e6; }
/* ---------- 登录页 ---------- */
/* 仿 Apple 账户页：接近纯白的底，仅在顶部有一层极淡的绿色晕染，
   让彩色徽标成为整页唯一的视觉重心。排版见 auth/login.html。 */
.body--login {
  background:
    radial-gradient(1100px 640px at 50% 8%,
                    #edf6f1 0%, rgba(237, 246, 241, 0)
72%),
    linear-gradient(180deg, #ffffff 0%, #fbfdfc 100%);
}
.body--login .app-main { padding: 0; }
.body--login .container { max-width: none; width: 100%;
}
/* 仅供读屏软件的隐藏文本 */
.visually-hidden {
  position: absolute !important;
  width: 1px;
  height: 1px;
  margin: -1px;
  padding: 0;
  overflow: hidden;
  clip: rect(0 0 0 0);
  white-space: nowrap;
  border: 0;
}
/* ---------- 工作台：页首横幅 ---------- */
.hero {
  position: relative;
  overflow: hidden;
  min-height: 176px;
  padding: 26px 28px;
  margin-bottom: 16px;
  border-radius: 12px;
  /* 左侧压深保证文字可读，右侧透出远山插画；
     背景按底部对齐，避免远山被居中裁掉 */
  background:
    linear-gradient(100deg, rgba(9, 40, 27, .92) 0%,
                            rgba(11, 48, 33, .68) 50%,
                            rgba(11, 48, 33, .06) 100%),
    url("{{ url_for('static', filename='hero.svg') }}")
    center bottom / cover no-repeat;
  color: #fff;
```

```
}
.hero__body { max-width: 720px; }
.hero__title {
  margin: 0 0 10px;
  font-size: 21px;
  font-weight: 700;
  line-height: 1.4;
  letter-spacing: .3px;
}
.hero__desc {
  margin: 0;
  max-width: 640px;
  font-size: 12.5px;
  line-height: 1.85;
  color: rgba(255, 255, 255, .86);
}
.hero__actions {
  display: flex;
  flex-wrap: wrap;
  gap: 10px;
  margin-top: 18px;
}
.hero-btn {
  display: inline-flex;
  align-items: center;
  gap: 6px;
  padding: 7px 15px;
  font-size: 13px;
  border-radius: 8px;
  text-decoration: none;
  transition: background-color .16s ease, border-color .16s ease,
              color .16s ease;
}
.hero-btn--solid {
  color: #14432f;
  background: rgba(255, 255, 255, .92);
  border: 1px solid rgba(255, 255, 255, .92);
}
.hero-btn--solid:hover { background: #fff; }
.hero-btn--ghost {
  color: #fff;
  background: rgba(255, 255, 255, .14);
  border: 1px solid rgba(255, 255, 255, .45);
}
.hero-btn--ghost:hover { background: rgba(255, 255, 255, .26); }
.hero-btn:focus-visible { outline: 2px solid #fff; outline-offset: 2px; }
/* ---------- 工作台：数据概览 ---------- */
```

```
.stat-cards {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(190px, 1fr));
  gap: 12px;
  margin-bottom: 22px;
}
.stat-card {
  display: block;
  padding: 15px 17px;
  background: var(--surface);
  border: 1px solid var(--line);
  border-radius: var(--radius);
  box-shadow: var(--shadow);
  color: var(--ink);
  text-decoration: none;
  transition: border-color .16s ease, box-shadow .16s ease;
}
.stat-card:hover {
  border-color: rgba(31, 107, 69, .45);
  box-shadow: 0 3px 12px rgba(19, 60, 43, .08);
}
.stat-card__label {
  display: block;
  font-size: 12.5px;
  color: var(--ink-soft);
}
.stat-card__value {
  display: block;
  margin: 3px 0 2px;
  font-size: 26px;
  font-weight: 700;
  line-height: 1.25;
  color: var(--accent);
  font-variant-numeric: tabular-nums;
}
.stat-card__unit {
  margin-left: 3px;
  font-size: 13px;
  font-weight: 500;
  color: var(--ink-soft);
}
.stat-card__note {
  display: block;
  font-size: 12px;
  color: var(--ink-soft);
}
/* ---------- 工作台：分区标题 ---------- */
.section { margin-bottom: 22px; }
```

```
.section__head {
  display: flex;
  align-items: baseline;
  flex-wrap: wrap;
  gap: 6px 14px;
  margin-bottom: 12px;
}
.section__title { margin: 0; font-size: 16px; }
.section__desc { margin: 0; font-size: 12.5px; color:
var(--ink-soft); }
/* ---------- 工作台：流程步骤卡 ---------- */
.step-grid {
  display: grid;
  grid-template-columns: repeat(auto-fill,
minmax(238px, 1fr));
  gap: 12px;
}
.step-card {
  position: relative;
  display: block;
  padding: 16px 16px 15px;
  background: var(--surface);
  border: 1px solid var(--line);
  border-radius: var(--radius);
  box-shadow: var(--shadow);
  color: var(--ink);
  text-decoration: none;
  transition: border-color .16s ease, box-shadow .16s
ease,
              transform .16s ease;
}
.step-card:hover {
  border-color: rgba(31, 107, 69, .55);
  box-shadow: 0 5px 16px rgba(19, 60, 43, .10);
  transform: translateY(-1px);
}
.step-card:focus-visible { outline: 2px solid
var(--accent); outline-offset: 2px; }
.step-card__no {
  position: absolute;
  top: 13px;
  right: 15px;
  font-size: 17px;
  font-weight: 700;
  color: #d7e3db;
  font-variant-numeric: tabular-nums;
}
.step-card__icon {
  display: grid;
  place-items: center;
```

```
    width: 32px;
    height: 32px;
    margin-bottom: 10px;
    border-radius: 9px;
    background: var(--accent-soft);
    color: var(--accent);
}
.step-card__title {
    display: block;
    margin-bottom: 5px;
    font-size: 14px;
    font-weight: 600;
    color: var(--accent);
}
.step-card__desc {
    display: block;
    font-size: 12.5px;
    line-height: 1.65;
    color: var(--ink-soft);
    text-wrap: pretty;
}
.step-card__badge {
    display: inline-block;
    margin-top: 10px;
    padding: 1px 9px;
    font-size: 11.5px;
    border-radius: 9px;
    background: #fdf1e2;
    color: #9a5a12;
    border: 1px solid #f0d3ab;
}
/* 前置数据不具备时置灰，并说明缺什么 */
.step-card--locked {
    background: #fafbfa;
    border-color: #eceff0;
    box-shadow: none;
    cursor: default;
}
.step-card--locked .step-card__no { color: #e6e9e8; }
.step-card--locked .step-card__icon {
    background: #f1f3f2;
    color: #a9b2ae;
}
.step-card--locked .step-card__title { color: #9aa3a0; }
.step-card--locked .step-card__desc { color: #b3bab7; }
.step-card__lock {
    display: inline-block;
    margin-top: 10px;
    padding: 1px 9px;
    font-size: 11.5px;
```

```
    border-radius: 9px;                                                    5051
    background: #f1f3f2;                                                   5052
    color: #8b948f;                                                        5053
}                                                                          5054
/* ---------- 工作台：整行入口 ---------- */                                5055
.row-card {                                                                5056
    display: flex;                                                         5057
    align-items: center;                                                   5058
    gap: 14px;                                                             5059
    padding: 15px 18px;                                                    5060
    background: var(--surface);                                            5061
    border: 1px solid var(--line);                                         5062
    border-radius: var(--radius);                                          5063
    box-shadow: var(--shadow);                                             5064
    color: var(--ink);                                                     5065
    text-decoration: none;                                                 5066
    transition: border-color .16s ease, box-shadow .16s                    5067
ease;                                                                      5068
}                                                                          5069
.row-card:hover {                                                          5070
    border-color: rgba(31, 107, 69, .45);                                  5071
    box-shadow: 0 3px 12px rgba(19, 60, 43, .08);                          5072
}                                                                          5073
.row-card__icon {                                                          5074
    display: grid;                                                         5075
    place-items: center;                                                   5076
    width: 36px;                                                           5077
    height: 36px;                                                          5078
    flex: none;                                                            5079
    border-radius: 10px;                                                   5080
    background: var(--accent-soft);                                        5081
    color: var(--accent);                                                  5082
}                                                                          5083
.row-card__main { flex: 1; min-width: 0; }                                 5084
.row-card__title {                                                         5085
    display: block;                                                        5086
    font-size: 14px;                                                       5087
    font-weight: 600;                                                      5088
    color: var(--accent);                                                  5089
}                                                                          5090
.row-card__desc {                                                          5091
    display: block;                                                        5092
    font-size: 12.5px;                                                     5093
    color: var(--ink-soft);                                                5094
}                                                                          5095
.row-card__arrow { color: #c3c9d0; flex: none; }                           5096
/* ---------- 结果摘要 ---------- */                                       5097
.stat-row {                                                                5098
    display: flex;                                                         5099
    flex-wrap: wrap;                                                       5100
```

```
  gap: 28px;
  margin: 4px 0 6px;
}
.stat__label { font-size: 12.5px; color:
var(--ink-soft); }
.stat__value { font-size: 20px; font-weight: 700;
font-variant-numeric: tabular-nums; }
/* ---------- 流程步骤条 ---------- */
.steps { display: flex; flex-wrap: wrap; gap: 8px;
margin: 4px 0 18px; }
.step {
  display: flex; align-items: center; gap: 7px;
  padding: 8px 12px; font-size: 13px;
  background: #fff; border: 1px solid var(--line);
  border-radius: var(--radius);
}
.step__no {
  display: inline-flex; align-items: center;
justify-content: center;
  width: 20px; height: 20px; border-radius: 50%;
  background: #eef1f5; color: var(--ink-soft);
  font-size: 12px; font-variant-numeric: tabular-nums;
}
.step__name { font-weight: 600; }
.step__state { font-size: 12.5px; color:
var(--ink-soft); }
.step--done { border-color: #b8ddc6; background:
var(--ok-soft); }
.step--done .step__no { background: var(--ok); color:
#fff; }
.step--done .step__state { color: var(--ok); }
.step--fail { border-color: #eec4c0; background:
var(--danger-soft); }
.step--fail .step__no { background: var(--danger);
color: #fff; }
.step--fail .step__state { color: var(--danger);
font-weight: 600; }
.step--next { border-color: var(--accent); }
.step--next .step__no { background: var(--accent);
color: #fff; }
/* ---------- 提示块 ---------- */
.notice {
  padding: 14px 16px; margin: 0 0 16px;
  border-radius: var(--radius); border-left: 4px solid;
}
.notice--warn {
  background: var(--danger-soft);
  border-left-color: var(--danger);
  color: #7a1c17;
}
```

```
.notice__title { margin: 0 0 8px; font-weight: 600; }
.notice__list { margin: 0; padding-left: 20px;
line-height: 1.9; }
.notice__list a { color: var(--accent); }
.notice__hint { margin: 8px 0 0; font-size: 12.5px;
color: var(--ink-soft); }
/* ---------- 图斑面板（按图斑分块展示，如指标数据采集） ---------- */
.patch-panel {
  border: 1px solid var(--line);
  border-radius: var(--radius);
  background: var(--surface);
  padding: 14px 16px;
  margin: 0 0 14px;
}
.patch-panel:last-child { margin-bottom: 0; }
.patch-panel__head {
  display: flex;
  align-items: baseline;
  flex-wrap: wrap;
  gap: 6px 14px;
  margin: 0 0 12px;
  padding-bottom: 9px;
  border-bottom: 1px solid var(--line);
}
.patch-panel__name {
  margin: 0;
  font-size: 15px;
  font-weight: 600;
}
.patch-panel__meta {
  font-size: 12.5px;
  color: var(--ink-soft);
}
.patch-panel__meta a { color: var(--accent); }
.patch-panel__tail { margin-left: auto; }
.patch-panel .notice { margin: 0; }
.patch-panel table { margin: 0; }
/* ---------- 窄屏适配 ---------- */
@media (max-width: 960px) {
  .app-shell { flex-direction: column; }
  .app-side { width: auto; flex: none; }
  .app-brand { padding: 12px 14px; }
  .app-nav { padding: 4px 10px 10px; }
  .app-nav ul { display: flex; flex-wrap: wrap; gap:
4px; }
  .app-nav a { margin-bottom: 0; }
  .app-nav__group { margin: 8px 0 4px; }
  .app-user { padding: 10px 14px; }
  .app-header { padding: 0 14px; gap: 10px; }
  .search-box { width: 180px; }
```

```
        .app-main { padding: 16px 14px 20px; }
        .app-foot { padding: 0 14px 14px; }
      }
      @media (max-width: 640px) {
        .crumb { display: none; }
        .search-box { width: auto; flex: 1; }
      }
    </style>
</head>
<body class="{% if not current_user.is_authenticated %}body--login{% endif %}">
    <a class="skip-link" href="#main">跳到主要内容</a>
    {# 取一次即可（同一请求内重复调用返回相同内容）。
       类别为 toast 的渲染成居中浮层，其余仍是页面顶部的提示条 ——
       错误提示需要停下来细读，不适合做成会消失的浮层。 #}
    {% set flash_msgs = get_flashed_messages(with_categories=true) %}
    {# 类别为 toast / ok 的渲染成页面正中的浮层（操作完成的即时反馈）；
       其余（尤其 error）仍是页面顶部的提示条 ——
       错误提示需要停下来细读，不适合做成会消失的浮层。
       没有浮层时这个容器是空的，不占任何位置。 #}
    <div class="toast-stack">
      {% for category, message in flash_msgs if category in ("toast", "ok") %}
      <div class="toast-center" role="status" aria-live="polite">{{ message }}</div>
      {% endfor %}
    </div>
    {% if current_user.is_authenticated %}
    <div class="app-shell">
      <aside class="app-side">
        <div class="app-brand">
          <span class="app-brand__mark">{{ ui.icon("mountain", 21) }}</span>
          <span>
            <span class="app-brand__name">矿山自然恢复评价</span><br>
            <span class="app-brand__sub">适宜性评价系统 Ｖ１．０</span>
          </span>
        </div>
        <nav class="app-nav" aria-label="功能导航">
          <ul>
            <li>
              <a href="/"{% if home_active %} aria-current="page"{% endif %}>
                {{ ui.icon("home") }}<span>工作台</span>
              </a>
            </li>
          </ul>
          {% for g in nav_groups %}
          <p class="app-nav__group">{{ g.title }}</p>
```

```
      <ul>                                                                5251
        {% for item in g.links %}                                         5252
        <li>                                                              5253
          <a href="{{ item.href }}"{% if item.active %}                   5254
aria-current="page"{% endif %}>                                           5255
            {{ ui.icon(item.icon) }}<span>{{ item.label                   5256
}}</span>                                                                 5257
            {% if item.badge %}<span                                      5258
class="app-nav__badge">{{ item.badge }}</span>{% endif                    5259
%}                                                                        5260
          </a>                                                            5261
        </li>                                                             5262
        {% endfor %}                                                      5263
      </ul>                                                               5264
      {% endfor %}                                                        5265
    </nav>                                                                5266
    <div class="app-user">                                                5267
      <a class="app-user__me" href="/profile"                             5268
title="个人设置">                                                         5269
        {% if current_user.avatar %}                                      5270
        <img class="app-user__avatar"                                     5271
             src="{{ url_for('static',                                    5272
filename=current_user.avatar) }}" alt="">                                 5273
        {% else %}                                                        5274
        <span class="app-user__avatar"                                    5275
aria-hidden="true">{{ current_user.initial }}</span>                      5276
        {% endif %}                                                       5277
        <span style="min-width:0;">                                       5278
          <span class="app-user__name">{{                                 5279
current_user.shown_name }}</span>                                         5280
          <span class="app-user__state">                                  5281
            <span class="app-user__dot"                                   5282
aria-hidden="true"></span>                                                5283
            {{ role_labels.get(current_user.role,                         5284
current_user.role) }}                                                     5285
          </span>                                                         5286
        </span>                                                           5287
      </a>                                                                5288
      <a class="app-user__out" href="/logout" title="退出登录"            5289
         aria-label="退出登录">{{ ui.icon("logout", 17) }}</a>            5290
    </div>                                                                5291
</aside>                                                                  5292
<div class="app-content">                                                 5293
  <header class="app-header">                                             5294
    <nav class="crumb" aria-label="当前位置">                             5295
      <a href="/">首页</a>                                                5296
      {{ ui.icon("chevron", 13) }}                                        5297
      <strong>{{ page_title }}</strong>                                   5298
    </nav>                                                                5299
    <span class="app-header__spacer"></span>                              5300
```

```
            <form class="search-box" method="get"                    5301
      action="/search" role="search">                                5302
              {{ ui.icon("search", 16) }}                            5303
              <label for="q" class="visually-hidden">搜索项目、图斑、标准<  5304
      /label>                                                        5305
              <input id="q" type="search" name="q"                   5306
      autocomplete="off"                                             5307
                       placeholder="搜索项目、图斑、标准" value="{{          5308
      search_q }}">                                                  5309
              <button type="submit" aria-label="搜索">{{             5310
      ui.icon("chevron", 15) }}</button>                             5311
            </form>                                                  5312
            <a class="icon-btn" href="/logs" title="操作日志"          5313
      aria-label="操作日志">                                         5314
              {{ ui.icon("bell", 18) }}                              5315
            </a>                                                     5316
          </header>                                                  5317
    {% else %}                                                       5318
    <div class="login-shell">                                        5319
    {% endif %}                                                      5320
      <main id="main" class="app-main">                              5321
        <div class="container">                                      5322
          {% for category, message in flash_msgs if category         5323
      not in ("toast", "ok") %}                                      5324
            <p class="flash {{ category }}" role="status">{{         5325
      message }}</p>                                                 5326
          {% endfor %}                                               5327
          {% block content %}{% endblock %}                          5328
        </div>                                                       5329
      </main>                                                        5330
    {% if current_user.is_authenticated %}                           5331
          <footer class="app-foot">                                  5332
            <span>历史遗留废弃矿山自然恢复适宜性评价系统 Ｖ1.0</span>          5333
            <span>数据截止 {{ today }}</span>                          5334
          </footer>                                                  5335
      </div>                                                         5336
    </div>                                                           5337
    {% else %}                                                       5338
    </div>                                                           5339
    {% endif %}                                                      5340
    {% block scripts %}{% endblock %}                                5341
</body>                                                              5342
</html>                                                              5343
# ==== app/templates/_macros.html ====                               5344
{# 共用宏：图标、流程状态面板 #}                                          5345
{# ========================================================================5346
   ==================                                                5347
   图标：内联 SVG 线性图标（24×24 视图框，随文字颜色走）                      5348
   不引图标字体或外部图片，避免额外请求与资源依赖；                           5349
   stroke 用 currentColor，在深色侧边栏与浅色内容区都能自适应。              5350
```

```
    ==========================================================5351
      ================ #}                                      5352
{% set ICONS = {                                              5353
  "dot": '<circle cx="12" cy="12" r="3"/>',                   5354
  "home": '<path d="M3 9.6 12 3l9 6.6V20a2 2 0 0 1-2 2H5a2    5355
    2 0 0 1-2-2z"/><path d="M9.5 22v-7h5v7"/>',               5356
  "folder": '<path d="M3 7.5A2 2 0 0 1 5 5.5h3.6l2 2.2H19a2   5357
    2 0 0 1 2 2v7.8a2 2 0 0 1-2 2H5a2 2 0 0 1-2-2z"/>',       5358
  "book": '<path d="M4 4.8A2.8 2.8 0 0 1 6.8                  5359
    2H20v17.5H6.8A2.8 2.8 0 0 0 4 22z"/><path d="M4           5360
    19.5A2.8 2.8 0 0 1 6.8 16.7H20"/>',                       5361
  "shield": '<path d="M12 22s8-4 8-10V5.2L12 2 4 5.2V12c0 6   5362
    8 10 8 10z"/><path d="m9 12 2.2 2.2L15.5 10"/>',          5363
  "layers": '<path d="m12 2.8 9 4.7-9 4.7-9-4.7z"/><path      5364
    d="m3 12.5 9 4.7 9-4.7"/><path d="m3 17 9 4.7 9-4.7"/>',5365
  "sliders": '<path d="M4 21v-6M4 11V3M12 21v-9M12 8V3M20     5366
    21v-4M20 13V3"/><path d="M1.5 15h5M9.5 12h5M17.5          5367
    17h5"/>',                                                 5368
  "database": '<ellipse cx="12" cy="5.5" rx="8"               5369
    ry="3.2"/><path d="M4 5.5v13c0 1.8 3.6 3.2 8 3.2s8-1.4    5370
    8-3.2v-13"/><path d="M4 12c0 1.8 3.6 3.2 8 3.2s8-1.4      5371
    8-3.2"/>',                                                5372
  "target": '<circle cx="12" cy="12" r="9"/><circle cx="12"   5373
    cy="12" r="4.6"/><circle cx="12" cy="12" r="1"/>',        5374
  "file": '<path d="M14 2.8H7a2 2 0 0 0-2 2v14.4a2 2 0 0 0    5375
    2 2h10a2 2 0 0 0 2-2V7.8z"/><path d="M14                  5376
    2.8V7.8h5"/><path d="M8.8 13h6.4M8.8 17h6.4"/>',          5377
  "users": '<circle cx="9.5" cy="8" r="3.6"/><path d="M3.5    5378
    20.5v-1.4A3.6 3.6 0 0 1 7.1 15.5h4.8a3.6 3.6 0 0 1 3.6    5379
    3.6v1.4"/><path d="M17 4.9a3.6 3.6 0 0 1 0 6.9"/><path    5380
    d="M19 15.8a3.6 3.6 0 0 1 2.4 3.4v1.3"/>',                5381
  "user": '<circle cx="12" cy="8" r="4"/><path d="M4.5        5382
    21v-1.6A4.4 4.4 0 0 1 8.9 15h6.2a4.4 4.4 0 0 1 4.4        5383
    4.4V21"/>',                                               5384
  "search": '<circle cx="11" cy="11" r="7"/><path d="m20.5    5385
    20.5-4-4"/>',                                             5386
  "bell": '<path d="M18 9.5a6 6 0 1 0-12 0c0 5-2.2 6.2-2.2    5387
    6.2h16.4S18 14.5 18 9.5"/><path d="M13.8 19.5a2 2 0 0     5388
    1-3.6 0"/>',                                              5389
  "plus": '<path d="M12 5v14M5 12h14"/>',                     5390
  "upload": '<path d="M12 16V4"/><path d="m7.5 8.5 4.5-4.5    5391
    4.5 4.5"/><path d="M4 16.5v2A2.5 2.5 0 0 0 6.5            5392
    21h11a2.5 2.5 0 0 0 2.5-2.5v-2"/>',                       5393
  "logout": '<path d="M15 4h3.5A2.5 2.5 0 0 1 21 6.5v11A2.5   5394
    2.5 0 0 1 18.5 20H15"/><path d="M10 16.5 5.5 12 10        5395
    7.5"/><path d="M5.5 12H15"/>',                            5396
  "chevron": '<path d="m9.5 5.5 6.5 6.5-6.5 6.5"/>',          5397
  "warn": '<path d="M12 3.5 21.5 20H2.5z"/><path d="M12       5398
    10v4.2"/><path d="M12 17.4h.01"/>',                       5399
  "mountain": '<path d="m3 20 6.5-11L14 17l2.5-4L21           5400
```

```
      20z"/><circle cx="17" cy="6" r="2.2"/>',
} %}
{% macro icon(name, size=18) %}
<svg class="ico" width="{{ size }}" height="{{ size }}"
      viewBox="0 0 24 24"
       fill="none" stroke="currentColor" stroke-width="1.7"
       stroke-linecap="round" stroke-linejoin="round"
       aria-hidden="true" focusable="false">{{
      ICONS.get(name, ICONS.dot)|safe }}</svg>
{% endmacro %}
{# 评价流程步骤条：按论文 4.1 节顺序显示各前置步骤的完成情况
    顺序为：限制性条件判定 → 划分微地质单元 → 录入指标数据 → 执行评价。
    论文原文：「先进行政策、位置约束地质灾害判断……对于满足限制性条件的
    废弃矿山，按照地形地貌划分……三种地质微单元类型」，故判定在前。 #}
{% macro workflow_steps(progress) %}
<div class="steps">
  <div class="step {% if progress.constraint.passed
    %}step--done{% elif progress.constraint.done
    %}step--fail{% else %}step--todo{% endif %}">
    <span class="step__no">1</span>
    <span class="step__name">限制性条件判定</span>
    <span class="step__state">
      {% if progress.constraint.passed %}已通过
      {% elif progress.constraint.done %}未通过
      {% else %}待完成{% endif %}
    </span>
  </div>
  <div class="step {% if progress.units.done %}step--done{%
    else %}step--todo{% endif %}">
    <span class="step__no">2</span>
    <span class="step__name">划分微地质单元</span>
    <span class="step__state">
      {% if progress.units.done %}已完成{% else %}待完成{% endif
    %}
    </span>
  </div>
  <div class="step {% if progress.data.ready %}step--done{%
    else %}step--todo{% endif %}">
    <span class="step__no">3</span>
    <span class="step__name">录入指标数据</span>
    <span class="step__state">
      {% if progress.data.ready %}已完成{% else %}待完成{% endif
    %}
    </span>
  </div>
  <div class="step step--next">
    <span class="step__no">4</span>
    <span class="step__name">执行适宜性评价</span>
    <span class="step__state">当前步骤</span>
  </div>
```

```
</div>
{% endmacro %}
{# 前置条件未满足时的提示块：只列缺什么，并给出处理入口 #}
{% macro workflow_blocking(progress) %}
{% if progress.blocking %}
<div class="notice notice--warn">
  <p class="notice__title">尚不能执行本步骤，需要先完成：</p>
  <ol class="notice__list">
    {% for b in progress.blocking %}
    <li>
      <strong>{{ b.label }}</strong>{% if b.reason %}：{{ b.reason }}{% endif %}
      <a class="link-btn" href="{{ b.fix_url }}">{{ b.fix_label }}</a>
    </li>
    {% endfor %}
  </ol>
</div>
{% endif %}
{% endmacro %}
{# 一行式状态摘要，用于列表页 #}
{% macro workflow_badge(progress) %}
{% if progress.ready %}
<span class="tag tag--ok">可执行评价</span>
{% elif progress.blocking %}
<span class="tag tag--warn">{{ progress.blocking[0].label }}</span>
<span class="muted">{{ progress.blocking[0].reason }}</span>
{% endif %}
{% endmacro %}
{# 层级引导：本系统是「项目 → 图斑」两级结构，评价针对的是图斑而非项目。
   下游各页按图斑组织数据，故只要存在「还没建图斑的项目」，就直接列出并给出入口。 #}
{% macro patch_guide(projects_without_patch) %}
{% if projects_without_patch %}
<div class="notice notice--warn">
  <p class="notice__title">
    以下 {{ projects_without_patch|length }}
    个项目下还没有图斑，本页不会出现它们：
  </p>
  <ul class="notice__list">
    {% for p in projects_without_patch %}
    <li>
      {{ p.name }}
      <a class="link-btn" href="/project/{{ p.id }}/patch/add">新建图斑</a>
    </li>
    {% endfor %}
  </ul>
</div>
{% endif %}
```

```
{% endmacro %}
# ==== app/templates/_login_badge.svg ====
<svg class="lp-badge__svg" xmlns="http://www.w3.org/2000/svg
     " viewBox="0 0 240 240" role="img"
     aria-label="历史遗留废弃矿山自然恢复适宜性评价系统徽标">
  <title>系统徽标</title>
  <g class="lp-ring" style="--k: 2.7">
    <circle cx="162.00" cy="120.00" r="2.60"
    fill="#c4f5ab" />
    <circle cx="157.84" cy="138.22" r="2.97"
    fill="#b8f5ab" />
    <circle cx="146.19" cy="152.84" r="3.06"
    fill="#acf5ab" />
    <circle cx="129.35" cy="160.95" r="2.80"
    fill="#abf5b7" />
    <circle cx="110.65" cy="160.95" r="2.40"
    fill="#abf5c2" />
    <circle cx="93.81" cy="152.84" r="2.14" fill="#abf5ce" />
    <circle cx="82.16" cy="138.22" r="2.23" fill="#abf5da" />
    <circle cx="78.00" cy="120.00" r="2.60" fill="#abf5e6" />
    <circle cx="82.16" cy="101.78" r="2.97" fill="#abf5da" />
    <circle cx="93.81" cy="87.16" r="3.06" fill="#abf5ce" />
    <circle cx="110.65" cy="79.05" r="2.80" fill="#abf5c2" />
    <circle cx="129.35" cy="79.05" r="2.40" fill="#abf5b7" />
    <circle cx="146.19" cy="87.16" r="2.14" fill="#acf5ab" />
    <circle cx="157.84" cy="101.78" r="2.23"
    fill="#b8f5ab" />
  </g>
  <g class="lp-ring" style="--k: 2.15">
    <circle cx="177.29" cy="129.07" r="3.92"
    fill="#a4e28d" />
    <circle cx="171.68" cy="146.33" r="4.01"
    fill="#9be28d" />
    <circle cx="161.01" cy="161.01" r="3.87"
    fill="#91e28d" />
    <circle cx="146.33" cy="171.68" r="3.55"
    fill="#8de292" />
    <circle cx="129.07" cy="177.29" r="3.17"
    fill="#8de29c" />
    <circle cx="110.93" cy="177.29" r="2.88"
    fill="#8de2a6" />
    <circle cx="93.67" cy="171.68" r="2.79" fill="#8de2af" />
    <circle cx="78.99" cy="161.01" r="2.93" fill="#8de2b9" />
    <circle cx="68.32" cy="146.33" r="3.25" fill="#8de2c2" />
    <circle cx="62.71" cy="129.07" r="3.63" fill="#8de2cc" />
    <circle cx="62.71" cy="110.93" r="3.92" fill="#8de2cc" />
    <circle cx="68.32" cy="93.67" r="4.01" fill="#8de2c2" />
    <circle cx="78.99" cy="78.99" r="3.87" fill="#8de2b9" />
    <circle cx="93.67" cy="68.32" r="3.55" fill="#8de2af" />
    <circle cx="110.93" cy="62.71" r="3.17" fill="#8de2a6" />
```

```
    <circle cx="129.07" cy="62.71" r="2.88" fill="#8de29c"/>
    <circle cx="146.33" cy="68.32" r="2.79" fill="#8de292"/>
    <circle cx="161.01" cy="78.99" r="2.93" fill="#91e28d"/>
    <circle cx="171.68" cy="93.67" r="3.25" fill="#9be28d"/>
    <circle cx="177.29" cy="110.93" r="3.63"
    fill="#a4e28d"/>
  </g>
  <g class="lp-ring" style="--k: 1.85">
    <circle cx="194.00" cy="120.00" r="4.94"
    fill="#91cf72"/>
    <circle cx="191.85" cy="137.71" r="4.92"
    fill="#88cf72"/>
    <circle cx="185.52" cy="154.39" r="4.73"
    fill="#80cf72"/>
    <circle cx="175.39" cy="169.07" r="4.42"
    fill="#78cf72"/>
    <circle cx="162.04" cy="180.90" r="4.06"
    fill="#72cf73"/>
    <circle cx="146.24" cy="189.19" r="3.73"
    fill="#72cf7b"/>
    <circle cx="128.92" cy="193.46" r="3.51"
    fill="#72cf83"/>
    <circle cx="111.08" cy="193.46" r="3.45"
    fill="#72cf8b"/>
    <circle cx="93.76" cy="189.19" r="3.56" fill="#72cf93"/>
    <circle cx="77.96" cy="180.90" r="3.81" fill="#72cf9c"/>
    <circle cx="64.61" cy="169.07" r="4.16" fill="#72cfa4"/>
    <circle cx="54.48" cy="154.39" r="4.51" fill="#72cfac"/>
    <circle cx="48.15" cy="137.71" r="4.80" fill="#72cfb4"/>
    <circle cx="46.00" cy="120.00" r="4.94" fill="#72cfbc"/>
    <circle cx="48.15" cy="102.29" r="4.92" fill="#72cfb4"/>
    <circle cx="54.48" cy="85.61" r="4.73" fill="#72cfac"/>
    <circle cx="64.61" cy="70.93" r="4.42" fill="#72cfa4"/>
    <circle cx="77.96" cy="59.10" r="4.06" fill="#72cf9c"/>
    <circle cx="93.76" cy="50.81" r="3.73" fill="#72cf93"/>
    <circle cx="111.08" cy="46.54" r="3.51" fill="#72cf8b"/>
    <circle cx="128.92" cy="46.54" r="3.45" fill="#72cf83"/>
    <circle cx="146.24" cy="50.81" r="3.56" fill="#72cf7b"/>
    <circle cx="162.04" cy="59.10" r="3.81" fill="#72cf73"/>
    <circle cx="175.39" cy="70.93" r="4.16" fill="#78cf72"/>
    <circle cx="185.52" cy="85.61" r="4.51" fill="#80cf72"/>
    <circle cx="191.85" cy="102.29" r="4.80"
    fill="#88cf72"/>
  </g>
  <g class="lp-ring" style="--k: 1.62">
    <circle cx="209.57" cy="128.82" r="5.68"
    fill="#76bb59"/>
    <circle cx="206.12" cy="146.13" r="5.40"
    fill="#6fbb59"/>
    <circle cx="199.37" cy="162.43" r="5.06"
```

```
    fill="#68bb59" />                                                    5601
    <circle cx="189.57" cy="177.10" r="4.71"                             5602
    fill="#61bb59" />                                                    5603
    <circle cx="177.10" cy="189.57" r="4.41"                             5604
    fill="#5abb59" />                                                    5605
    <circle cx="162.43" cy="199.37" r="4.19"                             5606
    fill="#59bb5f" />                                                    5607
    <circle cx="146.13" cy="206.12" r="4.10"                             5608
    fill="#59bb66" />                                                    5609
    <circle cx="128.82" cy="209.57" r="4.15"                             5610
    fill="#59bb6d" />                                                    5611
    <circle cx="111.18" cy="209.57" r="4.32"                             5612
    fill="#59bb73" />                                                    5613
    <circle cx="93.87" cy="206.12" r="4.60" fill="#59bb7a" />            5614
    <circle cx="77.57" cy="199.37" r="4.94" fill="#59bb81" />            5615
    <circle cx="62.90" cy="189.57" r="5.29" fill="#59bb88" />            5616
    <circle cx="50.43" cy="177.10" r="5.59" fill="#59bb8f" />            5617
    <circle cx="40.63" cy="162.43" r="5.81" fill="#59bb96" />            5618
    <circle cx="33.88" cy="146.13" r="5.90" fill="#59bb9d" />            5619
    <circle cx="30.43" cy="128.82" r="5.85" fill="#59bba4" />            5620
    <circle cx="30.43" cy="111.18" r="5.68" fill="#59bba4" />            5621
    <circle cx="33.88" cy="93.87" r="5.40" fill="#59bb9d" />             5622
    <circle cx="40.63" cy="77.57" r="5.06" fill="#59bb96" />             5623
    <circle cx="50.43" cy="62.90" r="4.71" fill="#59bb8f" />             5624
    <circle cx="62.90" cy="50.43" r="4.41" fill="#59bb88" />             5625
    <circle cx="77.57" cy="40.63" r="4.19" fill="#59bb81" />             5626
    <circle cx="93.87" cy="33.88" r="4.10" fill="#59bb7a" />             5627
    <circle cx="111.18" cy="30.43" r="4.15" fill="#59bb73" />            5628
    <circle cx="128.82" cy="30.43" r="4.32" fill="#59bb6d" />            5629
    <circle cx="146.13" cy="33.88" r="4.60" fill="#59bb66" />            5630
    <circle cx="162.43" cy="40.63" r="4.94" fill="#59bb5f" />            5631
    <circle cx="177.10" cy="50.43" r="5.29" fill="#5abb59" />            5632
    <circle cx="189.57" cy="62.90" r="5.59" fill="#61bb59" />            5633
    <circle cx="199.37" cy="77.57" r="5.81" fill="#68bb59" />            5634
    <circle cx="206.12" cy="93.87" r="5.90" fill="#6fbb59" />            5635
    <circle cx="209.57" cy="111.18" r="5.85"                             5636
    fill="#76bb59" />                                                    5637
  </g>                                                                   5638
  <g class="lp-ring" style="--k: 1.46">                                  5639
    <circle cx="226.00" cy="120.00" r="5.94"                             5640
    fill="#65a843" />                                                    5641
    <circle cx="224.55" cy="137.45" r="5.62"                             5642
    fill="#5fa843" />                                                    5643
    <circle cx="220.26" cy="154.42" r="5.29"                             5644
    fill="#59a843" />                                                    5645
    <circle cx="213.22" cy="170.45" r="5.00"                             5646
    fill="#53a843" />                                                    5647
    <circle cx="203.65" cy="185.11" r="4.77"                             5648
    fill="#4da843" />                                                    5649
    <circle cx="191.79" cy="197.99" r="4.63"                             5650
```

```
    fill="#47a843"/>                                                    5651
    <circle cx="177.98" cy="208.74" r="4.59"                            5652
    fill="#43a846"/>                                                    5653
    <circle cx="162.58" cy="217.07" r="4.67"                            5654
    fill="#43a84c"/>                                                    5655
    <circle cx="146.02" cy="222.76" r="4.85"                            5656
    fill="#43a852"/>                                                    5657
    <circle cx="128.75" cy="225.64" r="5.10"                            5658
    fill="#43a858"/>                                                    5659
    <circle cx="111.25" cy="225.64" r="5.42"                            5660
    fill="#43a85e"/>                                                    5661
    <circle cx="93.98" cy="222.76" r="5.75" fill="#43a864"/>            5662
    <circle cx="77.42" cy="217.07" r="6.06" fill="#43a86a"/>            5663
    <circle cx="62.02" cy="208.74" r="6.33" fill="#43a870"/>            5664
    <circle cx="48.21" cy="197.99" r="6.52" fill="#43a876"/>            5665
    <circle cx="36.35" cy="185.11" r="6.60" fill="#43a87c"/>            5666
    <circle cx="26.78" cy="170.45" r="6.58" fill="#43a882"/>            5667
    <circle cx="19.74" cy="154.42" r="6.45" fill="#43a888"/>            5668
    <circle cx="15.45" cy="137.45" r="6.23" fill="#43a88e"/>            5669
    <circle cx="14.00" cy="120.00" r="5.94" fill="#43a894"/>            5670
    <circle cx="15.45" cy="102.55" r="5.62" fill="#43a88e"/>            5671
    <circle cx="19.74" cy="85.58" r="5.29" fill="#43a888"/>             5672
    <circle cx="26.78" cy="69.55" r="5.00" fill="#43a882"/>             5673
    <circle cx="36.35" cy="54.89" r="4.77" fill="#43a87c"/>             5674
    <circle cx="48.21" cy="42.01" r="4.63" fill="#43a876"/>             5675
    <circle cx="62.02" cy="31.26" r="4.59" fill="#43a870"/>             5676
    <circle cx="77.42" cy="22.93" r="4.67" fill="#43a86a"/>             5677
    <circle cx="93.98" cy="17.24" r="4.85" fill="#43a864"/>             5678
    <circle cx="111.25" cy="14.36" r="5.10" fill="#43a85e"/>            5679
    <circle cx="128.75" cy="14.36" r="5.42" fill="#43a858"/>            5680
    <circle cx="146.02" cy="17.24" r="5.75" fill="#43a852"/>            5681
    <circle cx="162.58" cy="22.93" r="6.06" fill="#43a84c"/>            5682
    <circle cx="177.98" cy="31.26" r="6.33" fill="#43a846"/>            5683
    <circle cx="191.79" cy="42.01" r="6.52" fill="#47a843"/>            5684
    <circle cx="203.65" cy="54.89" r="6.60" fill="#4da843"/>            5685
    <circle cx="213.22" cy="69.55" r="6.58" fill="#53a843"/>            5686
    <circle cx="220.26" cy="85.58" r="6.45" fill="#59a843"/>            5687
    <circle cx="224.55" cy="102.55" r="6.23"                            5688
    fill="#5fa843"/>                                                    5689
  </g>                                                                  5690
  <g class="lp-badge__mark">                                            5691
    <circle cx="120" cy="120" r="27" fill="#14432f"/>                   5692
    <g transform="translate(120 120) scale(1.5)                         5693
    translate(-12 -12)" fill="none" stroke="#ffffff"                    5694
    stroke-width="1.9" stroke-linecap="round"                           5695
    stroke-linejoin="round">                                            5696
      <path d="m3 20 6.5-11L14 17l2.5-4L21 20z"/>                       5697
      <circle cx="17" cy="6" r="2.2"/>                                  5698
    </g>                                                                5699
  </g>                                                                  5700
```

```
</svg>
# ==== app/templates/error.html ====
{% extends "base.html" %}
{% block title %}{{ code }} · {{ message }}{% endblock %}
{% block content %}
<div class="card" style="max-width:560px;">
  <h2>{{ code }} · {{ message }}</h2>
  <p class="muted">
    请返回上一页重试；若确认该操作应当可用，请检查当前账号的角色权限。
  </p>
  <p class="actions" style="margin-top:16px;">
    <a class="btn" href="/">返回首页</a>
  </p>
</div>
{% endblock %}
# ==== app/templates/auth/login.html ====
{% extends "base.html" %}
{% from "_macros.html" import icon %}
{% block title %}登录 · 历史遗留废弃矿山自然恢复适宜性评价系统{% endblock %}
{% block content %}
<style>
  /* ================== 登录页（仿 Apple 账户页） ================== */
    纯白底、居中、大量留白；色彩只用在徽标与按钮上。
    首屏放徽标、标题、一句话说明与登录按钮；系统介绍在下方，
    随滚动逐条弹入、往回滚时弹回。 */
  .lp { color: #1d1d1f; }
  /* ---------- 首屏 ---------- */
  .lp-hero {
    display: flex;
    flex-direction: column;
    align-items: center;
    justify-content: center;
    text-align: center;
    min-height: 100dvh;
    padding: 56px 20px 72px;
  }
  /* 徽标：滚动时各圈向四周散开并淡出。
     --p 为 0~1 的滚动进度，由页面脚本写入；
     每圈自带 --k（散开倍数，由内向外递减），使各圈逐渐拉开距离 */
  .lp-badge {
    width: 212px;
    height: 212px;
    margin-bottom: 36px;
    --p: 0;
    animation: lp-in .7s ease both;
  }
  .lp-badge__svg {
    display: block;
    width: 100%;
    height: 100%;
```

```
    overflow: visible;                    /* 散开时圆点不被画布裁掉 */
    opacity: calc(1 - var(--p) * .94);
    filter: blur(calc(var(--p) * 2.6px));
  }
  .lp-ring {
    transform-box: fill-box;
    transform-origin: center;
    transform: scale(calc(1 + (var(--k) - 1) * var(--p)));
  }
  .lp-badge__mark {
    transform-box: fill-box;
    transform-origin: center;
    transform: scale(calc(1 + .18 * var(--p)));
  }
  .lp-title {
    margin: 0 0 18px;
    font-size: clamp(28px, 4.2vw, 42px);
    font-weight: 700;
    line-height: 1.3;
    letter-spacing: -.4px;
    animation: lp-in .7s ease .06s both;
  }
  .lp-cta {
    padding: 13px 52px;
    font-family: inherit;
    font-size: 17px;
    font-weight: 500;
    color: #fff;
    background: var(--accent);
    border: none;
    border-radius: 980px;
    cursor: pointer;
    animation: lp-in .7s ease .18s both;
    transition: background-color .2s ease, box-shadow .2s
    ease,
                transform .2s ease;
  }
  .lp-cta:hover {
    background: var(--accent-hover);
    box-shadow: 0 10px 26px rgba(31, 107, 69, .26);
  }
  .lp-cta:active { transform: translateY(1px); box-shadow:
    none; }
  .lp-cta:focus-visible { outline: 3px solid rgba(31, 107,
    69, .40); outline-offset: 3px; }
  .lp-more {
    margin-top: 38px;
    font-size: 13px;
    color: #86868b;
    text-decoration: none;
```

```
    animation: lp-in .7s ease .24s both;
  }
  .lp-more:hover { color: var(--accent); }
  @keyframes lp-in {
    from { opacity: 0; transform: translateY(12px); }
    to   { opacity: 1; transform: none; }
  }
  /* ---------- 滚动进场 ----------
     进入视口时按方向弹入，离开视口时弹回，可反复。
     交错延迟只加在「进入」的规则上，退出时不延迟，否则会显得拖沓。 */
  .lp-reveal {
    opacity: 0;
    transition: opacity .62s ease,
                transform .62s cubic-bezier(.16, .84, .34,
    1),
                border-color .62s ease, background-color
    .62s ease;
    transition-delay: 0s;
    will-change: opacity, transform;
  }
  .lp-reveal[data-dir="left"]  { transform:
    translate3d(-48px, 0, 0); }
  .lp-reveal[data-dir="right"] { transform:
    translate3d(48px, 0, 0); }
  .lp-reveal[data-dir="up"]    { transform: translate3d(0,
    36px, 0); }
  .lp-reveal.is-in {
    opacity: 1;
    transform: none;
    transition-delay: calc(var(--i, 0) * 85ms);
  }
  /* 标题颜色由淡到深，强化「浮现」的层次感 */
  .lp-reveal h2,
  .lp-reveal h3,
  .lp-reveal .lp-mod__name { color: #c2d2c8; transition:
    color .7s ease; }
  .lp-reveal.is-in h3,
  .lp-reveal.is-in .lp-mod__name { color: var(--accent); }
  .lp-reveal.is-in h2 { color: #1d1d1f; }
  /* ---------- 系统介绍 ---------- */
  .lp-intro {
    max-width: 980px;
    margin: 0 auto;
    padding: 20px 24px 72px;
  }
  .lp-intro__title {
    margin: 0 0 26px;
    font-size: clamp(22px, 2.6vw, 30px);
    text-align: center;
    letter-spacing: -.3px;
```

```
}
.lp-points {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(240px, 1fr));
  gap: 16px;
  margin-bottom: 46px;
}
.lp-point {
  padding: 24px 22px;
  background: #f5f5f7;
  border-radius: 18px;
}
.lp-point__icon {
  display: grid;
  place-items: center;
  width: 38px;
  height: 38px;
  margin-bottom: 14px;
  border-radius: 11px;
  background: #fff;
  color: var(--accent);
}
.lp-point h3 { margin: 0 0 8px; font-size: 16px; }
.lp-point p {
  margin: 0;
  font-size: 13.5px;
  line-height: 1.85;
  color: #6e6e73;
  text-wrap: pretty;
}
.lp-sub-title { margin: 0 0 20px; font-size: 17px; }
.lp-mods {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(258px, 1fr));
  gap: 12px;
  margin-bottom: 46px;
  align-items: start;
}
/*  模块卡：默认只显示编号与名称，鼠标悬停或键盘聚焦时才展开职责说明  */
.lp-mod {
  position: relative;
  padding: 17px 18px 17px 48px;
  background: #fff;
  border: 1px solid #e6e6e8;
  border-radius: 14px;
  outline: none;
  transition: border-color .3s ease, box-shadow .3s ease;
}
```

```
.lp-mod:hover,
.lp-mod:focus-within,
.lp-mod:focus {
  border-color: rgba(31, 107, 69, .42);
  box-shadow: 0 8px 22px rgba(19, 60, 43, .08);
}
.lp-mod__no {
  position: absolute;
  left: 17px;
  top: 17px;
  font-size: 13px;
  font-weight: 700;
  color: #c8d6cd;
  font-variant-numeric: tabular-nums;
}
.lp-mod__name {
  display: block;
  font-size: 14px;
  font-weight: 600;
}
/* 用 grid-template-rows 0fr→1fr 展开，高度过渡自然 */
.lp-mod__body {
  display: grid;
  grid-template-rows: 0fr;
  transition: grid-template-rows .34s ease, margin-top
  .34s ease;
}
.lp-mod:hover .lp-mod__body,
.lp-mod:focus-within .lp-mod__body {
  grid-template-rows: 1fr;
  margin-top: 7px;
}
.lp-mod__desc {
  overflow: hidden;
  margin: 0;
  font-size: 12.5px;
  line-height: 1.7;
  color: #6e6e73;
  opacity: 0;
  transition: opacity .3s ease;
  text-wrap: pretty;
}
.lp-mod:hover .lp-mod__desc,
.lp-mod:focus-within .lp-mod__desc { opacity: 1; }
/* 触摸设备没有悬停，直接把说明摊开 */
@media (hover: none) {
  .lp-mod__body { grid-template-rows: 1fr; margin-top:
  7px; }
  .lp-mod__desc { opacity: 1; }
}
```

```
.lp-foot {
  margin: 0;
  padding-top: 26px;
  border-top: 1px solid #e6e6e8;
  text-align: center;
  font-size: 12.5px;
  color: #86868b;
}
/* ---------- 登录面板：点击后渐变浮现 ---------- */
.lp-auth {
  position: fixed;
  inset: 0;
  z-index: 60;
  display: grid;
  place-items: center;
  padding: 24px;
  opacity: 0;
  visibility: hidden;
  transition: opacity .42s ease, visibility 0s linear .42s;
}
.lp-auth.is-open {
  opacity: 1;
  visibility: visible;
  transition: opacity .42s ease, visibility 0s;
}
/* 遮罩用绿色径向渐变，形成「渐变渗出来」的观感 */
.lp-auth__scrim {
  position: absolute;
  inset: 0;
  background:
    radial-gradient(1000px 640px at 50% 0%,
                    rgba(31, 107, 69, .22), rgba(31, 107, 69, 0) 70%),
    linear-gradient(180deg, rgba(255, 255, 255, .84),
                    rgba(246, 250, 247, .95));
  backdrop-filter: blur(16px) saturate(1.25);
  -webkit-backdrop-filter: blur(16px) saturate(1.25);
}
.lp-auth__panel {
  position: relative;
  width: 100%;
  max-width: 396px;
  padding: 36px 36px 26px;
  background: #fff;
  border-radius: 20px;
  box-shadow: 0 28px 80px rgba(14, 46, 32, .18);
  opacity: 0;
  transform: translateY(18px) scale(.975);
  transition: transform .55s cubic-bezier(.16, .84, .34,
```

```
    1) .05s,                                                          6001
                opacity .55s ease .05s;                               6002
  }                                                                   6003
  .lp-auth.is-open .lp-auth__panel { opacity: 1; transform:           6004
    none; }                                                           6005
  .lp-auth__close {                                                   6006
    position: absolute;                                               6007
    top: 14px;                                                        6008
    right: 14px;                                                      6009
    display: grid;                                                    6010
    place-items: center;                                              6011
    width: 30px;                                                      6012
    height: 30px;                                                     6013
    color: #86868b;                                                   6014
    background: transparent;                                          6015
    border: none;                                                     6016
    border-radius: 50%;                                               6017
    cursor: pointer;                                                  6018
    transition: background-color .18s ease, color .18s ease;          6019
  }                                                                   6020
  .lp-auth__close:hover { background: #f5f5f7; color:                 6021
    #1d1d1f; }                                                        6022
  .lp-auth__title {                                                   6023
    margin: 0 0 6px;                                                  6024
    text-align: center;                                               6025
    font-size: 24px;                                                  6026
    font-weight: 700;                                                 6027
    letter-spacing: -.2px;                                            6028
  }                                                                   6029
  .lp-auth__sub {                                                     6030
    margin: 0 0 26px;                                                 6031
    text-align: center;                                               6032
    font-size: 13px;                                                  6033
    color: #86868b;                                                   6034
  }                                                                   6035
  .lp-field { margin-bottom: 15px; }                                  6036
  .lp-field label {                                                   6037
    display: block;                                                   6038
    margin-bottom: 7px;                                               6039
    font-size: 13px;                                                  6040
    color: #6e6e73;                                                   6041
  }                                                                   6042
  .lp-field input {                                                   6043
    width: 100%;                                                      6044
    padding: 13px 16px;                                               6045
    font-family: inherit;                                             6046
    font-size: 16px;              /* 不小于 16px，避免手机端聚焦时自动放大 */  6047
    color: #1d1d1f;                                                   6048
    background: #f5f5f7;                                              6049
    border: 1px solid transparent;                                    6050
```

```
    border-radius: 12px;                                                    6051
    outline: none;                                                          6052
    transition: background-color .18s ease, border-color                    6053
    .18s ease,                                                              6054
                box-shadow .18s ease;                                       6055
  }                                                                         6056
  .lp-field input::placeholder { color: #a1a1a6; }                          6057
  .lp-field input:focus {                                                   6058
    background: #fff;                                                       6059
    border-color: rgba(31, 107, 69, .55);                                   6060
    box-shadow: 0 0 0 3px rgba(31, 107, 69, .14);                           6061
  }                                                                         6062
  .lp-remember {                                                            6063
    display: flex;                                                          6064
    align-items: center;                                                    6065
    gap: 8px;                                                               6066
    margin: 4px 0 20px;                                                     6067
    font-size: 13px;                                                        6068
    color: #6e6e73;                                                         6069
    cursor: pointer;                                                        6070
  }                                                                         6071
  .lp-remember input {                                                      6072
    width: 16px;                                                            6073
    height: 16px;                                                           6074
    margin: 0;                                                              6075
    accent-color: var(--accent);                                            6076
    cursor: pointer;                                                        6077
  }                                                                         6078
  .lp-submit {                                                              6079
    width: 100%;                                                            6080
    padding: 13px 20px;                                                     6081
    font-family: inherit;                                                   6082
    font-size: 16px;                                                        6083
    font-weight: 500;                                                       6084
    color: #fff;                                                            6085
    background: var(--accent);                                              6086
    border: none;                                                           6087
    border-radius: 980px;                                                   6088
    cursor: pointer;                                                        6089
    transition: background-color .2s ease, box-shadow .2s                   6090
    ease,                                                                   6091
                transform .2s ease;                                         6092
  }                                                                         6093
  .lp-submit:hover {                                                        6094
    background: var(--accent-hover);                                        6095
    box-shadow: 0 10px 24px rgba(31, 107, 69, .26);                         6096
  }                                                                         6097
  .lp-submit:active { transform: translateY(1px);                           6098
    box-shadow: none; }                                                     6099
  .lp-submit:focus-visible { outline: 3px solid rgba(31,                    6100
```

```
      107, 69, .40); outline-offset: 3px; }
   .lp-auth__hint {
      margin: 18px 0 0;
      text-align: center;
      font-size: 12.5px;
      color: #a1a1a6;
   }
   /* 系统设置里要求减少动态效果时，去掉位移、缩放与滚动特效 */
   @media (prefers-reduced-motion: reduce) {
      .lp-badge, .lp-title, .lp-cta, .lp-more { animation:
      none; }
      .lp-badge__svg { opacity: 1; filter: none; }
      .lp-ring, .lp-badge__mark { transform: none; }
      .lp-reveal { opacity: 1; transform: none !important; }
      .lp-reveal h2, .lp-reveal h3, .lp-reveal .lp-mod__name
      { color: inherit; }
      .lp-reveal.is-in h2 { color: #1d1d1f; }
      .lp-reveal.is-in h3, .lp-reveal.is-in .lp-mod__name {
      color: var(--accent); }
      .lp-auth, .lp-auth__panel { transition: opacity .2s
      linear; }
      .lp-auth__panel { transform: none; }
   }
   @media (max-width: 560px) {
      .lp-hero { padding: 36px 18px 48px; min-height: auto; }
      .lp-badge { width: 160px; height: 160px; margin-bottom:
      26px; }
      .lp-auth__panel { padding: 30px 24px 22px; }
   }
</style>
<div class="lp">
   {# ---------------- 首屏：徽标 + 标题 + 说明 + 登录入口
      ---------------- #}
   <section class="lp-hero">
      <div class="lp-badge" id="lp-badge" aria-hidden="true">
         {% include "_login_badge.svg" %}
      </div>
      <h1 class="lp-title">历史遗留废弃矿山<br>自然恢复适宜性评价系统</h1>
      <button type="button" class="lp-cta" id="lp-open"
              aria-expanded="false" aria-controls="lp-auth">登录
      </button>
      <a class="lp-more" href="#intro">
         向下滑动，了解系统 <span aria-hidden="true">↓</span>
      </a>
   </section>
   {# ---------------- 系统介绍：随滚动逐条弹入 ---------------- #}
   <section class="lp-intro" id="intro">
      <h2 class="lp-intro__title lp-reveal"
      data-dir="up">一套把评价流程固定下来的工具</h2>
      <div class="lp-points">
```

```
    {% set pdirs = ['left', 'up', 'right'] %}
    {% set points = [
      ('sliders', '方法可溯',
       '权重由层次分析法算出并做一致性检验，指标标准化、分单元赋分与面积加权合成的每一步都留有记录，可逐层回看。'),
      ('shield', '流程可控',
       '政策、区位约束、土壤污染、地质灾害、耕地属性五项限制性条件一票否决；未通过即终止评价，直接给出生态重建或人工辅助的结论。'),
      ('file', '结果可复算',
       '评价结果与逐层得分明细一并留存，报告取自已保存的评价记录，事后修改数据不会让报告与结论对不上。')
    ] %}
    {% for ic, title, text in points %}
    <div class="lp-point lp-reveal" data-dir="{{ pdirs[loop.index0 % 3] }}"
         style="--i: {{ loop.index0 }}">
      <span class="lp-point__icon">{{ icon(ic, 20) }}</span>
      <h3>{{ title }}</h3>
      <p>{{ text }}</p>
    </div>
    {% endfor %}
  </div>

  <h3 class="lp-sub-title lp-reveal" data-dir="up">九个功能模块</h3>
  <div class="lp-mods">
    {% set mdirs = ['left', 'up', 'right'] %}
    {% for no, name, desc in modules %}
    <div class="lp-mod lp-reveal" data-dir="{{ mdirs[loop.index0 % 3] }}"
         style="--i: {{ loop.index0 % 3 }}" tabindex="0">
      <span class="lp-mod__no">{{ no }}</span>
      <span class="lp-mod__name">{{ name }}</span>
      <div class="lp-mod__body"><p class="lp-mod__desc">{{ desc }}</p></div>
    </div>
    {% endfor %}
  </div>

  <p class="lp-foot lp-reveal" data-dir="up">
    历史遗留废弃矿山自然恢复适宜性评价系统 Ｖ1．0 ·
    运行于浏览器，数据保存在本机 ＳＱＬite 数据库
  </p>
</section>

{# ---------------- 登录面板：点击「登录」后渐变浮现 ---------------- #}
<div class="lp-auth" id="lp-auth" aria-hidden="true">
  <div class="lp-auth__scrim" data-lp-close></div>
  <div class="lp-auth__panel" role="dialog"
       aria-modal="true"
       aria-labelledby="lp-auth-title">
    <button type="button" class="lp-auth__close"
            data-lp-close
```

```
                aria-label="关闭登录框">                                    6201
          <svg width="16" height="16" viewBox="0 0 24 24"                   6202
      fill="none"                                                           6203
              stroke="currentColor" stroke-width="2"                        6204
      stroke-linecap="round"                                                6205
              aria-hidden="true"><path d="M6 6l12 12M18 6L6                 6206
      18"/></svg>                                                           6207
        </button>                                                           6208
        <h2 class="lp-auth__title" id="lp-auth-title">登录</h2>             6209
        <p class="lp-auth__sub">使用系统账号登录</p>                        6210
        <form method="post" novalidate>                                     6211
          {{ form.hidden_tag() }}                                           6212
          <div class="lp-field">                                            6213
            <label for="{{ form.username.id }}">用户名</label>              6214
            {{ form.username(id=form.username.id,                           6215
      autocomplete="username",                                              6216
                             placeholder="请输入用户名") }}                 6217
          </div>                                                            6218
          <div class="lp-field">                                            6219
            <label for="{{ form.password.id }}">密码</label>                6220
            {{ form.password(id=form.password.id,                           6221
      autocomplete="current-password",                                      6222
                             placeholder="请输入密码") }}                   6223
          </div>                                                            6224
          <label class="lp-remember" for="{{ form.remember.id               6225
      }}">                                                                  6226
            {{ form.remember(id=form.remember.id) }}                        6227
            <span>记住我（下次自动登录，一个月内有效）</span>               6228
          </label>                                                          6229
          {{ form.submit(class_="lp-submit") }}                             6230
        </form>                                                             6231
        <p class="lp-auth__hint">演示账号：admin 密码：admin123</p>        6232
      </div>                                                                6233
    </div>                                                                  6234
</div>                                                                      6235
{% endblock %}                                                              6236
{% block scripts %}                                                         6237
<script>                                                                    6238
(function () {                                                              6239
    var reduce = window.matchMedia &&                                       6240
      window.matchMedia('(prefers-reduced-motion:                           6241
      reduce)').matches;                                                    6242
    /* ---------- 登录面板 ---------- */                                    6243
    var btn = document.getElementById('lp-open');                           6244
    var auth = document.getElementById('lp-auth');                          6245
    var lastFocus = null;                                                   6246
    function openPanel() {                                                  6247
      lastFocus = document.activeElement;                                   6248
      auth.classList.add('is-open');                                        6249
      auth.setAttribute('aria-hidden', 'false');                            6250
```

```
    btn.setAttribute('aria-expanded', 'true');
    document.body.style.overflow = 'hidden';
    // 记到地址栏，直接访问 /login#login 也能展开登录框
    if (location.hash !== '#login') {
      history.replaceState(null, '', location.pathname +
    '#login');
    }
    var first = auth.querySelector('input');
    if (first) window.setTimeout(function () {
    first.focus(); }, 300);
  }
  function closePanel() {
    auth.classList.remove('is-open');
    auth.setAttribute('aria-hidden', 'true');
    btn.setAttribute('aria-expanded', 'false');
    document.body.style.overflow = '';
    if (location.hash === '#login') {
      history.replaceState(null, '', location.pathname);
    }
    if (lastFocus && lastFocus.focus) lastFocus.focus();
  }
  if (btn && auth) {
    btn.addEventListener('click', openPanel);
    Array.prototype.forEach.call(
      auth.querySelectorAll('[data-lp-close]'),
      function (el) { el.addEventListener('click',
    closePanel); }
    );
    document.addEventListener('keydown', function (e) {
      if (e.key === 'Escape' && auth.classList.contains('is-
    open')) closePanel();
    });
    // 登录失败回到本页、或直接带 #login 访问时，直接展开登录框
    if (location.hash === '#login' ||
    document.querySelector('.flash.error')) {
      openPanel();
    }
  }
  /* ---------- 内容进场 + 徽标散开：同一套滚动计算 ---------- */
     原先用 IntersectionObserver，实测在滚动到位置后回调并未生效，
     内容始终停在 opacity:0。改为直接按元素与视口的相对位置判断，
     与徽标用同一次滚动回调，行为可预期、也便于排查。 */
  var items = document.querySelectorAll('.lp-reveal');
  var badge = document.getElementById('lp-badge');
  var ticking = false;
  function applyScroll() {
    ticking = false;
    var vh = window.innerHeight || 800;
    /* 徽标：滚动前约三分之一屏内完成散开淡出 */
    if (badge) {
```

```
        var range = vh * 0.34;                                                  6301
        var p = range > 0 ? window.scrollY / range : 0;                         6302
        badge.style.setProperty('--p', Math.min(1,                              6303
      Math.max(0, p)).toFixed(3));                                              6304
      }                                                                         6305
      /* 内容：先批量测量，再改类名，避免读写交替触发反复重排 */                  6306
      if (reduce || !items.length) return;                                      6307
      var box = [];                                                             6308
      for (var i = 0; i < items.length; i++) {                                  6309
        box.push(items[i].getBoundingClientRect());                             6310
      }                                                                         6311
      for (var j = 0; j < items.length; j++) {                                  6312
        var r = box[j];                                                         6313
        // 上沿进入视口下部、下沿尚未越出视口上部 → 视为在视野内                6314
        var shown = r.top < vh * 0.88 && r.bottom > vh * 0.06;                  6315
        if (shown) items[j].classList.add('is-in');                             6316
        else items[j].classList.remove('is-in');                                6317
      }                                                                         6318
    }                                                                           6319
    function onScroll() {                                                       6320
      if (!ticking) {                                                           6321
        ticking = true;                                                         6322
        window.requestAnimationFrame(applyScroll);                              6323
      }                                                                         6324
    }                                                                           6325
    // 减少动态效果时不做滚动特效，内容直接显示                                6326
    if (reduce) {                                                               6327
      Array.prototype.forEach.call(items, function (el) {                       6328
        el.classList.add('is-in');                                              6329
      });                                                                       6330
      if (badge) badge.style.setProperty('--p', '0');                           6331
    } else {                                                                    6332
      window.addEventListener('scroll', onScroll, { passive:                    6333
      true });                                                                  6334
      window.addEventListener('resize', onScroll);                              6335
      applyScroll();                                                            6336
    }                                                                           6337
})();                                                                           6338
</script>                                                                       6339
{% endblock %}                                                                  6340
# ==== app/templates/auth/home.html ====                                        6341
{% extends "base.html" %}                                                       6342
{% from "_macros.html" import icon %}                                           6343
{% block title %}工作台 · 历史遗留废弃矿山自然恢复适宜性评价系统{% endblock %}  6344
{% block content %}                                                             6345
{# ---------- 页首横幅 ---------- #}                                           6346
<section class="hero">                                                          6347
  <div class="hero__body">                                                      6348
    <h1 class="hero__title">历史遗留废弃矿山自然恢复适宜性评价系统</h1>         6349
    <p class="hero__desc">                                                      6350
```

```
        以论文《历史遗留废弃矿山自然恢复适宜性评价指标体系建设及典型案例》
        建立的评价方法为算法内核。下方按论文　４．１　节的评价流程顺序排列，
        系统会在每一步检查前置条件，前置步骤未完成时不会放行后续计算。
      </p>
      <div class="hero__actions">
        <a class="hero-btn hero-btn--ghost"
      href="/indicator/import">
          {{ icon("upload", 16) }}批量导入
        </a>
        <a class="hero-btn hero-btn--solid"
      href="/project/add">
          {{ icon("plus", 16) }}新建评价项目
        </a>
      </div>
    </div>
</section>
{# ---------- 数据概览 ---------- #}
<section class="stat-cards">
    <a class="stat-card" href="/project/">
      <span class="stat-card__label">评价项目</span>
      <span class="stat-card__value">{{ dash.stats.projects
      }}<span class="stat-card__unit">个</span></span>
      <span class="stat-card__note">
        {%- if projects_without_patch -%}
          其中 {{ projects_without_patch|length }} 个项目尚未建立图斑
        {%- elif dash.progress.patches -%}
          已建立 {{ dash.progress.patches }} 个矿山图斑
        {%- else -%}
          尚未建立矿山图斑
        {%- endif -%}
      </span>
    </a>
    <a class="stat-card" href="/project/">
      <span class="stat-card__label">矿山图斑</span>
      <span class="stat-card__value">{{ dash.stats.patches
      }}<span class="stat-card__unit">个</span></span>
      <span class="stat-card__note">
        {% if dash.progress.ready %}其中 {{ dash.progress.ready
      }} 个具备评价条件{% else %}暂无具备评价条件的图斑{% endif %}
      </span>
    </a>
    <a class="stat-card" href="/unit/">
      <span class="stat-card__label">微地质单元</span>
      <span class="stat-card__value">{{ dash.stats.unit_types
      }}<span class="stat-card__unit">类</span></span>
      <span class="stat-card__note">
        {% if dash.progress.units %}共划分 {{
      dash.progress.units }} 个单元{% else %}尚未划分单元{% endif %}
      </span>
    </a>
```

```
  <a class="stat-card" href="/standard/">
    <span class="stat-card__label">标准与政策</span>
    <span class="stat-card__value">{{ dash.stats.standards
    }}<span class="stat-card__unit">项</span></span>
    <span class="stat-card__note">地方标准与政策文件</span>
  </a>
</section>
{# ---------- 评价流程 ---------- #}
<section class="section">
  <div class="section__head">
    <h2 class="section__title">评价流程</h2>
    <p class="section__desc">
      按论文 4.1 节顺序排列，前置步骤未完成时不会执行后续计算
    </p>
  </div>
  <div class="step-grid">
    {% for s in dash.steps %}
    {% if s.available %}
    <a class="step-card" href="{{ s.href }}">
      <span class="step-card__no">{{ s.no }}</span>
      <span class="step-card__icon">{{ icon(s.icon, 18)
    }}</span>
      <span class="step-card__title">{{ s.title }}</span>
      <span class="step-card__desc">{{ s.desc }}</span>
      {% if s.pending %}
      <span class="step-card__badge">待处理 {{ s.pending
    }}</span>
      {% endif %}
    </a>
    {% else %}
    <div class="step-card step-card--locked"
    aria-disabled="true">
      <span class="step-card__no">{{ s.no }}</span>
      <span class="step-card__icon">{{ icon(s.icon, 18)
    }}</span>
      <span class="step-card__title">{{ s.title }}</span>
      <span class="step-card__desc">{{ s.desc }}</span>
      <span class="step-card__lock">{{ s.note }}</span>
    </div>
    {% endif %}
    {% endfor %}
  </div>
</section>
{# ---------- 系统管理 ---------- #}
<section class="section">
  <div class="section__head">
    <h2 class="section__title">系统管理</h2>
  </div>
  <a class="row-card" href="/users">
    <span class="row-card__icon">{{ icon("users", 19)
```

```
      }}</span>                                                                    6451
      <span class="row-card__main">                                                6452
        <span class="row-card__title">用户与操作日志</span>                          6453
        <span class="row-card__desc">用户维护与角色权限；全部写操作自动留痕，可查询追溯 6454
      </span>                                                                      6455
      </span>                                                                      6456
      <span class="row-card__arrow">{{ icon("chevron", 17)                         6457
      }}</span>                                                                    6458
   </a>                                                                            6459
</section>                                                                        6460
{% endblock %}                                                                    6461
# ==== app/templates/auth/users.html ====                                         6462
{% extends "base.html" %}                                                         6463
{% block title %}用户与权限管理{% endblock %}                                     6464
{% block content %}                                                               6465
<div class="card">                                                                6466
   <h2>用户管理</h2>                                                              6467
   <p class="actions" style="margin-bottom:12px;">                                6468
      {% if current_user.role == "admin" %}                                       6469
      <a class="btn btn--sm" href="/users/add">新增用户</a>                       6470
      {% endif %}                                                                 6471
      <a class="btn btn--sm btn--ghost"                                           6472
      href="/logs">查看操作日志</a>                                               6473
   </p>                                                                           6474
   {% if current_user.role != "admin" %}                                          6475
   <p class="muted">                                                              6476
      当前角色为「{{ role_labels.get(current_user.role,                           6477
      current_user.role) }}」，                                                   6478
      仅管理员可新增、编辑或删除用户。                                            6479
   </p>                                                                           6480
   {% endif %}                                                                    6481
   <table>                                                                        6482
      <caption class="muted">共 {{ users|length }}                                6483
      个用户</caption>                                                            6484
      <thead>                                                                     6485
         <tr>                                                                     6486
            <th scope="col">ID</th>                                               6487
            <th scope="col">用户名</th>                                           6488
            <th scope="col">角色</th>                                             6489
            <th scope="col">状态</th>                                             6490
            {% if current_user.role == "admin" %}<th                              6491
      scope="col">操作</th>{% endif %}                                            6492
         </tr>                                                                    6493
      </thead>                                                                    6494
      <tbody>                                                                     6495
         {% for u in users %}                                                     6496
         <tr>                                                                     6497
            <td class="num">{{ u.id }}</td>                                       6498
            <td>                                                                  6499
               {{ u.username }}                                                   6500
```

```
            {% if u.id == current_user.id %}<span                    6501
    class="muted">（当前登录）</span>{% endif %}                      6502
          </td>                                                      6503
          <td>{{ role_labels.get(u.role, u.role) }}</td>             6504
          <td>                                                       6505
            {% if u.active %}<span class="tag                        6506
    tag--ok">启用</span>                                             6507
            {% else %}<span class="tag tag--muted">停用</span>{%     6508
    endif %}                                                         6509
          </td>                                                      6510
          {% if current_user.role == "admin" %}                      6511
          <td class="nowrap">                                        6512
            <span class="row-actions">                               6513
              <a class="link-btn" href="/users/{{ u.id               6514
    }}/edit">编辑</a>                                                6515
              <form method="post" action="/users/{{ u.id             6516
    }}/delete"                                                       6517
                    onsubmit="return confirm('确定删除用户「{{       6518
    u.username }}」吗？\n该操作无法恢复。');">                       6519
                {{ form.hidden_tag() }}                              6520
                <button type="submit" class="link-btn                6521
    link-btn--danger"                                                6522
                        {% if u.id == current_user.id                6523
    %}disabled                                                       6524
                        title="不能删除当前登录的账号"{% endif       6525
    %}>删除</button>                                                 6526
              </form>                                                6527
            </span>                                                  6528
          </td>                                                      6529
          {% endif %}                                                6530
        </tr>                                                        6531
        {% endfor %}                                                 6532
      </tbody>                                                       6533
    </table>                                                         6534
</div>                                                               6535
{% endblock %}                                                       6536
# ==== app/templates/auth/add_user.html ====                         6537
{% extends "base.html" %}                                            6538
{% block title %}新增用户{% endblock %}                              6539
{% block content %}                                                  6540
<div class="card" style="max-width:560px;">                          6541
  <h2>新增用户</h2>                                                  6542
  <form method="post">                                               6543
    {{ form.hidden_tag() }}                                          6544
    <div class="field">                                              6545
      <label class="field__label" for="username">用户名</label>      6546
      <input id="username" type="text" name="username"               6547
    required                                                         6548
             autocomplete="off" style="width:280px;">                6549
    </div>                                                           6550
```

```
      <div class="field">
        <label class="field__label" for="password">密码</label>
        <input id="password" type="password" name="password"
      required
                autocomplete="new-password"
      style="width:280px;">
      </div>
      <div class="field">
        <label class="field__label" for="role">角色</label>
        <select id="role" name="role">
          <option value="viewer">查看者（仅可查阅）</option>
          <option value="evaluator">评价员（可建项目、录数据、执行评价）</option
      >
          <option value="admin">管理员（可维护用户与基础数据）</option>
        </select>
      </div>
      <p class="actions" style="margin-top:18px;">
        <button type="submit" class="btn">保存</button>
        <a class="btn btn--ghost" href="/users">返回</a>
      </p>
    </form>
</div>
{% endblock %}
# ==== app/templates/auth/edit_user.html ====
{% extends "base.html" %}
{% block title %}编辑用户 · {{ user.username }}{% endblock %}
{% block content %}
<div class="card" style="max-width:560px;">
  <h2>编辑用户 · {{ user.username }}</h2>
  <form method="post">
    {{ form.hidden_tag() }}
    <div class="field">
      <label class="field__label" for="role">角色</label>
      <select id="role" name="role">
        {% for code, label in role_labels.items() %}
        <option value="{{ code }}" {% if user.role == code
    %}selected{% endif %}>{{ label }}</option>
        {% endfor %}
      </select>
    </div>
    <div class="field">
      <label class="checkbox" for="active">
        <input id="active" type="checkbox" name="active"
    value="1"
               {% if user.active %}checked{% endif %}>
        <span>启用该账号</span>
      </label>
      <p class="muted" style="margin:6px 0 0;">
        停用后该账号无法登录，但其历史操作记录会保留。
        系统至少需保留一名启用状态的管理员。
```

```
        </p>                                                                    6601
      </div>                                                                    6602
      <div class="field">                                                       6603
        <label class="field__label" for="password">重置密码（留空则不修改6604
      )</label>                                                                 6605
        <input id="password" type="password" name="password"                    6606
               autocomplete="new-password"                                      6607
      style="width:280px;"                                                      6608
               placeholder="留空表示保持原密码">                                6609
      </div>                                                                    6610
      <p class="actions" style="margin-top:18px;">                              6611
        <button type="submit" class="btn">保存</button>                         6612
        <a class="btn btn--ghost" href="/users">返回</a>                        6613
      </p>                                                                      6614
    </form>                                                                     6615
</div>                                                                          6616
{% endblock %}                                                                  6617
# ==== app/templates/auth/profile.html ====                                     6618
{% extends "base.html" %}                                                       6619
{% block title %}个人设置 · 历史遗留废弃矿山自然恢复适宜性评价系统{% endblock %}     6620
{% block content %}                                                             6621
<style>                                                                         6622
  .profile-grid {                                                               6623
    display: grid;                                                              6624
    grid-template-columns: 260px 1fr;                                           6625
    gap: 18px;                                                                  6626
    align-items: start;                                                         6627
  }                                                                             6628
  @media (max-width: 820px) {                                                   6629
    .profile-grid { grid-template-columns: 1fr; }                               6630
  }                                                                             6631
  /* 头像 */                                                                    6632
  .avatar-box {                                                                 6633
    display: flex;                                                              6634
    flex-direction: column;                                                     6635
    align-items: center;                                                        6636
    text-align: center;                                                         6637
  }                                                                             6638
  .avatar-lg {                                                                  6639
    display: grid;                                                              6640
    place-items: center;                                                        6641
    width: 132px;                                                               6642
    height: 132px;                                                              6643
    margin-bottom: 14px;                                                        6644
    border-radius: 50%;                                                         6645
    background: var(--accent-soft);                                             6646
    color: var(--accent);                                                       6647
    font-size: 48px;                                                            6648
    font-weight: 600;                                                           6649
    object-fit: cover;                                                          6650
```

```
      border: 3px solid #fff;                                                  6651
      box-shadow: 0 4px 16px rgba(19, 60, 43, .12);                            6652
    }                                                                          6653
    .avatar-lg--initial { user-select: none; }                                 6654
    .avatar-name {                                                             6655
      margin: 0 0 2px;                                                         6656
      font-size: 15px;                                                         6657
      font-weight: 600;                                                        6658
    }                                                                          6659
    .avatar-sub {                                                              6660
      margin: 0 0 16px;                                                        6661
      font-size: 12.5px;                                                       6662
      color: var(--ink-soft);                                                  6663
    }                                                                          6664
    .avatar-form { width: 100%; }                                              6665
    .avatar-form .avatar-pick { width: 100%; }                                 6666
    .avatar-filename {                                                         6667
      margin: 8px 0 10px;                                                      6668
      font-size: 12px;                                                         6669
      color: var(--ink-soft);                                                  6670
      word-break: break-all;                                                   6671
    }                                                                          6672
    /* 只读信息 */                                                             6673
    .kv {                                                                      6674
      display: grid;                                                           6675
      grid-template-columns: 96px 1fr;                                         6676
      gap: 6px 12px;                                                           6677
      margin: 0;                                                               6678
      font-size: 13.5px;                                                       6679
    }                                                                          6680
    .kv dt { color: var(--ink-soft); }                                         6681
    .kv dd { margin: 0; }                                                      6682
    .field__error {                                                            6683
      margin: 5px 0 0;                                                         6684
      font-size: 12.5px;                                                       6685
      color: var(--danger);                                                    6686
    }                                                                          6687
</style>                                                                       6688
<div class="profile-grid">                                                     6689
  {# ---------------- 左：头像 ---------------- #}                             6690
  <div class="card">                                                           6691
    <div class="avatar-box">                                                   6692
      {% if current_user.avatar %}                                             6693
      <img class="avatar-lg"                                                   6694
           src="{{ url_for('static', filename=current_user.a                   6695
    vatar) }}"                                                                 6696
           alt="{{ current_user.shown_name }} 的头像">                         6697
      {% else %}                                                               6698
      <span class="avatar-lg avatar-lg--initial"                               6699
    aria-hidden="true">{{ current_user.initial }}</span>                       6700
```

```
        {% endif %}
        <p class="avatar-name">{{ current_user.shown_name }}</p>
        <p class="avatar-sub">
          {{ role_labels.get(current_user.role, current_user.role) }} ·
          {{ current_user.username }}
        </p>
        <form class="avatar-form" method="post" action="/profile/avatar"
              enctype="multipart/form-data">
          {{ avatar_form.hidden_tag() }}
          <label class="btn btn--sm btn--ghost avatar-pick" for="avatar-file">
            选择图片
          </label>
          <input id="avatar-file" type="file" name="avatar" class="visually-hidden"
                 accept=".png,.jpg,.jpeg,.gif,.webp">
          <p class="avatar-filename" id="avatar-filename">未选择任何文件</p>
          <button type="submit" class="btn btn--sm">上传头像</button>
        </form>
        {% if current_user.avatar %}
        <form method="post" action="/profile/avatar/delete" style="margin-top:8px;"
              onsubmit="return confirm('确定移除当前头像吗？');">
          {{ avatar_form.hidden_tag() }}
          <button type="submit" class="btn btn--sm btn--ghost">移除头像</button>
        </form>
        {% endif %}
        <p class="muted" style="margin:12px 0 0;font-size:12px;">
          PNG / JPG / GIF / WebP，不超过 4 MB
        </p>
      </div>
    </div>

    {# ---------- 右：资料 + 密码 ---------- #}
    <div>
      <div class="card">
        <h2>个人资料</h2>
        <form method="post">
          {{ info_form.hidden_tag() }}
          <input type="hidden" name="action" value="info">
          <div class="field">
            <label class="field__label" for="{{ info_form.display_name.id }}">
              显示名称
```

```
        </label>                                                    6751
        {{ info_form.display_name(id=info_form.display_nam          6752
e.id,                                                               6753
                                          placeholder="留空则显示账号名") 6754
}}                                                                  6755
        {% for e in info_form.display_name.errors %}                6756
        <p class="field__error">{{ e }}</p>                         6757
        {% endfor %}                                                6758
      </div>                                                        6759
      <div class="field">                                           6760
        <label class="field__label" for="{{                         6761
info_form.title.id }}">职务</label>                                 6762
        {{ info_form.title(id=info_form.title.id,                   6763
placeholder="如：评价员、项目负责人") }}                            6764
        {% for e in info_form.title.errors %}                       6765
        <p class="field__error">{{ e }}</p>                         6766
        {% endfor %}                                                6767
      </div>                                                        6768
      <div class="field">                                           6769
        <label class="field__label" for="{{                         6770
info_form.email.id }}">邮箱</label>                                 6771
        {{ info_form.email(id=info_form.email.id) }}                6772
        {% for e in info_form.email.errors %}                       6773
        <p class="field__error">{{ e }}</p>                         6774
        {% endfor %}                                                6775
      </div>                                                        6776
      <div class="field">                                           6777
        <label class="field__label" for="{{                         6778
info_form.phone.id }}">手机号</label>                               6779
        {{ info_form.phone(id=info_form.phone.id) }}                6780
        {% for e in info_form.phone.errors %}                       6781
        <p class="field__error">{{ e }}</p>                         6782
        {% endfor %}                                                6783
      </div>                                                        6784
      <p class="actions" style="margin-top:18px;">                  6785
        {{ info_form.submit_info(class_="btn") }}                   6786
      </p>                                                          6787
    </form>                                                         6788
    <hr style="margin:20px 0;border:none;border-top:1px             6789
solid var(--line);">                                                6790
    <h3 style="margin:0 0 10px;font-size:15px;">账号信息</h3>       6791
    <dl class="kv">                                                 6792
      <dt>登录账号</dt><dd class="mono">{{                          6793
current_user.username }}</dd>                                       6794
      <dt>角色</dt>                                                 6795
      <dd>{{ role_labels.get(current_user.role,                     6796
current_user.role) }}</dd>                                          6797
      <dt>状态</dt>                                                 6798
      <dd>                                                          6799
        {% if current_user.active %}                                6800
```

```
          <span class="tag tag--ok">启用</span>                    6801
          {% else %}                                               6802
          <span class="tag tag--warn">停用</span>                  6803
          {% endif %}                                              6804
        </dd>                                                      6805
        <dt>创建时间</dt>                                          6806
        <dd class="mono">                                          6807
          {{ current_user.created_at.strftime("%Y-%m-%d            6808
%H:%M") if current_user.created_at else "—" }}                     6809
        </dd>                                                      6810
      </dl>                                                        6811
</div>                                                             6812
<div class="card">                                                 6813
  <h2>修改密码</h2>                                                6814
  <form method="post" style="max-width:420px;">                    6815
    {{ pwd_form.hidden_tag() }}                                    6816
    <input type="hidden" name="action" value="password">           6817
    <div class="field">                                            6818
      <label class="field__label" for="{{                          6819
pwd_form.old_password.id }}">                                      6820
        当前密码                                                   6821
      </label>                                                     6822
      {{ pwd_form.old_password(id=pwd_form.old_password.           6823
id,                                                                6824
                               autocomplete="current-pas           6825
sword") }}                                                         6826
      {% for e in pwd_form.old_password.errors %}                  6827
      <p class="field__error">{{ e }}</p>                          6828
      {% endfor %}                                                 6829
    </div>                                                         6830
    <div class="field">                                            6831
      <label class="field__label" for="{{                          6832
pwd_form.new_password.id }}">                                      6833
        新密码                                                     6834
      </label>                                                     6835
      {{ pwd_form.new_password(id=pwd_form.new_password.           6836
id,                                                                6837
                               autocomplete="new-passwor           6838
d") }}                                                             6839
      {% for e in pwd_form.new_password.errors %}                  6840
      <p class="field__error">{{ e }}</p>                          6841
      {% endfor %}                                                 6842
    </div>                                                         6843
    <div class="field">                                            6844
      <label class="field__label" for="{{                          6845
pwd_form.confirm.id }}">                                           6846
        确认新密码                                                 6847
      </label>                                                     6848
      {{ pwd_form.confirm(id=pwd_form.confirm.id,                  6849
                          autocomplete="new-password")             6850
```

```
      }}                                                                    6851
            {% for e in pwd_form.confirm.errors %}                          6852
            <p class="field__error">{{ e }}</p>                             6853
            {% endfor %}                                                    6854
          </div>                                                            6855
          <p class="actions" style="margin-top:18px;">                      6856
            {{ pwd_form.submit_pwd(class_="btn") }}                         6857
          </p>                                                              6858
        </form>                                                             6859
      </div>                                                                6860
   </div>                                                                   6861
</div>                                                                      6862
{% endblock %}                                                              6863
{% block scripts %}                                                         6864
<script>                                                                    6865
// 原生文件选择框样式难以统一，这里换成自定义按钮 + 显示所选文件名                6866
(function () {                                                              6867
   var input = document.getElementById('avatar-file');                      6868
   var label = document.getElementById('avatar-filename');                  6869
   if (!input || !label) return;                                            6870
   input.addEventListener('change', function () {                           6871
      label.textContent = input.files && input.files.length                 6872
        ? input.files[0].name : '未选择任何文件';                             6873
   });                                                                      6874
})();                                                                       6875
</script>                                                                   6876
{% endblock %}                                                              6877
# ==== app/templates/auth/logs.html ====                                    6878
{% extends "base.html" %}                                                   6879
{% block title %}操作日志{% endblock %}                                      6880
{% block content %}                                                         6881
<div class="card">                                                          6882
   <h2>操作日志</h2>                                                        6883
   <p class="actions" style="margin-bottom:12px;">                          6884
      <a class="btn btn--sm btn--ghost"                                     6885
      href="/users">返回用户管理</a>                                        6886
   </p>                                                                     6887
   {% if not records %}                                                     6888
   <p class="muted">暂无操作记录。</p>                                      6889
   {% else %}                                                               6890
   <table>                                                                  6891
      <caption class="muted">共 {{ total }} 条记录，当前第 {{ page }}       6892
      / {{ pages }} 页</caption>                                            6893
      <thead>                                                               6894
         <tr>                                                               6895
            <th scope="col">时间</th>                                       6896
            <th scope="col">操作人</th>                                     6897
            <th scope="col">操作</th>                                       6898
            <th scope="col">对象</th>                                       6899
            <th scope="col">说明</th>                                       6900
```

```
        </tr>                                                                    6901
      </thead>                                                                   6902
      <tbody>                                                                    6903
        {% for r in records %}                                                   6904
        <tr>                                                                     6905
          <td class="mono nowrap">{{ r.created_at.strftime("%Y                   6906
      -%m-%d %H:%M:%S") if r.created_at else "—" }}</td>                         6907
          <td>{{ users_map.get(r.user_id, "—") }}</td>                           6908
          <td style="text-align:left;">{{ r.action }}</td>                       6909
          <td>                                                                   6910
            {% if r.target_type %}{{ r.target_type }}{% if                       6911
      r.target_id %}#{{ r.target_id }}{% endif %}                                6912
            {% else %}—{% endif %}                                               6913
          </td>                                                                  6914
          <td style="text-align:left;">{{ r.detail or "—"                        6915
      }}</td>                                                                    6916
        </tr>                                                                    6917
        {% endfor %}                                                             6918
      </tbody>                                                                   6919
    </table>                                                                     6920
    {% if pages > 1 %}                                                           6921
    <nav class="actions" aria-label="日志分页">                                  6922
      {% if page > 1 %}                                                          6923
      <a class="btn btn--sm btn--ghost" href="/logs?page={{                      6924
      page - 1 }}">上一页</a>                                                    6925
      {% endif %}                                                                6926
      <span class="muted">第 {{ page }} / {{ pages }} 页</span>                  6927
      {% if page < pages %}                                                      6928
      <a class="btn btn--sm btn--ghost" href="/logs?page={{                      6929
      page + 1 }}">下一页</a>                                                    6930
      {% endif %}                                                                6931
    </nav>                                                                       6932
    {% endif %}                                                                  6933
    {% endif %}                                                                  6934
</div>                                                                           6935
{% endblock %}                                                                   6936
#  ==== app/templates/project/index.html ====                                    6937
{% extends "base.html" %}                                                        6938
{% block title %}项目与矿山档案{% endblock %}                                    6939
{% block content %}                                                              6940
<div class="card">                                                               6941
    <h2>项目与矿山档案</h2>                                                     6942
    <p class="actions" style="margin-bottom:12px;">                              6943
      <a class="btn btn--sm" href="/project/add">新建项目</a>                    6944
    </p>                                                                         6945
    {% if not projects %}                                                        6946
    <p class="muted">                                                            6947
      暂无项目。点击上方「新建项目」开始建立评价项目，                          6948
      或运行 <span class="mono">python seed_demo.py</span>                       6949
      载入论文案例。                                                             6950
```

```
</p>
{% else %}
<table>
  <caption class="muted">共 {{ projects|length }} 个项目</caption>
  <thead>
    <tr>
      <th scope="col">项目名称</th>
      <th scope="col">编号</th>
      <th scope="col">状态</th>
      <th scope="col">图斑数</th>
      <th scope="col">操作</th>
    </tr>
  </thead>
  <tbody>
    {% for p in projects %}
    <tr>
      <td style="text-align:left;">{{ p.name }}</td>
      <td class="mono">{{ p.code or "—" }}</td>
      <td><span class="tag tag--muted">{{ p.status }}</span></td>
      <td class="num">
        {% if p.patches.count() %}
          {{ p.patches.count() }}
        {% else %}
          <span class="tag tag--warn">0</span>
        {% endif %}
      </td>
      <td class="nowrap">
        <span class="row-actions">
          <a class="link-btn" href="/project/{{ p.id }}/patches">{% if p.patches.count() %}矿山图斑{% else %}＋新建图斑{% endif %}</a>
          <a class="link-btn" href="/project/{{ p.id }}/edit">编辑</a>
          <form method="post" action="/project/{{ p.id }}/delete"
                onsubmit="return confirm('确定删除项目「{{ p.name }}」吗？\n该项目下的图斑、微地质单元与评价数据将一并删除，且无法恢复。');">
            {{ form.hidden_tag() }}
            <button type="submit" class="link-btn link-btn--danger">删除</button>
          </form>
        </span>
      </td>
    </tr>
    {% endfor %}
  </tbody>
</table>
{% endif %}
```

```
</div>                                                                  7001
{% endblock %}                                                          7002
# ==== app/templates/project/edit.html ====                             7003
{% extends "base.html" %}                                               7004
{% block title %}{{ "编辑项目" if project else "新建项目" }}{%           7005
   endblock %}                                                          7006
{% block content %}                                                     7007
<div class="card" style="max-width:640px;">                             7008
  <h2>{{ "编辑项目" if project else "新建项目" }}</h2>                  7009
  <form method="post">                                                  7010
    {{ form.hidden_tag() }}                                             7011
    <div class="field">                                                 7012
      <label class="field__label" for="name">项目名称</label>           7013
      <input id="name" type="text" name="name"                          7014
    style="width:100%;" required                                        7015
             value="{{ project.name if project else '' }}">             7016
    </div>                                                              7017
    <div class="field">                                                 7018
      <label class="field__label" for="code">项目编号</label>           7019
      <input id="code" type="text" name="code"                          7020
    style="width:240px;"                                                7021
             value="{{ project.code or '' if project else               7022
    '' }}">                                                             7023
    </div>                                                              7024
    {% if project %}                                                    7025
    <div class="field">                                                 7026
      <label class="field__label" for="status">状态</label>             7027
      <select id="status" name="status">                                7028
        {% for s in ["进行中", "已完成", "已归档"] %}                   7029
        <option value="{{ s }}" {% if project.status == s               7030
    %}selected{% endif %}>{{ s }}</option>                              7031
        {% endfor %}                                                    7032
      </select>                                                         7033
    </div>                                                              7034
    {% endif %}                                                         7035
    <div class="field">                                                 7036
      <label class="field__label" for="description">项目说明</la        7037
    bel>                                                                7038
      <textarea id="description" name="description">{{                  7039
    project.description or '' if project else ''                        7040
    }}</textarea>                                                       7041
    </div>                                                              7042
    <p class="actions" style="margin-top:18px;">                        7043
      <button type="submit" class="btn">保存</button>                   7044
      <a class="btn btn--ghost" href="/project/">返回</a>               7045
    </p>                                                                7046
  </form>                                                               7047
</div>                                                                  7048
{% endblock %}                                                          7049
# ==== app/templates/project/patches.html ====                          7050
```

```
{% extends "base.html" %}
{% block title %}矿山图斑 · {{ project.name }}{% endblock %}
{% block content %}
<div class="card">
  <h2>矿山图斑 · {{ project.name }}</h2>
  <p class="actions" style="margin-bottom:12px;">
    <a class="btn btn--sm" href="/project/{{ project.id
    }}/patch/add">新建图斑</a>
    <a class="btn btn--sm btn--ghost"
    href="/project/">返回项目列表</a>
  </p>
  {% if not project.patches.count() %}
  <p class="muted">该项目下暂无图斑。点击上方「新建图斑」建立第一个矿山图斑。</p>
  {% else %}
  <table>
    <caption class="muted">共 {{ project.patches.count() }}
    个图斑</caption>
    <thead>
      <tr>
        <th scope="col">图斑名称</th>
        <th scope="col">行政区</th>
        <th scope="col">面积（公顷）</th>
        <th scope="col">开采矿种</th>
        <th scope="col">微地质单元</th>
        <th scope="col">操作</th>
      </tr>
    </thead>
    <tbody>
      {% for p in project.patches %}
      <tr>
        <td style="text-align:left;">{{ p.name }}</td>
        <td>{{ p.admin_region or "—" }}</td>
        <td class="num">{{ p.area_ha if p.area_ha is not
    none else "—" }}</td>
        <td>{{ p.mineral_type or "—" }}</td>
        <td class="num">{{ p.units.count() }}</td>
        <td class="nowrap">
          <span class="row-actions">
            <a class="link-btn" href="/unit/{{ p.id
    }}">微地质单元</a>
            <a class="link-btn" href="/constraint/{{ p.id
    }}">限制性判定</a>
            <a class="link-btn" href="/project/patch/{{
    p.id }}/edit">编辑</a>
            <form method="post" action="/project/patch/{{
    p.id }}/delete"
                  onsubmit="return confirm('确定删除图斑「{{
    p.name }}」吗？\n该图斑的微地质单元、指标数据与评价结果将一并删除，且无法恢复。');">
              {{ form.hidden_tag() }}
              <button type="submit" class="link-btn
```

```
      link-btn--danger">删除</button>                                    7101
                </form>                                                  7102
              </span>                                                    7103
            </td>                                                        7104
          </tr>                                                          7105
          {% endfor %}                                                   7106
        </tbody>                                                         7107
      </table>                                                           7108
      {% endif %}                                                        7109
</div>                                                                   7110
{% endblock %}                                                           7111
# ==== app/templates/project/patch_edit.html ====                        7112
{% extends "base.html" %}                                                7113
{% block title %}{{ "编辑图斑" if patch else "新建图斑" }}{%              7114
     endblock %}                                                         7115
{% block content %}                                                      7116
<div class="card" style="max-width:720px;">                              7117
  <h2>{{ "编辑图斑" if patch else "新建图斑" }} · {{ project.name        7118
    }}</h2>                                                              7119
  <form method="post">                                                   7120
    {{ form.hidden_tag() }}                                              7121
    <div class="field">                                                  7122
      <label class="field__label" for="patch-name">图斑名称</lab7123
    el>                                                                  7124
      <input id="patch-name" type="text" name="name"                     7125
    style="width:100%;" required                                         7126
             value="{{ patch.name if patch else '' }}">                  7127
    </div>                                                               7128
    <div class="field">                                                  7129
      <label class="field__label" for="admin-region">所在行政区</ 7130
    label>                                                               7131
      <input id="admin-region" type="text"                               7132
    name="admin_region" style="width:100%;"                              7133
             value="{{ patch.admin_region or '' if patch                 7134
    else '' }}">                                                         7135
    </div>                                                               7136
    <div class="field">                                                  7137
      <label class="field__label" for="area-ha">图斑面积（公顷）</la7138
    bel>                                                                 7139
      <input id="area-ha" type="number" step="any"                       7140
    name="area_ha"                                                       7141
             value="{{ patch.area_ha if patch and                        7142
    patch.area_ha is not none else '' }}">                               7143
    </div>                                                               7144
    <div class="field">                                                  7145
      <label class="field__label" for="mineral-type">开采矿种</l7146
    abel>                                                                7147
      <input id="mineral-type" type="text"                               7148
    name="mineral_type" style="width:240px;"                             7149
             value="{{ patch.mineral_type or '' if patch                 7150
```

```
else '' }}">
</div>
<div class="field">
  <label class="field__label" for="mining-method">开采方式</label>
  <input id="mining-method" type="text" name="mining_method" style="width:240px;"
         value="{{ patch.mining_method or '' if patch else '' }}">
</div>
<div class="field">
  <label class="field__label" for="stop-years">停产年限</label>
  <input id="stop-years" type="number" step="any" name="stop_years"
         value="{{ patch.stop_years if patch and patch.stop_years is not none else '' }}">
</div>
<div class="field">
  <label class="checkbox" for="responsible-missing">
    <input id="responsible-missing" type="checkbox" name="responsible_entity_missing" value="1"
           {% if not patch or patch.responsible_entity_missing %}checked{% endif %}>
    <span>责任主体缺失</span>
  </label>
</div>
<h3>参照生态系统（指标标准化基准）</h3>
<p class="muted">
  论文 4.2.2 节取值：植被覆盖度 75%、Shannon-Wiener 多样性指数 0.7。
  该基准与图斑绑定管理，可按不同参照生态系统分别设定。
</p>
<div class="field">
  <label class="field__label" for="coverage-benchmark">植被覆盖度基准（%）</label>
  <input id="coverage-benchmark" type="number" step="any" name="coverage_benchmark"
         value="{{ reference.coverage_benchmark if reference else 75 }}">
</div>
<div class="field">
  <label class="field__label" for="shannon-benchmark">Shannon-Wiener 多样性指数基准</label>
  <input id="shannon-benchmark" type="number" step="any" name="shannon_benchmark"
         value="{{ reference.shannon_benchmark if reference else 0.7 }}">
</div>
<div class="field">
  <label class="field__label" for="dominant-species">建群种
```

```
      </label>
        <input id="dominant-species" type="text"
      name="dominant_species" style="width:100%;"
              value="{{ reference.dominant_species or '' if
      reference else '' }}">
      </div>
      <p class="actions" style="margin-top:18px;">
        <button type="submit" class="btn">保存</button>
        <a class="btn btn--ghost" href="/project/{{
      project.id }}/patches">返回</a>
      </p>
    </form>
</div>
{% if patch %}
<div class="card" style="max-width:720px;">
    <h2>遥感影像挂接</h2>
    <p class="muted">
      挂接遥感底图或无人机正射影像，用于辅助微地质单元划分与植被覆盖度判读。
      支持 PNG、JPG、GIF、WebP、BMP、TIFF，单个文件不超过 32 MB。
    </p>
    {% if patch.image_path %}
    <figure style="margin:0 0 16px;">
      <img src="{{ url_for('static', filename=patch.image_path
      ) }}"
           alt="图斑「{{ patch.name }}」已挂接的遥感影像"
           style="max-width:100%;border:1px solid
      var(--line);border-radius:6px;">
      <figcaption class="muted" style="margin-top:6px;">
        {{ patch.image_note or "遥感影像" }}
        {% if patch.image_uploaded_at %}
        · 上传于 {{ patch.image_uploaded_at.strftime("%Y-%m-%d
      %H:%M") }}
        {% endif %}
      </figcaption>
    </figure>
    <form method="post" action="/project/patch/{{ patch.id
      }}/image/delete"
          onsubmit="return confirm('确定移除该遥感影像吗？');">
      {{ form.hidden_tag() }}
      <button type="submit" class="btn btn--sm
      btn--danger">移除影像</button>
    </form>
    <hr style="margin:16px 0;border:none;border-top:1px solid
      var(--line);">
    {% endif %}
    <form method="post" action="/project/patch/{{ patch.id
      }}/image"
          enctype="multipart/form-data">
      {{ form.hidden_tag() }}
      <div class="field">
```

```
        <label class="field__label" for="image-file">
          {{ "替换影像文件" if patch.image_path else "选择影像文件" }}
        </label>
        <input id="image-file" type="file" name="image"
               accept=".png,.jpg,.jpeg,.gif,.webp,.bmp,.tif,.tiff" required>
      </div>
      <div class="field">
        <label class="field__label" for="image-note">影像说明（如来源、时相）</label>
        <input id="image-note" type="text" name="image_note"
               placeholder="如：2025 年 6 月无人机正射影像"
               value="{{ patch.image_note or '' }}">
      </div>
      <p class="actions">
        <button type="submit" class="btn">{{ "替换影像" if patch.image_path else "上传并挂接" }}</button>
      </p>
    </form>
</div>
{% endif %}
{% endblock %}
# ==== app/templates/unit/list.html ====
{% extends "base.html" %}
{% from "_macros.html" import patch_guide %}
{% block title %}微地质单元{% endblock %}
{% block content %}
<div class="card">
  <h2>微地质单元</h2>
  {{ patch_guide(projects_without_patch) }}
  {% if not rows %}
  <p class="muted">
    暂无矿山图斑。请先到「项目与矿山档案」建立图斑。
  </p>
  {% else %}
  <table>
    <thead>
      <tr>
        <th scope="col">所属项目</th>
        <th scope="col">矿山图斑</th>
        <th scope="col">已划分单元</th>
        <th scope="col">占比合计</th>
        <th scope="col">操作</th>
      </tr>
    </thead>
    <tbody>
      {% for r in rows %}
      <tr>
        <td>{{ r.patch.project.name if r.patch.project else
```

```
      "—" }}</td>                                                    7301
          <td style="text-align:left;">{{ r.patch.name }}</td>       7302
          <td class="num">{{ r.count }} 个</td>                      7303
          <td class="num">                                           7304
            {% if r.count %}                                         7305
              {{ "%.2f%%"|format(r.ratio_sum * 100) }}               7306
              {% if r.ratio_sum < 0.999 or r.ratio_sum >             7307
      1.001 %}                                                       7308
              <span class="warn">（不等于 100%）</span>              7309
              {% endif %}                                            7310
            {% else %}—{% endif %}                                   7311
          </td>                                                      7312
          <td class="nowrap">                                        7313
            <span class="row-actions">                               7314
              <a class="link-btn" href="/unit/{{ r.patch.id          7315
      }}">划分与维护</a>                                             7316
              <a class="link-btn" href="/indicator/">录入指标数据</a 7317
      >                                                              7318
            </span>                                                  7319
          </td>                                                      7320
        </tr>                                                        7321
        {% endfor %}                                                 7322
      </tbody>                                                       7323
    </table>                                                         7324
    {% endif %}                                                      7325
</div>                                                               7326
{% endblock %}                                                       7327
# ==== app/templates/unit/index.html ====                            7328
{% extends "base.html" %}                                            7329
{% block title %}微地质单元 · {{ patch.name }}{% endblock %}         7330
{% block content %}                                                  7331
<div class="card">                                                   7332
  <h2>微地质单元 · {{ patch.name }}</h2>                             7333
  {% if units %}                                                     7334
  <table>                                                            7335
    <caption class="muted">                                          7336
      共 {{ units|length }} 个单元；                                 7337
      面积合计 <span class="num">{{ "%.4f"|format(total_area)        7338
    }}</span>；                                                      7339
      占比合计 <span class="num">{{ "%.2f%%"|format(ratio_sum *      7340
    100) }}</span>                                                   7341
      {% if ratio_sum and (ratio_sum < 0.999 or ratio_sum >          7342
    1.001) %}                                                        7343
      <span class="warn">（占比合计不等于 100%，请检查面积录入）</span>7344
      {% endif %}                                                    7345
    </caption>                                                       7346
    <thead>                                                          7347
      <tr>                                                           7348
        <th scope="col">微地质单元</th>                              7349
        <th scope="col">面积</th>                                    7350
```

```
          <th scope="col">面积占比</th>
          <th scope="col">说明</th>
          <th scope="col">操作</th>
        </tr>
      </thead>
      <tbody>
        {% for u in units %}
        <tr>
          <td>{{ u.unit_type }}</td>
          <td class="num">{{ u.area if u.area is not none
      else "—" }}</td>
          <td class="num">
            {% if u.ratio is not none %}{{
      "%.2f%%"|format(u.ratio * 100) }}{% else %}—{% endif %}
          </td>
          <td style="text-align:left;">{{ u.description or
      "—" }}</td>
          <td class="nowrap">
            <span class="row-actions">
              <a class="link-btn" href="/unit/edit/{{ u.id
      }}">编辑</a>
              <form method="post" action="/unit/delete/{{
      u.id }}"
                    onsubmit="return confirm('确定删除微地质单元「{{
      u.unit_type }}」吗？\n该单元已录入的指标数据将一并删除，且无法恢复。');">
                {{ form.hidden_tag() }}
                <button type="submit" class="link-btn
      link-btn--danger">删除</button>
              </form>
            </span>
          </td>
        </tr>
        {% endfor %}
      </tbody>
    </table>
    {% else %}
    <p class="muted">尚未划分微地质单元。请在下方添加。</p>
    {% endif %}
    <h3>添加微地质单元</h3>
    <form method="post" action="/unit/{{ patch.id }}/add">
      {{ form.hidden_tag() }}
      <div class="actions">
        <label class="field__label" for="add-unit-type"
      style="margin:0;">类型</label>
        <select id="add-unit-type" name="unit_type">
          {% for t in unit_types %}<option value="{{ t }}">{{
      t }}</option>{% endfor %}
        </select>
        <label class="field__label" for="add-unit-area"
      style="margin:0;">面积</label>
```

```
        <input id="add-unit-area" type="number" step="any"
      name="area">
        <label class="field__label" for="add-unit-desc"
      style="margin:0;">说明</label>
        <input id="add-unit-desc" type="text"
      name="description" style="width:220px;">
        <button type="submit" class="btn btn--sm">添加</button>
      </div>
    </form>
    <p class="actions" style="margin-top:20px;">
      <a class="btn btn--sm btn--ghost" href="/project/{{
      patch.project_id }}/patches">返回图斑列表</a>
      <a class="btn btn--sm btn--ghost"
      href="/indicator/">去录入指标数据</a>
    </p>
</div>
{% endblock %}
# ==== app/templates/unit/edit.html ====
{% extends "base.html" %}
{% block title %}编辑微地质单元{% endblock %}
{% block content %}
<div class="card">
    <h2>编辑微地质单元 · {{ patch.name }}</h2>
    <p class="muted">修改面积后，系统会自动重算本图斑各单元的面积占比。</p>
    <form method="post">
      {{ form.hidden_tag() }}
      <div class="field">
        <label class="field__label" for="unit-type">微地质单元类型</l
      abel>
        <select id="unit-type" name="unit_type">
          {% for t in unit_types %}
          <option value="{{ t }}" {% if unit.unit_type == t
      %}selected{% endif %}>{{ t }}</option>
          {% endfor %}
        </select>
      </div>
      <div class="field">
        <label class="field__label" for="unit-area">面积</label>
        <input id="unit-area" type="number" step="any"
      name="area"
               value="{{ unit.area if unit.area is not none
      else '' }}">
      </div>
      <div class="field">
        <label class="field__label" for="unit-desc">说明</label>
        <input id="unit-desc" type="text" name="description"
      style="width:420px;"
               value="{{ unit.description or '' }}">
      </div>
      <p class="actions" style="margin-top:16px;">
```

```
        <button type="submit" class="btn">保存</button>
        <a class="btn btn--ghost" href="/unit/{{
    unit.patch_id }}">返回</a>
    </p>
  </form>
</div>
{% endblock %}
# ==== app/templates/standard/index.html ====
{% extends "base.html" %}
{% block title %}政策与标准知识库{% endblock %}
{% block content %}
<div class="card">
  <h2>政策与标准知识库</h2>  <p class="actions"
    style="margin-bottom:14px;">
    <a class="btn btn--sm" href="/standard/add">添加标准</a>
    <a class="btn btn--sm btn--ghost"
    href="/standard/compare">省际口径比对</a>
  </p>
  <h3>地方标准</h3>
  {% if local %}
  <table>
    <thead>
      <tr>
        <th scope="col">省份</th>
        <th scope="col">名称</th>
        <th scope="col">编号</th>
        <th scope="col">规则数</th>
        <th scope="col">操作</th>
      </tr>
    </thead>
    <tbody>
      {% for s in local %}
      <tr>
        <td>{{ s.province }}</td>
        <td style="text-align:left;">{{ s.name }}</td>
        <td class="mono">{{ s.doc_number or "—" }}</td>
        <td class="num">{{ s.rules|length }}</td>
        <td class="nowrap">
          <span class="row-actions">
            <a class="link-btn" href="/standard/{{ s.id
    }}/rules">规则</a>
            <a class="link-btn" href="/standard/{{ s.id
    }}/edit">编辑</a>
            <form method="post" action="/standard/{{ s.id
    }}/delete"
                  onsubmit="return confirm('确定删除「{{
    s.province }} {{ s.name }}」吗？\n该标准下的 {{ s.rules|length
    }} 条规则将一并删除，且无法恢复。');">
              {{ form.hidden_tag() }}
              <button type="submit" class="link-btn
```

```
    link-btn--danger">删除</button>
            </form>
          </span>
        </td>
      </tr>
      {% endfor %}
    </tbody>
  </table>
  {% else %}
  <p class="muted">暂无地方标准。运行 <span class="mono">python
    seed_demo.py</span> 可载入论文表 1 的 11 项标准与政策文件。</p>
  {% endif %}
  <h3>政策文件</h3>
  {% if policy %}
  <table>
    <thead>
      <tr>
        <th scope="col">省份</th>
        <th scope="col">名称</th>
        <th scope="col">文号</th>
        <th scope="col">规则数</th>
        <th scope="col">操作</th>
      </tr>
    </thead>
    <tbody>
      {% for s in policy %}
      <tr>
        <td>{{ s.province }}</td>
        <td style="text-align:left;">{{ s.name }}</td>
        <td class="mono">{{ s.doc_number or "—" }}</td>
        <td class="num">{{ s.rules|length }}</td>
        <td class="nowrap">
          <span class="row-actions">
            <a class="link-btn" href="/standard/{{ s.id
    }}/rules">规则</a>
            <a class="link-btn" href="/standard/{{ s.id
    }}/edit">编辑</a>
            <form method="post" action="/standard/{{ s.id
    }}/delete"
                  onsubmit="return confirm('确定删除「{{
    s.province }} {{ s.name }}」吗？\n该文件下的 {{ s.rules|length
    }} 条规则将一并删除，且无法恢复。');">
              {{ form.hidden_tag() }}
              <button type="submit" class="link-btn
    link-btn--danger">删除</button>
            </form>
          </span>
        </td>
      </tr>
      {% endfor %}
```

```
      </tbody>                                                          7551
    </table>                                                            7552
    {% else %}                                                          7553
    <p class="muted">暂无政策文件。</p>                                  7554
    {% endif %}                                                         7555
</div>                                                                  7556
{% endblock %}                                                          7557
# ==== app/templates/standard/edit.html ====                            7558
{% extends "base.html" %}                                               7559
{% block title %}{{ "编辑标准" if standard else "添加标准" }}{%          7560
    endblock %}                                                         7561
{% block content %}                                                     7562
<div class="card" style="max-width:800px;">                             7563
  <h2>{{ "编辑标准" if standard else "添加标准" }}</h2>                 7564
  <form method="post">                                                  7565
    {{ form.hidden_tag() }}                                             7566
    <div class="field">                                                 7567
      <label class="field__label" for="province">省份</label>           7568
      <input id="province" type="text" name="province"                  7569
    style="width:160px;" required                                       7570
             value="{{ standard.province if standard else               7571
    '' }}">                                                             7572
    </div>                                                              7573
    <div class="field">                                                 7574
      <label class="field__label" for="std-name">名称</label>           7575
      <input id="std-name" type="text" name="name"                      7576
    style="width:100%;" required                                        7577
             value="{{ standard.name if standard else ''                7578
    }}">                                                                7579
    </div>                                                              7580
    <div class="field">                                                 7581
      <label class="field__label" for="doc-number">标准编号 /           7582
    文号</label>                                                        7583
      <input id="doc-number" type="text" name="doc_number"              7584
    style="width:320px;"                                                7585
             value="{{ standard.doc_number or '' if                     7586
    standard else '' }}">                                               7587
    </div>                                                              7588
    <div class="field">                                                 7589
      <label class="field__label" for="category">类别</label>           7590
      <select id="category" name="category">                            7591
        {% for c in ["地方标准", "政策文件"] %}                         7592
        <option value="{{ c }}" {% if standard and                      7593
    standard.category == c %}selected{% endif %}>{{ c                   7594
    }}</option>                                                         7595
        {% endfor %}                                                    7596
      </select>                                                         7597
    </div>                                                              7598
    <div class="field">                                                 7599
      <label class="field__label" for="publish-date">发布时间</l        7600
```

```
      abel>                                                                7601
        <input id="publish-date" type="text"                               7602
      name="publish_date" style="width:200px;"                             7603
                placeholder="如 2023-03-30"                                7604
                value="{{ standard.publish_date or '' if                   7605
      standard else '' }}">                                                7606
      </div>                                                               7607
      <div class="field">                                                  7608
        <label class="field__label" for="implement-date">实施时间<         7609
      /label>                                                              7610
        <input id="implement-date" type="text"                             7611
      name="implement_date" style="width:200px;"                           7612
                placeholder="如 2023-04-30（未明确可留空）"                7613
                value="{{ standard.implement_date or '' if                 7614
      standard else '' }}">                                                7615
      </div>                                                               7616
      <div class="field">                                                  7617
        <label class="field__label" for="summary">特点 /                   7618
      判定标准</label>                                                     7619
        <textarea id="summary" name="summary"                              7620
      style="min-height:96px;">{{ standard.summary or '' if                7621
      standard else '' }}</textarea>                                       7622
      </div>                                                               7623
      <p class="actions" style="margin-top:18px;">                         7624
        <button type="submit" class="btn">保存</button>                    7625
        {% if standard %}                                                  7626
        <form method="post" action="/standard/{{ standard.id               7627
      }}/delete" style="display:inline;"                                   7628
              onsubmit="return confirm('确定删除「{{                       7629
      standard.province }} {{ standard.name                                7630
      }}」吗？\n该标准下的规则将一并删除，且无法恢复。');">                7631
          {{ form.hidden_tag() }}                                          7632
          <button type="submit" class="btn                                 7633
      btn--danger">删除该标准</button>                                     7634
        </form>                                                            7635
        {% endif %}                                                        7636
        <a class="btn btn--ghost" href="/standard/">返回</a>               7637
      </p>                                                                 7638
    </form>                                                                7639
</div>                                                                     7640
{% endblock %}                                                             7641
# ==== app/templates/standard/rules.html ====                              7642
{% extends "base.html" %}                                                  7643
{% block title %}结构化规则 · {{ standard.province }}{% endblock           7644
    %}                                                                     7645
{% block content %}                                                        7646
<div class="card">                                                         7647
    <h2>结构化规则 · {{ standard.province }} {{ standard.name              7648
      }}</h2>                                                              7649
    {% if rules %}                                                         7650
```

```
<table>
  <caption class="muted">共 {{ rules|length }} 条规则</caption>
  <thead>
    <tr>
      <th scope="col">判定项</th>
      <th scope="col">条件</th>
      <th scope="col">条件类型</th>
      <th scope="col">适用分区</th>
      <th scope="col">备注</th>
      <th scope="col">操作</th>
    </tr>
  </thead>
  <tbody>
    {% for r in rules %}
    <tr>
      <td style="text-align:left;">{{ r.item }}</td>
      <td class="mono nowrap">{{ r.operator }} {{ r.threshold or "" }}</td>
      <td>{{ r.condition_type }}</td>
      <td>{{ r.region or "—" }}</td>
      <td style="text-align:left;">{{ r.note or "—" }}</td>
      <td>
        <form method="post" action="/standard/rule/{{ r.id }}/delete"
              onsubmit="return confirm('确定删除规则「{{ r.item }} {{ r.operator }} {{ r.threshold or '' }}」吗？');">
          {{ form.hidden_tag() }}
          <button type="submit" class="btn btn--sm btn--danger">删除</button>
        </form>
      </td>
    </tr>
    {% endfor %}
  </tbody>
</table>
{% else %}
<p class="muted">该标准暂无规则。请在下方添加第一条规则。</p>
{% endif %}

<h3>添加规则</h3>
<form method="post">
  {{ form.hidden_tag() }}
  <div class="actions">
    <div>
      <label class="field__label" for="r-item">判定项</label>
      <input id="r-item" type="text" name="item"
             placeholder="如：植被覆盖度" required>
    </div>
    <div>
```

```
          <label class="field__label" for="r-op">比较符</label>
          <select id="r-op" name="operator">
            {% for o in [">=", ">", "<=", "<", "=="]
    %}<option value="{{ o }}">{{ o }}</option>{% endfor %}
          </select>
        </div>
        <div>
          <label class="field__label" for="r-thr">阈值</label>
          <input id="r-thr" type="text" name="threshold"
    placeholder="如：50" style="width:110px;">
        </div>
        <div>
          <label class="field__label" for="r-cond">条件类型</label
    >
          <select id="r-cond" name="condition_type">
            <option value="必要条件">必要条件</option>
            <option value="充分条件">充分条件</option>
          </select>
        </div>
        <div>
          <label class="field__label" for="r-region">适用分区</lab
    el>
          <input id="r-region" type="text" name="region"
    placeholder="如宁夏分区" style="width:130px;">
        </div>
        <div>
          <label class="field__label" for="r-note">备注</label>
          <input id="r-note" type="text" name="note"
    style="width:220px;">
        </div>
        <div>
          <button type="submit" class="btn">添加规则</button>
        </div>
      </div>
    </form>
    <p class="actions" style="margin-top:20px;">
      <a class="btn btn--sm btn--ghost"
      href="/standard/">返回标准列表</a>
      <a class="btn btn--sm btn--ghost"
      href="/standard/compare">省际口径比对</a>
    </p>
</div>
{% endblock %}
# ==== app/templates/standard/compare.html ====
{% extends "base.html" %}
{% block title %}省际口径比对{% endblock %}
{% block content %}
<div class="card">
  <h2>省际口径比对</h2>
  {% if not groups %}
```

```
  <p class="muted">
    暂无规则数据。请先在「标准知识库」中为标准录入规则，
    或运行 <span class="mono">python seed_demo.py</span> 载入论文表 1 数据。
  </p>
  {% else %}
  {% for item, rs in groups.items() %}
  <h3>{{ item }}</h3>
  <table>
    <thead>
      <tr>
        <th scope="col">省份</th>
        <th scope="col">标准 / 文件</th>
        <th scope="col">条件</th>
        <th scope="col">条件类型</th>
        <th scope="col">备注</th>
      </tr>
    </thead>
    <tbody>
      {% for r in rs %}
      <tr>
        <td>{{ r.standard.province }}</td>
        <td style="text-align:left;">{{ r.standard.name }}</td>
        <td class="mono nowrap">{{ r.operator }} {{ r.threshold or "" }}</td>
        <td>{{ r.condition_type }}</td>
        <td style="text-align:left;">{{ r.note or "—" }}</td>
      </tr>
      {% endfor %}
    </tbody>
  </table>
  {% endfor %}
  {% endif %}
  <p class="actions" style="margin-top:16px;">
    <a class="btn btn--ghost" href="/standard/">返回标准列表</a>
  </p>
</div>
{% endblock %}
# ==== app/templates/constraint/index.html ====
{% extends "base.html" %}
{% from "_macros.html" import patch_guide %}
{% block title %}限制性指标判定{% endblock %}
{% block content %}
<div class="card">
  <h2>限制性指标判定</h2>
  {{ patch_guide(projects_without_patch) }}
  {% if not patches %}
  <p class="muted">暂无矿山图斑。请先在「项目与矿山档案」中建立图斑。</p>
```

```
  {% else %}
  <table>
    <thead>
      <tr>
        <th scope="col">所属项目</th>
        <th scope="col">矿山图斑</th>
        <th scope="col">判定结果</th>
        <th scope="col">操作</th>
      </tr>
    </thead>
    <tbody>
      {% for p in patches %}
      {% set r = latest.get(p.id) %}
      <tr>
        <td>{{ p.project.name if p.project else "—" }}</td>
        <td style="text-align:left;">{{ p.name }}</td>
        <td style="text-align:left;">
          {% if r %}
            {% if r.passed %}
            <span class="tag tag--ok">通过</span>
            {% else %}
            <span class="tag tag--warn">不通过</span>
            {% endif %}
            <span class="muted">{{ r.reason }}</span>
          {% else %}
          <span class="tag tag--muted">未判定</span>
          {% endif %}
        </td>
        <td class="nowrap">
          <span class="row-actions">
            <a class="link-btn" href="/constraint/{{ p.id
    }}">执行判定</a>
          </span>
        </td>
      </tr>
      {% endfor %}
    </tbody>
  </table>
  {% endif %}
</div>
{% endblock %}
# ==== app/templates/constraint/check.html ====
{% extends "base.html" %}
{% block title %}限制性条件判定 · {{ patch.name }}{% endblock %}
{% block content %}
<div class="card" style="max-width:760px;">
  <h2>限制性条件判定 · {{ patch.name }}</h2>
  <form method="post">
    {{ form.hidden_tag() }}
    {% for field, name, val in field_states %}
```

```
      <fieldset style="border:1px solid                                    7851
      var(--line);border-radius:6px;padding:10px                           7852
      14px;margin:0 0 10px;">                                              7853
        <legend style="padding:0 6px;font-weight:600;">{{                  7854
      name }}</legend>                                                     7855
        <div class="actions">                                              7856
          <label class="checkbox">                                         7857
            <input type="radio" name="{{ field }}" value="1"               7858
                   {% if val is none or val %}checked{% endif              7859
      %}>                                                                  7860
            <span>通过</span>                                              7861
          </label>                                                         7862
          <label class="checkbox">                                         7863
            <input type="radio" name="{{ field }}" value="0"               7864
                   {% if val is not none and not val                       7865
      %}checked{% endif %}>                                                7866
            <span>不通过</span>                                            7867
          </label>                                                         7868
        </div>                                                             7869
      </fieldset>                                                          7870
      {% endfor %}                                                         7871
      <p class="actions" style="margin-top:16px;">                         7872
        <button type="submit" class="btn">保存判定</button>                7873
        <a class="btn btn--ghost" href="/constraint/">返回</a>             7874
      </p>                                                                 7875
    </form>                                                                7876
    {% if record %}                                                        7877
    <div style="margin-top:18px;padding:14px;border-radius:6px7878
      ;border-left:4px solid;                                              7879
                 {% if record.passed %}background:var(--ok-soft7880
      );border-left-color:var(--ok);                                       7881
                 {% else %}background:var(--danger-soft);border7882
      -left-color:var(--danger);{% endif %}">                              7883
      <p style="margin:0 0 4px;font-weight:600;">                          7884
        判定结论：                                                         7885
        {% if record.passed %}                                             7886
        <span class="ok">通过，可进入定量评价</span>                       7887
        {% else %}                                                         7888
        <span class="warn">不通过 —— 建议生态重建或人工辅助</span>         7889
        {% endif %}                                                        7890
      </p>                                                                 7891
      <p class="muted" style="margin:0;">{{ record.reason                  7892
      }}</p>                                                               7893
    </div>                                                                 7894
    {% endif %}                                                            7895
</div>                                                                     7896
{% endblock %}                                                             7897
# ==== app/templates/ahp/index.html ====                                   7898
{% extends "base.html" %}                                                  7899
{% block title %}AHP 权重计算{% endblock %}                                7900
```

```
{% block content %}
<div class="card" style="max-width:720px;">
  <h2>AHP 权重计算 · 选择指标层级</h2>
  <form method="post">
    {{ form.hidden_tag() }}
    <div class="field">
      <label class="field__label" for="{{ form.unit_type.id
    }}">微地质单元类型</label>
      {{ form.unit_type(id=form.unit_type.id) }}
    </div>
    <div class="field">
      <label class="field__label" for="{{ form.level.id
    }}">指标层级</label>
      {{ form.level(id=form.level.id) }}
    </div>
    <div class="field">
      <label class="field__label" for="{{ form.l1.id }}">
        一级指标（层级选「二级指标」时需选择）
      </label>
      {{ form.l1(id=form.l1.id) }}
    </div>
    <p class="actions" style="margin-top:16px;">
      {{ form.submit(class_="btn") }}
      <a class="btn btn--ghost" href="/ahp/indicators">指标定义维
    护</a>
      <a class="btn btn--ghost" href="/ahp/experts">多专家矩阵</a
    >
    </p>
  </form>
</div>
<div class="card">
  <h2>各微地质单元当前生效权重</h2>
  <table>
    <thead>
      <tr>
        <th scope="col">微地质单元</th>
        <th scope="col">当前权重来源</th>
        <th scope="col">操作</th>
      </tr>
    </thead>
    <tbody>
      {% for r in weight_rows %}
      <tr>
        <td style="text-align:left;">{{ r.unit_type }}</td>
        <td style="text-align:left;">
          {% if r.custom %}
          <span class="tag tag--warn">{{ r.label }}</span>
          {% else %}
          <span class="tag tag--muted">{{ r.label }}</span>
          {% endif %}
```

```
            </td>
            <td class="nowrap">
              <span class="row-actions">
                <a class="link-btn" href="/ahp/matrix?unit_type=
    {{ r.unit_type }}&level=1">一级指标权重</a>
                <a class="link-btn" href="/ahp/matrix?unit_type=
    {{ r.unit_type }}&level=1&calc=1">查看一致性</a>
                {% if r.custom %}
                <form method="post" action="/ahp/weights/reset"
                      onsubmit="return confirm('确定把「{{
    r.unit_type }}」恢复为论文表 3 的默认权重吗？');">
                  <input type="hidden" name="csrf_token"
    value="{{ csrf_token() }}">
                  <input type="hidden" name="unit_type"
    value="{{ r.unit_type }}">
                  <button type="submit" class="link-btn
    link-btn--danger">恢复默认</button>
                </form>
                {% endif %}
              </span>
            </td>
          </tr>
          {% endfor %}
        </tbody>
      </table>
    </div>
    {% endblock %}
    # ==== app/templates/ahp/matrix.html ====
    {% extends "base.html" %}
    {% block title %}判断矩阵录入{% endblock %}
    {% block content %}
    <div class="card">
      <h2>
        判断矩阵录入
        · {{ unit_type }}
        · {% if level == 1 %}一级指标{% else %}二级指标（{{ l1 }}）{%
        endif %}
      </h2>
      <p class="muted">
        采用 1–9 标度：1 同等重要、3 稍微重要、5 明显重要、7 强烈重要、9 极端重要，
        2、4、6、8 为相邻判断的中间值。只需录入上三角，对角线为 1，下三角自动取倒数。
        矩阵已按论文表 3 的权重预填。
      </p>
      <form method="post">
        {{ form.hidden_tag() }}
        <table style="width:auto;">
          <thead>
            <tr>
              <th></th>
              {% for name in names %}<th scope="col">{{ name
```

```
    }}</th>{% endfor %}                                                    8001
        </tr>                                                              8002
      </thead>                                                             8003
      <tbody>                                                              8004
        {% for i in range(names|length) %}                                 8005
        <tr>                                                               8006
          <th scope="row">{{ names[i] }}</th>                              8007
          {% for j in range(names|length) %}                               8008
            {% if i == j %}                                                8009
            <td class="muted num">1</td>                                   8010
            {% elif i < j %}                                               8011
            <td>                                                           8012
              <label class="muted" style="position:absolute;               8013
    left:-9999px;"                                                         8014
                     for="m-{{ i }}-{{ j }}">{{ names[i] }}                8015
    相对 {{ names[j] }} 的重要性</label>                                    8016
              <input id="m-{{ i }}-{{ j }}" type="number"                  8017
    step="0.01" min="0.01"                                                 8018
                     name="m_{{ i }}_{{ j }}" value="{{                    8019
    default[i][j] }}">                                                     8020
            </td>                                                          8021
            {% else %}                                                     8022
            <td class="muted num">{{ default[i][j] }}</td>                 8023
            {% endif %}                                                    8024
          {% endfor %}                                                     8025
        </tr>                                                              8026
        {% endfor %}                                                       8027
      </tbody>                                                             8028
    </table>                                                               8029
    <p class="actions">                                                    8030
      <button type="submit" class="btn" name="action"                      8031
    value="calc">计算权重</button>                                         8032
      {% if result and result.consistent %}                                8033
      <button type="submit" class="btn" name="action"                      8034
    value="save">保存并生效</button>                                       8035
      {% endif %}                                                          8036
      <a class="btn btn--ghost" href="/ahp/">重新选择</a>                  8037
    </p>                                                                   8038
  </form>                                                                  8039
  <p class="muted" style="margin-top:10px;">                               8040
    当前「{{ unit_type }}」生效权重：<strong>{{ source_label               8041
    }}</strong>。                                                          8042
    点击「保存并生效」后，本单元类型的适宜性评价将采用此处算出的权重；   8043
    如需回到论文取值，可在 <a href="/ahp/">权重首页</a> 恢复默认。        8044
  </p>                                                                     8045
  {% if result %}                                                          8046
  <hr style="margin:22px 0;border:none;border-top:1px solid                8047
    var(--line);">                                                         8048
  <h3>计算结果</h3>                                                        8049
  <table style="width:auto;">                                              8050
```

```
    <thead>                                                                  8051
      <tr>                                                                   8052
        <th scope="col">指标</th>                                            8053
        {% for name in names %}<th scope="col">{{ name                       8054
    }}</th>{% endfor %}                                                      8055
      </tr>                                                                  8056
    </thead>                                                                 8057
    <tbody>                                                                  8058
      <tr>                                                                   8059
        <th scope="row">权重</th>                                            8060
        {% for w in result.weights %}<td class="mono">{{ w                   8061
    }}</td>{% endfor %}                                                      8062
      </tr>                                                                  8063
    </tbody>                                                                 8064
  </table>                                                                   8065
  <div class="stat-row">                                                     8066
    <div>                                                                    8067
      <div class="stat__label">最大特征值 λmax</div>                         8068
      <div class="stat__value mono">{{ result.lambda_max                     8069
    }}</div>                                                                 8070
    </div>                                                                   8071
    <div>                                                                    8072
      <div class="stat__label">一致性指标 CI</div>                           8073
      <div class="stat__value mono">{{                                       8074
    "%.4f"|format(result.ci) }}</div>                                        8075
    </div>                                                                   8076
    <div>                                                                    8077
      <div class="stat__label">一致性比例 CR</div>                           8078
      <div class="stat__value mono">{{                                       8079
    "%.4f"|format(result.cr) }}</div>                                        8080
    </div>                                                                   8081
    <div>                                                                    8082
      <div class="stat__label">一致性检验（要求 CR &lt; 0.1）</div>          8083
      <div class="stat__value">                                              8084
        {% if result.consistent %}<span class="ok">通过</span>               8085
        {% else %}<span class="warn">不通过</span>{% endif %}                8086
      </div>                                                                 8087
    </div>                                                                   8088
  </div>                                                                     8089
  {% if not result.consistent and result.most_inconsistent                   8090
    %}                                                                       8091
  <p style="margin-top:14px;padding:12px;border-radius:6px;b                 8092
    ackground:var(--danger-soft);                                            8093
              border-left:4px solid var(--danger);">                         8094
    <strong class="warn">建议优先调整：</strong>                             8095
    第 {{ result.most_inconsistent.i + 1 }} 行、第 {{                        8096
    result.most_inconsistent.j + 1 }} 列的比较                               8097
    （当前 <span class="mono num">{{ result.most_inconsistent.v              8098
    alue|round(3) }}</span>，                                                8099
    理想约 <span class="mono num">{{ result.most_inconsistent.e              8100
```

```
      xpected|round(3) }}</span>)                                          8101
    </p>                                                                   8102
    {% endif %}                                                            8103
    {% endif %}                                                            8104
</div>                                                                     8105
{% endblock %}                                                             8106
# ==== app/templates/ahp/indicators.html ====                              8107
{% extends "base.html" %}                                                  8108
{% block title %}指标定义维护{% endblock %}                                  8109
{% block content %}                                                        8110
<div class="card">                                                         8111
    <h2>指标体系维护</h2>                                                   8112
    <p class="actions" style="margin-bottom:14px;">                        8113
      <a class="btn btn--sm btn--ghost"                                    8114
      href="/ahp/">返回权重计算</a>                                         8115
      <a class="btn btn--sm btn--ghost"                                    8116
      href="/ahp/experts">多专家矩阵</a>                                    8117
      <form method="post" action="/ahp/indicators/reset"                   8118
      style="display:inline;"                                              8119
            onsubmit="return confirm('将按论文取值重置全部指标定义，自定义修改会丢失。 8120
      确定继续？');">                                                       8121
        {{ form.hidden_tag() }}                                            8122
        <button type="submit" class="btn btn--sm                           8123
      btn--danger">按论文取值重置</button>                                  8124
      </form>                                                              8125
    </p>                                                                   8126
    {% for level, title in [(1, "一级指标"), (2, "二级指标"), (3,          8127
      "三级指标")] %}                                                      8128
    {% set level_items = all_ind|selectattr("level",                       8129
      "equalto", level)|list %}                                            8130
    {% if level_items %}                                                   8131
    <h3>{{ title }}（{{ level_items|length }} 项）</h3>                     8132
    <table>                                                                8133
      <thead>                                                              8134
        <tr>                                                               8135
          <th scope="col">指标</th>                                        8136
          <th scope="col">上级指标</th>                                    8137
          <th scope="col">单位</th>                                        8138
          <th scope="col">标准化类型</th>                                  8139
          <th scope="col">理想取值</th>                                    8140
          <th scope="col">启用单元</th>                                    8141
          <th scope="col">限制性</th>                                      8142
          <th scope="col">操作</th>                                        8143
        </tr>                                                              8144
      </thead>                                                             8145
      <tbody>                                                              8146
        {% for i in level_items %}                                         8147
        <tr>                                                               8148
          <td style="text-align:left;">                                    8149
            <span class="mono muted">{{ i.code                             8150
```

```
      }}</span><br>{{ i.name }}
          </td>
          <td>{{ (all_ind|selectattr("code", "equalto",
      i.parent_code)|list|first).name if i.parent_code else
      "—" }}</td>
          <td>{{ i.unit or "—" }}</td>
          <td>{{ type_label(i.ideal_type) }}</td>
          <td class="mono">{{ i.ideal_value or "—" }}</td>
          <td style="text-align:left;">{{ i.enabled_units or
      "—" }}</td>
          <td>
            {% if i.is_constraint %}<span class="tag
      tag--warn">是</span>
            {% else %}<span class="tag tag--muted">否</span>{%
      endif %}
          </td>
          <td class="nowrap">
            <span class="row-actions">
              <a class="link-btn" href="/ahp/indicators/{{
      i.code }}/edit">编辑</a>
            </span>
          </td>
        </tr>
        {% endfor %}
      </tbody>
    </table>
    {% endif %}
    {% endfor %}
</div>
{% endblock %}
# ==== app/templates/ahp/indicator_edit.html ====
{% extends "base.html" %}
{% block title %}编辑指标 · {{ ind.name }}{% endblock %}
{% block content %}
<div class="card" style="max-width:680px;">
    <h2>编辑指标 · {{ ind.name }}</h2>
    <p class="muted">
      指标编码 <span class="mono">{{ ind.code }}</span>、层级 {{
      ind.level }} 级，
      由论文指标体系确定，不可修改。
    </p>
    <form method="post">
      {{ form.hidden_tag() }}
      <div class="field">
        <label class="field__label" for="name">指标名称</label>
        <input id="name" type="text" name="name"
      style="width:100%;" required
               value="{{ ind.name }}">
      </div>
      <div class="field">
```

```
  <label class="field__label" for="unit">计量单位</label>
  <input id="unit" type="text" name="unit"
style="width:180px;"
         value="{{ ind.unit or '' }}">
</div>
<div class="field">
  <label class="field__label" for="ideal-type">标准化类型</la
bel>
  <select id="ideal-type" name="ideal_type">
    {% for code, label in [("benefit","越大越优"),("cost","越
小越优"),("mid","适中型"),("discrete","离散赋值")] %}
    <option value="{{ code }}" {% if ind.ideal_type ==
code %}selected{% endif %}>{{ label }}</option>
    {% endfor %}
  </select>
</div>
<div class="field">
  <label class="field__label" for="ideal-value">理想取值</la
bel>
  <input id="ideal-value" type="text"
name="ideal_value" style="width:100%;"
         value="{{ ind.ideal_value or '' }}">
  <p class="muted" style="margin:4px 0 0;">
    如土石比 0.8、土层厚度 20、pH 5~9、植被覆盖度 75。
  </p>
</div>
<fieldset style="border:1px solid
var(--line);border-radius:6px;padding:10px
14px;margin:0 0 12px;">
  <legend style="padding:0 6px;font-weight:600;">启用该指标的微
地质单元</legend>
  {% for t in unit_types %}
  <label class="checkbox" style="margin-right:18px;">
    <input type="checkbox" name="enabled_units"
value="{{ t }}"
           {% if t in ind.units_list() %}checked{%
endif %}>
    <span>{{ t }}</span>
  </label>
  {% endfor %}
  <p class="muted" style="margin:8px 0 0;">
    论文设定：岩质边坡不启用土壤指标，平台不启用地形指标。
  </p>
</fieldset>
<div class="field">
  <label class="checkbox" for="is-constraint">
    <input id="is-constraint" type="checkbox"
name="is_constraint" value="1"
           {% if ind.is_constraint %}checked{% endif %}>
    <span>同时作为限制性指标（参与一票否决判定）</span>
```

```
        </label>                                                                    8251
      </div>                                                                        8252
      <div class="field">                                                           8253
        <label class="field__label" for="description">取值依据说明</              8254
      label>                                                                        8255
        <textarea id="description" name="description"                               8256
      style="min-height:88px;">{{ ind.description or ''                             8257
      }}</textarea>                                                                 8258
      </div>                                                                        8259
      <p class="actions" style="margin-top:16px;">                                  8260
        <button type="submit" class="btn">保存</button>                             8261
        <a class="btn btn--ghost" href="/ahp/indicators">返回指标列                  8262
      表</a>                                                                        8263
      </p>                                                                          8264
    </form>                                                                         8265
</div>                                                                              8266
{% endblock %}                                                                      8267
# ==== app/templates/ahp/experts.html ====                                          8268
{% extends "base.html" %}                                                           8269
{% block title %}多专家判断矩阵{% endblock %}                                       8270
{% block content %}                                                                 8271
<div class="card">                                                                  8272
  <h2>多专家判断矩阵 · 录入与汇总</h2>                                             8273
  <form method="get" class="actions" style="margin-bottom:14                        8274
    px;">                                                                           8275
    <div>                                                                           8276
      <label class="field__label" for="unit-type">微地质单元类型</l                 8277
    abel>                                                                           8278
      <select id="unit-type" name="unit_type"                                       8279
    onchange="this.form.submit()">                                                  8280
        {% for t in ["岩质边坡", "碎石边坡", "平台"] %}                             8281
        <option value="{{ t }}" {% if unit_type == t                                8282
    %}selected{% endif %}>{{ t }}</option>                                          8283
        {% endfor %}                                                                8284
      </select>                                                                     8285
    </div>                                                                          8286
    <div>                                                                           8287
      <label class="field__label" for="level">指标层级</label>                     8288
      <select id="level" name="level"                                               8289
    onchange="this.form.submit()">                                                  8290
        <option value="1" {% if level == 1 %}selected{%                             8291
    endif %}>一级指标</option>                                                      8292
        <option value="2" {% if level == 2 %}selected{%                             8293
    endif %}>二级指标</option>                                                      8294
      </select>                                                                     8295
    </div>                                                                          8296
  </form>                                                                           8297
  <p class="muted">                                                                 8298
    当前范围：<strong>{{ unit_type }}</strong> ·                                    8299
    {% if level == 1 %}一级指标{% else %}二级指标（{{ l1 }}）{% endif              8300
```

```
    %}。                                                                    8301
    若选二级指标，请先到                                                    8302
    <a href="/ahp/">权重计算</a> 选择一级指标后进入本页。                   8303
</p>                                                                        8304
{% if rows %}                                                               8305
<table>                                                                     8306
    <caption class="muted">已录入 {{ rows|length }}                         8307
    位专家的矩阵</caption>                                                  8308
    <thead>                                                                 8309
      <tr>                                                                  8310
        <th scope="col">专家</th>                                           8311
        <th scope="col">矩阵规模</th>                                       8312
        <th scope="col">该矩阵 CR</th>                                      8313
        <th scope="col">一致性</th>                                         8314
        <th scope="col">录入时间</th>                                       8315
        <th scope="col">操作</th>                                           8316
      </tr>                                                                 8317
    </thead>                                                                8318
    <tbody>                                                                 8319
      {% for r in rows %}                                                   8320
      <tr>                                                                  8321
        <td>{{ r.expert_name }}</td>                                        8322
        <td class="num">{{ n }} × {{ n }}</td>                              8323
        <td class="mono">{{ "%.4f"|format(r.cr) if r.cr is                  8324
    not none else "—" }}</td>                                               8325
        <td>                                                                8326
          {% if r.cr is not none and r.cr < 0.1 %}<span                     8327
    class="tag tag--ok">通过</span>                                         8328
          {% else %}<span class="tag tag--warn">不通过</span>{%             8329
    endif %}                                                                8330
        </td>                                                               8331
        <td class="mono nowrap">{{ r.created_at.strftime("%Y                8332
    -%m-%d %H:%M") if r.created_at else "—" }}</td>                         8333
        <td>                                                                8334
          <form method="post" action="/ahp/experts/{{ r.id                  8335
    }}/delete"                                                              8336
                onsubmit="return confirm('确定删除专家「{{                  8337
    r.expert_name }}」的矩阵吗？');">                                       8338
            {{ form.hidden_tag() }}                                         8339
            <button type="submit" class="btn btn--sm                        8340
    btn--danger">删除</button>                                              8341
          </form>                                                           8342
        </td>                                                               8343
      </tr>                                                                 8344
      {% endfor %}                                                          8345
    </tbody>                                                                8346
</table>                                                                    8347
<p class="actions" style="margin-bottom:10px;">                             8348
    <a class="btn" href="/ahp/experts/aggregate?unit_type={{                8349
    unit_type }}&level={{ level }}{% if l1 %}&l1={{ l1 }}{%                 8350
```

```
    endif %}">                                                          8351
      汇总计算（几何平均）                                              8352
    </a>                                                                8353
  </p>                                                                  8354
  {% else %}                                                            8355
  <p class="muted">尚未录入任何专家矩阵。请在下方录入第一位专家的判断矩阵。</p>  8356
  {% endif %}                                                           8357
  <h3>录入一位专家的判断矩阵</h3>                                       8358
  <form method="post">                                                  8359
    {{ form.hidden_tag() }}                                             8360
    <input type="hidden" name="action" value="add">                     8361
    <input type="hidden" name="unit_type" value="{{                     8362
    unit_type }}">                                                      8363
    <input type="hidden" name="level" value="{{ level }}">              8364
    {% if l1 %}<input type="hidden" name="l1" value="{{ l1              8365
    }}">{% endif %}                                                     8366
    <div class="field">                                                 8367
      <label class="field__label" for="expert-name">专家姓名</label>    8368
    >                                                                   8369
      <input id="expert-name" type="text" name="expert_name"            8370
             style="width:220px;" placeholder="如：张工">               8371
    </div>                                                              8372
    <table style="width:auto;">                                         8373
      <thead>                                                           8374
        <tr>                                                            8375
          <th></th>                                                     8376
          {% for name in names %}<th scope="col">{{ name                8377
    }}</th>{% endfor %}                                                 8378
        </tr>                                                           8379
      </thead>                                                          8380
      <tbody>                                                           8381
        {% for i in range(n) %}                                         8382
        <tr>                                                            8383
          <th scope="row">{{ names[i] }}</th>                           8384
          {% for j in range(n) %}                                       8385
            {% if i == j %}                                             8386
            <td class="muted num">1</td>                                8387
            {% elif i < j %}                                            8388
            <td>                                                        8389
              <label class="muted" style="position:absolute;            8390
    left:-9999px;"                                                      8391
                     for="e-{{ i }}-{{ j }}">{{ names[i] }}             8392
    相对 {{ names[j] }} 的重要性</label>                                8393
              <input id="e-{{ i }}-{{ j }}" type="number"               8394
    step="0.01" min="0.01"                                              8395
                     name="m_{{ i }}_{{ j }}" value="{{                 8396
    default[i][j] }}">                                                  8397
            </td>                                                       8398
            {% else %}                                                  8399
            <td class="muted num">{{ default[i][j] }}</td>              8400
```

```
                {% endif %}                                                    8401
              {% endfor %}                                                     8402
            </tr>                                                              8403
            {% endfor %}                                                       8404
          </tbody>                                                             8405
        </table>                                                               8406
        <p class="actions" style="margin-top:12px;">                           8407
          <button type="submit" class="btn">录入该专家矩阵</button>            8408
          <a class="btn btn--ghost" href="/ahp/indicators">指标维护</a>        8409
        </p>                                                                   8410
          <a class="btn btn--ghost" href="/ahp/">返回权重计算</a>              8411
        </p>                                                                   8412
    </form>                                                                    8413
</div>                                                                         8414
{% endblock %}                                                                 8415
# ==== app/templates/ahp/experts_aggregate.html ====                           8416
{% extends "base.html" %}                                                      8417
{% block title %}多专家矩阵汇总{% endblock %}                                  8418
{% block content %}                                                            8419
<div class="card">                                                             8420
    <h2>多专家矩阵汇总 · {{ unit_type }} · {% if level == 1 %}一级指标{%      8421
      else %}二级指标（{{ l1 }}）{% endif %}</h2>                              8422
    {% if not result %}                                                        8423
    <p class="muted">                                                          8424
      该范围内尚未录入专家矩阵。                                               8425
      <a href="/ahp/experts?unit_type={{ unit_type                             8426
      }}&level={{ level }}{% if l1 %}&l1={{ l1 }}{% endif                      8427
      %}">去录入</a>                                                           8428
    </p>                                                                       8429
    {% else %}                                                                 8430
    <div class="stat-row">                                                     8431
      <div>                                                                    8432
        <div class="stat__label">参与汇总的专家</div>                          8433
        <div class="stat__value">{{ result.count }} 位</div>                   8434
      </div>                                                                   8435
      <div>                                                                    8436
        <div class="stat__label">λmax</div>                                    8437
        <div class="stat__value mono">{{ result.lambda_max                     8438
      }}</div>                                                                 8439
      </div>                                                                   8440
      <div>                                                                    8441
        <div class="stat__label">CI</div>                                      8442
        <div class="stat__value mono">{{                                       8443
      "%.4f"|format(result.ci) }}</div>                                        8444
      </div>                                                                   8445
      <div>                                                                    8446
        <div class="stat__label">CR（要求 &lt; 0.1）</div>                     8447
        <div class="stat__value mono">{{                                       8448
      "%.4f"|format(result.cr) }}</div>                                        8449
      </div>                                                                   8450
```

```
    <div>                                                                    8451
      <div class="stat__label">一致性检验</div>                              8452
      <div class="stat__value">                                              8453
        {% if result.consistent %}<span class="ok">通过</span>               8454
        {% else %}<span class="warn">不通过</span>{% endif %}                8455
      </div>                                                                 8456
    </div>                                                                   8457
  </div>                                                                     8458
  <h3>汇总后权重</h3>                                                        8459
  <table style="width:auto;">                                                8460
    <thead>                                                                  8461
      <tr>                                                                   8462
        <th scope="col">指标</th>                                            8463
        {% for name in names %}<th scope="col">{{ name                       8464
    }}</th>{% endfor %}                                                      8465
      </tr>                                                                  8466
    </thead>                                                                 8467
    <tbody>                                                                  8468
      <tr>                                                                   8469
        <th scope="row">权重</th>                                            8470
        {% for w in result.weights %}<td class="mono">{{ w                   8471
    }}</td>{% endfor %}                                                      8472
      </tr>                                                                  8473
    </tbody>                                                                 8474
  </table>                                                                   8475
  <h3>各专家矩阵的一致性</h3>                                                8476
  <table style="width:auto;">                                                8477
    <thead>                                                                  8478
      <tr><th scope="col">专家</th><th scope="col">该矩阵                    8479
    CR</th><th scope="col">判定</th></tr>                                    8480
    </thead>                                                                 8481
    <tbody>                                                                  8482
      {% for r in rows %}                                                    8483
      <tr>                                                                   8484
        <td>{{ r.expert_name }}</td>                                         8485
        <td class="mono">{{ "%.4f"|format(r.cr) if r.cr is                   8486
    not none else "—" }}</td>                                                8487
        <td>                                                                 8488
          {% if r.cr is not none and r.cr < 0.1 %}<span                      8489
    class="ok">通过</span>                                                   8490
          {% else %}<span class="warn">不通过</span>{% endif %}              8491
        </td>                                                                8492
      </tr>                                                                  8493
      {% endfor %}                                                           8494
    </tbody>                                                                 8495
  </table>                                                                   8496
  {% if not result.consistent and result.most_inconsistent                   8497
    %}                                                                       8498
  <p style="padding:12px;border-radius:6px;background:var(--                 8499
    danger-soft);                                                            8500
```

```
                border-left:4px solid var(--danger);">
      <strong class="warn">汇总矩阵一致性不通过，建议优先调整：</strong>
      第 {{ result.most_inconsistent.i + 1 }} 行、第 {{
      result.most_inconsistent.j + 1 }} 列的比较
      （当前 <span class="mono">{{ result.most_inconsistent.value
      |round(3) }}</span>，
      理想约 <span class="mono">{{ result.most_inconsistent.expec
      ted|round(3) }}</span>）。
      该偏差通常来自个别专家打分，可先核对一致性较差的专家矩阵。
    </p>
    {% endif %}
    {% endif %}
    <p class="actions" style="margin-top:16px;">
      <form method="post"
            action="/ahp/experts?unit_type={{ unit_type
      }}&level={{ level }}{% if l1 %}&l1={{ l1 }}{% endif %}"
            style="display:inline;"
            onsubmit="return confirm('保存后，「{{ unit_type
      }}」的适宜性评价将改用这组汇总权重计算。确定继续吗？');">
        {{ form.hidden_tag() }}
        <input type="hidden" name="action" value="save_agg">
        <button type="submit" class="btn" {% if not
      result.consistent %}disabled{% endif %}>
          保存为生效权重
        </button>
      </form>
      <a class="btn btn--ghost" href="/ahp/experts?unit_type={
      { unit_type }}&level={{ level }}{% if l1 %}&l1={{ l1
      }}{% endif %}">返回录入</a>
      <a class="btn btn--ghost" href="/ahp/">返回权重计算</a>
    </p>
</div>
{% endblock %}
# ==== app/templates/indicator/index.html ====
{% extends "base.html" %}
{% from "_macros.html" import patch_guide %}
{% block title %}指标数据采集{% endblock %}
{% block content %}
<div class="card">
    <h2>指标数据采集与管理</h2>
    <p class="actions" style="margin-bottom:12px;">
      <a class="btn btn--sm" href="/indicator/import">批量导入（CSV
      / Excel）</a>
    </p>
    {{ patch_guide(projects_without_patch) }}
    {% if not patches %}
    <p class="muted">
      暂无矿山图斑。请先在「项目与矿山档案」中建立图斑，
      或运行 <span class="mono">python seed_demo.py</span>
      载入论文案例。
```

```
</p>
{% else %}
{% for p in patches %}
<div class="patch-panel">
  <div class="patch-panel__head">
    <h3 class="patch-panel__name">{{ p.name }}</h3>
    <span class="patch-panel__meta">
      所属项目：{{ p.project.name if p.project else "—" }}
    </span>
    <span class="patch-panel__meta">
      微地质单元：{{ p.units.count() }} 个
    </span>
    {% if p.units.count() %}
    <span class="patch-panel__meta patch-panel__tail">
      <a class="link-btn" href="/indicator/import">批量导入本图斑数据</a>
    </span>
    {% endif %}
  </div>
  {% if not p.units.count() %}
  <div class="notice notice--warn">
    <p class="notice__title">该图斑尚未划分微地质单元，暂不能录入指标数据。</p>
    <p style="margin:0;">
      指标数据是按微地质单元（岩质边坡 / 碎石边坡 / 平台）分别组织的，
      请先完成划分，再回到本页录入。
    </p>
    <p style="margin:10px 0 0;">
      <a class="btn btn--sm" href="/unit/{{ p.id }}">去划分微地质单元</a>
    </p>
  </div>
  {% else %}
  <table>
    <thead>
      <tr>
        <th scope="col">微地质单元</th>
        <th scope="col">面积占比</th>
        <th scope="col">数据状态</th>
        <th scope="col">操作</th>
      </tr>
    </thead>
    <tbody>
      {% for u in p.units %}
      <tr>
        <td>{{ u.unit_type }}</td>
        <td class="num">
          {% if u.ratio is not none %}{{ "%.2f%%"|format(u.ratio * 100) }}{% else %}—{% endif %}
        </td>
        <td>
```

```
                {% if u.indicator_data %}                                    8601
                <span class="tag tag--ok">已录入</span>                      8602
                {% else %}                                                   8603
                <span class="tag tag--muted">未录入</span>                   8604
                {% endif %}                                                  8605
              </td>                                                          8606
              <td class="nowrap">                                            8607
                <span class="row-actions">                                   8608
                  <a class="link-btn" href="/indicator/{{ u.id               8609
    }}">录入 / 编辑</a>                                                      8610
                </span>                                                      8611
              </td>                                                          8612
            </tr>                                                            8613
            {% endfor %}                                                     8614
          </tbody>                                                           8615
        </table>                                                             8616
        {% endif %}                                                          8617
      </div>                                                                 8618
      {% endfor %}                                                           8619
      {% endif %}                                                            8620
    </div>                                                                   8621
    {% endblock %}                                                           8622
    # ==== app/templates/indicator/edit.html ====                            8623
    {% extends "base.html" %}                                                8624
    {% block title %}指标数据录入 · {{ unit.unit_type }}{% endblock %}       8625
    {% block content %}                                                      8626
    <div class="card" style="max-width:820px;">                              8627
      <h2>指标数据录入 · {{ unit.unit_type }}</h2>                           8628
      <p class="muted">                                                      8629
        输入框留空表示该指标「不适用」，与「实测值为 0」含义不同，计算时会被跳过。  8630
        岩质边坡不设土壤指标、平台不设地形指标，系统已按单元类型隐藏不相关字段。    8631
      </p>                                                                   8632
      <form method="post">                                                   8633
        {{ form.hidden_tag() }}                                              8634
        {% for group_name, fields, has_hm in groups %}                       8635
        <h3>{{ group_name }}</h3>                                            8636
        {% for fname, cname, unit_ in fields %}                              8637
        <div class="field">                                                  8638
          <label class="field__label" for="f-{{ fname }}">                   8639
            {{ cname }}{% if unit_ %} ({{ unit_ }}) {% endif %}              8640
          </label>                                                           8641
          {% if fname == "aspect" %}                                         8642
          <select id="f-{{ fname }}" name="aspect">                          8643
            <option value="">— 不适用 —</option>                             8644
            {% for a in aspect_types %}                                      8645
            <option value="{{ a }}" {% if current.get("aspect")              8646
        == a %}selected{% endif %}>{{ a }}</option>                          8647
            {% endfor %}                                                     8648
          </select>                                                          8649
          {% else %}                                                         8650
```

```
        <input id="f-{{ fname }}" type="number" step="any"
      name="{{ fname }}"
             value="{{ current.get(fname) if
      current.get(fname) is not none else '' }}">
        {% endif %}
      </div>
      {% endfor %}
      {% endfor %}
      {% for group_name, fields, has_hm in groups %}
        {% if has_hm %}
        <h3>重金属含量（mg/kg）</h3>
        <p class="muted">未检测的金属可留空，系统按论文注释视为未受污染处理。</p>
        <table style="width:auto;">
          <thead>
            <tr>
              {% for m in heavy_metals %}<th scope="col">{{ m
      }}</th>{% endfor %}
            </tr>
          </thead>
          <tbody>
            <tr>
              {% for m in heavy_metals %}
              <td>
                <label class="muted" style="position:absolute;
      left:-9999px;"
                       for="hm-{{ m }}">{{ m }} 含量</label>
                <input id="hm-{{ m }}" type="number"
      step="any" name="hm_{{ m }}"
                       value="{{ current.get('heavy',
      {}).get(m) if current.get('heavy', {}).get(m) is not
      none else '' }}">
              </td>
              {% endfor %}
            </tr>
          </tbody>
        </table>
        {% endif %}
      {% endfor %}
      <p class="actions" style="margin-top:18px;">
        <button type="submit" class="btn">保存</button>
        <a class="btn btn--ghost" href="/indicator/">返回</a>
      </p>
    </form>
</div>
{% endblock %}
# ==== app/templates/indicator/import.html ====
{% extends "base.html" %}
{% block title %}批量导入指标数据{% endblock %}
{% block content %}
<div class="card" style="max-width:860px;">
```

```
<h2>批量导入指标数据</h2>
<p class="muted">
  支持 CSV 与 Excel（.xlsx/.xls）。文件每行对应一个微地质单元，
  系统按「单元类型」匹配图斑内已划分的单元后写入数据。
  先下载模板按格式填写，可减少出错。
</p>
<p class="actions" style="margin-bottom:14px;">
  <a class="btn btn--sm btn--ghost"
  href="/indicator/import/template">下载导入模板（CSV）</a>
  <a class="btn btn--sm btn--ghost"
  href="/indicator/">返回指标数据</a>
</p>
<form method="post" enctype="multipart/form-data">
  {{ form.hidden_tag() }}
  <div class="field">
    <label class="field__label" for="patch-id">目标矿山图斑</lab
  el>
    <select id="patch-id" name="patch_id" required>
      <option value="">— 请选择 —</option>
      {% for p in patches %}
      <option value="{{ p.id }}">{{ p.name }}（{{
  p.units.count() }} 个单元）</option>
      {% endfor %}
    </select>
  </div>
  <div class="field">
    <label class="field__label" for="datafile">数据文件</label
  >
    <input id="datafile" type="file" name="datafile"
           accept=".csv,.xlsx,.xls" required>
  </div>
  <p class="actions" style="margin-top:16px;">
    <button type="submit" class="btn">开始导入</button>
  </p>
</form>
<h3>模板列说明</h3>
<p class="muted">
  共 {{ headers|length }} 列。留空表示该指标「不适用」。
  坡向须填「阳坡 / 阴阳坡 / 阴坡」之一。未检测的重金属留空即可
  （按论文注释视为未受污染）。
</p>
<p class="mono muted" style="line-height:1.9;">{{
  headers|join(" , ") }}</p>
<h3>导入规则</h3>
<ul class="muted" style="padding-left:20px;line-height:1.9
  ;">
  <li>「单元类型」必须与图斑内已划分的单元一致，否则该行会被跳过并在结果中提示。</li>
  <li>同一单元类型若在文件中出现多行，以最后一行为准。</li>
  <li>已有数据的单元会被覆盖更新；没有数据的单元将新建记录。</li>
  <li>数值列请填纯数字，不要带单位（如填 <span class="mono">10</span> 而非
```

```
      <span class="mono">10cm</span>)。</li>
    </ul>
</div>
{% if report %}
<div class="card" style="max-width:860px;">
    <h2>导入结果</h2>
    <div class="stat-row">
      <div>
        <div class="stat__label">目标图斑</div>
        <div style="font-size:15px;">{{ report.patch }}</div>
      </div>
      <div>
        <div class="stat__label">成功导入</div>
        <div class="stat__value ok">{{ report.saved }}
      个单元</div>
      </div>
      <div>
        <div class="stat__label">问题行</div>
        <div class="stat__value {% if report.errors %}warn{%
      endif %}">{{ report.errors|length }}</div>
      </div>
    </div>
    {% if report.errors %}
    <h3>未导入的行</h3>
    <ul class="muted" style="padding-left:20px;line-height:1.9
      ;">
      {% for e in report.errors %}<li>{{ e }}</li>{% endfor %}
    </ul>
    {% else %}
    <p class="ok" style="margin-top:12px;">全部数据行均已成功导入。</p>
    {% endif %}
    <p class="actions" style="margin-top:16px;">
      <a class="btn btn--ghost" href="/indicator/">查看指标数据</a>
    </p>
</div>
{% endif %}
{% endblock %}
# ==== app/templates/evaluate/index.html ====
{% extends "base.html" %}
{% from "_macros.html" import patch_guide %}
{% block title %}适宜性定量评价{% endblock %}
{% block content %}
<div class="card">
    <h2>适宜性定量评价</h2>
    {{ patch_guide(projects_without_patch) }}
    {% if not patches %}
    <p class="muted">暂无矿山图斑。请先到「项目与矿山档案」建立图斑。</p>
    {% else %}
    <table>
      <thead>
```

```
        <tr>                                                                    8801
          <th scope="col">所属项目</th>                                          8802
          <th scope="col">矿山图斑</th>                                          8803
          <th scope="col">前置条件</th>                                          8804
          <th scope="col">最近评价结果</th>                                      8805
          <th scope="col">操作</th>                                              8806
        </tr>                                                                   8807
      </thead>                                                                  8808
      <tbody>                                                                   8809
        {% for p in patches %}                                                  8810
        {% set pr = progress.get(p.id) %}                                       8811
        <tr>                                                                    8812
          <td>{{ p.project.name if p.project else "—" }}</td>                   8813
          <td style="text-align:left;">{{ p.name }}</td>                        8814
          <td style="text-align:left;">                                         8815
            {% if pr and pr.ready %}                                            8816
            <span class="tag tag--ok">已具备</span>                             8817
            {% elif pr %}                                                       8818
            <span class="tag tag--warn">缺：{{                                  8819
      pr.blocking[0].label }}</span>                                            8820
            <span class="muted">{{ pr.blocking[0].reason                        8821
      }}</span>                                                                 8822
            {% endif %}                                                         8823
          </td>                                                                 8824
          <td>                                                                  8825
            {% if latest.get(p.id) %}                                           8826
            F = <span class="mono num">{{                                       8827
      latest[p.id].final_score }}</span>,                                       8828
            <span class="tag tag--ok">{{ latest[p.id].grade                     8829
      }}</span>                                                                 8830
            {% else %}                                                          8831
            <span class="tag tag--muted">未评价</span>                          8832
            {% endif %}                                                         8833
          </td>                                                                 8834
          <td class="nowrap">                                                   8835
            <span class="row-actions">                                          8836
              <a class="link-btn" href="/evaluate/{{ p.id                       8837
      }}">执行评价</a>                                                          8838
              {% if latest.get(p.id) %}                                         8839
              <a class="link-btn" href="/evaluate/result/{{                     8840
      latest[p.id].id }}">查看结果</a>                                          8841
              {% endif %}                                                       8842
            </span>                                                             8843
          </td>                                                                 8844
        </tr>                                                                   8845
        {% endfor %}                                                            8846
      </tbody>                                                                  8847
    </table>                                                                    8848
    {% endif %}                                                                 8849
</div>                                                                          8850
```

```
{% endblock %}
# ==== app/templates/evaluate/run.html ====
{% extends "base.html" %}
{% import "_macros.html" as ui %}
{% block title %}执行评价 · {{ patch.name }}{% endblock %}
{% block content %}
<div class="card" style="max-width:880px;">
  <h2>执行适宜性评价 · {{ patch.name }}</h2>
  {{ ui.workflow_steps(progress) }}
  {{ ui.workflow_blocking(progress) }}
  {% if progress.ready %}
  <form method="post">
    {{ form.hidden_tag() }}
    <p class="actions">
      <button type="submit" class="btn">开始评价</button>
      <a class="btn btn--ghost" href="/evaluate/">返回</a>
    </p>
  </form>
  {% else %}
  <p class="actions">
    <button type="button" class="btn" disabled>开始评价</button>
    <a class="btn btn--ghost" href="/evaluate/">返回</a>
  </p>
  {% endif %}
</div>
<div class="card" style="max-width:880px;">
  <h2>当前流程状态</h2>
  <table>
    <thead>
      <tr><th scope="col">步骤</th><th scope="col">状态</th><th
    scope="col">说明</th></tr>
    </thead>
    <tbody>
      <tr>
        <td>限制性条件判定</td>
        <td>
          {% if progress.constraint.passed %}<span
    class="tag tag--ok">已通过</span>
          {% elif progress.constraint.done %}<span
    class="tag tag--warn">未通过</span>
          {% else %}<span class="tag tag--muted">待完成</span>{
    % endif %}
        </td>
        <td style="text-align:left;">{{
    progress.constraint.detail }}</td>
      </tr>
      <tr>
        <td>划分微地质单元</td>
        <td>
          {% if progress.units.done %}<span class="tag
```

```
      tag--ok">已完成</span>                                                    8901
            {% else %}<span class="tag tag--muted">待完成</span>{ 8902
      % endif %}                                                               8903
          </td>                                                                8904
          <td style="text-align:left;">{{                                     8905
      progress.units.detail }}</td>                                            8906
        </tr>                                                                  8907
        <tr>                                                                   8908
          <td>录入指标数据</td>                                                 8909
          <td>                                                                 8910
            {% if progress.data.ready %}<span class="tag                       8911
      tag--ok">已完成</span>                                                   8912
            {% else %}<span class="tag tag--muted">待完成</span>{ 8913
      % endif %}                                                               8914
          </td>                                                                8915
          <td style="text-align:left;">{{                                     8916
      progress.data.detail }}</td>                                             8917
        </tr>                                                                  8918
      </tbody>                                                                 8919
    </table>                                                                   8920
</div>                                                                         8921
{% if weight_sources %}                                                        8922
<div class="card" style="max-width:880px;">                                    8923
    <h2>本次将采用的指标权重</h2>  <table style="max-width:720px;">            8924
      <thead>                                                                  8925
        <tr><th scope="col">微地质单元</th><th                                 8926
      scope="col">权重来源</th></tr>                                           8927
      </thead>                                                                 8928
      <tbody>                                                                  8929
        {% for ut, label in weight_sources.items() %}                          8930
        <tr>                                                                   8931
          <td style="text-align:left;">{{ ut }}</td>                           8932
          <td style="text-align:left;">                                        8933
            {% if "自定义" in label %}                                         8934
            <span class="tag tag--warn">{{ label }}</span>                     8935
            {% else %}                                                         8936
            <span class="tag tag--muted">{{ label }}</span>                    8937
            {% endif %}                                                        8938
          </td>                                                                8939
        </tr>                                                                  8940
        {% endfor %}                                                           8941
      </tbody>                                                                 8942
    </table>                                                                   8943
</div>                                                                         8944
{% endif %}                                                                    8945
{% endblock %}                                                                 8946
# ==== app/templates/evaluate/result.html ====                                 8947
{% extends "base.html" %}                                                      8948
{% block title %}评价结果{% endblock %}                                        8949
{% block content %}                                                            8950
```

```
<div class="card">
  <h2>适宜性评价结果 · {{ patch.name if patch else "" }}</h2>
  <div class="stat-row">
    <div>
      <div class="stat__label">最终评价因子 F</div>
      <div class="stat__value mono">{{ r.final_score }}</div>
    </div>
    <div>
      <div class="stat__label">适宜性等级</div>
      <div class="stat__value ok">{{ r.grade }}</div>
    </div>
    <div style="flex:1;min-width:240px;">
      <div class="stat__label">推荐措施</div>
      <div style="font-size:15px;">{{ r.measures }}</div>
    </div>
  </div>
  {% if unit_scores %}
  <h3>各微地质单元得分</h3>
  <table style="max-width:860px;">
    <thead>
      <tr>
        <th scope="col">微地质单元</th>
        <th scope="col">面积占比</th>
        <th scope="col">单元评价因子 F′</th>
        <th scope="col">加权贡献</th>
        <th scope="col">权重来源</th>
      </tr>
    </thead>
    <tbody>
      {% for u in unit_scores %}
      <tr>
        <td>{{ u.unit_type }}</td>
        <td class="num">{{ "%.2f%%"|format(u.ratio * 100) }}</td>
        <td class="mono">{{ u.unit_score }}</td>
        <td class="mono">{{ "%.4f"|format(u.ratio * u.unit_score) }}</td>
        <td class="muted">{{ u.weight_source or "论文表 3 默认权重" }}</td>
      </tr>
      {% endfor %}
    </tbody>
  </table>
  {% endif %}
  <p class="actions" style="margin-top:18px;">
    <a class="btn btn--ghost" href="/report/{{ r.patch_id }}">查看得分明细</a>
    <a class="btn btn--ghost" href="/evaluate/">返回评价列表</a>
  </p>
```

```
</div>                                                
```

```
            </td>                                                    9051
          </tr>                                                      9052
          {% endfor %}                                               9053
        </tbody>                                                     9054
      </table>                                                       9055
      {% endif %}                                                    9056
</div>                                                               9057
{% endblock %}                                                       9058
# ==== app/templates/report/detail.html ====                         9059
{% extends "base.html" %}                                            9060
{% block title %}得分明细 · {{ patch.name }}{% endblock %}           9061
{% block content %}                                                  9062
<div class="card">                                                   9063
  <h2>得分明细 · {{ patch.name }}</h2>                               9064
  <div class="stat-row">                                             9065
    <div>                                                            9066
      <div class="stat__label">最终评价因子 F</div>                  9067
      <div class="stat__value mono">{{                              9068
    "%.4f"|format(result.final_score) }}</div>                       9069
    </div>                                                           9070
    <div>                                                            9071
      <div class="stat__label">适宜性等级</div>                      9072
      <div class="stat__value ok">{{ result.grade }}</div>           9073
    </div>                                                           9074
    <div style="flex:1;min-width:240px;">                            9075
      <div class="stat__label">推荐措施</div>                        9076
      <div style="font-size:15px;">{{                                9077
    result.grade_info.measure }}</div>                               9078
    </div>                                                           9079
  </div>                                                             9080
  <p class="muted" style="margin-top:12px;">                         9081
    本页与导出报告均取自                                             9082
    <strong>{% if record %}{{ record.created_at.strftime("%Y         9083
    -%m-%d %H:%M:%S") }}{% endif %}                                  9084
    记录的那次评价结果</strong>（评价记录编号 {{ record.id if record else  9085
    "—" }}），                                                       9086
    不会因事后修改指标数据而变化。                                   9087
  </p>                                                               9088
  <p class="actions" style="margin-top:16px;">                       9089
    <a class="btn" href="/report/{{ patch.id }}/pdf">导出 PDF        9090
    报告</a>                                                         9091
    <a class="btn btn--ghost" href="/report/{{ patch.id              9092
    }}/word">导出 Word 报告</a>                                      9093
    <a class="btn btn--ghost" href="/report/">返回</a>               9094
  </p>                                                               9095
</div>                                                               9096
<div class="card">                                                   9097
  <h2>结果图表</h2>                                                  9098
  <h3>各微地质单元得分对比</h3>                                      9099
  <div>{{ chart_units|safe }}</div>                                  9100
```

```
    <h3>各一级指标平均得分</h3>
    <p class="muted">反映各一级指标对评价结果的实际影响，便于识别关键控制项。</p>
    <div>{{ chart_weight|safe }}</div>
</div>
{% for u in result.unit_results %}
<div class="card">
    <h2>{{ u.unit_type }} · 占比 {{ "%.2f%%"|format(u.ratio *
      100) }} · 单元得分 {{ "%.4f"|format(u.unit_score) }}</h2>
    {% if u.weight_source %}
    <p class="muted" style="margin-top:-6px;">权重来源：{{
      u.weight_source }}</p>
    {% endif %}
    <table>
      <thead>
        <tr>
          <th scope="col">一级指标</th>
          <th scope="col">二级指标</th>
          <th scope="col">实测值</th>
          <th scope="col">标准化值</th>
          <th scope="col">权重</th>
          <th scope="col">贡献</th>
        </tr>
      </thead>
      <tbody>
        {% for d in u.details %}
        <tr>
          <td>{{ d.l1 }}</td>
          <td style="text-align:left;">{{ d.l2 }}</td>
          <td class="mono" style="text-align:left;">{{ "—" if
      d.raw is none else d.raw }}</td>
          <td class="mono">{{ "—" if d.std_value is none else
      "%.4f"|format(d.std_value) }}</td>
          <td class="mono">{{ "%.4f"|format(d.weight) }}</td>
          <td class="mono">{{ "—" if d.contribution is none
      else "%.4f"|format(d.contribution) }}</td>
        </tr>
        {% endfor %}
      </tbody>
    </table>
    <p class="muted" style="margin:0;">
      一级指标得分：
      {% for k, v in u.l1_scores.items() %}{{ k }} = <span
      class="num">{{ "%.4f"|format(v) }}</span>{% if not
      loop.last %};{% endif %}{% endfor %}
    </p>
</div>
{% endfor %}
{% endblock %}
# ==== app/templates/report/print.html ====
<!DOCTYPE html>
```

```
<html lang="zh-CN">
<head>
<meta charset="UTF-8">
<title>自然恢复适宜性评价报告 · {{ patch.name }}</title>
<style>
  @page { size: A4; margin: 18mm 16mm; }
  * { box-sizing: border-box; }
  body {
    margin: 0; color: #23262b; background: #fff;
    font-family: "Microsoft YaHei", "SimSun", sans-serif;
    font-size: 10.5pt; line-height: 1.65;
  }
  h1 { font-size: 17pt; text-align: center; margin: 0 0
    6px; }
  h2 { font-size: 13pt; color: #1f3a5f; margin: 20px 0 8px;
       border-bottom: 1px solid #dfe3e8; padding-bottom:
    4px; }
  h3 { font-size: 11.5pt; color: #1f3a5f; margin: 16px 0
    6px; }
  .sub { text-align: center; color: #6b7280; font-size:
    9.5pt; margin-bottom: 18px; }
  table { width: 100%; border-collapse: collapse; margin:
    8px 0 14px;
            font-variant-numeric: tabular-nums; }
  th, td { border: 1px solid #c9ccd1; padding: 4px 8px;
    text-align: center; }
  th { background: #eef2f7; font-weight: 600; }
  .left { text-align: left; }
  .mono { font-family: Consolas, "Courier New", monospace; }
  .kv { width: auto; }
  .kv th { width: 30%; text-align: left; }
  .kv td { text-align: left; }
  ul { margin: 6px 0 12px; padding-left: 20px; }
  .note { color: #6b7280; font-size: 9.5pt; }
  .chart { margin: 10px 0 16px; }
  .sig { margin-top: 28px; font-size: 10pt; }
  .avoid-break { break-inside: avoid; }
  svg { max-width: 100%; }
</style>
</head>
<body>
  <h1>历史遗留废弃矿山自然恢复适宜性评价报告</h1>
  <p class="sub">
    {{ patch.name }}
    {% if patch.admin_region %} · {{ patch.admin_region
    }}{% endif %}
    <br>历史遗留废弃矿山自然恢复适宜性评价系统 Ｖ１.０ ·
    生成时间 {{ generated_at }}
  </p>
  <h2>一、图斑基本信息</h2>
```

```
<table class="kv">
  <tr><th>图斑名称</th><td>{{ patch.name }}</td></tr>
  <tr><th>所在行政区</th><td>{{ patch.admin_region or "—"
  }}</td></tr>
  <tr><th>图斑面积（公顷）</th><td>{{ patch.area_ha if
  patch.area_ha is not none else "—" }}</td></tr>
  <tr><th>开采矿种</th><td>{{ patch.mineral_type or "—"
  }}</td></tr>
  <tr><th>开采方式</th><td>{{ patch.mining_method or "—"
  }}</td></tr>
  <tr><th>责任主体缺失</th><td>{{ "是" if patch.responsible_entit
  y_missing else "否" }}</td></tr>
  {% if reference %}
  <tr><th>参照生态系统</th>
      <td>植被覆盖度基准 {{ reference.coverage_benchmark }}%,
          多样性指数基准 {{ reference.shannon_benchmark
  }}</td></tr>
  {% endif %}
</table>
<h2>二、评价结论</h2>
<table class="kv">
  <tr><th>最终评价因子 F</th><td class="mono"><strong>{{
  "%.4f"|format(result.final_score) }}</strong></td></tr>
  <tr><th>适宜性等级</th><td><strong>{{ result.grade
  }}</strong></td></tr>
  <tr><th>判定依据</th><td>{{ result.grade_info.meaning
  }}</td></tr>
  <tr><th>推荐措施</th><td>{{ result.grade_info.measure
  }}</td></tr>
</table>
<h2>三、各微地质单元得分</h2>
<table>
  <thead>
    <tr><th>微地质单元</th><th>面积占比</th><th>单元评价因子
  F'</th><th>加权贡献</th></tr>
  </thead>
  <tbody>
    {% for u in result.unit_results %}
    <tr>
      <td>{{ u.unit_type }}</td>
      <td class="mono">{{ "%.2f%%"|format(u.ratio * 100)
  }}</td>
      <td class="mono">{{ "%.4f"|format(u.unit_score)
  }}</td>
      <td class="mono">{{ "%.4f"|format(u.weighted)
  }}</td>
    </tr>
    {% endfor %}
  </tbody>
</table>
```

```
<div class="chart">{{ chart_units|safe }}</div>
<h2>四、一级指标权重分布</h2>
<div class="chart">{{ chart_weight|safe }}</div>
<h2>五、得分明细</h2>
{% for u in result.unit_results %}
<h3>{{ u.unit_type }}（占比 {{ "%.2f%%"|format(u.ratio * 100) }}，单元得分 {{ "%.4f"|format(u.unit_score) }}）</h3>
<table>
  <thead>
    <tr>
      <th>一级指标</th><th>二级指标</th><th>实测值</th>
      <th>标准化值</th><th>权重</th><th>贡献</th>
    </tr>
  </thead>
  <tbody>
    {% for d in u.details %}
    <tr>
      <td>{{ d.l1 }}</td>
      <td class="left">{{ d.l2 }}</td>
      <td class="mono left">{{ "—" if d.raw is none else d.raw }}</td>
      <td class="mono">{{ "—" if d.std_value is none else "%.4f"|format(d.std_value) }}</td>
      <td class="mono">{{ "%.4f"|format(d.weight) }}</td>
      <td class="mono">{{ "—" if d.contribution is none else "%.4f"|format(d.contribution) }}</td>
    </tr>
    {% endfor %}
  </tbody>
</table>
<p class="note">
  一级指标得分：
  {% for k, v in u.l1_scores.items() %}{{ k }} = {{ "%.4f"|format(v) }}{% if not loop.last %}；{% endif %}{% endfor %}
</p>
{% endfor %}
<h2>六、评价依据说明</h2>
<ul class="note">
  <li>指标标准化按论文式（8） 执行，按指标性质分为越大越优、越小越优、适中型与离散赋值四类。</li>
  <li>单元评价因子按论文式（9） 由二级指标权重加权求得；整矿评价因子由各微地质单元按面积占比加权合成。</li>
  <li>适宜性等级按论文表 4 划分：F&lt;0.3 不适宜、0.3≤F≤0.7 基本适宜、F≥0.7 适宜。</li>
  <li>指标权重取自论文表 3，按岩质边坡、碎石边坡、平台三类微地质单元分别确定。</li>
</ul>
<p class="sig">
  评价人（签字）：_____　　日期：_____
</p>
```

```
</body>                                                                    9301
</html>                                                                    9302
# ==== app/templates/search.html ====                                      9303
{% extends "base.html" %}                                                  9304
{% from "_macros.html" import icon %}                                      9305
{% block title %}搜索 · 历史遗留废弃矿山自然恢复适宜性评价系统{% endblock %}      9306
{% block content %}                                                        9307
<div class="card">                                                         9308
  <h2>搜索结果</h2>                                                        9309
  {% if not q %}                                                           9310
  <p class="muted">请输入关键词。可检索项目名称与编号、图斑名称与行政区/矿种、   9311
    标准名称与文号。</p>                                                   9312
  {% else %}                                                               9313
  <p class="muted">                                                        9314
    关键词「<strong>{{ q }}</strong>」共找到                               9315
    <span class="num">{{ total }}</span> 条结果。                          9316
  </p>                                                                     9317
  {% if not total %}                                                       9318
  <p class="muted">没有匹配的内容。换个词试试，或检查是否有错别字。</p>      9319
  {% endif %}                                                              9320
  {% endif %}                                                              9321
</div>                                                                     9322
{% if projects %}                                                          9323
<div class="card">                                                         9324
  <h2>评价项目（{{ projects|length }}）</h2>                               9325
  <table>                                                                  9326
    <thead>                                                                9327
      <tr>                                                                 9328
        <th scope="col">项目名称</th>                                      9329
        <th scope="col">编号</th>                                          9330
        <th scope="col">状态</th>                                          9331
        <th scope="col">图斑数</th>                                        9332
        <th scope="col">操作</th>                                          9333
      </tr>                                                                9334
    </thead>                                                               9335
    <tbody>                                                                9336
      {% for p in projects %}                                              9337
      <tr>                                                                 9338
        <td style="text-align:left;">{{ p.name }}</td>                     9339
        <td class="mono">{{ p.code or "—" }}</td>                          9340
        <td><span class="tag tag--muted">{{ p.status                       9341
  }}</span></td>                                                           9342
        <td class="num">{{ p.patches.count() }}</td>                       9343
        <td class="nowrap">                                                9344
          <span class="row-actions">                                       9345
            <a class="link-btn" href="/project/{{ p.id                     9346
  }}/patches">矿山图斑</a>                                                 9347
            <a class="link-btn" href="/project/{{ p.id                     9348
  }}/edit">编辑</a>                                                        9349
          </span>                                                          9350
```

```
            </td>                                                                                    9351
          </tr>                                                                                      9352
          {% endfor %}                                                                               9353
        </tbody>                                                                                     9354
      </table>                                                                                       9355
</div>                                                                                               9356
{% endif %}                                                                                          9357
{% if patches %}                                                                                     9358
<div class="card">                                                                                   9359
  <h2>矿山图斑（{{ patches|length }}）</h2>                                                          9360
  <table>                                                                                            9361
    <thead>                                                                                          9362
      <tr>                                                                                           9363
        <th scope="col">所属项目</th>                                                                9364
        <th scope="col">图斑名称</th>                                                                9365
        <th scope="col">行政区</th>                                                                  9366
        <th scope="col">矿种</th>                                                                    9367
        <th scope="col">操作</th>                                                                    9368
      </tr>                                                                                          9369
    </thead>                                                                                         9370
    <tbody>                                                                                          9371
      {% for m in patches %}                                                                         9372
      <tr>                                                                                           9373
        <td>{{ m.project.name if m.project else "—" }}</td>                                          9374
        <td style="text-align:left;">{{ m.name }}</td>                                               9375
        <td>{{ m.admin_region or "—" }}</td>                                                         9376
        <td>{{ m.mineral_type or "—" }}</td>                                                         9377
        <td class="nowrap">                                                                          9378
          <span class="row-actions">                                                                 9379
            <a class="link-btn" href="/constraint/{{ m.id                                            9380
  }}">限制性判定</a>                                                                               9381
            <a class="link-btn" href="/unit/{{ m.id                                                  9382
  }}">微地质单元</a>                                                                               9383
            <a class="link-btn" href="/evaluate/{{ m.id                                              9384
  }}">执行评价</a>                                                                                 9385
          </span>                                                                                    9386
        </td>                                                                                        9387
      </tr>                                                                                          9388
      {% endfor %}                                                                                   9389
    </tbody>                                                                                         9390
  </table>                                                                                           9391
</div>                                                                                               9392
{% endif %}                                                                                          9393
{% if standards %}                                                                                   9394
<div class="card">                                                                                   9395
  <h2>标准与政策文件（{{ standards|length }}）</h2>                                                  9396
  <table>                                                                                            9397
    <thead>                                                                                          9398
      <tr>                                                                                           9399
        <th scope="col">类别</th>                                                                    9400
```

```
            <th scope="col">省份</th>                                    9401
            <th scope="col">名称</th>                                    9402
            <th scope="col">文号</th>                                    9403
            <th scope="col">操作</th>                                    9404
          </tr>                                                          9405
        </thead>                                                         9406
        <tbody>                                                          9407
          {% for s in standards %}                                       9408
          <tr>                                                           9409
            <td>{{ s.category }}</td>                                    9410
            <td>{{ s.province }}</td>                                    9411
            <td style="text-align:left;">{{ s.name }}</td>               9412
            <td class="mono">{{ s.doc_number or "—" }}</td>              9413
            <td class="nowrap">                                          9414
              <span class="row-actions">                                 9415
                <a class="link-btn" href="/standard/{{ s.id              9416
        }}/rules">规则</a>                                               9417
              </span>                                                    9418
            </td>                                                        9419
          </tr>                                                          9420
          {% endfor %}                                                   9421
        </tbody>                                                         9422
      </table>                                                           9423
    </div>                                                               9424
    {% endif %}                                                          9425
    {% endblock %}                                                       9426
    # ==== seed_demo.py ====                                             9427
    # -*- coding: utf-8 -*-                                              9428
    """内置玉田县案例：把论文第 5 节案例数据写入数据库。                 9429
    用于功能验证与操作演示。数据逐字取自论文表 5 与第 5 节。             9430
    运行：python seed_demo.py                幂等，只补写缺失的演示数据，9431
                                                 不会删除用户自建的项目与图斑9432
          python seed_demo.py --reset        连同标准库与指标定义一并重建9433
    """                                                                  9434
    import json                                                          9435
    import sys                                                           9436
    sys.stdout.reconfigure(encoding="utf-8")                             9437
    from app import create_app                                           9438
    from app.models import (db, User, Project, MinePatch,                9439
        ReferenceEcosystem,                                              9440
                                MicroUnit, IndicatorData, Standard,      9441
        StandardRule,                                                    9442
                                ConstraintCheck, EvaluationResult,       9443
        ScoreDetail)                                                     9444
    app = create_app()                                                   9445
    # 内置演示案例的项目编号，用于区分「演示数据」与「用户自建数据」    9446
    DEMO_CODE = "DEMO-001"                                               9447
    # 论文表 1：11 项地方标准与政策文件                                  9448
    STANDARDS = [                                                        9449
        dict(category="地方标准", province="辽宁",                       9450
```

```
    name="矿山废弃地植被自然恢复技术规程",
        doc_number="DB21/T 3702—2023",
    publish_date="2023-03-30",
        implement_date="2023-04-30",
        summary="首个正式发布的矿山自然恢复地方标准；提出必要条件和充分条件",
        rules=[("植被覆盖度", ">=", "50", "必要条件", None),
               ("无地质灾害", "==", "是", "必要条件", None),
               ("无水土污染", "==", "是", "必要条件", None),
               ("与参照系差距", "<", "20", "充分条件", "%")]),
    dict(category="地方标准", province="湖北",
    name="废弃露天矿山生态环境自然恢复标准",
        doc_number="DB42/T 2421—2025",
    publish_date="2025-08-20",
        implement_date="2025-10-20",
        summary="结合矿山生态需求进行自然恢复适宜性分级判断",
        rules=[("自然恢复适宜性分级", "==", "是", "必要条件", None)]),
    dict(category="地方标准", province="内蒙古",
    name="历史遗留废弃矿山自然恢复判定规范",
        doc_number="DB15/T 4280—2026",
    publish_date="2026-01-30",
        implement_date="2026-03-01",
        summary="调查内容包含参照生态系统调查",
        rules=[("地质灾害发育程度", "<=", "弱发育", "必要条件", None),
               ("地形地貌与周边协调", "==", "是", "必要条件", None),
               ("有利于植被恢复的表层土壤", "==", "是", "必要条件", None),
               ("已形成稳定植被覆盖", "==", "是", "必要条件", None)]),
    dict(category="地方标准", province="江西",
    name="废弃矿山自然恢复技术规范",
        doc_number="DB36/T 2221—2026",
    publish_date="2026-02-02",
        implement_date="2026-08-01",
        summary="对参照生态系统进行了要求；提出自然恢复适宜性指数",
        rules=[("参照生态系统调查", "==", "是", "必要条件", None),
               ("自然恢复适宜性指数", ">=", "阈值", "充分条件", None)]),
    dict(category="地方标准", province="宁夏", name="矿山自然恢复认定规程",
        doc_number=None, publish_date="2026-06-18",
    implement_date=None,
        summary="分区差异化：三个一级生态修复分区及 22 个二级分区",
        rules=[("无地质灾害风险", "==", "是", "必要条件", None),
               ("植被覆盖度趋势", "==", "稳定或正增长", "必要条件", None),
               ("与参照系差距", "<", "20", "充分条件", "%")]),
    dict(category="政策文件", province="安徽",
        name="安徽省自然资源厅关于规范废弃露天矿山采取自然恢复方式进行治理的通知",
        doc_number="皖自然资修函[2020]34号",
    publish_date="2020-03-03", implement_date=None,
        summary="自然恢复条件：责任主体为政府、无地质灾害、无水土污染、不在三线三边、水土条件良好",
        rules=[("责任主体", "==", "政府", "必要条件", None),
               ("无地质灾害", "==", "是", "必要条件", None),
               ("无水土污染", "==", "是", "必要条件", None)]),
    dict(category="政策文件", province="山东",
```

```
            name="山东省自然资源厅关于进一步规范历史遗留废弃露天开采矿山采取自然恢复方式进行治理的通知",
            doc_number="鲁自然资字[２０２２]１０４号",
        publish_date="２０２２-０８-０２", implement_date=None,
            summary="条件与安徽省相同，另增加自然恢复认定标准",
            rules=[("无地质灾害", "==", "是", "必要条件", None),
                   ("无水土污染", "==", "是", "必要条件", None)]),
        dict(category="政策文件", province="湖南",
            name="关于印发《湖南省历史遗留矿山自然恢复指导意见（试行）》的通知",
            doc_number="湘自资办发[２０２３]５８号",
        publish_date="２０２３-０７-２０", implement_date=None,
            summary="自然恢复认定标准：无地质灾害或无威胁对象、无水土污染、设备拆除、符合规划、植被覆盖度
        >７５%、连续３年正向增长",
            rules=[("植被覆盖度", ">", "７５", "必要条件", "%"),
                   ("植被覆盖率趋势", "==", "连续３年正向增长", "必要条件", None),
                   ("无地质灾害", "==", "是", "必要条件", None),
                   ("符合国土空间规划", "==", "是", "必要条件", None)]),
        dict(category="政策文件", province="新疆",
            name="自治区历史遗留废弃矿山自然恢复方式销号图斑工作指南（试行）",
            doc_number="新自然资源办函〔２０２４〕２１９号",
        publish_date="２０２４-１２-２０", implement_date=None,
            summary="荒漠区沙地/裸土地/裸岩石砾地满足无地质灾害、地形协调、无水土污染即可判定",
            rules=[("植被覆盖度", ">=", "周边３０%", "必要条件", None),
                   ("无地质灾害", "==", "是", "必要条件", None),
                   ("地形平缓地貌稳定", "==", "是", "必要条件", None)]),
        dict(category="政策文件", province="广西",
            name="广西壮族自治区历史遗留矿山自然恢复技术指南",
            doc_number="桂自然资办〔２０２４〕２５０号",
        publish_date="２０２４-１２-３０", implement_date=None,
            summary="除高陡边坡和裸露岩体外，水土水热条件良好、植被自然恢复趋势明显",
            rules=[("无地质灾害", "==", "是", "必要条件", None),
                   ("植被自然恢复趋势", "==", "明显", "必要条件", None),
                   ("生产设备拆除完毕", "==", "是", "必要条件", None)]),
        dict(category="政策文件", province="西藏",
            name="西藏自治区历史遗留废弃矿山生态自然恢复认定指导意见（试行）",
            doc_number="藏自然资发〔２０２５〕１９号",
        publish_date="２０２５-０９-２４", implement_date=None,
            summary="含修复成本高、生态修复效益不显著等适用条件",
            rules=[("责任主体", "==", "政府", "必要条件", None),
                   ("地形地貌完整与周边协调", "==", "是", "必要条件", None),
                   ("无地质灾害", "==", "是", "必要条件", None)]),
]
CASE = {
    "name": "玉田县典型历史遗留废弃矿山案例",
    "patch_name": "玉田县孤树镇白云岩采石场",
    "area_ha": 5.6,
    "mineral_type": "白云岩",
    "units": [
        {
            "unit_type": "岩质边坡", "ratio": 0.1386,
            "data": {"coverage": 68.88, "shannon": 0.00,
```

```
                    "slope": 63.33, "aspect": "阴阳坡"},
            },
            {
                "unit_type": "碎石边坡", "ratio": 0.6295,
                "data": {"soil_ratio": 0.34, "thickness": 10,
        "ph": 9.24,
                    "organic_matter": 4.89,
                    "heavy_metals": {"Hg": 0.05, "As":
        5.17, "Ni": 12.60,
                                     "Cu": 11.70, "Cd":
        0.04, "Pb": 6.50},
                    "coverage": 87.92, "shannon": 0.60,
                    "slope": 28.75, "aspect": "阳坡"},
            },
            {
                "unit_type": "平台", "ratio": 0.2319,
                "data": {"soil_ratio": 0.56, "thickness": 30,
        "ph": 8.72,
                    "organic_matter": 10.92,
                    "heavy_metals": {"Hg": 0.02, "As":
        3.13, "Ni": 12.33,
                                     "Cu": 11.14, "Cd":
        0.09, "Pb": 9.98},
                    "coverage": 58.42, "shannon": 0.72},
            },
        ],
    }
    def clean_case(reset=False):
        """清理数据。
        默认只清理内置演示案例（code = DEMO-001）自身的数据，**不触碰
        用户自己建立的项目与图斑**。加 --reset 时才额外清空标准库与指标定义，
        以便把演示环境恢复到初始状态。
        注意：不要在此处执行 Project.query.delete()，否则用户新建的项目
        会在每次启动时被删除。
        """
        from app.models import EvaluationResult, ScoreDetail,
        ConstraintCheck
        demo_projects = Project.query.filter_by(code=DEMO_CODE).
        all()
        demo_ids = [p.id for p in demo_projects]
        patch_ids = []
        if demo_ids:
            patch_ids = [p.id for p in MinePatch.query.filter(
                MinePatch.project_id.in_(demo_ids)).all()]
        if patch_ids:
            unit_ids = [u.id for u in MicroUnit.query.filter(
                MicroUnit.patch_id.in_(patch_ids)).all()]
            if unit_ids:
                IndicatorData.query.filter(
                    IndicatorData.unit_id.in_(unit_ids)).delete(
```

```
                synchronize_session=False)
        result_ids = [r.id for r in EvaluationResult.query.f
    ilter(
            EvaluationResult.patch_id.in_(patch_ids)).all()]
        if result_ids:
            ScoreDetail.query.filter(
                ScoreDetail.result_id.in_(result_ids)).delet
    e(
                synchronize_session=False)
        EvaluationResult.query.filter(
            EvaluationResult.patch_id.in_(patch_ids)).delete
    (
            synchronize_session=False)
        ConstraintCheck.query.filter(
            ConstraintCheck.patch_id.in_(patch_ids)).delete(
            synchronize_session=False)
        ReferenceEcosystem.query.filter(
            ReferenceEcosystem.patch_id.in_(patch_ids)).dele
    te(
            synchronize_session=False)
        MicroUnit.query.filter(MicroUnit.patch_id.in_(patch_
    ids)).delete(
            synchronize_session=False)
        MinePatch.query.filter(MinePatch.id.in_(patch_ids)).
    delete(
            synchronize_session=False)
    if demo_ids:
        Project.query.filter(Project.id.in_(demo_ids)).delet
    e(
            synchronize_session=False)
    if reset:
        StandardRule.query.delete()
        Standard.query.delete()
    #  兜底：清理指向已不存在评价记录的孤儿得分明细，
    #  避免长期运行后 score_details 表堆积无效数据。
    _purge_orphan_details()
    db.session.commit()
    return len(demo_ids)
def demo_patch():
    """取内置演示案例的图斑。
    库中可能存在用户自建项目或验证算例（编号 VERIFY-），
    故不能简单用 MinePatch.query.first()，否则取到的对象不确定。
    没有演示案例时才退回第一个图斑。
    """
    proj = Project.query.filter_by(code=DEMO_CODE).first()
    if proj is not None:
        m = MinePatch.query.filter_by(project_id=proj.id).fi
    rst()
        if m is not None:
            return m
```

```
    return MinePatch.query.first()
def _purge_orphan_details():
    """清理指向已不存在评价记录的孤儿得分明细。
    长期运行后 score_details 表可能堆积无效明细，启动时顺带清一次。
    """
    live_ids = [r.id for r in EvaluationResult.query.all()]
    if live_ids:
        ScoreDetail.query.filter(~ScoreDetail.result_id.in_(
    live_ids)).delete(
            synchronize_session=False)
    else:
        ScoreDetail.query.delete()
    db.session.commit()
def seed(reset=False):
    with app.app_context():
        db.create_all()        # 全新环境下先建表，保证本脚本可独立运行
        _purge_orphan_details()
        admin = User.query.filter_by(username="admin").first
    ()
        if not admin:
            from werkzeug.security import
    generate_password_hash
            admin = User(username="admin",
                         password_hash=generate_password_has
    h("admin123"),
                         role="admin")
            db.session.add(admin)
            db.session.commit()
        # 幂等：演示案例已存在就不重复写入，更不清空用户自己的项目
        existing = Project.query.filter_by(code=DEMO_CODE).f
    irst()
        if existing and not reset:
            print("演示案例已存在，跳过写入（用户自建项目不受影响）")
            _report()
            _seed_reference_data(reset=False)
            return
        clean_case(reset=reset)
        project = Project(name=CASE["name"], code=DEMO_CODE,
                          description="论文第 5 节典型案例",
    created_by=admin.id)
        db.session.add(project)
        db.session.flush()
        patch = MinePatch(project_id=project.id,
    name=CASE["patch_name"],
                          admin_region="河北省唐山市玉田县",
                          area_ha=CASE["area_ha"],
                          mineral_type=CASE["mineral_type"],
                          mining_method="露天开采",
    stop_years=None,
                          responsible_entity_missing=True)
```

```
        db.session.add(patch)                                                  9701
        db.session.flush()                                                     9702
        ref = ReferenceEcosystem(patch_id=patch.id,                            9703
                                 coverage_benchmark=75.0,                      9704
    shannon_benchmark=0.7)                                                     9705
        db.session.add(ref)                                                    9706
        for u in CASE["units"]:                                                9707
            unit = MicroUnit(patch_id=patch.id,                                9708
    unit_type=u["unit_type"],                                                  9709
                             ratio=u["ratio"], area=None)                      9710
            db.session.add(unit)                                               9711
            db.session.flush()                                                 9712
            d = u["data"]                                                      9713
            ind = IndicatorData(                                               9714
                unit_id=unit.id,                                               9715
                soil_ratio=d.get("soil_ratio"),                                9716
                thickness=d.get("thickness"),                                  9717
                ph=d.get("ph"),                                                9718
                organic_matter=d.get("organic_matter"),                        9719
                heavy_metals=json.dumps(d["heavy_metals"],                     9720
    ensure_ascii=False)                                                        9721
                             if d.get("heavy_metals") else                     9722
    None,                                                                      9723
                coverage=d.get("coverage"),                                    9724
                shannon=d.get("shannon"),                                      9725
                slope=d.get("slope"),                                          9726
                aspect=d.get("aspect"),                                        9727
            )                                                                  9728
            db.session.add(ind)                                                9729
        #  限制性条件判定记录：演示案例按论文第 5 节描述属「无地质灾害隐患点」、                  9730
        #  具备自然恢复条件，故五项全部通过。缺此记录会使后续评价被流程门禁拦住。                  9731
        db.session.add(ConstraintCheck(                                        9732
            patch_id=patch.id,                                                 9733
            policy_pass=True, location_pass=True,                              9734
    pollution_pass=True,                                                       9735
            geo_pass=True, farmland_pass=True,                                 9736
            passed=True, reason="全部通过"))                                    9737
        db.session.commit()                                                    9738
        print("已写入案例：", CASE["name"])                                      9739
        print("   图斑：", CASE["patch_name"])                                  9740
        print("   微地质单元：岩质边坡 13.86% / 碎石边坡 62.95% / 平台                 9741
    23.19%")                                                                   9742
        print("   管理员账号：admin / admin123")                                9743
        _seed_reference_data(reset=reset)                                      9744
        _report()                                                              9745
def _seed_reference_data(reset=False):                                         9746
    """写入标准库与指标定义。                                                      9747
    参考资料只在库为空（或显式 --reset）时写入，避免覆盖用户自行维护                       9748
    的标准条目与指标定义。                                                         9749
    """                                                                        9750
```

```
    if reset or Standard.query.count() == 0:                    9751
        seed_standards()                                        9752
    from app.services.indicator_service import                  9753
    seed_indicators                                             9754
    n = seed_indicators()                                       9755
    print("指标定义：%d 项" % n)                                  9756
def _report():                                                  9757
    total_p = Project.query.count()                             9758
    total_patch = MinePatch.query.count()                       9759
    print()                                                     9760
    print("当前库内：项目 %d 个、图斑 %d 个" % (total_p, total_patch))  9761
    if total_p > total_patch:                                   9762
        print("  提示：项目是归档单位，评价针对的是图斑。")              9763
        print("        新项目需先建立图斑，才会出现在后续各模块中。")     9764
def seed_standards():                                           9765
    """写入论文表 1 的 11 项标准与政策文件及其结构化规则。"""              9766
    StandardRule.query.delete()                                 9767
    Standard.query.delete()                                     9768
    db.session.commit()                                         9769
    for s in STANDARDS:                                         9770
        std = Standard(category=s["category"],                  9771
    province=s["province"],                                     9772
                       name=s["name"],                          9773
    doc_number=s["doc_number"],                                 9774
                       publish_date=s["publish_date"],          9775
                       implement_date=s["implement_date"],      9776
                       summary=s["summary"])                    9777
        db.session.add(std)                                     9778
        db.session.flush()                                      9779
        for item, op, thr, ctype, note in s["rules"]:           9780
            db.session.add(StandardRule(standard_id=std.id,     9781
    item=item,                                                  9782
                                        operator=op,            9783
    threshold=thr,                                              9784
                                                                9785
    condition_type=ctype, note=note))                           9786
    db.session.commit()                                         9787
    print("已写入标准/政策文件：%d 项，规则 %d 条"                  9788
          % (len(STANDARDS), sum(len(s["rules"]) for s in       9789
    STANDARDS)))                                                9790
if __name__ == "__main__":                                      9791
    #  --reset：把演示环境恢复到初始状态（清空标准库与指标定义后重建）。    9792
    #  无论是否 --reset，都不会删除用户自建的项目与图斑。              9793
    seed(reset="--reset" in sys.argv[1:])                       9794
#  ==== 测试/test_ahp.py ====                                   9795
#  -*- coding: utf-8 -*-                                       9796
"""地基自检：验证 AHP 引擎与应用可启动。"""                           9797
import sys                                                      9798
sys.stdout.reconfigure(encoding="utf-8")                        9799
import numpy as np                                              9800
```

```
import os
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app.services import ahp_service as ahp
LINE = "=" * 66
# ------------------------------------------------------------------
# 1） 完全一致矩阵 → 权重应精确还原，CR=0
# ------------------------------------------------------------------
print(LINE)
print("测试 1  完全一致判断矩阵的权重还原与一致性")
print(LINE)
# 由权重构造完全一致矩阵：a_ij = w_i / w_j
w_true = np.array([0.44, 0.31, 0.25])
A = np.outer(w_true, 1.0 / w_true)
r = ahp.weights_and_consistency(A)
print("  判断矩阵 A：")
for row in A:
    print("    " + "  ".join(f"{v:8.4f}" for v in row))
print()
print("  计算权重 :", "  ".join(f"{v:.4f}" for v in
    r["weights"]))
print("  真实权重 :", "  ".join(f"{v:.4f}" for v in w_true))
print("  λmax = %.6f   CI = %.3e   CR = %.3e   一致: %s"
      % (r["lambda_max"], r["ci"], r["cr"],
    r["consistent"]))
assert np.allclose(r["weights"], w_true, atol=1e-6),
    "权重未精确还原"
assert r["consistent"], "完全一致矩阵应通过检验"
# ------------------------------------------------------------------
# 2） 上三角输入 → 完整矩阵 → 一致性定位
# ------------------------------------------------------------------
print()
print(LINE)
print("测试 2  上三角录入与最不一致项定位")
print(LINE)
# 一个不太一致的 3 阶矩阵：上三角 [2, 5, 3]（a12=2, a13=5, a23=3）
M = ahp.matrix_from_upper([2, 5, 3])
r2 = ahp.weights_and_consistency(M)
bad = ahp.find_most_inconsistent(M)
print("  由 [a12=2, a13=5, a23=3] 构造矩阵：")
for row in M:
    print("    " + "  ".join(f"{v:8.4f}" for v in row))
```

```
print("   权重:", "   ".join(f"{v:.4f}" for v in r2["weights"]),
      "   CR = %.3f   一致: %s" % (r2["cr"], r2["consistent"]))
print("   最不一致比较项: 第(%d,%d) 项,当前 %.3f,理想 %.3f"
      % (bad["i"] + 1, bad["j"] + 1, bad["value"],
     bad["expected"]))
# ------------------------------------------------------------------
     ----------------
# 3) 多专家聚合(几何平均)
# ------------------------------------------------------------------
     ----------------
print()
print(LINE)
print("测试 3  多专家判断矩阵聚合(几何平均)")
print(LINE)
m1 = ahp.matrix_from_upper([2, 4, 3])
m2 = ahp.matrix_from_upper([3, 5, 2])
agg = ahp.aggregate_matrices([m1, m2])
print("   专家1 上三角 [2,4,3],专家2 上三角 [3,5,2]")
print("   聚合后上三角: a12=%.4f a13=%.4f a23=%.4f"
      % (agg[0, 1], agg[0, 2], agg[1, 2]))
assert abs(agg[0, 1] - np.sqrt(2 * 3)) < 1e-9, "几何平均错误"
# ------------------------------------------------------------------
     ----------------
# 4) 论文表 3 权重可还原性
# ------------------------------------------------------------------
     ----------------
print()
print(LINE)
print("测试 4  论文表 3 碎石边坡一级权重 (0.44, 0.31, 0.25)")
print(LINE)
j = ahp.judge_from_weights([0.44, 0.31, 0.25])
print("   λmax = %.6f   CR = %.3e   一致: %s" %
     (j["lambda_max"], j["cr"], j["consistent"]))
assert j["consistent"], "论文权重应能作为完全一致矩阵主特征向量"
# ------------------------------------------------------------------
     ----------------
# 5) 应用可启动 + 建表
# ------------------------------------------------------------------
     ----------------
print()
print(LINE)
print("测试 5  Flask 应用启动与建表")
print(LINE)
from app import create_app
from app.models import db
app = create_app()
with app.app_context():
     db.create_all()
     tables = sorted(db.metadata.tables.keys())
print("   已创建 %d 张表:" % len(tables))
```

```
for t in tables:
    print("  -", t)
print()
print(LINE)
print("全部测试通过 √")
print(LINE)
# ==== 测试/test_evaluate.py ====
# -*- coding: utf-8 -*-
"""评价引擎自检：用论文玉田县案例跑通完整计算链。"""
import sys
sys.stdout.reconfigure(encoding="utf-8")
import os
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app.services import evaluation_service as ev
from app.services import standardization as std
LINE = "=" * 70
# ------------------------------------------------------------------
# 论文第 5 节案例数据（表 5，逐字抄录）
# ------------------------------------------------------------------
# 三类单元面积占比：岩质边坡 13.86%、碎石边坡 62.95%、平台 23.19%
# 注：岩质边坡无土壤指标、平台无地形指标，对应"不适用"。
case = [
    {
        "unit_type": "岩质边坡",
        "ratio": 0.1386,
        "data": {
            "coverage": 68.88, "shannon": 0.00,
            "slope": 63.33, "aspect": "阴阳坡",
        },
    },
    {
        "unit_type": "碎石边坡",
        "ratio": 0.6295,
        "data": {
            "soil_ratio": 0.34, "thickness": 10, "ph":
9.24, "organic_matter": 4.89,
            "heavy_metals": {"Hg": 0.05, "As": 5.17, "Ni":
12.60,
                             "Cu": 11.70, "Cd": 0.04, "Pb":
6.50},
            "coverage": 87.92, "shannon": 0.60,
            "slope": 28.75, "aspect": "阳坡",
        },
```

```
    },
    {
        "unit_type": "平台",
        "ratio": 0.2319,
        "data": {
            "soil_ratio": 0.56, "thickness": 30, "ph":
8.72, "organic_matter": 10.92,
            "heavy_metals": {"Hg": 0.02, "As": 3.13, "Ni":
12.33,
                             "Cu": 11.14, "Cd": 0.09, "Pb":
9.98},
            "coverage": 58.42, "shannon": 0.72,
        },
    },
]
# ----------------------------------------------------------------
    ----------------
# 逐单元输出标准化与得分
# ----------------------------------------------------------------
    ----------------
print(LINE)
print("玉田县案例 · 评价引擎完整计算")
print(LINE)
for u in case:
    r = ev.score_unit(u["unit_type"], u["data"])
    print()
    print("  【%s】  (面积占比 %.2f%%)" % (u["unit_type"],
    u["ratio"] * 100))
    print("  " + "-" * 60)
    for l1, score in r["l1_scores"].items():
        print("    %s 得分 = %.4f" % (l1, score))
    print("    单元评价因子 F' = %.4f" % r["unit_score"])
print()
print(LINE)
print("面积加权合成（式(9)）")
print(LINE)
res = ev.score_mine(case)
total = res["final_score"]
print()
for ur in res["unit_results"]:
    print("  %s  %.2f%% × %.4f = %.4f"
          % (ur["unit_type"], ur["ratio"] * 100,
    ur["unit_score"], ur["weighted"]))
print("  " + "-" * 60)
print("  最终评价因子 F = %.4f" % total)
print()
print("  适宜性等级：%s" % res["grade"])
print("  判定依据  ：%s" % res["grade_info"]["meaning"])
print("  推荐措施  ：%s" % res["grade_info"]["measure"])
print()
```

```
print(LINE)
print("结论")
print(LINE)
print("  - 评价引擎按论文式(8)(9) 与表 3 权重完整跑通")
print("  - 坡度标准化采用论文 25° 分界的线性折减默认值（见 standardization.py）")
print("  - 最终 F ≈ %.2f，判定为「%s」，与论文「最终适宜自然恢复」的结论一致"
      % (total, res["grade"]))
print("  - 论文表 5 所载最终得分为 0.76；差异集中在坡度、坡向的标准化口径，")
print("    该口径在 standardization.std_slope / aspect_score 中集中定义、可配置")
print(LINE)
# ==== 测试/test_web_ahp.py ====
# -*- coding: utf-8 -*-
"""AHP 网页界面自检：走通选择→录入→计算 全流程。"""
import sys
sys.stdout.reconfigure(encoding="utf-8")
import os
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
LINE = "=" * 66
# 测试配置：关闭 CSRF 以简化测试客户端（真实运行仍开启 CSRF）
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///:memory:"
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
client = app.test_client()
# 建立表并创建测试用户，登录后再访问业务页面
with app.app_context():
    from app.models import db, User
    from werkzeug.security import generate_password_hash
    db.create_all()
    if not User.query.filter_by(username="admin").first():
        db.session.add(User(username="admin",
                            password_hash=generate_password_hash("admin123"),
                            role="admin"))
        db.session.commit()
client.post("/login", data={"username": "admin",
    "password": "admin123"})
print(LINE)
print("测试 1  首页与 AHP 入口")
print(LINE)
```

```
r = client.get("/")
print("  首页状态码：", r.status_code, "（应 200）")
assert r.status_code == 200
r = client.get("/ahp/")
print("  AHP 入口状态码：", r.status_code, "（应 200）")
assert r.status_code == 200
assert "微地质单元类型" in r.get_data(as_text=True)
print()
print(LINE)
print("测试 2  选择碎石边坡一级指标 → 跳转矩阵页")
print(LINE)
r = client.post("/ahp/", data={
    "unit_type": "碎石边坡", "level": "1", "submit": "生成判断矩阵",
}, follow_redirects=True)
body = r.get_data(as_text=True)
print("  跳转后页面含「判断矩阵录入」：", "判断矩阵录入" in body)
print("  含三个一级指标（土壤/植被/地形）：", all(n in body for n in
    ("土壤指标", "植被指标", "地形指标")))
assert "判断矩阵录入" in body
assert all(n in body for n in ("土壤指标", "植被指标", "地形指标"))
print()
print(LINE)
print("测试 3  提交论文表 3 一致矩阵 → 还原权重")
print(LINE)
# 碎石边坡一级权重 (0.44, 0.31, 0.25) 对应的完全一致矩阵上三角：
# a12 = 0.44/0.31 = 1.4194, a13 = 0.44/0.25 = 1.76, a23 =
    0.31/0.25 = 1.24
r = client.post("/ahp/matrix?unit_type=碎石边坡&level=1", data={
    "m_0_1": "1.4194", "m_0_2": "1.76", "m_1_2": "1.24",
}, follow_redirects=True)
body = r.get_data(as_text=True)
print("  结果页含「计算结果」：", "计算结果" in body)
print("  含权重 0.44：", "0.44" in body, " 含 0.31：", "0.31" in
    body, " 含 0.25：", "0.25" in body)
print("  含一致性检验通过：", "通过" in body)
assert "计算结果" in body
assert "0.44" in body and "0.31" in body and "0.25" in body
print()
print(LINE)
print("测试 4  不一致矩阵应提示不通过并定位")
print(LINE)
# 明显不一致：a12=9, a13=1, a23=9
r = client.post("/ahp/matrix?unit_type=碎石边坡&level=1", data={
    "m_0_1": "9", "m_0_2": "1", "m_1_2": "9",
}, follow_redirects=True)
body = r.get_data(as_text=True)
print("  含「不通过」：", "不通过" in body)
print("  含「建议优先调整」：", "建议优先调整" in body)
assert "不通过" in body
assert "建议优先调整" in body
```

```
print()
print(LINE)
print("网页界面全部测试通过 √")
print(LINE)
#  ==== 测试/test_web_modules.py ====
#  -*- coding: utf-8 -*-
"""模块自检：登录、项目档案、微地质单元、标准库、限制性判定。"""
import os
import sys
sys.stdout.reconfigure(encoding="utf-8")
#  本脚本位于 测试/  子目录，先把仓库根目录加入模块搜索路径，
#  使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
LINE = "=" * 66
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" + os.path.join(
        ROOT, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
client = app.test_client()
print(LINE)
print("测试 1  未登录应被重定向到登录页")
print(LINE)
r = client.get("/project/")
print("  /project/ 状态码:", r.status_code, " 重定向到:",
    r.headers.get("Location"))
assert r.status_code == 302 and "login" in
    r.headers.get("Location", "")
print()
print(LINE)
print("测试 2  登录")
print(LINE)
r = client.post("/login", data={"username": "admin",
    "password": "admin123"},
                follow_redirects=True)
body = r.get_data(as_text=True)
#  工作台：页首横幅 + 评价流程分区 + 数据概览
print("  登录后含「评价流程」:", "评价流程" in body)
print("  含用户名 admin:", "admin" in body)
print("  含页首横幅:", 'class="hero"' in body)
print("  含数据概览:", 'class="stat-card"' in body)
assert "评价流程" in body and 'class="hero"' in body
print()
```

```
print(LINE)
print("测试 3  项目与矿山档案")
print(LINE)
r = client.get("/project/")
body = r.get_data(as_text=True)
print("  项目列表状态码：", r.status_code)
print("  含案例项目：", "玉田县典型历史遗留废弃矿山案例" in body)
assert "玉田县典型历史遗留废弃矿山案例" in body
# 新建项目
client.post("/project/add", data={"name": "测试项目A", "code": "T-001"})
r = client.get("/project/")
print("  新建项目后含「测试项目A」：", "测试项目A" in r.get_data(as_text=True))
assert "测试项目A" in r.get_data(as_text=True)
# 清理测试数据，避免污染演示库
with app.app_context():
    from app.models import db, Project
    for p in Project.query.filter_by(name="测试项目A").all():
        db.session.delete(p)
    db.session.commit()
print("  已清理测试项目")
print()
print(LINE)
print("测试 4  微地质单元（面积占比自动计算）")
print(LINE)
from app.models import MinePatch
with app.app_context():
    patch = MinePatch.query.filter_by(name="玉田县孤树镇白云岩采石场").first()
    pid = patch.id
r = client.get("/unit/%d" % pid)
body = r.get_data(as_text=True)
print("  单元列表含三类单元：", all(t in body for t in ("岩质边坡", "碎石边坡", "平台")))
print("  含占比合计：", "占比合计" in body)
assert "岩质边坡" in body and "占比合计" in body
print()
print(LINE)
print("测试 5  标准知识库与省际口径比对")
print(LINE)
r = client.get("/standard/")
body = r.get_data(as_text=True)
print("  含辽宁/湖南/新疆标准：", all(p in body for p in ("辽宁", "湖南", "新疆")))
assert "辽宁" in body
r = client.get("/standard/compare")
body = r.get_data(as_text=True)
print("  口径比对页含「植被覆盖度」分组：", "植被覆盖度" in body)
print("  含多省份并列：", "湖南" in body and "新疆" in body)
```

```
assert "植被覆盖度" in body
print()
print(LINE)
print("测试 6  限制性指标判定（一票否决）")
print(LINE)
r = client.get("/constraint/%d" % pid)
body = r.get_data(as_text=True)
print("  含五项条件:", all(x in body for x in ("政策约束", "区位约束",
    "土壤污染", "地质灾害", "耕地属性")))
assert all(x in body for x in ("政策约束", "区位约束", "土壤污染",
    "地质灾害", "耕地属性"))
# 全部通过
client.post("/constraint/%d" % pid, data={
    "policy_pass": "1", "location_pass": "1",
    "pollution_pass": "1",
    "geo_pass": "1", "farmland_pass": "1"})
r = client.get("/constraint/%d" % pid)
body = r.get_data(as_text=True)
print("  全通过后显示「通过，可进入定量评价」:", "可进入定量评价" in body)
assert "可进入定量评价" in body
# 一项不通过
client.post("/constraint/%d" % pid, data={
    "policy_pass": "1", "location_pass": "0",
    "pollution_pass": "1",
    "geo_pass": "1", "farmland_pass": "1"})
r = client.get("/constraint/%d" % pid)
body = r.get_data(as_text=True)
print("  一项不通过后显示「不通过」:", "不通过" in body)
print("  含失败项「区位约束」:", "区位约束" in body)
assert "不通过" in body
# 恢复为全部通过，避免把演示数据留在「不通过」状态影响后续步骤
client.post("/constraint/%d" % pid, data={
    "policy_pass": "1", "location_pass": "1",
    "pollution_pass": "1",
    "geo_pass": "1", "farmland_pass": "1"})
r = client.get("/constraint/%d" % pid)
print("  已恢复为全部通过:", "可进入定量评价" in r.get_data(as_text=True))
print()
print(LINE)
print("全部测试通过 √")
print(LINE)
# ==== 测试/test_web_evaluate.py ====
# -*- coding: utf-8 -*-
"""端到端自检：录数据 → 执行评价 → 查看结果。"""
import os
import sys
sys.stdout.reconfigure(encoding="utf-8")
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__fi
```

```
    e__)))                                                          10251
if ROOT not in sys.path:                                            10252
    sys.path.insert(0, ROOT)                                        10253
from app import create_app                                          10254
LINE = "=" * 66                                                     10255
# 与真实配置共用同一个 app.db，但关闭 CSRF 便于测试客户端                  10256
class TestConfig:                                                   10257
    SECRET_KEY = "test"                                             10258
    SQLALCHEMY_DATABASE_URI = "sqlite:///" + os.path.join(          10259
        ROOT, "app", "app.db")                                      10260
    SQLALCHEMY_TRACK_MODIFICATIONS = False                          10261
    WTF_CSRF_ENABLED = False                                        10262
    TESTING = True                                                  10263
app = create_app(TestConfig)                                        10264
client = app.test_client()                                          10265
# 登录（案例种子已创建 admin 用户）                                     10266
client.post("/login", data={"username": "admin",                    10267
    "password": "admin123"})                                        10268
print(LINE)                                                         10269
print("测试 1  指标数据列表")                                          10270
print(LINE)                                                         10271
r = client.get("/indicator/")                                       10272
body = r.get_data(as_text=True)                                     10273
print("  状态码：", r.status_code)                                     10274
print("  含案例图斑：", "玉田县孤树镇白云岩采石场" in body)                 10275
print("  含三个单元：", all(n in body for n in ("岩质边坡", "碎石边坡",    10276
    "平台")))                                                       10277
assert r.status_code == 200                                         10278
assert "岩质边坡" in body and "碎石边坡" in body and "平台" in body        10279
print()                                                             10280
print(LINE)                                                         10281
print("测试 2  指标数据录入页（按单元类型显示字段）")                      10282
print(LINE)                                                         10283
from app.models import MicroUnit, MinePatch                         10284
with app.app_context():                                             10285
    unit_rock = MicroUnit.query.filter_by(unit_type="岩质边坡")10286
    first()                                                         10287
    unit_gravel = MicroUnit.query.filter_by(unit_type="碎石边坡"10288
    ).first()                                                       10289
r = client.get("/indicator/%d" % unit_rock.id)                      10290
b_rock = r.get_data(as_text=True)                                   10291
print("  岩质边坡页：无「土石比」字段：", "土石比" not in b_rock, "（应 True）")10292
print("  岩质边坡页：含「植被覆盖度」：", "植被覆盖度" in b_rock)          10293
assert "土石比" not in b_rock and "植被覆盖度" in b_rock                 10294
r = client.get("/indicator/%d" % unit_gravel.id)                    10295
b_gravel = r.get_data(as_text=True)                                 10296
print("  碎石边坡页：含「土石比」字段：", "土石比" in b_gravel)           10297
print("  碎石边坡页：含 8 种重金属：", all(m in b_gravel for m in       10298
    ("Cd", "Hg", "Pb", "Cr", "As", "Ni", "Cu", "Zn")))              10299
assert "土石比" in b_gravel and all(m in b_gravel for m in          10300
```

```
    ("Cd", "Hg", "Pb", "Cr", "As", "Ni", "Cu", "Zn"))
print()
print(LINE)
print("测试 3  执行评价")
print(LINE)
with app.app_context():
    import seed_demo
    patch = seed_demo.demo_patch()
r = client.post("/evaluate/%d" % patch.id, data={},
    follow_redirects=True)
body = r.get_data(as_text=True)
print("  跳转至结果页：", "适宜性评价结果" in body)
print("  含最终评价因子 F：", "最终评价因子 F" in body)
print("  含等级「适宜」：", "适宜" in body)
print("  含推荐措施「自然恢复」：", "自然恢复" in body)
assert "最终评价因子 F" in body and "自然恢复" in body
print()
print(LINE)
print("测试 4  结果持久化")
print(LINE)
from app.models import EvaluationResult
with app.app_context():
    res = EvaluationResult.query.order_by(EvaluationResult.i
    d.desc()).first()
    print("  最终得分 F =", res.final_score)
    print("  等级 =", res.grade)
    assert res.grade == "适宜"
    assert 0.7 <= res.final_score < 1.0
print()
print(LINE)
print("端到端全部测试通过 √")
print(LINE)
# ==== 测试/test_web_report.py ====
# -*- coding: utf-8 -*-
"""报告输出自检：得分明细页 + Word 导出。"""
import os
import sys
import io
sys.stdout.reconfigure(encoding="utf-8")
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__fi
    e__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
LINE = "=" * 66
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" + os.path.join(
```

```
        ROOT, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
client = app.test_client()
# 登录（案例种子已创建 admin 用户）
client.post("/login", data={"username": "admin",
    "password": "admin123"})
from app.models import MinePatch, EvaluationResult
with app.app_context():
    import seed_demo
    patch = seed_demo.demo_patch()
    pid = patch.id
# 报告取自已保存的评价记录，因此先按网页路径真实执行一次评价
from app.views.evaluate import load_patch_input,
    save_evaluation
with app.app_context():
    units = load_patch_input(pid)
    rec = save_evaluation(pid, units)
    rid = rec.id
    print("前置：已执行评价并保存记录 id=%d，F=%.4f（%s）"
          % (rid, rec.final_score, rec.grade))
    from app.models import ScoreDetail
    n_detail = ScoreDetail.query.filter_by(result_id=rid).co
    unt()
    print("        逐层得分明细已落库：%d 条" % n_detail)
print()
print(LINE)
print("测试 1  报告列表")
print(LINE)
r = client.get("/report/")
print("  状态码：", r.status_code)
body = r.get_data(as_text=True)
print("  含「所属项目」列：", "所属项目" in body)
assert r.status_code == 200
print()
print(LINE)
print("测试 2  得分明细页")
print(LINE)
r = client.get("/report/%d" % pid)
body = r.get_data(as_text=True)
print("  状态码：", r.status_code)
print("  含「得分明细」：", "得分明细" in body)
print("  含「最终评价因子 F」：", "最终评价因子 F" in body)
print("  含二级指标明细（植被覆盖度 / 坡度 / 坡向）：", all(n in body for n in
    ("植被覆盖度", "坡度", "坡向")))
print("  含权重 / 标准化值列：", "标准化值" in body and "贡献" in body)
assert r.status_code == 200
assert "植被覆盖度" in body and "标准化值" in body
```

```
print()
print(LINE)
print("测试 3  Word 报告导出")
print(LINE)
r = client.get("/report/%d/word" % pid)
print("  状态码:", r.status_code)
print("  Content-Type:", r.headers.get("Content-Type",
    "")[:60])
data = r.data
print("  文件大小:", len(data), "字节")
print("  是否为有效 docx(PK 头):", data[:2] == b"PK")
assert r.status_code == 200 and data[:2] == b"PK"
# 用 python-docx 打开验证内容
import docx
d = docx.Document(io.BytesIO(data))
text = "\n".join(p.text for p in d.paragraphs)
print("  含报告标题:", "自然恢复适宜性评价报告" in text)
print("  含评价结论章节:", "评价结论" in text)
print("  含得分明细章节:", "得分明细" in text)
print("  段落数:", len(d.paragraphs), " 表格数:", len(d.tables))
assert "自然恢复适宜性评价报告" in text and len(d.tables) >= 3
print()
print(LINE)
print("报告输出全部测试通过 √")
print(LINE)
# ==== 测试/test_ui_fixes.py ====
# -*- coding: utf-8 -*-
"""界面修复自检:「记住我」与各模块删除功能。"""
import os
import re
import sys
# 本脚本位于 测试/ 子目录,先把仓库根目录加入模块搜索路径,
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file
    __)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
sys.stdout.reconfigure(encoding="utf-8")
LINE = "=" * 70
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" + os.path.join(
        ROOT, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
    REMEMBER_COOKIE_DURATION = __import__("datetime").timede
    lta(days=30)
app = create_app(TestConfig)
```

```
client = app.test_client()
# ------------------------------------------------------------------
    ----------------
print(LINE)
print("一、「记住我」")
print(LINE)
html = client.get("/login").get_data(as_text=True)
# 1） 复选框必须有可点击关联的 label（这是原先失效的根因）
#     只判断「复选框与文字是否落在同一个 <label> 内」，不绑定具体类名 ——
#     否则换个样式类就会误报，而真正的回归点其实与类名无关。
labels = re.findall(r"<label[^>]*>(.*?)</label>", html,
    re.S)
wrapped = any(('name="remember"' in b and "记住我" in b) for b
    in labels)
print("  %s 「记住我」文字与复选框包在同一 label 内" % ("[符合]" if wrapped
    else "[不符合]"))
assert wrapped, "复选框未与文字关联，点击文字不会勾选"
has_for = 'for="remember"' in html
print("  %s label 带 for 属性指向复选框" % ("[符合]" if has_for else
    "[不符合]"))
# 2） 勾选后应下发带过期时间的 remember_token
c = app.test_client()
r = c.post("/login", data={"username": "admin", "password":
    "admin123",
                           "remember": "y"},
    follow_redirects=False)
cookies = r.headers.getlist("Set-Cookie")
rt = [h for h in cookies if h.startswith("remember_token=")]
print("  %s 勾选后下发 remember_token" % ("[符合]" if rt else
    "[不符合]"))
assert rt, "未下发 remember_token"
exp = re.search(r"Expires=([^;]+)", rt[0])
print("  %s 该 cookie 带过期时间：%s"
      % ("[符合]" if exp else "[不符合]", exp.group(1) if exp
    else "无（会话 cookie）"))
assert exp, "remember_token 无过期时间，关浏览器即失效"
# 3） 不勾选时不应下发 remember_token
c2 = app.test_client()
r2 = c2.post("/login", data={"username": "admin",
    "password": "admin123"},
             follow_redirects=False)
rt2 = [h for h in r2.headers.getlist("Set-Cookie") if
    h.startswith("remember_token=")]
print("  %s 未勾选时不下发 remember_token" % ("[符合]" if not rt2
    else "[不符合]"))
assert not rt2
# 4） 携带 remember_token 的新会话应自动登录
c3 = app.test_client()
c3.set_cookie("remember_token",
              rt[0].split(";")[0].split("=", 1)[1],
```

```
    domain="localhost")                                                    10501
r3 = c3.get("/", follow_redirects=False)                                   10502
print("  %s 仅凭 remember_token 访问首页可自动登录（状态码 %d）"                  10503
      % ("[符合]" if r3.status_code == 200 else "[不符合]",                 10504
    r3.status_code))                                                       10505
assert r3.status_code == 200                                               10506
# ------------------------------------------------------------------------10507
    ----------------                                                       10508
print()                                                                    10509
print(LINE)                                                                10510
print("二、删除功能")                                                          10511
print(LINE)                                                                10512
client.post("/login", data={"username": "admin",                           10513
    "password": "admin123"})                                               10514
# 项目：列表页须有删除按钮                                                     10515
html = client.get("/project/").get_data(as_text=True)                      10516
has_del = "/delete" in html and "btn--danger" in html                      10517
print("  %s 项目列表含删除按钮" % ("[符合]" if has_del else "[不符合]"))          10518
assert has_del, "项目列表缺少删除按钮"                                         10519
print("  %s 删除前有二次确认" % ("[符合]" if "confirm(" in html else          10520
    "[不符合]"))                                                           10521
assert "confirm(" in html                                                  10522
# 图斑列表                                                                 10523
with app.app_context():                                                    10524
    from app.models import MinePatch, Project                              10525
    import seed_demo                                                       10526
    patch = seed_demo.demo_patch()                                         10527
    patch_id = patch.id                                                    10528
html = client.get("/project/%d/patches" %                                  10529
    patch_id).get_data(as_text=True)                                       10530
print("  %s 图斑列表含删除按钮" % ("[符合]" if "/project/patch/" in           10531
    html and "btn--danger" in html else "[不符合]"))                        10532
assert "btn--danger" in html                                               10533
# 微地质单元                                                               10534
html = client.get("/unit/%d" % patch_id).get_data(as_text=Tr10535
    ue)                                                                    10536
print("  %s 微单元列表含删除按钮" % ("[符合]" if "/unit/delete/" in           10537
    html else "[不符合]"))                                                 10538
assert "/unit/delete/" in html                                             10539
print("  %s 微单元编辑为独立页面（表单不再嵌在表格行内）"                          10540
      % ("[符合]" if "/unit/edit/" in html else "[不符合]"))                10541
# 标准                                                                     10542
html = client.get("/standard/").get_data(as_text=True)                     10543
print("  %s 标准列表含删除按钮" % ("[符合]" if "btn--danger" in html         10544
    else "[不符合]"))                                                      10545
assert "btn--danger" in html                                               10546
# 实际执行一次删除：新建项目 → 删除 → 确认消失                                  10547
client.post("/project/add", data={"name": "待删除测试项目",                  10548
    "code": "DEL-1"})                                                      10549
with app.app_context():                                                    10550
```

```
    from app.models import db, Project
    tmp = Project.query.filter_by(name="待删除测试项目").first()
    tmp_id = tmp.id
html = client.get("/project/").get_data(as_text=True)
print("  %s 新建项目成功" % ("[符合]" if "待删除测试项目" in html else
    "[不符合]"))
r = client.post("/project/%d/delete" % tmp_id,
    follow_redirects=True)
html = r.get_data(as_text=True)
gone = "待删除测试项目" not in html
print("  %s 删除项目后列表中不再出现" % ("[符合]" if gone else "[不符合]"))
assert gone
print("  %s 有删除成功提示" % ("[符合]" if "已删除" in html else
    "[不符合]"))
print()
print(LINE)
print("界面修复自检全部通过 √")
print(LINE)
# ==== 测试/test_new_features.py ====
# -*- coding: utf-8 -*-
"""补齐模块自检：操作日志与用户维护、遥感影像、指标与多专家矩阵、
批量导入、图表与 PDF 导出。
对应计划书功能清单与系统实现的差异核对结论。
"""
import io
import json
import os
import re
import sys
sys.stdout.reconfigure(encoding="utf-8")
import fitz
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__
    e__))))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
from app.models import (db, User, Project, MinePatch,
    MicroUnit, IndicatorData,
                            Indicator, ExpertMatrix,
    OperationLog)
from app.utils.schema import ensure_schema
from app.utils import browser, charts
LINE = "=" * 72
BASE = ROOT
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" +
    os.path.join(BASE, "app", "app.db")
```

```
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
    UPLOAD_FOLDER = os.path.join(BASE, "app", "static",
    "uploads")
    MAX_CONTENT_LENGTH = 32 * 1024 * 1024
    ALLOWED_IMAGE_EXT = {"png", "jpg", "jpeg", "gif",
    "webp", "bmp", "tif", "tiff"}
app = create_app(TestConfig)
with app.app_context():
    db.create_all()
    ensure_schema(db)
    ExpertMatrix.query.delete()
    db.session.commit()
    # 固定取内置演示案例的图斑，避免受库中用户自建项目 / 验证算例的影响
    import seed_demo
    patch_id = seed_demo.demo_patch().id
c = app.test_client()
c.post("/login", data={"username": "admin", "password":
    "admin123"})
def ok(flag, label, extra=""):
    print("  %s %-44s %s" % ("[符合]" if flag else "[不符合]",
    label, extra))
    assert flag, label
# ------------------------------------------------------------------
    ------------------
print(LINE)
print("操作日志与用户维护")
print(LINE)
before = None
with app.app_context():
    before = OperationLog.query.count()
c.post("/project/add", data={"name": "日志核对项目", "code":
    "LOG-T"})
with app.app_context():
    after = OperationLog.query.count()
    last = OperationLog.query.order_by(OperationLog.id.desc(
    )).first()
ok(after == before + 1, "写操作被写入操作日志", "%d -> %d" % (before,
    after))
ok(last.action == "新建项目", "日志记录了可读动作名", last.action)
ok("project" in (last.target_type or ""), "日志记录了对象类型",
    last.target_type)
r = c.get("/logs")
body = r.get_data(as_text=True)
ok(r.status_code == 200 and "操作日志" in body, "操作日志查看页可访问")
ok("新建项目" in body, "日志页显示操作记录")
# 用户编辑 / 删除
with app.app_context():
    u = User(username="tmp_user", role="viewer",
```

```
                password_hash=User.query.first().password_hash)
    db.session.add(u); db.session.commit()
    uid = u.id
r = c.get("/users/%d/edit" % uid)
ok(r.status_code == 200, "用户编辑页可访问")
c.post("/users/%d/edit" % uid, data={"role": "evaluator",
    "active": "1"})
with app.app_context():
    ok(db.session.get(User, uid).role == "evaluator",
    "角色已修改并保存")
# 最后一个管理员保护
with app.app_context():
    admin_id = User.query.filter_by(role="admin").first().id
c.post("/users/%d/edit" % admin_id, data={"role": "viewer",
    "active": "1"})
with app.app_context():
    ok(db.session.get(User, admin_id).role == "admin",
    "拒绝把最后一个管理员降级")
c.post("/users/%d/delete" % uid, follow_redirects=True)
with app.app_context():
    ok(db.session.get(User, uid) is None, "用户删除生效")
# 清理日志核对用的项目
with app.app_context():
    for p in Project.query.filter_by(name="日志核对项目").all():
        db.session.delete(p)
    db.session.commit()
# ------------------------------------------------------------
    ------------------
print()
print(LINE)
print("遥感影像挂接")
print(LINE)
doc = fitz.open()
pg = doc.new_page(width=320, height=200)
pg.insert_text((20, 60), "RS image", fontname="china-s",
    fontsize=16)
png = doc[0].get_pixmap(dpi=100).tobytes("png")
doc.close()
r = c.post("/project/patch/%d/image" % patch_id,
           data={"image": (io.BytesIO(png), "影像.png"),
                 "image_note": "2025年6月无人机正射影像"},
           content_type="multipart/form-data",
    follow_redirects=True)
body = r.get_data(as_text=True)
ok("已挂接" in body and "<img" in body, "影像上传并显示")
with app.app_context():
    rel = db.session.get(MinePatch, patch_id).image_path
ok(bool(rel), "数据库记录影像路径", rel)
abs_path = os.path.join(TestConfig.UPLOAD_FOLDER,
    rel.replace("uploads/", "", 1))
```

```
ok(os.path.isfile(abs_path), "影像文件落盘")
ok(c.get("/static/" + rel).status_code == 200, "影像可经 URL
    访问")
r = c.post("/project/patch/%d/image" % patch_id,
           data={"image": (io.BytesIO(b"x"), "bad.txt")},
           content_type="multipart/form-data",
    follow_redirects=True)
ok("不支持" in r.get_data(as_text=True), "拒绝非图片格式")
c.post("/project/patch/%d/image/delete" % patch_id,
    follow_redirects=True)
with app.app_context():
    ok(db.session.get(MinePatch, patch_id).image_path is
    None, "影像移除后字段清空")
ok(not os.path.isfile(abs_path), "影像文件已删除")
# ------------------------------------------------------------
    ----------------
print()
print(LINE)
print("指标定义维护与多专家矩阵")
print(LINE)
r = c.get("/ahp/indicators")
body = r.get_data(as_text=True)
with app.app_context():
    n_ind = Indicator.query.count()
ok(r.status_code == 200 and n_ind > 0, "指标定义已入库并可查看", "%d
    项" % n_ind)
ok(all(k in body for k in ("一级指标", "二级指标", "三级指标")),
   "指标树按层级展示")
ok("岩质边坡" in body and "平台" in body, "显示各指标启用的微地质单元")
c.post("/ahp/indicators/coverage/edit",
       data={"name": "植被覆盖度", "unit": "%", "ideal_type":
    "benefit",
             "ideal_value": "75", "enabled_units": ["岩质边坡"],
             "description": "自检修改"})
with app.app_context():
    ok(Indicator.query.filter_by(code="coverage").first().de
    scription == "自检修改",
       "指标定义可编辑")
c.post("/ahp/indicators/reset")
with app.app_context():
    ok("自检修改" not in (Indicator.query.filter_by(code="covera
    ge").first().description or ""),
       "可按论文取值重置")
base = {"m_0_1": "1.4194", "m_0_2": "1.76", "m_1_2": "1.24"}
for name in ("张工", "李工", "王工"):
    d = dict(base); d.update({"action": "add", "unit_type":
    "碎石边坡",
                              "level": "1", "expert_name":
    name})
    c.post("/ahp/experts", data=d)
```

```
with app.app_context():
    ok(ExpertMatrix.query.filter_by(unit_type="碎石边坡",
    level=1).count() == 3,
       "录入 3 位专家的判断矩阵")
r = c.get("/ahp/experts/aggregate?unit_type=碎石边坡&level=1")
body = r.get_data(as_text=True)
ok("汇总后权重" in body and "各专家矩阵的一致性" in body,
   "多专家矩阵几何平均汇总可用")
ws = re.search(r"<th scope=\"row\">权重</th>\s*(.*?)</tr>",
    body, re.S)
vals = re.findall(r">([0-9.]+)</td>", ws.group(1)) if ws
    else []
ok(len(vals) == 3 and abs(sum(map(float, vals)) - 1.0) <
    0.01,
   "汇总权重归一", "、".join(vals))
ok(abs(float(vals[0]) - 0.44) < 0.02, "汇总权重接近论文表 3 取值",
    vals[0] if vals else "")
# ------------------------------------------------------------
    ----------------
print()
print(LINE)
print("批量导入")
print(LINE)
r = c.get("/indicator/import/template")
tpl = r.data.decode("utf-8-sig")
ok(r.status_code == 200 and "单元类型" in tpl, "导入模板可下载")
ok("土石比" in tpl and "植被覆盖度" in tpl, "模板包含指标列")
csv_text = (
    "单元类型,土石比,土层厚度,pH,有机质,Hg,As,Ni,Cu,Cd,Pb,植被覆盖度,Shannon,坡度
    ,坡向\n"
    "碎石边坡,0.34,10,9.24,4.89,0.05,5.17,12.60,11.70,0.04,6.50,
    87.92,0.60,28.75,阳坡\n"
    "平台,0.56,30,8.72,10.92,0.02,3.13,12.33,11.14,0.09,9.98,5
    8.42,0.72,,,\n"
    "不存在单元,1,1,7,1,,,,,,,,,\n"
)
r = c.post("/indicator/import",
           data={"patch_id": str(patch_id),
                 "datafile": (io.BytesIO(csv_text.encode("ut
    f-8")), "d.csv")},
           content_type="multipart/form-data",
    follow_redirects=True)
body = r.get_data(as_text=True)
ok("导入结果" in body, "导入结果页展示")
ok("不存在单元" in body, "错误行被报告")
with app.app_context():
    u = MicroUnit.query.filter_by(patch_id=patch_id,
    unit_type="碎石边坡").first()
    ind = IndicatorData.query.filter_by(unit_id=u.id).first(
    )
```

```
    ok(abs(ind.soil_ratio - 0.34) < 1e-6 and
    abs(ind.coverage - 87.92) < 1e-6,
        "导入数据已写入数据库")
r = c.post("/indicator/import",
           data={"patch_id": str(patch_id),
                 "datafile": (io.BytesIO(b"x"), "d.pdf")},
           content_type="multipart/form-data",
    follow_redirects=True)
ok("仅支持" in r.get_data(as_text=True), "拒绝不支持的导入格式")
# ------------------------------------------------------------------
    ------------------
print()
print(LINE)
print("图表与 PDF 导出")
print(LINE)
svg = charts.hbars([("岩质边坡", 0.54, "0.5401"), ("平台", 0.84,
    "0.8376")])
ok("<svg" in svg and "0.8376" in svg, "条形图生成且带数值标签")
svg2 = charts.donut([("植被指标", 0.6, "0.6000"), ("土壤指标", 0.4,
    "0.4000")])
ok("<svg" in svg2 and "占比" in svg2, "环形图生成且带图例")
# 报告是评价结论的呈现，需先真实执行一次评价才能打开
with app.app_context():
    from app.views.evaluate import load_patch_input,
    save_evaluation
    save_evaluation(patch_id, load_patch_input(patch_id))
r = c.get("/report/%d" % patch_id)
body = r.get_data(as_text=True)
ok(body.count("<svg") >= 2, "得分明细页含图表", "%d 个 SVG" %
    body.count("<svg"))
exe = browser.browser_path()
if exe is None:
    print("  [跳过] 未找到浏览器，PDF 导出无法验证")
else:
    r = c.get("/report/%d/pdf" % patch_id)
    data = r.data
    ok(data[:4] == b"%PDF", "PDF 报告导出成功", "%d 字节" %
    len(data))
    d = fitz.open(stream=data, filetype="pdf")
    txt = "".join(d[i].get_text() for i in
    range(d.page_count))
    ok("自然恢复适宜性评价报告" in txt, "PDF 含报告标题")
    ok("评价结论" in txt and "得分明细" in txt, "PDF 含结论与明细章节")
    ok(d.page_count >= 2, "PDF 页数合理", "%d 页" % d.page_count)
    d.close()
print()
print(LINE)
print("补齐模块自检全部通过 √")
print(LINE)
# ==== 测试/test_workflow.py ====
```

```
#  -*- coding: utf-8 -*-
"""流程门禁自检：前置步骤未完成时，后续模块必须拦得住。
论文 4.1 节确定的评价流程存在先后关系，缺任何一环都不应产出评价结论。
本测试逐个构造「缺前置」的场景，验证系统不会放行。
"""
import os
import sys
sys.stdout.reconfigure(encoding="utf-8")
#  本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
from app.models import (db, Project, MinePatch, MicroUnit,
    IndicatorData,
                        ConstraintCheck, EvaluationResult)
from app.services import workflow as wf
LINE = "=" * 72
BASE = ROOT
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" +
    os.path.join(BASE, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
c = app.test_client()
c.post("/login", data={"username": "admin", "password":
    "admin123"})
_fails = []
def ok(flag, label, extra=""):
    print("  %s %-46s %s" % ("[符合]" if flag else "[不符合]",
    label, extra))
    if not flag:
        _fails.append(label)
def new_patch(tag):
    with app.app_context():
        p = Project(name="门禁-" + tag, code="WF-" + tag)
        db.session.add(p); db.session.flush()
        patch = MinePatch(project_id=p.id, name="门禁图斑-" +
    tag)
        db.session.add(patch); db.session.flush()
        pid = patch.id
        db.session.commit()
    return pid
def add_units(pid):
    with app.app_context():
```

```
        db.session.add(MicroUnit(patch_id=pid,
    unit_type="碎石边坡",
                                 area=1.0, ratio=1.0))
        db.session.commit()
def add_constraint(pid, passed=True):
    with app.app_context():
        db.session.add(ConstraintCheck(
            patch_id=pid, policy_pass=True,
    location_pass=passed,
            pollution_pass=True, geo_pass=True,
    farmland_pass=True,
            passed=passed, reason="全部通过" if passed else
    "未通过：区位约束"))
        db.session.commit()
def add_data(pid):
    with app.app_context():
        u = MicroUnit.query.filter_by(patch_id=pid).first()
        db.session.add(IndicatorData(unit_id=u.id,
    soil_ratio=0.34, thickness=10,
                                     ph=9.24,
    organic_matter=4.89, coverage=87.92,
                                     shannon=0.60,
    slope=28.75, aspect="阳坡"))
        db.session.commit()
def cleanup(pid, tag):
    with app.app_context():
        from app.models import ScoreDetail
        # 先删得分明细，再删评价记录，避免留下孤儿明细
        rids = [r.id for r in EvaluationResult.query.filter_
    by(patch_id=pid)]
        if rids:
            ScoreDetail.query.filter(ScoreDetail.result_id.i
    n_(rids)).delete(
                synchronize_session=False)
        EvaluationResult.query.filter_by(patch_id=pid).delet
    e()
        ConstraintCheck.query.filter_by(patch_id=pid).delete
    ()
        uids = [u.id for u in MicroUnit.query.filter_by(patc
    h_id=pid)]
        if uids:
            IndicatorData.query.filter(IndicatorData.unit_id
    .in_(uids)).delete(
                synchronize_session=False)
        MicroUnit.query.filter_by(patch_id=pid).delete()
        m = db.session.get(MinePatch, pid)
        if m:
            db.session.delete(m)
        pr = Project.query.filter_by(name="门禁-" +
    tag).first()
```

```
            if pr:
                db.session.delete(pr)
            db.session.commit()
def results_of(pid):
    with app.app_context():
        return EvaluationResult.query.filter_by(patch_id=pid
    ).count()
def post_eval(pid):
    return c.post("/evaluate/%d" % pid, data={},
    follow_redirects=False)
def get_report(pid):
    return c.get("/report/%d" % pid, follow_redirects=False)
# ----------------------------------------------------------------
    ------------------
print(LINE)
print("一、前置条件缺失时，评价必须被拦住")
print(LINE)
print()
print("场景 1：连微地质单元都没划分")
pid = new_patch("1")
r = post_eval(pid)
n = results_of(pid)
ok(r.status_code == 302 and n == 0, "未划分单元 → 不产生评价结果",
   "HTTP %d，结果数 %d" % (r.status_code, n))
with app.app_context():
    pr = wf.patch_progress(pid)
steps = [b["step"] for b in pr["blocking"]]
ok(not pr["ready"] and "units" in steps,
   "拦原因包含「划分微地质单元」", "、".join(steps))
ok(steps == ["constraint", "units"],
   "拦截顺序符合论文 4.1 节：限制性判定在前，划分单元在后",
   "、".join(steps))
cleanup(pid, "1")
print()
print("场景 2：有单元，但未做限制性条件判定")
pid = new_patch("2"); add_units(pid); add_data(pid)
r = post_eval(pid)
n = results_of(pid)
ok(r.status_code == 302 and n == 0, "未做限制性判定 → 不产生评价结果",
   "结果数 %d" % n)
with app.app_context():
    pr = wf.patch_progress(pid)
ok(any(b["step"] == "constraint" for b in pr["blocking"]),
   "拦原因指向「执行限制性条件判定」")
cleanup(pid, "2")
print()
print("场景 3：限制性条件判定【未通过】")
pid = new_patch("3"); add_units(pid); add_data(pid);
    add_constraint(pid, passed=False)
r = post_eval(pid)
```

```
    n = results_of(pid)                                                    11001
    ok(r.status_code == 302 and n == 0, "判定未通过 → 不进入定量评价",        11002
       "结果数 %d" % n)                                                    11003
    with app.app_context():                                                11004
        pr = wf.patch_progress(pid)                                        11005
    ok(not pr["constraint"]["passed"], "流程状态显示判定未通过")             11006
    blocked = [b for b in pr["blocking"] if b["step"] ==                   11007
        "constraint"]                                                      11008
    # 提示只写「缺什么」：label 说明未通过，reason 直接引用判定记录里没过的项   11009
    ok(bool(blocked) and "未通过" in blocked[0]["label"],                  11010
       "拦提示指明限制性条件判定未通过", blocked[0]["label"] if blocked else  11011
        "")                                                                11012
    ok(bool(blocked) and "区位约束" in (blocked[0]["reason"] or ""),       11013
       "并把具体没通过的条件带出来", blocked[0]["reason"] if blocked else "")  11014
    cleanup(pid, "3")                                                      11015
    print()                                                                11016
    print("场景 4：判定通过，但指标数据全空")                               11017
    pid = new_patch("4"); add_units(pid); add_constraint(pid,              11018
        passed=True)                                                       11019
    r = post_eval(pid)                                                     11020
    n = results_of(pid)                                                    11021
    ok(r.status_code == 302 and n == 0, "数据未录入 → 不产生评价结果",        11022
       "结果数 %d" % n)                                                    11023
    with app.app_context():                                                11024
        pr = wf.patch_progress(pid)                                        11025
    ok(any(b["step"] == "data" for b in pr["blocking"]),                   11026
       "拦原因指向「录入指标数据」")                                        11027
    ok(pr["data"]["missing"] == ["碎石边坡"],                              11028
       "指出具体缺哪个单元的数据", "、".join(pr["data"]["missing"]))        11029
    cleanup(pid, "4")                                                      11030
    print()                                                                11031
    print("场景 5：单元面积占比合计不为 100%")                              11032
    pid = new_patch("5"); add_constraint(pid, passed=True)                 11033
    with app.app_context():                                                11034
        u = MicroUnit(patch_id=pid, unit_type="碎石边坡", area=1.0,        11035
        ratio=0.5)                                                         11036
        db.session.add(u); db.session.commit()                             11037
        db.session.add(IndicatorData(unit_id=u.id,                         11038
        soil_ratio=0.3, thickness=10,                                      11039
                                              ph=8.0,                      11040
        organic_matter=5.0, coverage=80.0,                                 11041
                                              shannon=0.6, slope=20.0,     11042
        aspect="阳坡"))                                                    11043
        db.session.commit()                                                11044
    r = post_eval(pid)                                                     11045
    ok(results_of(pid) == 0, "占比不足 100% → 不产生评价结果")              11046
    cleanup(pid, "5")                                                      11047
    print()                                                                11048
    print(LINE)                                                            11049
    print("二、前置条件齐备时才放行")                                       11050
```

```
print(LINE)                                                                  11051
pid = new_patch("OK"); add_units(pid); add_constraint(pid,                   11052
      passed=True); add_data(pid)                                            11053
with app.app_context():                                                      11054
      pr = wf.patch_progress(pid)                                            11055
ok(pr["ready"], "单元 + 判定通过 + 数据齐备 → 判定为可执行")                  11056
r = post_eval(pid)                                                           11057
n = results_of(pid)                                                          11058
ok(n == 1, "此时执行评价成功产生结果", "结果数 %d" % n)                      11059
print("   重定向到:", r.headers.get("Location"))                             11060
print()                                                                      11061
print(LINE)                                                                  11062
print("三、报告不得先于评价产生")                                           11063
print(LINE)                                                                  11064
pid2 = new_patch("R")     # 什么都不做                                       11065
r = get_report(pid2)                                                         11066
ok(r.status_code == 302 and "/report/" ==                                    11067
      (r.headers.get("Location") or ""),                                     11068
    "前置缺失时 /report/<id> 被退回列表", "HTTP %d" % r.status_code)         11069
for suffix in ("/pdf", "/word"):                                             11070
      rr = c.get("/report/%d%s" % (pid2, suffix),                            11071
      follow_redirects=False)                                                11072
      ok(rr.status_code == 302, "导出 %s 同样被拦住" % suffix, "HTTP11073
      %d" % rr.status_code)                                                  11074
cleanup(pid2, "R")                                                           11075
cleanup(pid, "OK")                                                           11076
print()                                                                      11077
print(LINE)                                                                  11078
print("四、前置齐备但从未执行评价时，报告仍不得产生")                       11079
print(LINE)                                                                  11080
# 报告是评价结论的呈现：即使前置步骤都做完了，只要没真正执行过定量评价，    11081
# 也不应凭空推算出结论。                                                     11082
pid3 = new_patch("N")                                                        11083
add_units(pid3); add_constraint(pid3, passed=True);                          11084
      add_data(pid3)                                                         11085
with app.app_context():                                                      11086
      pr = wf.report_progress(pid3)                                          11087
ok(not pr["ready"], "无评价记录 → 报告流程判定为未就绪")                     11088
ok(bool(pr["blocking"]) and pr["blocking"][0]["step"] ==                     11089
      "evaluate",                                                            11090
    "拦原因指向「执行适宜性评价」")                                          11091
r = get_report(pid3)                                                         11092
ok(r.status_code == 302 and results_of(pid3) == 0,                           11093
    "直接访问 /report/<id> 被退回，且不产生任何评价结果",                    11094
    "HTTP %d" % r.status_code)                                               11095
# 真正执行评价之后，报告才放行，且取值与评价记录一致                         11096
r = post_eval(pid3)                                                          11097
ok(results_of(pid3) == 1, "先执行评价 → 产生评价记录")                       11098
with app.app_context():                                                      11099
      pr = wf.report_progress(pid3)                                          11100
```

```
    rec = pr["result"]                                                  11101
ok(pr["ready"] and rec is not None, "有评价记录后，报告流程就绪")          11102
rr = get_report(pid3)                                                   11103
body = rr.get_data(as_text=True)                                        11104
ok(rr.status_code == 200, "此时可打开得分明细页", "HTTP %d" %           11105
    rr.status_code)                                                     11106
ok(("%.4f" % rec.final_score) in body,                                  11107
   "报告数值取自评价记录，而非重新计算",                                11108
   "F=%.4f" % rec.final_score)                                          11109
cleanup(pid3, "N")                                                      11110
print()                                                                 11111
print(LINE)                                                             11112
print("五、内置演示数据的清理不得波及用户自建数据")                      11113
print(LINE)                                                             11114
# 回归保护：seed_demo.clean_case() 曾执行 Project.query.delete()，      11115
# 导致用户新建的项目在每次启动时被连带删除。                            11116
import seed_demo as sd                                                  11117
with app.app_context():                                                 11118
    keep_p = Project(name="用户自建项目-守护", code="KEEP-001")          11119
    db.session.add(keep_p); db.session.flush()                          11120
    keep_patch = MinePatch(project_id=keep_p.id,                        11121
    name="用户自建图斑-守护")                                           11122
    db.session.add(keep_patch); db.session.flush()                      11123
    keep_pid = keep_p.id                                                11124
    keep_patch_id = keep_patch.id                                       11125
    db.session.commit()                                                 11126
with app.app_context():                                                 11127
    sd.clean_case(reset=False)                                          11128
with app.app_context():                                                 11129
    still_p = db.session.get(Project, keep_pid)                         11130
    still_patch = db.session.get(MinePatch, keep_patch_id)              11131
ok(still_p is not None, "清理演示数据后，用户自建项目仍存在")            11132
ok(still_patch is not None, "清理演示数据后，用户自建图斑仍存在")        11133
# 演示数据清掉后应能重新写入，保证演示环境可恢复                        11134
sd.seed()                                                               11135
with app.app_context():                                                 11136
    demo_back = Project.query.filter_by(code=sd.DEMO_CODE).f            11137
    irst()                                                              11138
ok(demo_back is not None, "演示案例可重新写入（幂等恢复）")              11139
with app.app_context():                                                 11140
    m = db.session.get(MinePatch, keep_patch_id)                        11141
    if m:                                                               11142
        db.session.delete(m)                                            11143
    p = db.session.get(Project, keep_pid)                               11144
    if p:                                                               11145
        db.session.delete(p)                                            11146
    db.session.commit()                                                 11147
print()                                                                 11148
print(LINE)                                                             11149
print("六、同一图斑不得重复划分同类型微地质单元")                        11150
```

```
print(LINE)
#  论文 4.1 节把图斑分为平台、碎石边坡、岩质边坡三类，每类至多一个。
#  允许重复会让面积占比、按类型组织的指标数据与批量导入对不上号。
pid4 = new_patch("D")
c.post("/unit/%d/add" % pid4, data={"unit_type": "平台",
      "area": "1.0"},
          follow_redirects=True)
with app.app_context():
      n1 = MicroUnit.query.filter_by(patch_id=pid4).count()
ok(n1 == 1, "首次添加「平台」成功", "单元数 %d" % n1)
r = c.post("/unit/%d/add" % pid4, data={"unit_type": "平台",
      "area": "2.0"},
                follow_redirects=True)
with app.app_context():
      n2 = MicroUnit.query.filter_by(patch_id=pid4).count()
ok(n2 == 1, "重复添加同类型被拒绝", "单元数 %d" % n2)
ok("同一类型只能划分一个" in r.get_data(as_text=True), "给出重复提示")
r = c.post("/unit/%d/add" % pid4, data={"unit_type": "火山口",
      "area": "1.0"},
                follow_redirects=True)
with app.app_context():
      n3 = MicroUnit.query.filter_by(patch_id=pid4).count()
ok(n3 == 1, "非法单元类型被拒绝", "单元数 %d" % n3)
c.post("/unit/%d/add" % pid4, data={"unit_type": "碎石边坡",
      "area": "3.0"},
          follow_redirects=True)
with app.app_context():
      uid = MicroUnit.query.filter_by(patch_id=pid4,
                                      unit_type="碎石边坡").first()
      ).id
r = c.post("/unit/edit/%d" % uid,
                data={"unit_type": "平台", "area": "3.0"},
      follow_redirects=True)
with app.app_context():
      types = sorted(u.unit_type for u in
                     MicroUnit.query.filter_by(patch_id=pid4)
      all())
ok(types == ["平台", "碎石边坡"], "改类型时同样拦截重复", "、".join(types))
cleanup(pid4, "D")
print()
print(LINE)
if _fails:
      print("存在未通过项：")
      for f in _fails:
           print("  -", f)
      sys.exit(1)
print("流程门禁自检全部通过 √")
print(LINE)
#  ==== 测试/test_weights.py ====
#  -*- coding: utf-8 -*-
```

```
"""权重生效链路自检：AHP 算出的权重必须真正参与评价计算。
此前 weight_schemes / weight_items 两张表从未被写入，评价引擎也只用
代码里的论文表 3 默认权重，相当于权重模块算完即弃。本测试守住这条链路：
    默认 → 论文表 3
    保存 AHP 权重 → 评价改用新权重，结果随之变化
    恢复默认 → 结果回到论文取值
"""
import os
import sys
sys.stdout.reconfigure(encoding="utf-8")
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
from app.models import (db, Project, MinePatch, MicroUnit,
    IndicatorData,
                        ConstraintCheck, EvaluationResult,
    WeightScheme,
                        WeightItem)
from app.services import workflow as wf
from app.services import weight_service as ws
LINE = "=" * 72
BASE = ROOT
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" +
    os.path.join(BASE, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
c = app.test_client()
c.post("/login", data={"username": "admin", "password":
    "admin123"})
_fails = []
def ok(flag, label, extra=""):
    print("  %s %-46s %s" % ("[符合]" if flag else "[不符合]",
    label, extra))
    if not flag:
        _fails.append(label)
TAG = "W"
UNIT_TYPE = "碎石边坡"
def build_case():
    """建一个只含单一微地质单元的图斑，便于观察权重变化对得分的影响。"""
    with app.app_context():
        p = Project(name="权重测试-%s" % TAG, code="WEIGHT-%s"
    % TAG)
```

```
        db.session.add(p); db.session.flush()
        patch = MinePatch(project_id=p.id, name="权重测试图斑-%s"
    % TAG)
        db.session.add(patch); db.session.flush()
        db.session.add(MicroUnit(patch_id=patch.id,
    unit_type=UNIT_TYPE,
                                 area=1.0, ratio=1.0))
        db.session.flush()
        u = MicroUnit.query.filter_by(patch_id=patch.id).fir
    st()
        db.session.add(IndicatorData(
            unit_id=u.id, soil_ratio=0.34, thickness=10,
    ph=9.24,
            organic_matter=4.89, coverage=87.92,
    shannon=0.60,
            slope=28.75, aspect="阳坡"))
        db.session.add(ConstraintCheck(
            patch_id=patch.id, policy_pass=True,
    location_pass=True,
            pollution_pass=True, geo_pass=True,
    farmland_pass=True,
            passed=True, reason="全部通过"))
        db.session.commit()
        return patch.id
def cleanup(pid):
    with app.app_context():
        from app.models import ScoreDetail
        rids = [r.id for r in EvaluationResult.query.filter_
    by(patch_id=pid)]
        if rids:
            ScoreDetail.query.filter(ScoreDetail.result_id.i
    n_(rids)).delete(
                synchronize_session=False)
        EvaluationResult.query.filter_by(patch_id=pid).delet
    e()
        ConstraintCheck.query.filter_by(patch_id=pid).delete
    ()
        uids = [u.id for u in MicroUnit.query.filter_by(pat
    h_id=pid)]
        if uids:
            IndicatorData.query.filter(IndicatorData.unit_id
    .in_(uids)).delete(
                synchronize_session=False)
        MicroUnit.query.filter_by(patch_id=pid).delete()
        m = db.session.get(MinePatch, pid)
        if m:
            db.session.delete(m)
        p = Project.query.filter_by(code="WEIGHT-%s" %
    TAG).first()
        if p:
```

```
                db.session.delete(p)                                    11301
            db.session.commit()                                         11302
def run_eval(pid):                                                      11303
    c.post("/evaluate/%d" % pid, data={},                               11304
    follow_redirects=False)                                             11305
    with app.app_context():                                             11306
        return EvaluationResult.query.filter_by(patch_id=pid11307
    ) \                                                                 11308
            .order_by(EvaluationResult.id.desc()).first().fi11309
    nal_score                                                           11310
print(LINE)                                                             11311
print("一、默认应使用论文表 3 的权重")                                  11312
print(LINE)                                                             11313
with app.app_context():                                                 11314
    ws.reset_weights(UNIT_TYPE)                                         11315
    w, label, scheme = ws.effective_weights(UNIT_TYPE)                  11316
ok(scheme is None and "论文表" in label, "未配置时权重来源为论文表 3",  11317
    label)                                                              11318
ok(abs(w["l1"]["土壤指标"] - 0.44) < 1e-9, "一级权重取自论文表 3",      11319
   "土壤指标 %.2f" % w["l1"]["土壤指标"])                               11320
pid = build_case()                                                      11321
base_score = run_eval(pid)                                              11322
print("  默认权重下的评价值 F = %.4f" % base_score)                     11323
print()                                                                 11324
print(LINE)                                                             11325
print("二、保存 AHP 算出的权重后，评价应改用新权重")                    11326
print(LINE)                                                             11327
with app.app_context():                                                 11328
    # 模拟 AHP 计算结果：把土壤指标权重提高、植被与地形降低（仍和为 1）  11329
    ws.save_weights(UNIT_TYPE, [("soil", 0.70), ("veg",                 11330
    0.20),                                                              11331
                                ("terrain", 0.10)])                     11332
    n_items = WeightItem.query.count()                                  11333
ok(n_items >= 3, "weight_items 表已写入权重条目", "%d 条" % n_items)    11334
with app.app_context():                                                 11335
    w2, label2, scheme2 = ws.effective_weights(UNIT_TYPE)               11336
ok(scheme2 is not None and "自定义" in label2, "权重来源变为自定义方案", 11337
    label2)                                                             11338
ok(abs(w2["l1"]["土壤指标"] - 0.70) < 1e-9, "新权重已生效",             11339
   "土壤指标 %.2f" % w2["l1"]["土壤指标"])                              11340
ok(abs(w2["l1"]["植被指标"] - 0.20) < 1e-9, "同批保存的其他项也已生效")  11341
ok(abs(w2["l2"]["土壤指标"]["土石比"] - 0.25) < 1e-9,                   11342
   "未覆盖的二级权重仍保留论文取值",                                    11343
   "土石比 %.2f" % w2["l2"]["土壤指标"]["土石比"])                      11344
new_score = run_eval(pid)                                               11345
print("  新权重下的评价值 F = %.4f" % new_score)                        11346
ok(abs(new_score - base_score) > 1e-4,                                  11347
   "权重变化确实改变了评价结果",                                        11348
   "%.4f → %.4f" % (base_score, new_score))                             11349
print()                                                                 11350
```

```
print(LINE)
print("三、结果应记录所用权重来源")
print(LINE)
import json
with app.app_context():
    rec = EvaluationResult.query.filter_by(patch_id=pid) \
        .order_by(EvaluationResult.id.desc()).first()
    stored = json.loads(rec.unit_scores)
ok(any(u.get("weight_source") for u in stored),
   "评价记录中标注了权重来源",
   stored[0].get("weight_source") or "")
print()
print(LINE)
print("四、恢复默认后应回到论文取值")
print(LINE)
with app.app_context():
    ws.reset_weights(UNIT_TYPE)
    w3, label3, scheme3 = ws.effective_weights(UNIT_TYPE)
ok(scheme3 is None and "论文表" in label3, "已恢复为论文表 3 权重",
   label3)
back_score = run_eval(pid)
print("  恢复后评价值 F = %.4f（默认时 %.4f）" % (back_score,
   base_score))
ok(abs(back_score - base_score) < 1e-9, "评价结果与默认时一致")
print()
print(LINE)
print("五、权重首页概览")
print(LINE)
with app.app_context():
    rows = ws.summary()
ok(len(rows) == 3, "三种微地质单元均有权重来源说明", "%d 行" % len(rows))
r = c.get("/ahp/")
body = r.get_data(as_text=True)
ok(r.status_code == 200 and "当前生效权重" in body, "权重首页正常展示")
print()
print(LINE)
print("六、一致性不通过的矩阵不得保存生效")
print(LINE)
with app.app_context():
    ws.reset_weights(UNIT_TYPE)
n = 3
# 明显矛盾的矩阵：a12=9, a13=1/9, a23=9 —— CR 必然超限
mats = [(0, 1, "9"), (0, 2, "0.1111"), (1, 2, "9")]
data = {"csrf_token": "", "action": "save"}
for i, j, v in mats:
    data["m_%d_%d" % (i, j)] = v
r = c.post("/ahp/matrix?unit_type=%s&level=1" % UNIT_TYPE,
    data=data,
           follow_redirects=True)
with app.app_context():
```

```
        _, label4, scheme4 = ws.effective_weights(UNIT_TYPE)
ok(scheme4 is None, "CR 超限的矩阵未被保存生效", label4)
ok("一致性" in r.get_data(as_text=True), "页面给出了未通过一致性的提示")
cleanup(pid)
print()
print(LINE)
if _fails:
    print("存在未通过项：")
    for f in _fails:
        print("  -", f)
    sys.exit(1)
print("权重生效链路自检全部通过 √")
print(LINE)
# ==== app/services/__init__.py ====
# -*- coding: utf-8 -*-
# ==== app/utils/__init__.py ====
# -*- coding: utf-8 -*-
# ==== app/views/__init__.py ====
# -*- coding: utf-8 -*-
```